

FUJITSU 1FINITY L100 USER MANUAL



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1FINITY™ L100

Dynamic ROADM
Functionality

FUJITSU

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Richardson, Texas 75082

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About 1FINITY

Fujitsu's 1FINITY™ platform gives network operators an open, simple, and scalable architecture to meet escalating bandwidth demand. The disaggregated platform delivers flexibility, scalability, and efficiency. Unlike converged systems, the programmable, modular, blade-centric design of 1FINITY allows expansion only as demand requires. Additional features include high rack space usage, evergreen technology design, operational convergence, open pluggable optics, open application programming interfaces, and open protocols.

Disaggregation of the 1FINITY platform gives its product families integrated access, transport, switch, and lambda functions. A software framework provides logical re-aggregation to provide programmable software management and control. The modular blades operate both independently or in combination with other Fujitsu or competitor optical platforms.

The 1FINITY platform uses the YANG data modeling language to model configuration and state data. YANG data models are made up of modules and sub modules that are managed by a command line interface (CLI). A YANG module defines a data model through the hierarchical organization of and constraints on its data. Each Yang module, uniquely identified by a name space URL, provides the data model, remote procedural calls, and notifications for a particular part of the system.

CLI provides the mechanism for the user to enter commands that configure and manage entities. For example, interfaces, access control lists, and maintenance endpoints and perform operations, administrative, and maintenance tasks on the NE. The CLI is hierarchical, which means that it has nested layers of commands.

Note: For more detailed use of CLI with 1FINITY information, refer to *1FINITY CLI User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

Compliance Information

General Industry

FCC

This equipment has been tested and found to comply with the limits for Class A digital devices, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interferences when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio energy and if not installed in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, which the user will be required to correct at his/her expense.

Industry Canada

This digital apparatus does not exceed the Class A limits for noise emissions from digital apparatus set out in the Radio Interference Regulations of the Canadian Department of Communications to ICES-003.

Cet appareil numérique ne dépasse pas les limites d'émission de bruit pour les appareils numériques de Classe A tel que définit dans le document NMB-003 publié par le Département Canadien de Communications.

CE Markings

CE Certification



All 1FINITY systems meet all requirements for CE marking. It is compliant with the following standards:

- 93/68/EEC (CE Marking)
- 2014/30/EU (EMC Directive)
- 2015/863 (RoHS Directive)
- 2014/35/EU (Low Voltage Directive)
- 2012/19/EU (WEEE Directive)
- IEC 60950-1, IEC 60825-1, EN 60950-1, EN 60825-1 and all related requirement
- Laser IEC/EN 60825-2 Standards
- 21 CFR Chapter 1, Sub chapter J, Class 1 Laser Product
- Class A Emission limits for the following standards:
 - FCC Part 15, Class A (U.S.A.)
 - ICES-003, Class A (Canada)
 - CISPR Publication 32 Class A (Europe)
- Product safety regulatory standards:
 - UL 60950-1, 2nd Edition, 2014-10-14 (Information Technology Equipment-Safety—Part 1: General Requirements)
 - CAN/CSA C22.2 No. 60950-1-07, 2nd Edition, 2014-10 (Information Technology Equipment-Safety—Part 1: General Requirements)
 - IEC 60950-1:2005 (Second Edition): Am1:2009 + Am2:2013
 - IEC/EN 60825-1 (Safety of laser products—Part 1: Equipment classification and requirements)
 - IEC/EN 60825-2 (Safety of laser products—Part 2: Optical fibre communication systems [OFCS])

- RCM Mark (formally known as cTick) (Australia)



- NOM Compliance Mark for Mexico



Compliance Information

L100/L110-Specific

L100 UL Labels

NC0001281 Rev 01

L110 UL Labels

NC001282 Rev 01

UL Fuse Ratings

Model/Type Reference	Rating
L100 (BDL1-3R11)	−48 V DC, 7.5 A MAX
L110 (BDL1-3B11)	−48 V DC, 6 A MAX

General Compliance

- NEBS Level 3:
 - GR-63-CORE Issue 04 (including Seismic Zone 4 requirements)
 - GR-1089-Electromagnetic Compatibility (EMC) and Electrical Safety
 - GR-3108-Class 1 - Equipment in the Outside Plant (OSP)
 - GR-1089, Section 9 (Bonding and Grounding) required installation standards:
 - Installed in a Common Bonding Network (CBN)
 - Installed in NEBS Level 3 environments, for example, Central Offices as well as Customer Premise Environment (CPE)
 - Installed in accordance with a DC-C or DC-I configuration



Warning: All metallic signaling cables should be shielded with the shield grounded at both ends to meet intra-building lightning requirements.

Warning: The intra-building port(s) (USER1, USER2, LMP, UTLP, and MAINT) of the equipment or sub assembly is suitable for connection to intra-building or unexposed wiring or cabling only. The intra-building port(s) of the equipment

or assembly **MUST NOT** be metallically connected to interfaces that connect to the OSP or its wiring. These interfaces are designed for use as intra-building interfaces only (Type 2 or Type 4 ports) as described in GR-1089 and require isolation from the exposed ODP cabling. The addition of Primary Protectors is not sufficient protection in order to connect these interfaces metallically to OSP wiring.

- The 1FINITY L100 and L110 blades and the PIL1-3PS2 PIU are compliant with the Laser IEC/EN 60825-1, 2 standards.
- The 1FINITY L100 and L110 blades and the PIL1-3PS2 PIU comply with 21 CFR Chapter 1, sub-chapter J, Class 1 Laser Product.
- The 1FINITY L100 and L110 blades and the PIL1-3PS2 PIU are compliant with the Class A Emission limits of the following standards:
 - FCC Part 15, Class A
 - ICES-003
 - CISPR Publication 32 Class A (Europe)
- The 1FINITY L100 Series blades are approved for the following product safety regulatory standards:
 - UL 60950-1, 2nd Edition, 2014-10-14 (Information Technology Equipment-Safety—Part 1: General Requirements)
 - CAN/CSA C22.2 No. 60950-1-07, 2nd Edition, 2014-10 (Information Technology Equipment-Safety—Part 1: General Requirements)

- IEC 60950-1:2005 (Second Edition); Am1:2009 + Am2:2013
- UL 62368-1 2nd Edition: Audio/video, information and communication technology equipment - Part 1: Safety requirements
- CAN/CSA C22.2 NO. 62368-1-14 Audio/video, information and communication technology equipment - Part 1: Safety requirements
- IEC 62368-1 2nd Edition: Audio/video, information and communication technology equipment - Part 1: Safety requirements
- IEC/EN 60825-1 (Safety of laser products—Part 1: Equipment classification and requirements)
- IEC/EN 60825-2 (Safety of laser products—Part 2: Optical fibre communication systems [OFCS])

- The PoE application and cabling is suitable for indoor use only, and is not suitable for outside applications.
- The following symbol identifies any terminal intended for connection to an external conductor for protection against electrical shock in case of a fault, or the terminal of a protective earth (ground) electrode.

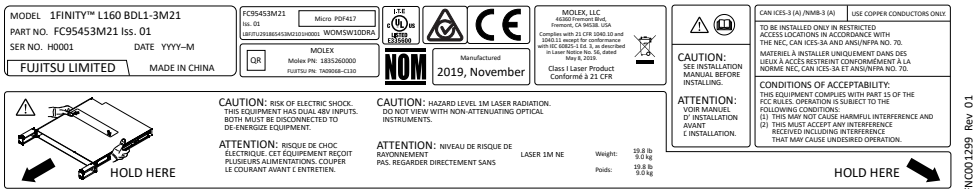


Figure 1
Protective Earth (Ground) Symbol

Compliance Information

L160-Specific

L160 UL Labels



UL Fuse Ratings

Model/Type Reference	Rating
L160 (BDL1-3M21)	–48 V DC

General Compliance

- NEBS Level 3:
 - GR-63-CORE Issue 05 (including Seismic Zone 4 requirements)
 - GR-1089-Electromagnetic Compatibility (EMC) and Electrical Safety
 - GR-3108-Class 1-Equipment in the Outside Plant (OSP)
- GR-1089, Section 9 (Bonding and Grounding) required installation standards:
 - Installed in a Common Bonding Network (CBN)
 - Installed in NEBS Level 3 environments, for example, Central Offices as well as Customer Premise Environment (CPE)
 - Installed in accordance with a DC-C or DC-I configuration
- GR-1089, R4-16, R4-18 (Intra-Building Lightning Protection)
 - **Warning:** All metallic signaling cables should be shielded with the shield grounded at both ends to meet intra-building lightning requirements.
 - **Warning:** The intra-building port(s) (UTLP1, UTLP2, and MAINT) of the equipment or subassembly is suitable for connection to intra-building or unexposed wiring or cabling only. The intra-building port(s) of the equipment or assembly MUST NOT be metallically connected to interfaces that connect to the OSP or its wiring. These interfaces are designed for use as intra-building interfaces only (Type 2 or Type 4 ports) as described in GR-1089 and required isolation from the exposed ODP cabling. The addition of Primary Protectors is not sufficient protection to connect these interfaces metallically to OSP wiring.
- UL/CSA 62368-1
 - The 1FINITY L160 Blade is compliant with the Laser IEC/EN 60825-1, 2 standards.
 - The 1FINITY L160 Blade is compliant with 21 CFR 1040.10 and 1040.11 except for conformance with IEC 60825-1 Ed. 3., as described in Laser Notice No. 56, dated May 8, 2019., Class 1 Laser Product.
 - The 1FINITY L160 Blade is compliant with the Class A Emission limits of the following standards:
 - FCC Part 15, Class A
 - ICES-003
 - CISPR Publication 32 Class A (Europe)
 - The 1FINITY L160 Blade is approved for the following product safety regulatory standards:
 - CAN/CSA-C22.2 No. 62368-1/UL 62368-1, 2nd Edition, 2014-12-01 (Audio/video, information and communication technology equipment—Part 1: Safety requirements)
 - IEC 60950-1:2005 (Second Edition); Am1:2009 + Am2:2013
 - IEC 62368-1:2014 (Modified) (Audio/video, information and communication technology equipment—Part 1: Safety requirements)

- EN 62368-1:2014 (Audio/video, information and communication technology equipment—Part 1: Safety requirements (IEC 62368-1:2014, modified))
- IEC/EN 60825-1 (Safety of laser products—Part 1: Equipment classification and requirements)
- IEC/EN 60825-2 (Safety of laser products—Part 2: Optical fibre communication systems [OFCS])

- The following symbol identifies the terminal intended for connection to an external conductor for protection against electrical shock in case of a fault, or the terminal of a protective earth (ground) electrode.



Figure 2
Protective Earth (Ground) Symbol

Laser Safety Classification and Caution

The 1FINITY L160 Blade is compliant with the Laser IEC/EN 60825-1, 2 standards.

Class 1 Caution

CLASS 1 LASER PRODUCT

Hazard Level 1M Caution

HAZARD LEVEL 1M LASER RADIATION

DO NOT VIEW DIRECTLY WITH NON-ATTENUATING OPTICAL INSTRUMENTS

Laser Safety Information

The L160 (BDL1-3M21) C-Band Raman uses Laser Class 1M and Hazard Level 1M optical interface units in accordance with the following standards:

- IEC 60825-1:2007(Ed.2)/EN 60825-1:2007 (Laser class information)
- IEC 60825-1:2014(Ed.3)/EN 60825-1:2014 (Laser class information)
- IEC 60825-2:2010(Ed.3.2)/EN 60825-2:2004/A2:2010 (Hazard level information)

Radiation output of classification information:

Output of laser radiation: 0.1 mW (Value can be changed after laser safety compliance test)

Note: The use of an aperture specified as condition 3 limits the amount of radiation that is collected from a large diameter (measurement distance = 100 mm).

Emitted wavelengths: 1504.5...1566.93 nm

Beam divergence: 0.25 rad

Important Warnings and Cautions

Observe all warnings in the text or on equipment labels regarding high-voltage or high-temperature conditions. The following warnings and figures apply to most Fujitsu products.

Plug-in Unit Cautions

Observe the following precautions when handling plug-in units:

- Hold the plug-in unit only by its edges.
- Slowly and firmly push a plug-in unit into its slot to avoid damaging the unit. Do not force the unit. A slow insertion method minimizes power surges during installation. The longer ground connector pins ensure the ground circuit is connected first.

ESD Cautions

Units are stamped with anti-electrostatic markings (shown at right). Observe the following precautions to avoid damage from ESD:

- Always transport and store the unit in an ESD approved shipping bag.
- Always wear an ESD wrist strap, with a minimum 1-megohm resistance, that is connected to safety ground. Do not use a damaged wrist strap.



Figure 3
Anti-electrostatic Markings

Warnings

Fiber Warnings

Danger: Never handle exposed fiber with bare hands or touch it to bare skin. Fiber fragments can enter the skin and are difficult to detect and remove. Follow local safety precautions regarding fiber.

Avertissements de Fibre

Danger: Ne jamais manipuler une fibre exposée à mains nues et ne jamais la faire entrer directement en contact avec une partie du corps. Des fragments de fibre peuvent pénétrer la peau et leur détection et extraction sont extrêmement difficile.

Laser Safety Precautions

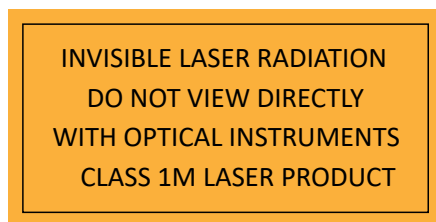
1FINITY systems consist of IEC/EN 60825-1 Class 1 and Class 1M optical interface units.

Avertissements Relié au Laser

Les systèmes 1FINITY contient des interfaces optiques basées sur des lasers de IEC/EN 60825-1 Classe 1 et Classe 1M.

Laser Safety Precautions

Avertissements Relié au Laser



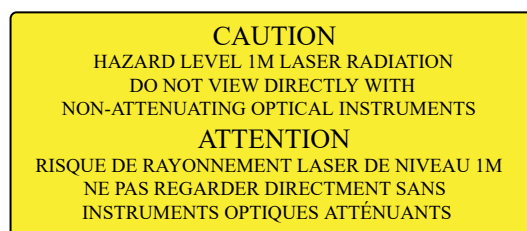
FNCO00951_Rev_01

1FINITY systems are WDM Network compliant with IEC/EN 60825-1, 2 (OFCS).

1FINITY systems consist of IEC/EN 60825-1, 2 Hazard Level 1M Optical interface units.

Le réseau WDM pour 1FINITY est compatible avec la norme IEC/EN 60825-1, 2 (OFCS).

1FINITY est constitué de modules d'interface optique compatibles avec la norme de sécurité IEC/EN 60825-1, 2 niveau 1M.



Danger: 1FINITY systems generate invisible laser radiation. Observe the following precautions:

- Avoid direct exposure to the beam.
- Never look into the end of a fiber, fiber cord, or fiber pigtail. Permanent eye damage or blindness can occur quickly when laser radiation is present.
- Viewing the laser output with certain optical instruments designed for use at a distance (for example, telescopes and binoculars) may pose an eye hazard.
- Use of controls, adjustments, or procedures other than those specified may result in hazardous laser radiation exposure.

Installation Restrictions

The following guidelines apply to the installation of this equipment:

- Systems should be installed only in restricted access areas (for example, dedicated equipment rooms or equipment closets) in accordance with the National Electrical Code, ANSI/NFPA 70.

Danger: Les systèmes 1FINITY génèrent du rayonnement laser invisible. Veuillez toujours observer les précautions suivantes:

- Eviter toute exposition directe au rayon.
- Ne jamais regarder dans l'extrémité ouverte d'une optique ou dans son connecteur. La présence, même très brève, d'un signal optique invisible peut engendrer des dommages permanents à l'œil et à la vision.
- L'observation du signal laser avec certains instruments optiques conçus pour l'observation à longue distance (par exemple, les télescopes et les jumelles) peuvent causer des dommages sévères à l'œil et à la vision.
- L'application de procédures, calibration ou ajustements autres que celles dans ce manuel peuvent causer une exposition dangereuse aux radiations optiques transmises par cet équipement.

Restrictions d'Installation

Les mesures de sécurité suivantes s'appliquent lors de l'installation de cet équipement:

- Le système devrait seulement être installé dans une pièce avec un accès contrôlé (par exemple, une pièce dédiée ou un placard d'équipement) conforme au Code Électrique Canadien (CEC) Partie 1 CAN/CSA C22.1.

Installation Restrictions

- **Elevated Operating Ambient**—If installed in a closed or multi-unit rack assembly, the operating ambient temperature of the rack environment may be greater than room ambient. Therefore, consideration should be given to installing the equipment in an environment compatible with the maximum ambient temperature (Tma) specified by the manufacturer.
- **Reduced Air Flow**—Installation of the equipment in a rack should be such that the amount of air flow required for safe operation of the equipment is not compromised.
- A suitable disconnect device must be provided for the equipment. Caution: To de-energize the equipment, all input power circuits (–48 V DC battery) must be removed prior to maintenance/servicing or upgrading.
- **Mechanical Loading**—Mounting of the equipment in the rack should be such that a hazardous condition is not achieved due to uneven mechanical loading.
- **Circuits Overloading**—Consideration should be given to the connection of the equipment to the supply circuit and the effect that overloading of the circuits might have on overcurrent protection and supply wiring. Appropriate consideration of equipment nameplate ratings should be used when addressing this concern.
- **Reliable Earthing**—Reliable earthing of rack-mounted equipment should be maintained. Particular attention should be given to supply connections other than direct connections to the branch circuit (e.g. use of power strips).
- Use UL Listed Lugs suitable for the wire size and bolt patterns/size.
- For the input power harness, use only UL-listed, closed-loop connectors.
- Use copper conductors only for power circuits.
- The unit must be connected to an earthed (grounded) outlet to comply with international safety standards.
- This equipment is not suitable for use in locations where children are likely to be present.
- This equipment is not likely to connect to an Ethernet network with outside plant routing.
- Any copper cable greater than 10 feet must be shielded.

Restrictions d'Installation

- **Opération à température ambiante élevée**—Si cet équipement est installé dans un cabinet fermé ou dans une étagère contenant plusieurs pièces d'équipement, la température de l'équipement peut devenir plus élevée que la température ambiante. Toute installation de ce type devrait prendre en considération que l'installation doit être compatible avec la température ambiante maximum (Tma) spécifiée par le fabricant.
- **Déplacement d'air réduit**—Toute installation de cet équipement dans une étagère ne doit pas compromettre la quantité de déplacement d'air requis pour le fonctionnement sécuritaire de cet équipement.
- Cet équipement devrait être accompagné d'un dispositif adéquat de déconnexion. Avertissement: S'assurer que toute source d'alimentation électrique, sous la forme de batterie ou courant direct a –48 V DC, est proprement déconnecté avant d'entreprendre toute opération de maintenance sur l'équipement.
- **Installation Mécanique**—Veuillez-vous assurer que des conditions dangereuses liées à l'équilibre du poids de l'équipement ne soient pas créées lors de l'installation.
- **Surcharge de Circuits**—L'installation de l'équipement doit prendre en considération le risque et l'impact d'une surcharge électrique sur le circuit d'alimentation et le dispositif de protection de surcharge de courant. Veuillez considérer les directives de classification électrique fournies sur l'étiquette d'identification de l'équipement.
- **Mise à Terre Adéquate**—Une mise à terre appropriée doit être maintenue pour cet équipement ainsi que pour l'étagère d'installation. L'installateur devrait spécifiquement s'assurer de ne pas directement connecter l'équipement au circuit d'alimentation (e. g. utilisation de barres d'alimentation)
- Utilisez seulement des écrous et terminaisons approuvées par Underwriters Laboratoire (UL) qui sont compatibles avec les fils d'alimentation ainsi que la grosseur et le filetage des boulons utilisés avec cet équipement.
- Pour le faisceau d'alimentation d'entrée, utilisez uniquement une boucle fermée répertoriée UL connecteurs.
- Utilisez des conducteurs en cuivre pour les circuits d'alimentation.
- L'unité doit être connectée à une prise mise à la terre pour être conforme aux normes de sécurité internationales.
- Cet équipement ne convient pas pour une utilisation dans des endroits où des enfants sont susceptibles d'être présents.
- Il est peu probable que cet équipement se connecte à un réseau Ethernet avec un routage externe.
- Tout câble de cuivre de plus de 10 pieds doit être blindé.

Flammable Liquids Warning

Danger: Do not use flammable liquids or sprays around telecommunications equipment. Electrical fan motors and other potential ignition sources within the equipment might ignite the flammable material and cause personal injury or damage to the equipment. If uncertain about whether a liquid or spray is flammable, contact the manufacturer.

Mise en Garde Concernant les Liquides Inflammables

Danger: Ne pas utiliser de liquides ou aérosols volatils près de l'équipement de télécommunication. Les ventilateurs électriques ainsi que d'autres sources d'allumage à l'intérieur de l'équipement pourraient enflammer les liquides ou aérosols et causer des blessures corporelles ou du dommage à l'équipement. Veuillez consulter le fabricant de tout liquide ou aérosols utilisés près de l'équipement si vous êtes incertains de son degré d'inflammabilité.

BSMI Warning

警告使用者：

此為甲類資訊技術設備，於居住環境中使用時，可能會造成射頻擾動，

在此種情況下，使用者會被要求採取某些適當的對策。

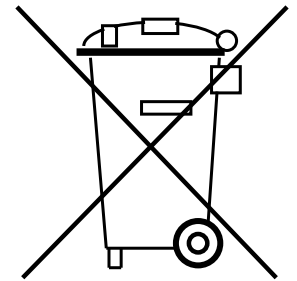
Proper Disposal Procedures for Information Technology Equipment within the EU

All electrical and electronic equipment or units must be disposed of in accordance with European Directive 2012/19/EU for Waste Electrical and Electronic Equipment (WEEE).

Network communications or commercial equipment customers can dispose of Fujitsu equipment by contacting the Fujitsu regional office in their country. Customers can locate regional offices at <http://www.fujitsu.com/global/>.

Fujitsu will ensure the proper recycling, reuse, and other forms of recovery of WEEE. Potential hazardous substances in WEEE can harm the environment. You can contribute to the elimination of these effects through your cooperation.

Additional WEEE disposal information may be found at: <http://www.fujitsu.com/global/about/environment/approach/policy/>



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Document Change Notice

Issue 1.1

Location of Change	Description of Change
Preface	■ Updated Contact Information section
Unit Description	■ Added L160 Fan Unit section
Ordering Information	■ Updated Blade and PIU section ■ Updated Software section
System Administration	■ Updated FTP/SFTP File Transfer, Client Session section
System Maintenance	■ Updated Alarms and Conditions section ■ Updated Upgrade System Software and Firmware section

Issue 1

Location of Change	Description of Change
Throughout	■ Initial release

Supporting Documentation

The following supporting documents are available at <https://partners.fnc.fujitsu.com/documentation>. To be sure of obtaining the latest information, the user should always access the online documentation.

Supporting Documentation	Type
Product Change Documents	■ Customer Information Releases (CIR)
	■ Technical Information Bulletins (TIB)
	■ Product Change Notices (PCN)
	■ Method of Procedure Documents (MOP)

1

System Description and Engineering

In this chapter:

- 1.1 System Profile
- 1.2 ROADM Applications
- 1.3 ROADM Features
- 1.4 System Features
- 1.5 Management Interfaces
- 1.6 Management Protocols
- 1.7 System and Network Services
- 1.8 Engineering Specifications

1.1 System Profile

In this section:

1.1.1 Overview

1.1.1 Overview

The 1FINITY L-Series solution is Fujitsu next-generation ROADM network solution, which consists of an L100 ROADM configuration, an L130 ROADM configuration, an L160 Raman configuration, and an L200 ILA configuration. The L100 ROADM configuration is based on Colorless, Directionless (CD) and flexible grid add/drop architecture. The L130 ROADM configuration is based on Colorless, Directionless, Contentionless (CDC) and flexible grid architecture. A state-of-the-art ROADM network can be constructed with the L100 configuration, or the L130 configuration, or a combination of both configurations. The L-Series solution provides the optimal level of flexibility at the optical layer to support rapid bandwidth and capacity provisioning, topology flexibility, and simplified operations.

Colorless

Classic ROADMs are limited by fixed add/drop transceiver and wavelength assignments. Wavelengths selected in these deployments rely on transceivers that are manually connected to the right mux/demux port at the add/drop side. All add/drop sides must be physically wired and rewired whenever a change is made.

A colorless ROADM automates the assignment of add/drop wavelength functionality through the use of additional Wavelength Selectable Switches (WSSs) in place of different mux/demux ports. Any wavelength can be assigned to any port at the add/drop side without a technician.

Directionless

Classic ROADMs are directionally dependent, which means add/drop port pairs and the transponders connected to them are fixed to an outgoing direction. Directionless ROADMs allow any wavelength to route in any direction served by the node.

Contentionless

Wavelength blocking occurs when two wavelengths of the same color converge at the same WSS at the same time, creating network contention. Classic ROADMs require operators to manually partition the add/drop structures so that different colored wavelengths are associated with different structures.

This approach reduces flexibility and requires extra add/drop structures to accommodate particular channels. Contentionless architecture, by contrast, allows multiple copies on the same wavelength on a single add/drop structure with no partitioning restrictions.

Flexible ITU Grid

The concept of a flexible spectrum feeds into the CDC solution as a way for operators to future-proof their networks to contend with transport speeds beyond 100G. The CDC solution supports flexible grid that operates from 37.5 to 500 GHz in increments of 12.5 GHz spacing, for example, 75 GHz spacing or 125 GHz spacing.

1FINITY L-Series Solution

The 1FINITY L-Series solution has the following benefits:

- Operating Expenses (OPEX) and Capital Expenditures (CAPEX) reduction
- Rapid bandwidth/capacity provisioning
- Topology flexibility and simplified operations

Note: For more detailed L130 ROADM configuration information, refer to *1FINITY L130 User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

Note: For more detailed L200 ILA configuration information, refer to *1FINITY L200 ILA User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.2

ROADM Applications

In this section:

- 1.2.1 Colorless Directionless ROADM Application
- 1.2.2 Colorless (Direct Access) ROADM Application
- 1.2.3 Mix Configuration ROADM Application
- 1.2.4 Virtual Network Element using vMCU Hosted on Main Blade
- 1.2.5 Interoperability with FLASHWAVE 9500
- 1.2.6 Interoperability with 1FINITY L130

1.2.1

Colorless Directionless ROADM Application

Up to four degrees of CD ROADM is supported. Each degree of ROADM requires an L100 (BDL1-3R11) 1x9 WSS blade. Each L110 (BDL1-3B11) Optical Mux Base blade provides up to 32 add/drop channels to any degree of ROADM.

The L110 Blade houses up to two 4x16 splitter/coupler PIUs [PIL1-3PS2]. Each PIL1-3PS2 provides client ports (C1...C16) for up to 16 add/drop signals. Each signal from each client port can be routed to edge ports E1 through E4, for example, signals routed to the E1 port are passed to degree 1 and the signals routed to the E2 port are passed to degree 2.

At each L100 Blade, the cable from the appropriate edge port on the L110 Blade is connected to one of the client ports (C1...C9) on the L100 Blade. The add/drop signals from the client ports are mapped to a channel on the edge port (E1) of the L100 Blade. Pass-through signals between ROADM degrees can be passed through a client port (C1...C9) on the L100 Blade at each degree to a client port (C1...C9) on the L100 Blade at the other degree.

The following figure shows an example of add/drop cabling for a 4-degree CD ROADM configuration.

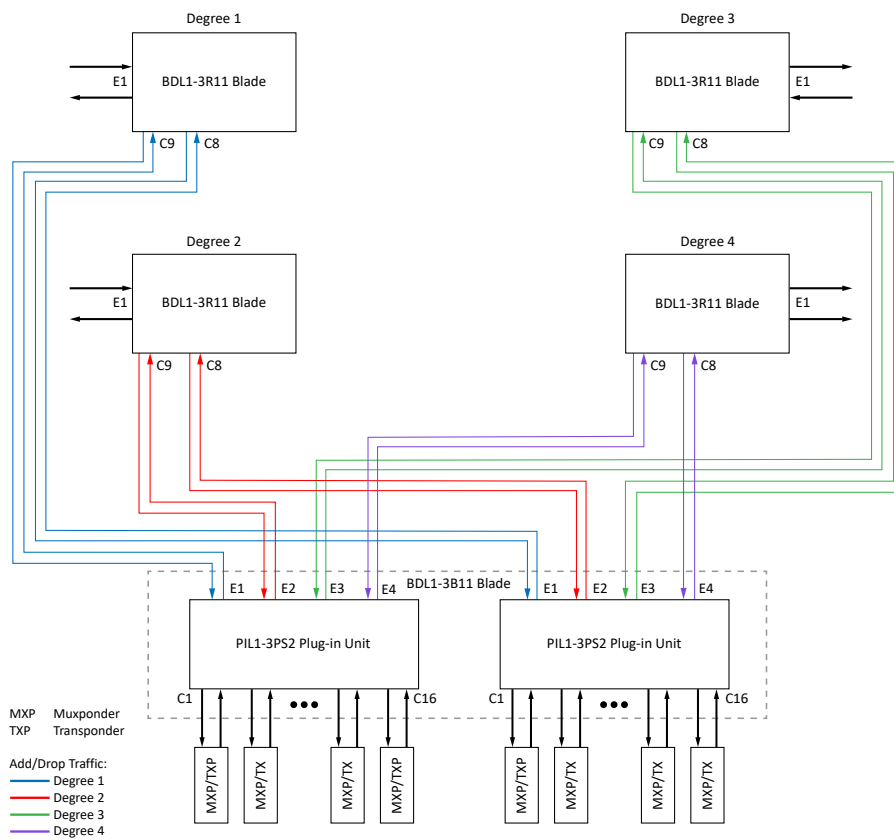


Figure 4
Example of 4-Degree CD ROADM Add/Drop Cabling (32-Channels)

The following figure shows an example of pass-through (hub) cabling for a 4-degree CD ROADM configuration.

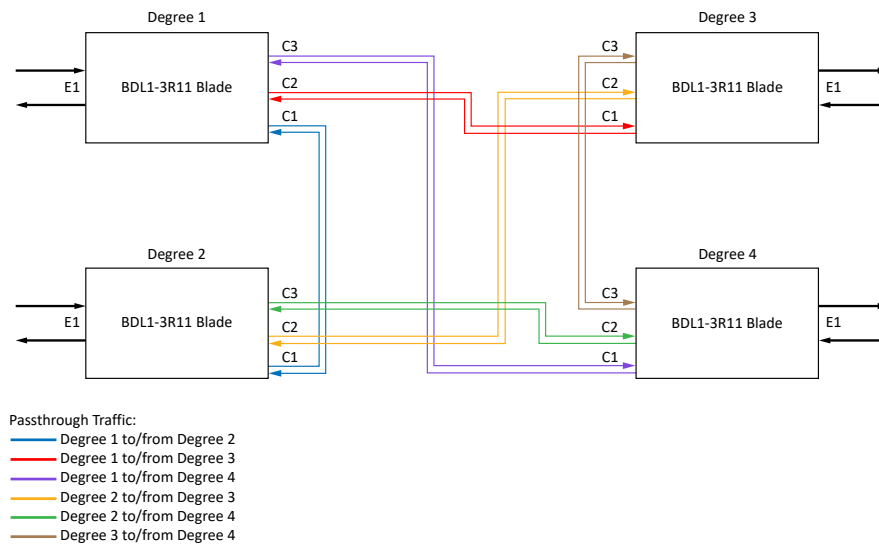


Figure 5
Example of 4-Degree CD ROADM Pass-Through (Hub) Cabling

Related Links

[L100 Blade](#)
[L110 Blade](#)
[4x16 Splitter/Coupler PIU](#)

1.2.2 Colorless (Direct Access) ROADM Application

Up to four degrees of direct access Reconfigurable Optical Add/Drop Multiplexer (ROADM) are supported. Each degree of ROADM requires an L100 (BDL1-3R11) 1x9 WSS ROB.

With the direct access feature, narrowband signals can be directly connected to the client ports (C1...C9) of each L100 Blade. The add/drop signals from the client ports are mapped to a channel on the edge port (E1) of the L100 Blade. Pass-through signals between ROADM degrees can pass through a client port (C1...C9) on the L100 Blade at each degree to a client port (C1...C9) on the L100 Blade at the other degree.

The following figure shows an example of add/drop cabling for a 4-degree colorless ROADM configuration.

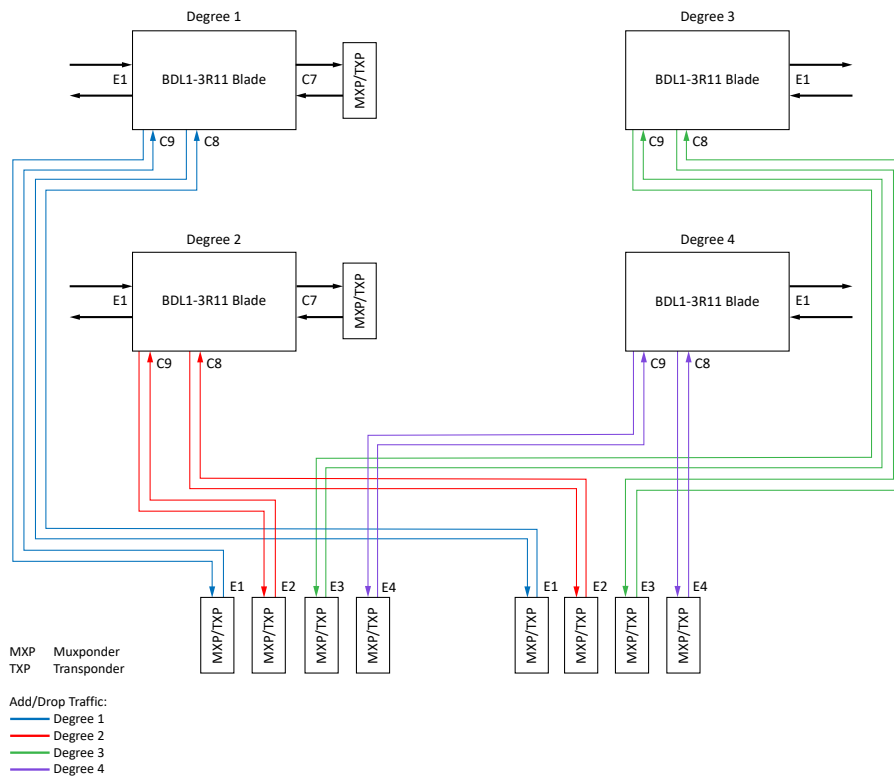


Figure 6
Example of 4-Degree Colorless ROADM Add/Drop Cabling

The direct access feature can be used with CD add/drop ROADM connections.

1.2.3 Mix Configuration ROADM Application

Mix colorless and CD add/drop configuration is supported. Each degree of ROADM requires an L100 (BDL1-3R11) 1x9 WSS ROB.

The following cross-connection types are supported individually and in mix configuration:

- Colorless (direct access)
- CD (colorless and directionless)

The following figure shows an example of mixture of colorless, and CD add/drop port connections.

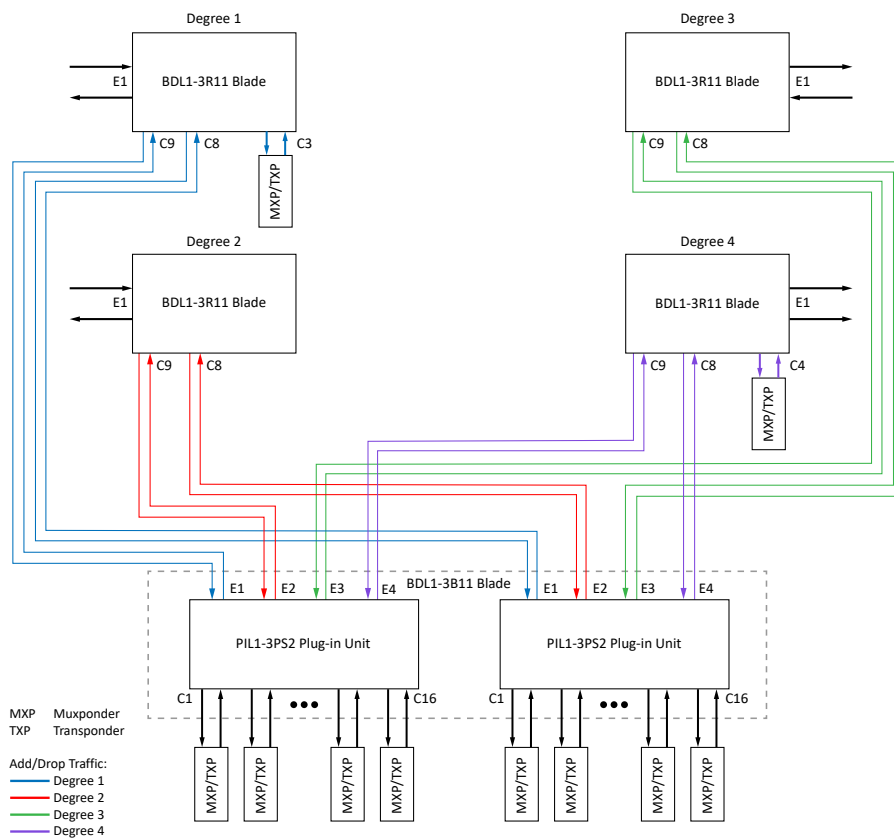


Figure 7
Mixture of Colorless and CD Add/Drop Port Connections

1.2.4 Virtual Network Element using vMCU Hosted on Main Blade

A virtual NE combines a set of L100 Blades to provide a particular solution. In a virtual NE, virtual Management and Control Unit (vMCU) software component is hosted on the Main L100 Blade. The vMCU provides the management complex for the virtual NE and terminates the external management interface. The vMCU provides hardware-independent common control software for the components of the virtual NE.

As shown in the following figure, an external Software Defined Network (SDN) controller connects to the Main L100 Blade through a Data Communications Network (DCN) that is connected to the LCN1 port on the Main L100 Blade. The LCN3 port on the Main L100 Blade is connected to the LCN4 port on the first tributary blade. The LCN3 port on each succeeding tributary blade is connected to the LCN4 port on the next tributary blade. Finally, the LCN3 port on the last tributary blade is connected to the LCN4 port on the Main L100 Blade. These connections provide management and control signaling between each blade in the virtual NE to and from the vMCU on the Main L100 Blade.



L100 Blade
L110 Blade
4x16 Splitter/Coupler PIU

1.2.5

Interoperability with FLASHWAVE 9500

From 1FINITY L100 Release 2.4, interoperability with the FLASHWAVE® 9500 Multi-Haul ROADM is supported on an optical span. 1FINITY L100 ROADM interoperability is compatible with FLASHWAVE 9500 Release 9.4. The following optical span configurations between the 1FINITY L100 ROADM and the FLASHWAVE 9500 Multi-Haul ROADM are supported:

1FINITY L100	FLASHWAVE 9500 MH AMP—WMP5-ASC2
1FINITY L100	FLASHWAVE 9500 MH AMP—WMP5-ALC2
1FINITY L100	FLASHWAVE 9500 ILA AMP—WMP5-AES1
1FINITY L100	FLASHWAVE 9500 ILA AMP—WMP5-AEL1

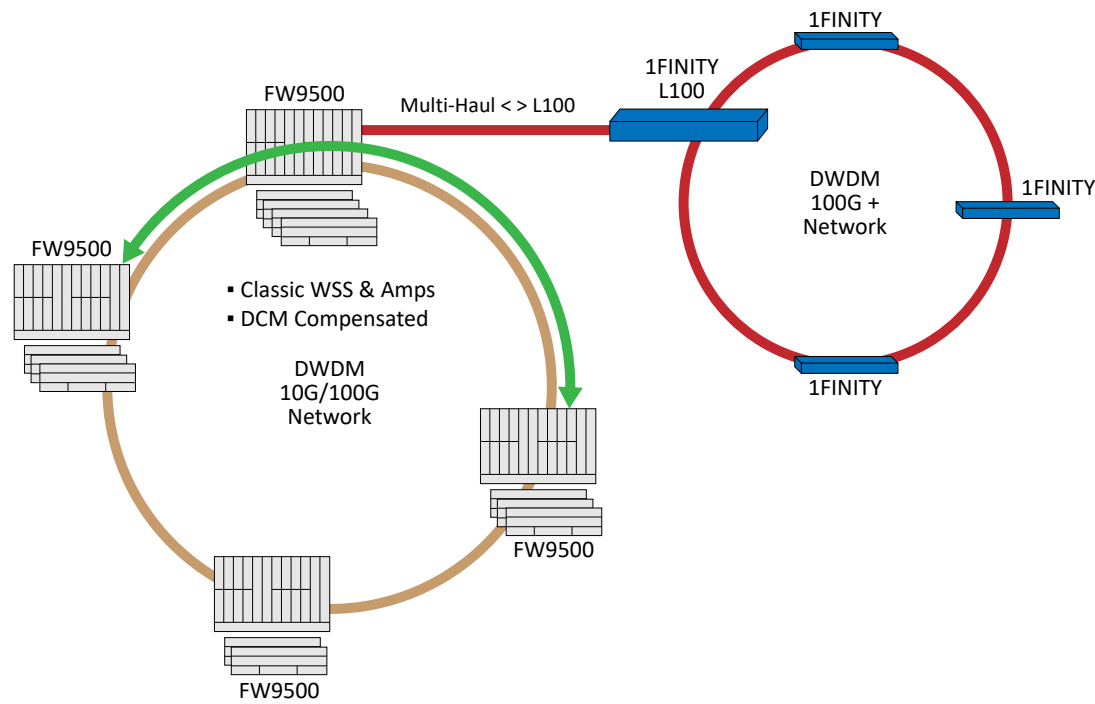


Figure 9

L100 ROADM Interoperability with FLASHWAVE 9500 Multi-Haul ROADM

Note: In the preceding figure, 1FINITY refers to L100 or L200 Blade.

1.2.6

Interoperability with 1FINITY L130

From Release 19.1.1, the functionality of interoperability with the L130 Blade is introduced.

Note: If L100 Blade interoperates with L130 Release 19.1.1, only an optical span with Raman amplifiers (1FINITY L160) is supported.

Note: When L100 Blade interoperates with L130 Blade, the OSNR setting on the L100 must be set to enable. Otherwise, an outstanding *osnrModeMismatch* alarm is raised on the L100 Blade.

1.3

ROADM Features

Automatic In-Service State

When equipment or facilities or both are preprovisioned, before equipment installation, the equipment or facilities or both are placed in Automatic In-Service (AINS) state. When the equipment is installed, the equipment or facilities or both are placed In-Service (IS).

Note: If the installed equipment does not match the preprovisioned equipment type, an MEA condition is raised.

Automatic Line Control

The Automatic Line Control (ALC) feature automatically adjusts the total optical output power and keeps the power of each channel constant and at the appropriate level. This feature ensures that manual adjustments are not required when channels are added to or dropped from the WDM facility. ALC ensures that other channels do not suffer performance losses (traffic hits) when a channel is added to or deleted from the system.

Automatic Gain Control

The Automatic Gain Control (AGC) feature automatically adjusts the gain of each channel to a constant and appropriate level. The optical amplifier in the L100 Blade continuously monitors the composite input and output signal power. A control circuit maintains a constant target gain using preamplifier and postamplifier variable optical attenuators.

Automatic Laser Shutdown

The Automatic Laser Shutdown (ALS) feature automatically shuts down the laser when the postamplifier detects a signal loss and an Optical Supervisory Channel (OSC) loss.

Automatic Power Reduction

The Automatic Power Reduction (APR) feature automatically reduces the laser power when the postamplifier detects high reflection on its output port. APR is triggered when the reported Output Return Loss (ORL) is less than 17.6 dB. Normal operation restarts when ORL exceeds 20.6 dB. APR mode limits output power to 3 dBm regardless of current provisioned settings and input power. It does not prevent operating at lower output powers.

Simple Network Management Protocol Trap

A Simple Network Management Protocol (SNMP) trap is an event notification that is sent from an SNMP agent to one or more SNMP managers when an extraordinary event, for example, an alarm, occurs. The SNMP agent generates SNMP traps for all standing conditions unless the standing condition disables the agent.

Link Layer Discovery Protocol

Link Layer Discovery Protocol (LLDP) is an industry standard protocol that allows network devices to advertise their identities to their neighbor devices. A network controller can query the information gathered through LLDP to create the network topology. The ability to query each device allows standard network management to build the network and enables multivendor interoperability.

The L100 Blades use LLDP on their OSC ports to discover their neighbors and their IDs. LLDP discovers information for each OSC, for example, remote node system name, remote node management IP address, remote OSC port ID, and remote chassis ID.

When an OSC port and its corresponding Ethernet facility are provisioned, LLDP is automatically provisioned on the OSC port. No manual provisioning is required for LLDP. After LLDP is provisioned, a periodic message is sent to neighboring devices.

If any information changes on the local device, for example, the system IP address, LLDP notifies the neighboring devices. When the neighboring devices receive the change information, they notify the network controller.

Internal Rapid Spanning Tree Protocol

Rapid Spanning Tree Protocol (RSTP) has been added to allow the addition of alternate paths for DCN signaling without the issue of looping. RSTP provides a mechanism to rapidly respond to any changes in the network, for example, failed spans, addition of components, and removal of components. RSTP typically responds to any of these events in less than 1 second.

For earlier releases, no alternate path was provided for DCN signals between the blades in the L100 Lambda Series VNE. All blades were simply daisy-chained using the LCN3 and LCN4 ports on each blade.

Flexible Grid

Flexible grid capability allows flexible spectrum allocation and improves spectrum usage for the CD ROADM network. Flexible grid supports moving from a fixed, 50 GHz channel spaced system to a flexible channel spaced system. This finer-grained frequency grid supports mixed channel sizes in increments of 12.5 GHz spectral slices with a minimum slot width of 37.5 GHz and maximum slot width of 500 GHz. Any combinations of frequency slots are allowed, as long as no two frequency slots overlap.

Flexible grid application using only a 37.5 GHz channel plan supports a maximum 128 channels with central frequencies from 196.10625 to 191.34375 THz. The default slot width is 50 GHz. An Individual Licensed Feature (ILF) is needed to enable the flexible grid capability.

Note: The maximum slot-width of 500 GHz is supported, but the maximum tested slot-width is 100 GHz.

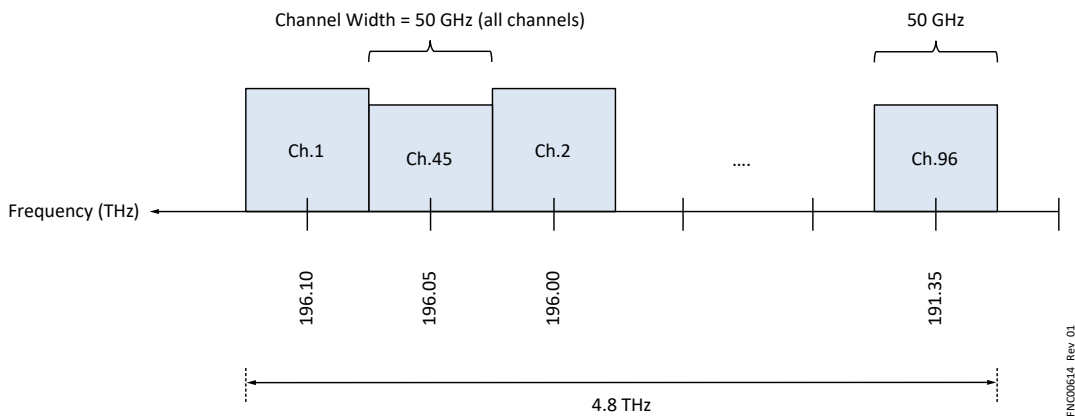


Figure 10
L100 Fixed Grid

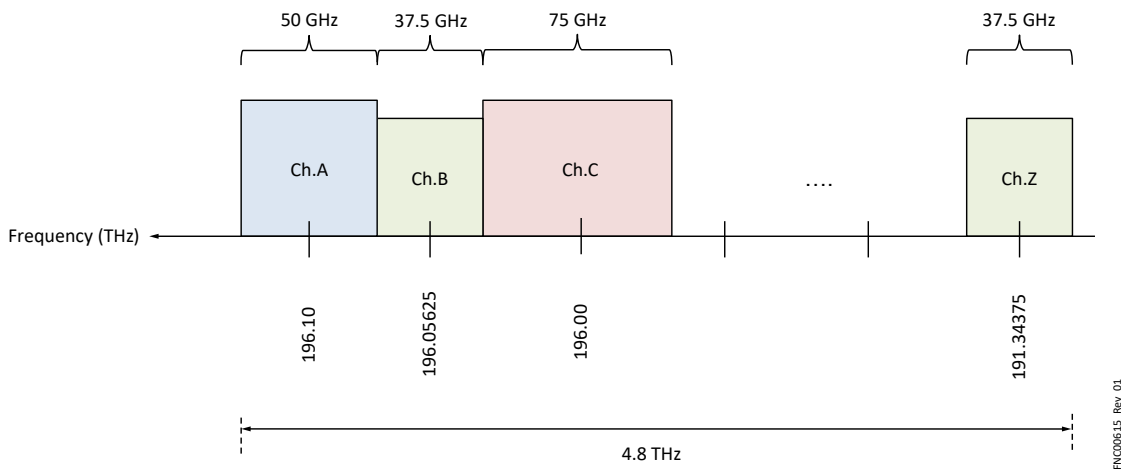


Figure 11
L100 Flexible Grid

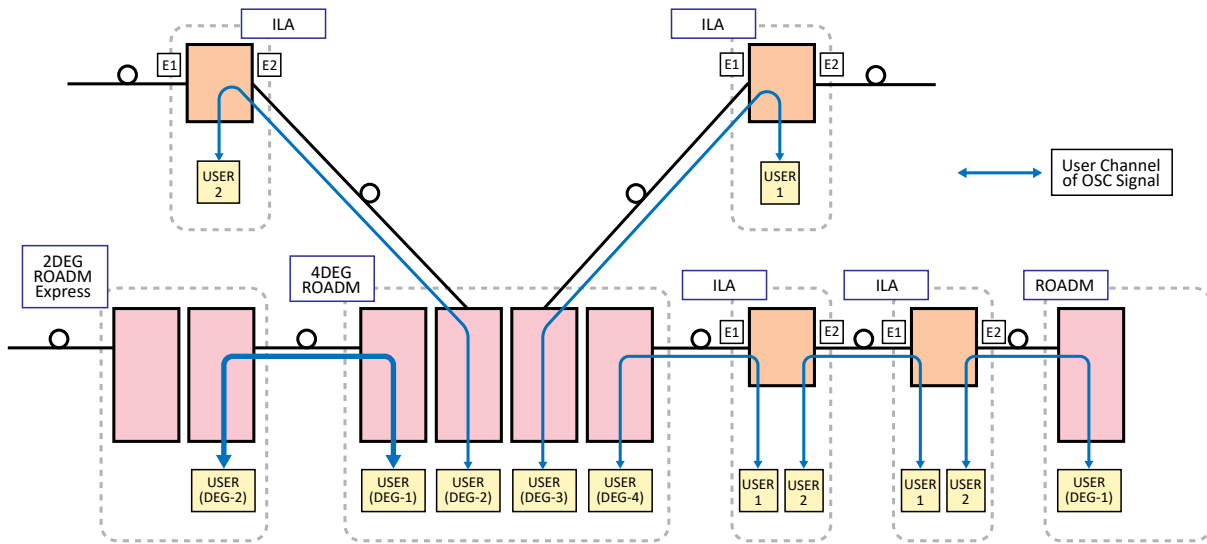
User Channel

The user channel allows the user to set up an Ethernet communication connection between two devices (either ASCII terminals or computers) located at adjacent or nonadjacent L100 Lambda node. The user channel communications are transmitted through the L100 as part of the OSC payload. A user channel is functionally equivalent to a hardwired, back-to-back Ethernet connection between the devices.

Data packets are received through Ethernet management port (RJ-45 connector), wrapped in High-Level Data Link Control (HDLC) frame, and transferred across the networks or NEs. No capability exists for the interface to communicate with the user network from a logical perspective; therefore, no MAC address is assigned to the interface. It is merely a 10 Mb/s Ethernet data pipe between NEs.

The user channel LEDs are hardware controlled similar to Local Management Port (LMP). Provisioning is not required to enable user channel. Autonegotiation advertisement for user channel ports is 10 Mb full/half and MDIX occurs during autonegotiation. User channel supports IP MTU of a maximum value of 1500 bytes.

The following figure shows an example of user channel between ROADM and ROADM Ethernet connections.



- All the traffic on user channel is untagged.
- IP Connectivity is available among all sites over user channel.

FNC000897_Rev_01

Figure 12
User Channel between ROADM and ROADM Ethernet Connections

1.4 System Features

In this section:

- | | | | |
|-------|-----------------------------------|--------|--|
| 1.4.1 | Autoprovisioning | 1.4.10 | Configurable Log Files |
| 1.4.2 | Multicommand Configuration Commit | 1.4.11 | Jumbo Frame Support with IP MTU Maximum 9574 Bytes |
| 1.4.3 | In-Service Upgrade | 1.4.12 | Postoutadj |
| 1.4.4 | Proxy ARP | 1.4.13 | OSC Power Adjustment |
| 1.4.5 | Scripting Automation—Python | 1.4.14 | Critical Data Storage Feature |
| 1.4.6 | Secure Copy | 1.4.15 | Automatic Software Download |
| 1.4.7 | System Zeroization | 1.4.16 | Tributary Auto Shelf ID Assignment |
| 1.4.8 | Root Access | | |
| 1.4.9 | CONSOLE Port | | |

1.4.1 Autoprovisioning

If autoprovisioning (autoP) is enabled, the system automatically configures all unprovisioned hardware (including plug-in units and pluggable modules) when that hardware is first installed in the system. The autoprovisioning includes hardware present at initial system turn-up and hardware that is installed after turn-up. Autoprovisioning configures hardware to default settings and creates default facilities, as required, which are configured to default settings.

Note: Autoprovisioning is enabled by factory default.

Note: Fujitsu recommends using the autoprovisioning feature first and afterwards editing the provisioning manually using CLI commands, if required.

1.4.2 Multicommand Configuration Commit

The L100 Blade supports commit command after executing multiple commands without having to save each configuration change performed. User can execute a set of commands at any statement hierarchy level and save the configuration. Multicommand configuration commit is supported in both NETCONF and CLI.

Same Hierarchy—A set of commands used for same type of entities.

Command Examples (Same Hierarchy):

```
% set fabric och-connection XCON-ADD-CH1 src-if och-7/1/0/C1 dst-if ots-1/0/0/E1
dst-och 1
```

```
% set fabric och-connection XCON-DROP-CH1 src-if ots-1/0/0/E1 src-och 1 dst-if
och-7/1/0/C1
```

```
% set fabric och-connection XCON-ADD-CH2 src-if och-7/1/0/C2 dst-if ots-1/0/0/E1
dst-och 2
```

```
% set fabric och-connection XCON-DROP-CH2 src-if ots-1/0/0/E1 src-och 2 dst-if
och-7/1/0/C2
```

```
% commit
```

Different Hierarchy—A set of commands used for different types of entities.

Command Examples (Different Hierarchy):

```
% set interfaces interface och-2/1/0/C1 type opticalChannel enabled och remote-tp-
type fujitsu-tp
```

```
% set interfaces interface ots-1/0/0/E1 ots oms och 1 center-frequency 196.1 admin-
status up slot-width 50.0
```

```
% set fabric och-connection XCON-ADD-CH1 src-if och-2/1/0/C1 dst-if ots-1/0/0/E1
dst-och 1
```

```
% set fabric och-connection XCON-DROP-CH1 src-if ots-1/0/0/E1 src-och 1 dst-if
och-2/1/0/C1
```

```
% commit
```

1.4.3

In-Service Upgrade

In-service upgrade ensures that no traffic impact and database upgrade issues occur during software upgrade.

The following table lists the supported software release upgrade paths.

Table 1
Upgrade Paths

FROM		TO Release					
Release	G-ISSUE	Direct Upgrade					
		2.4	2.4.2	2.5	2.5.2	2.6.2	19.1.1
2.3	02-03-1	Yes	—	—	—	—	—
2.3.2	02-03-2	—	—	Yes	—	—	—
2.4	02-04-1	—	Yes	—	—	—	—
2.4.2	02-04-2	—	—	Yes	—	Yes	—
2.5	02-05-1	—	—	—	—	Yes	—
2.5.2	02-05-2	—	—	—	—	—	Yes
2.6.2	02-06-2	—	—	—	—	—	Yes

1.4.4

Proxy ARP

Proxy Address Resolution Protocol (Proxy-ARP) is enabled on a Gateway Network Element (GNE). The GNE serves as a proxy for IPv4 communication between remote subtended nodes and a DCN router. When enabled, the GNE uses its Ethernet address in all management frames and allows NEs to be managed remotely by forwarding management frames between itself and the DCN. The GNE and the subtended nodes must be on the same IPv4 subnet connected through an embedded data communications channel, for example, OSC. The GNE is provisioned with a default or static route and OSPF is enabled on all embedded communication channels with static route distribution enabled.

1.4.5

Scripting Automation

Python

The scripting automation feature allows Level-4 users to run their own Python scripts to simplify system administration tasks. The Python scripts can provision and monitor a device using the Python CLI library.

Scripts can be executed in four ways:

On Demand	Immediately, through CLI
Bootup	Upon start-up
Periodic	At specified times/dates
Event	When an alarm is raised/cleared or an event notification occurs

Bootup, Periodic, and Event scripts require the provisioning of a Bootup profile, a Periodic profile, and an Event profile, respectively. An *On Demand* script does not require provisioning.

Note: For more detailed scripting automation information, refer to *Scripting Automation Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.4.6

Secure Copy

Secure Copy (SCP) is a command line utility that is used to securely copy files and directories between a local machine and a remote NE over an SSH connection. This utility is included in most Linux® and UNIX® environments. When transferring data with SCP, both the files and passwords are encrypted.

The following commands are used for SCP file transfer. The user can execute the commands through the bash shell started from CLI or through shell on the CONSOLE port.

Command (Send File):

```
scp <SourceFile> <user>@<host>:<directory>/<TargetFile>
```

Command (Receive File):

```
scp <user>@<host>:<directory>/<SourceFile> <TargetFile>
```

Command (Recursive Transfer):

```
scp -r <user>@<host>:<directory>/<SourceFolder> <TargetFolder>
```

Command (Specifying Remote Server Port):

```
scp -p 2222 <user>@<host>:<directory>/SourceFile> <TargetFile>
```

1.4.7

System Zeroization

The L100 Blade allows a user to securely delete the Critical Security Parameters (CSPs) and log files from the Main and tributary blades. This feature zeroizes all the partitions by writing zeros to the disk. After the zeroization is done, the system is loaded back with factory default settings. Log files, user data files, user configuration files, CSPs, and Remote Authentication Dial-In User Service (RADIUS) files are the subsets of items that get zeroized.

Note: The user must set the tributary blades into shelf provisioning mode to perform zeroization.

1.4.8

Root Access

Root access provides users with superuser privileges, allowing them to perform system debugging and to activate native Linux features. The root access features can be used only by users with Level-4 privileges. Root access is disabled by default.

Note: For more detailed root access information, refer to *Root Access Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.4.9

CONSOLE Port

The CONSOLE port is a serial port that allows the user to connect to the blade with an RJ-45 connector for access when required. The console port is located on the front of the blade and is labeled as CONSOLE. The CONSOLE port supports a rate of minimum 9600 b/s to maximum 115,200 b/s. The CLI session launched through the CONSOLE port follows the same session policies that are enforced on a CLI session launched through an IP-based management port. The idle time-out console session is 15 minutes by default.

Note: L100 Blade supports CONSOLE port in Issue 14, FC95453R11-I14 or later.

1.4.10

Configurable Log Files

1FINITY enhanced logging allows users to provision the generation of new log files based on the file size and number. This feature allows users to control the rotation of the log files, maximizing file space, and storage efficiency. Only users with Level-4 privileges can configure these log files.

1.4.11

Jumbo Frame Support with IP MTU Maximum 9574 Bytes

The NEs support *ETH max-frame-size* of 9600 bytes for the following interfaces:

- LCN1
- LCN2
- LCN3
- LCN4
- OSC

Note: In Release 19.1.1, *ETH max-frame-size* is a read-only attribute and the default value is 9600 bytes.

From Release 19.1.1, the *IP MTU* for these ports can be provisioned up to 9574 bytes. Before Release 19.1.1, *IP MTU* can only be provisioned to maximum of 1500 bytes. The default size of these ports is 1500 bytes.

Note: In Release 19.1.1, *ETH Interface MTU* is used for backward compatibility and if a user edits the *ETH Interface MTU* value, it does not affect the system. In later releases, *ETH Interface MTU* is removed and cannot be used.

The NEs support *IP MTU* of a maximum value of 1500 bytes for the following interfaces:

- LMP
- User channel

Note: The default is 1500 bytes.

1.4.12

Postoutadj

L100 Blade supports postoutadj value setting feature that is used to perform the following functionalities:

- Reducing noise in an optical network
- Enhancing the signal transmission distance

The *postoutadj* design value for the network is calculated during network design and planning process. Postoutadj value range is within $-15.0 \dots 0.0$ dB. The default value is 0.0 dB. Setting should be implemented in the initial system turn-up.

1.4.13

OSC Power Adjustment

Release 19.1.1 introduces *OSC power adjustment* feature to reduce the impact of an OSC signal on the traffic signal transmit distance in a network. An OSC signal may cause Cross Phase Mixing (XPM) that is one of the root causes of reduction in the signal transmit distance.

The *OSC power adjustment* feature enables the L100 Blade to place an external PAD between the 100 Mb/s OSC SFP (FC9581B62F) and the OSC port. This PAD reduces the impact of an OSC on the signal transmit distance. The effectiveness of the PAD depends on the span loss and the fiber type.



Important: Fujitsu does not recommend using third-party OSC SFP module because the power of third-party OSC module is not guaranteed.

1.4.14

Critical Data Storage Feature

The purpose of Critical Data Storage Feature (CDSF) is to provide for the automatic recovery of specific NE management interfaces when both the primary and secondary NE databases are found to be corrupted after an NE reset. Using the recovered NE management interfaces, the user may address the corrupted database by restoring it from an externally stored backup file. 1FINITY management interfaces can be recovered as long as NE is not power reset.

CDSF is a critical data segment feature which allows DCN connectivity up for the L100 Blade even if the NE database is corrupted. The user can still log on to the NE remotely and restore database instead of doing truck roll.

If the primary and secondary databases are corrupted, the NE raises a database corruption alarm.

1.4.15

Automatic Software Download

The automatic software download feature automatically updates a tributary blade software when the tributary is spared or in-service upgrade is performed. The automatic download is supported when the Main blade with Release 2.5 or later is connected to the tributary blade with Release 2.4.1 or later.

1.4.16

Tributary Auto Shelf ID Assignment

Before Release 2.6, a new tributary blade must be manually preprovisioned with the tributary ID and shelf mode as tributary before it is connected to the shelf. With the tributary auto shelf ID assignment feature enabled in Release 2.6 and later, no manual preprovision of the new tributary blade is required before connecting to the system for provisioning.

For more information, refer to [Enable Auto Shelf ID Assignment for Tributary Blade Feature](#) and [Set Up Tributary Blade with Auto Shelf ID Assignment Enabled](#).

1.5

Management Interfaces

In this section:

1.5.1 Command Line Interface

1.5.2 WebGUI

1.5.3 Virtuora NC

1.5.1

Command Line Interface

Command Line Interface (CLI) provides a mechanism for users to enter commands that configure and manage the NE and perform operations, administrative, and maintenance tasks on the NE. The CLI is rendered using data models defined in YANG modules. The CLI is hierarchical, which means that the configuration commands have nested layers. CLI uses the SSH® server service.

Note: Refer to [About CLI](#) for more information.

Note: For more detailed CLI information, refer to *1FINITY CLI User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.5.2

WebGUI

Similar to the CLI, the WebGUI provides users with an interface to configure and manage the NE. However, the WebGUI user interface is graphical, rather than command line. The WebGUI is supported through web browsers, including Microsoft® Internet Explorer®, Mozilla® Firefox®, Google Chrome™, and Safari®. The WebGUI is designed to work on Android™ mobile and iPad Air™, iPad mini™, and iPad Pro™ mobile digital devices.

Note: The WebGUI does not provide the same level of operation support as CLI or NETCONF. Fujitsu recommends to use CLI or NETCONF for provisioning and status collection.

Note: For more detailed WebGUI information, refer to *1FINITY WebGUI User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.5.3

Virtuora NC

Virtuora® NC is Fujitsu general software platform for SDN/NFV-based network management and service delivery. Although Virtuora NC is designed for much more, it includes graphical user interface that operators can use to configure, monitor, and manage 1FINITY NEs in a traditional (non-SDN) fashion.

Virtuora NC leverages open software, for example, OpenDaylight (a Linux Foundation®), and is designed to be deployed (and scale) on virtual, cloud-based, infrastructure.

Virtuora NC interacts directly with network devices through southbound interfaces. Operators and northbound software systems interact indirectly with the network through Virtuora NC. Virtuora NC provides operators and northbound software systems with a higher-level abstract view of the network and tools for configuring, monitoring, and managing services within the network.

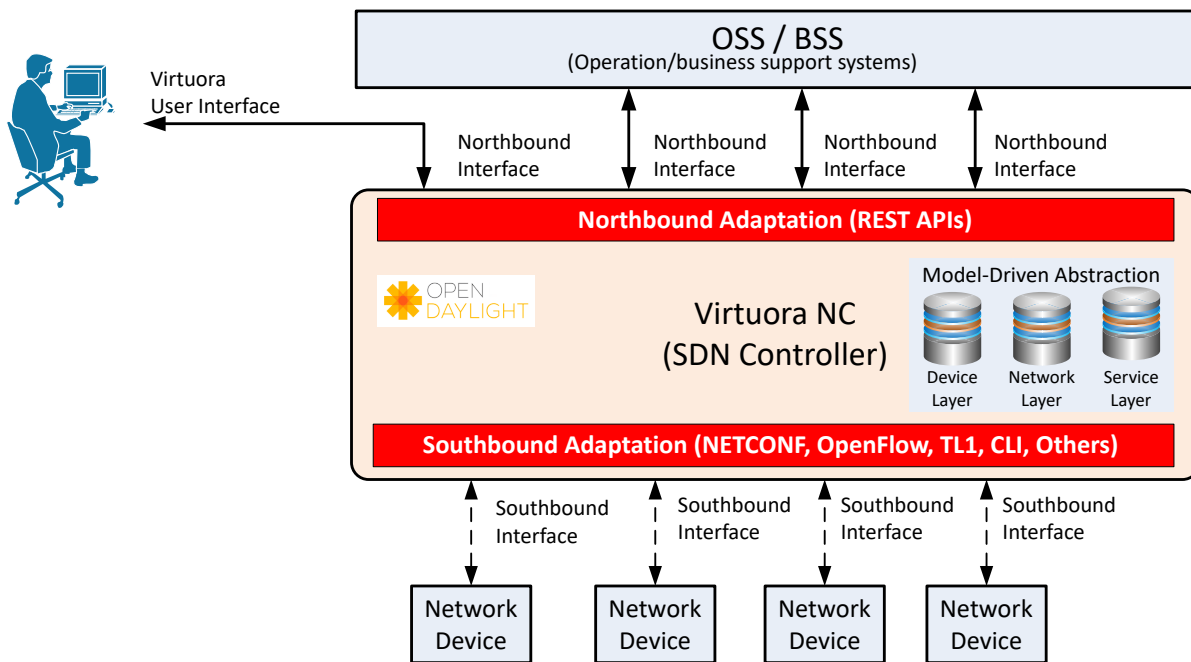


Figure 13
Virtuora NC (SDN Controller)

Virtuora NC configures, monitors, and manages network devices (both legacy and non-legacy) through southbound interfaces using a variety of protocols, including NETCONF, OpenFlow, TL1, and CLI. Virtuora NC manages the 1FINITY devices with the NETCONF protocol.

Virtuora NC northbound interface REST APIs allow it to be integrated with the service provider local operations and business support systems (OSS/BSS). Northbound REST APIs allow Virtuora NC to team with local software systems to realize the service provider vision for automated SDN-based service delivery.

Virtuora NC provides a graphical user interface that operators can use to configure, monitor, and manage the network in a traditional (non-SDN) fashion.

Note: For more detailed Virtuora NC information, refer to *Virtuora NC User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

1.6

Management Protocols

In this section:

1.6.1 NETCONF

1.6.2 Simple Network Management Protocol

1.6.1

NETCONF

NETCONF is a Remote Procedure Call (RPC)-based protocol used to manage network devices in vendor-agnostic fashion. NETCONF is designed to support machine-to-machine communication, typically between network devices and an SDN controller, for example, Fujitsu Virtuora NC.

1FINITY NEs support the NETCONF protocol, using YANG data models in combination with NETCONF to communicate device capability, configuration, and state information. Both NETCONF and YANG use Extensible Markup Language (XML) encoding.

SDN controllers, for example, Virtuora NC, manage 1FINITY NEs over TCP/IP network connections using the NETCONF protocol. Information is exchanged in XML format between the NETCONF client (the SDN controller) and the NETCONF server (the 1FINITY NE) on a secure connection-oriented session. State data is separated from configuration data. NETCONF provides RPCs that allow the client to query the server configuration and state and allow the client to edit the server configuration. The client can subscribe to notifications from the server.

Note: For more detailed NETCONF information, refer to *1FINITY NETCONF Protocol User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

1.6.2

Simple Network Management Protocol

SNMP is an application-layer protocol defined by the Internet Architecture Board (IAB) in RFC 1157 for exchanging management information between network devices. It is a part of the Transmission Control Protocol/Internet Protocol (TCP/IP) suite. SNMP is one of the widely accepted protocols used to manage and monitor network elements. Most of the professional-grade network elements come bundled with an SNMP agent. These agents have to be enabled and configured to communicate with the Network Management System (NMS).

SNMP Traps

An SNMP trap is an event notification that is sent from an SNMP agent to one or more SNMP managers when an extraordinary event, for example, an alarm, occurs. The SNMP agent generates SNMP traps for all standing conditions unless the standing condition disables the agent.

Note: For more detailed SNMP information, refer to *1FINITY SNMP User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

1.7

System and Network Services

In this section:

- | | | | |
|--------------|--|---------------|---|
| 1.7.1 | FTP and SFTP | 1.7.8 | Syslog (RFC 5424) |
| 1.7.2 | HTTP and HTTPS | 1.7.9 | Zero Touch Provisioning |
| 1.7.3 | Network Timing Protocol | 1.7.10 | Access Control List |
| 1.7.4 | Remote Authentication Dial-In User Service | 1.7.11 | Telnet |
| 1.7.5 | REST API | 1.7.12 | Terminal Access Controller Access-Control System Plus |
| 1.7.6 | Secure Shell | | |
| 1.7.7 | Streaming Telemetry | | |

1.7.1

FTP and SFTP

FTP is a client-server network protocol for transferring files between hosts. SFTP is a secure version of FTP that encrypts commands and data transfers using OpenSSH. 1FINITY NEs support both protocols and can participate as either client or server.

The FTP server service is disabled by default. When it is enabled, it listens on TCP port 21. The SFTP server service is disabled by default. When it is enabled, the TCP port is configurable (port 2202 is used by default).

1.7.2

HTTP and HTTPS

Hypertext Transfer Protocol (HTTP) is a client-server network protocol for data communication over the World Wide Web. HTTPS is a secure version of HTTP that encrypts commands and data transfers.

1FINITY NEs support HTTP and HTTPS web server services. Both services are disabled by default. The TCP port is configurable for both services. By default, HTTP listens on TCP port 80 and HTTPS listens on TCP port 443.

The WebGUI service and the REST API service both use the HTTP/HTTPS web sever service.

1.7.3

Network Timing Protocol

Network Timing Protocol (NTP) is a networking protocol that allows users to synchronize the NE system clock to a clock on an external NTP server. By default, NTP is enabled.

1.7.4

Remote Authentication Dial-In User Service

RADIUS is a remote authentication method which allows users to provision user account information (username and passwords) for authentication on a server external to the network element. It provides the ability for users to log accounting information (for example: login time and IP) on an external server.

Note: For more detailed RADIUS information, refer to *1FINITY RADIUS User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

1.7.5 REST API

The 1FINITY REST API web service allows users or software systems to interact with and configure the 1FINITY system through a YANG-data-model-based REST API. The REST API web service is disabled by default and uses the HTTP/HTTP web server service.

Note: For more detailed REST API information, refer to *1FINITY REST API User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

1.7.6 Secure Shell

SSH is a client-server network protocol for secure communication sessions between hosts over TCP/IP network connections. 1FINITY NEs can participate as either client or server. The SSH server service is enabled by default and listens on TCP port 22. The TCP port is configurable. CLI uses the SSH server service.

1.7.7 Streaming Telemetry

1FINITY devices support streaming telemetry. Streaming telemetry provides a periodic push mechanism to send operational state data from a network element to a monitoring server (network management system). A simple configuration mechanism consists of uploading a configuration file that describes the destination(s) (for example, address and port), and separate command file(s) which consist of NETCONF XML get requests for data of interest. The command files can select the entire tree or focus on a particular monitoring point.

Note: For more detailed streaming telemetry information, refer to *1FINITY Telemetry User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

1.7.8 Syslog (RFC 5424)

Syslog is a standard protocol used to send event notification messages to a logging server. Syslog is enabled by default on 1FINITY NEs to capture logs generated by different applications. Syslog uses the standard BSD syslog protocol, specified in RFC 3164. The logs cover information, for example, login success/failures, config change notifications, and application errors.

The 1FINITY NEs support syslog filtering, saving syslogs to local syslog files, and sending syslogs to the remote syslog server. The filtering can be done based on facility and severity. When the filtered syslogs are saved locally, a maximum of four such files can be created.

Note: For more detailed syslog information, refer to *1FINITY Syslog User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.7.9 Zero Touch Provisioning

Zero Touch Provisioning (ZTP) allows 1FINITY NEs to be autoconfigured by a controller when the device is first installed. ZTP eliminates the need for the installation technician to connect to the device and manually configure it.

After hardware is installed, the installation technician connects the NE to the DCN. ZTP, using a standard DHCP server service, announces the NE to one or more controllers. The controllers get a REST notification that identifies the network element. With this information, the controller can associate the NE to a configuration profile and automatically configure the NE.

All network communication is fully secure. The announcement method allows the controller to discover the NE automatically.

Note: For more detailed ZTP information, refer to *1FINITY Zero Touch Provisioning User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.7.10 Access Control List

Access Control List (ACL) is used for packet filtering that controls the Internet Protocol (IP) traffic passing through a network. A set of rules is defined in the ACL to decide whether the IP traffic is allowed or denied. 1FINITY ACL supports both IPv4 and IPv6 addresses to allow or deny IP traffic.

Note: For more detailed ACL information, refer to *1FINITY Access Control List User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.7.11 Telnet

Telnet is a client-server network protocol for text-oriented communication between hosts over TCP/IP network connections. 1FINITY NEs can participate as either client or server.

1.7.12 Terminal Access Controller Access-Control System Plus

1FINITY NEs support Authentication, Authorization, and Accounting (AAA) using Terminal Access Controller Access-Control System Plus (TACACS+). The TACACS+ is an open, distributed client-server protocol that is widely supported and provides independent control over AAA. An AAA server provides authentication, authorization, and accounting services. The TACACS+ services are maintained on TACACS+ servers independent of 1FINITY NEs. Key benefits of using TACACS+ are user accounts, user authorization, session authorization and accounting can be centrally maintained in a distributed environment.

Note: For more detailed TACACS+ protocol information, refer to *1FINITY TACACS+ Protocol User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.8 Engineering Specifications

In this section:

- | | | | |
|-------|--|-------|------------------------------------|
| 1.8.1 | Optical Signal-to-Noise Ratio Specifications | 1.8.6 | Pluggable Transceiver Modules |
| 1.8.2 | L100 Specifications | 1.8.7 | Power Specifications |
| 1.8.3 | L110 Specifications | 1.8.8 | 1FINITY Housing |
| 1.8.4 | L160 Specifications | 1.8.9 | C-Band Frequencies and Wavelengths |
| 1.8.5 | 4x16 Splitter/Coupler Specifications | | |

1.8.1 Optical Signal-to-Noise Ratio Specifications

From Release 2.6, L100 Blade introduces an enhanced Optical Signal-to-Noise Ratio (OSNR) performance feature. This feature provides improved levels of optical power for both flexible grid and fixed 50-GHz applications with the following benefits:

- Reachability is enhanced especially for signals greater than standard 50 GHz.
- Signal quality is improved. The signal quality improves the Pre-FEC BER and provides enhanced network resiliency to network transients.

The OSNR enhancement setting is at the NE level and is mainly for a new build network. System alarms against any span level interface if the OSNR setting is different between spans (adjacent nodes). By default, OSNR enhancement setting is disabled. In some cases, changing the OSNR setting on a running system may impact the signal traffic.

Note: Fujitsu recommends any change to the OSNR setting on a running system be performed during a maintenance window.

Note: OSNR setting must be enabled in L100 Blade to support interoperability with L130 Blade.

Table 2
Example of OSNR Enhancement at 200 Gb/s Signal

Signal Type and Modulation	Reach	
	OSNR Setting OFF (Default)	OSNR Setting ON
75 GHz, 2A8PSK	■ Maximum reach: 1368 km ■ At 19-dB spans: 1368 km	■ Maximum reach: 1850 km ■ At 19-dB spans: 1824 km
50 GHz, 16QAM	■ Maximum reach: 770 km ■ At 19-dB spans: 760 km	■ Maximum reach: 840 km ■ At 19-dB spans: 836 km

1.8.2 L100 Specifications

Table 3
General Specifications

Property	Specification
Weight	21.164 lb (9.6 kg)
Dimensions (H x W x D)	1.75 x 19.00 x 17.72 in. (44.45 x 483.00 x 450.00 mm) Note: The depth dimension is without the fiber management. The depth dimension with fiber management is less than 23.62 in. (600 mm).
Operating temperature range	5...40 °C
Storage temperature range	–40...70 °C
Short-term temperature range	–5...50 °C (96-hour period, maximum of 15 days/year)
Operating humidity range	5...85%
Short-term humidity range	5...93% (96-hour period, maximum of 15 days/year)
Number of ports/circuits	12 optical ports (LC with shutters) 5 electrical LAN ports 1 USB 2.0 Standard A port 2 SFP cage ports (for optical SFP) 4 SFP cage ports (for Ethernet copper or optical SFP)
Connector type	LC optical port RJ-45 LAN port USB 2.0 Standard A port
Power supply	Dual feed, fixed
Input voltage	■ Minimum: –40 V DC ■ Maximum: –57 V DC
Power consumption	■ Typical: 217.32 W (calculated) ■ Maximum: 293.75 W (calculated) Note: The power consumption includes the power consumption for the blade with 2 Fan units installed.

Table 4
Optical Specifications

Property	Optical Power (dBm)		Channels Per Port
	Minimum	Maximum	
C1–C9 IN Optical input power	–7.7	18.7	Maximum 96 channels per port
C1–C9 OUT Optical output power	–9.5	14.6	Maximum 96 channels per port
E1-IN Optical input power	–19.3	13.0	Maximum 96 channels per port
E1-OUT Optical output power	–1.7	23.4	Maximum 96 channels per port

1.8.3 L110 Specifications

Table 5
General Specifications

Property	Specification
Weight	14.771 lb (6.7 kg)
Dimensions (H x W x D)	1.75 x 19.00 x 17.72 in. (44.45 x 483.00 x 450.00 mm) Note: The depth dimension is without the fiber management. The depth dimension with fiber management is less than 23.62 in. (600 mm).
Operating temperature range	5...40 °C
Storage temperature range	–40...70 °C
Short-term temperature range	–5...50 °C (96-hour period, maximum of 15 days/year)
Operating humidity range	5...85%
Short-term humidity range	5...93% (96-hour period, maximum of 15 days/year)
Number of ports/circuits	3 electrical LAN ports 4 SFP cage ports (for Ethernet copper or optical SFP)
Connector type	LC optical port RJ-45 LAN port
Power supply	Dual feed, fixed
Input voltage	–40...–57 V DC

Table 5 (Cont.)
General Specifications

Property	Specification
Power consumption	■ Typical: 177.50 W (calculated) ■ Maximum: 237.30 W (calculated) Note: The power consumption includes the power consumption for the blade with 2 Fan units and 2 PIUs installed.

1.8.4 L160 Specifications

Table 6
Engineering Specifications

Property	Specification
Weight	19.8 lb (9.0 kg)
Dimensions (H x W x D)	1.75 x 19.00 x 17.72 in. (44.45 x 483.00 x 450.00 mm) Note: The depth dimension is without the fiber management. The depth dimension with fiber management is less than 23.62 in. (600 mm).
Operating temperature range	5...40 °C
Storage temperature range	–40...70 °C
Short-term temperature range	–5...50 °C (96-hour period, maximum of 15 days/year)
Operating humidity range	5...85%
Short-term humidity range	5...93% (96-hour period, maximum of 15 days/year)
Input voltage	■ Minimum: –40 V DC ■ Maximum: –57 V DC
Power consumption	■ Typical: 127.4 W (calculated) ■ Maximum: 230.7 W (calculated)

L160 Raman Blade Operational Requirement Considerations

- Minimum outside plant fiber distance is 40 km
- Minimum span loss is 15 dB
- Optical return loss is >17.6 dB

- Total loss (bulk loss) between L160 Raman blade and the outside plan transmission line is <1 dB
- Accumulated loss in the fiber (splices and connectors loss), particularly in the first 20 km, should be kept to a minimum as this could affect Raman gain. Performance is best with the cleanest possible transmission line.

1.8.5

4x16 Splitter/Coupler Specifications

Table 7
General Specifications

Property	Specification
Weight	3.483 lb (1.58 kg)
Dimensions (H x W x D)	1.52 x 6.89 x 9.67 in. (38.7 x 175.1 x 245.5 mm)
Operating temperature range	−5...40 °C
Storage temperature range	−40...70 °C
Humidity range	5...93%
Number of ports/circuits	20 pairs of optical ports (transmit/receive)
Connector type	LC optical port
Power consumption	■ Typical: 33 W (estimated) ■ Maximum: 58 W (estimated)

Table 8
Optical Specifications

Property	Optical Power (dBm)		Comment
	Minimum	Maximum	
C1–C16 IN Optical input power	−6.0	7.6	Maximum 1 channel per port
C1–C16 OUT Optical output power	−10.0	10.1	Maximum 1 channel per port
E1–E4 IN Optical input power	−9.5	5.0	Maximum 16 channels per port
E1–E4 OUT Optical output power	−7.4	10.8	Maximum 16 channels per port

1.8.6

Pluggable Transceiver Modules

The SFP modules provide the following features:

- Flexible, reconfigurable optical or data interfaces
- Optical-to-electrical and electrical-to-optical signal conversion
- Incoming optical signal monitoring and alarm reporting

For more detailed SFP modules information, refer to *1FINITY Pluggable Transceiver Modules Reference Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.8.7

Power Specifications

The L100 Blades operate at –48 V DC with a maximum current draw of 5 A. The UL fuse/breaker recommendation for the L100 Blades is 10 A.

The typical current draw is under 5 A across the input voltage range of the L100 Blades.

The following table describes the power cabling for the L100 Blades.

Table 9
Power Specifications

Connector	Wire Gauge	Wire Color
Main A (–48 V DC)	14 to 12 AWG	Red
GA (Return A)	14 to 12 AWG	Black
Main B (–48 V DC)	14 to 12 AWG	Red/white
GB (Return B)	14 to 12 AWG	Black/white
FG (Frame Ground)	12 AWG	Green/yellow

1.8.8

1FINITY Housing

The 1FINITY Housing can power and support up to six (6) 1FINITY Blades. The power terminals on the Housing supplies redundant A and B –48 V DC power input feeds to each Blade.

For more detailed Housing specification information, refer to *1FINITY Housing User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

1.8.9

C-Band Frequencies and Wavelengths

The standard 96-channel C-Band frequencies and wavelengths, with a 50-GHz grid, for the L100 Blades and PIUs are shown in [Frequency Wavelength](#).

Note: The channel and frequency values are not restricted to the values are shown in [Frequency Wavelength](#). For example, channel 1 can be assigned a center frequency of 196.10 THz with a slot width of 50 GHz and channel 96 can be assigned a center frequency of 196.05 THz with a slot width of 50 GHz. The channel and frequency values shown in the table are used to prevent a traffic hit during a software and Firmware (FW) upgrade.

Note: The L100 Blades support flexible grid.

2

Unit Description

In this chapter:

- 2.1 Technical Specifications
- 2.2 L100 Blade
- 2.3 L110 Blade
- 2.4 L160 Raman Amplifier
- 2.5 Fan Unit
- 2.6 Power Terminal Adapter
- 2.7 CD ROADM Cabling

The L100 Blades provide a low-cost direct access (Colorless (C), Colorless/Directionless (CD), mix configuration ROADM. Up to four degrees of C/CD ROADM are supported with up to 128 add/drop 100 Gb/s or higher (including 150 and 200 Gb/s) channels.

The L100 Series consists of two types of blades:

- L100 (BDL1-3R11) 1x9 WSS ROADM on a Blade (ROB)
- L110 (BDL1-3B11) Optical Mux Base blade

These blades can be mounted in any of the following configurations:

- 19-inch, 2-post rack
- 23-inch, 2-post rack
- 19-inch, 4-post rack
- 23-inch, 4-post rack
- Six-slot Housing

Blades installed in a rack are referred to as the disaggregated option. Blades installed in a six-slot Housing are referred to as the aggregated option.

2.1 Technical Specifications

Table 10
Base System

System Configuration	Local Management Port	Local Communication Network	Front LEDs	Fan Units	Power Supply
Modular 1RU blade	One 10/100 Mb/s Ethernet RJ-45	Four Gigabit Ethernet SFP (T, SX, LX, EX, ZX)	System status, severity, <i>FIND ME</i> , and port	Two replaceable Fan units	Dual fixed power supplies

Table 11
Line Optics

Line Ports Per Blade	Line Rate	Optical Module	Tx Wavelength	Rx Wavelength
1	96 optical channels (100 and 200 Gb/s), 128 optical channels	Fixed, flexible	1528.77...1566.77 nm	1528.77...1566.77 nm

Table 12
Performance Monitoring

Service PMs	OTN PMs	Thresholds and TCA
24-Hour, 15-Minute	n/a	Supported (user assignable)

Table 13
Management

Virtuora NC	WebGUI	CLI	NETCONF/YANG	SNMP	NTP, SFTP, Telnet, and FTP	In-Band Management	Telcordia OSMINE Support
Yes	Yes	Yes	Yes	SNMPv2, SNMPv3	Telnet, FTP and SFTP Release 1.1, NTP	OSC	Telcordia® CLEI™ code

Table 14
Physical Characteristics

Dimensions (H x W x D)	Weight	Rack Compatibility
1.75 x 19.00 x 17.72 in. (44.45 x 483.00 x 450.00 mm) W=19 or 23 in. with mounting rails Note: The depth dimension is without the fiber management. The depth dimension with fiber management is less than 23.62 in. (600 mm).	■ L100 chassis: 8.10 kg (17.857 lb) ■ L110 chassis: 6.08 kg (13.404 lb) ■ L130 chassis: 10.3 kg (21 lb) ■ 4x16 PIU: 1.58 kg (3.483 lb)	19 and 23 in.

Table 15
Operating Environment

Operating Temperature	Short-Term Temperature	Normal Operating Humidity	Short-Term Humidity
5...40 °C	−5...50 °C	5...85% noncondensing	5...93% noncondensing

Table 16
Power

Power Supply	120 V AC	−48 V DC	Power Dissipation
Dual feed, fixed power supplies	No	−40...−57 V DC	■ L100: 217.32 W (typical) ■ L110: 177.5 W (typical) ■ L130: 217.32 W (typical)

Table 17
ROADM Capacity and Functions

Configuration	ROADM Degrees	Topology	Wavelengths	Wavelength Range	Maximum Channels Per Degree	Maximum System Capacity	Span Loss	OSC
CD, direct access (C), mix configuration ROADM	Up to 4	P-to-P, Linear, Ring, Mesh	100G, 200G	1528.77...1566.77 nm	96 (50 GHz ITU-T fixed grid), 128 (37.5 GHz ITU-T flexible grid)	25.6T (200Gx128 channels) per degree	0...35 dB L130 Blade without L160 Blade: 0...34 dB (SMF), 0...35 dB (TWRS/TWC, ELEAF/OLEAF DSF, EXTCORE) L130 Blade with L160 Blade: 16...35.5 dB (SMF), 16...36 dB (TWRS/TWC), 16...36.5 dB (ELEAF/OLEAF, DSF)	OC-3, 100 Mb/s Ethernet, Gigabit Ethernet

2.2

L100 Blade

In this section:

- 2.2.1 Connector Descriptions
- 2.2.2 LED Descriptions

The L100 (BDL1-3R11) 9-port ROADM RoB has nine client ports and one edge port and includes a built-in universal amplifier. Up to four degrees of C/CD ROADM and 128 100 Gb/s or higher (including 150 and 200 Gb/s) channels are supported.

One L100 Blade is required for each degree of ROADM. The following figures show front and rear views of the L100 Blade.

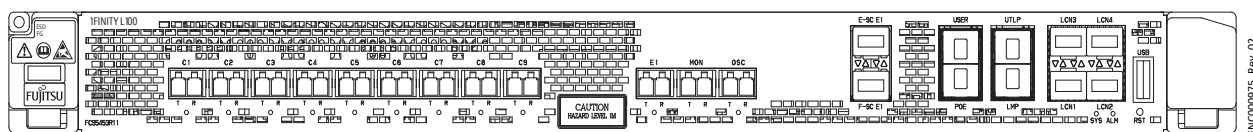


Figure 14
Front View of 1x9 WSS L100 6-port ROADM ROB (Issue 13 and earlier)

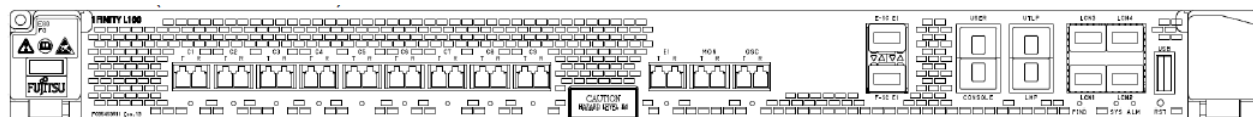


Figure 15
Front View of 1x9 WSS L100 6-port ROADM ROB (Issue 14 and later)

Note: Issue 05 and earlier units have optical port direction labels O/I instead of T/R.

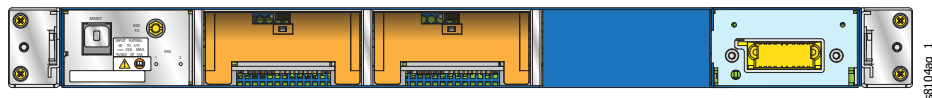


Figure 16
Rear View of L100 Blade 1x9 WSS ROB

Related Links

[Blade and PIU](#)
[L100 Specifications](#)
[L100 Blade LED Indicators](#)
[SFP Module](#)
[Install Blade in Housing](#)
[Remove Blade from Rack](#)
[Remove Blade from Housing](#)
[C-Band Frequencies and Wavelengths](#)

2.2.1

Connector Descriptions

Table 18
Connector and Function

Connector	Connector Type	Location	Function
C1...C9	LC with shutter	Front panel	Pass-through (hub) connection to/from client port on the L100 Blade of other ROADM degree or add/drop connection to/from the 4x16 splitter/coupler edge ports (E1...E4)
E1	LC with shutter	Front panel	Connection to/from line fiber
E-SC-E1	SFP cage populated with Ethernet SFP module	Front panel	100 Mb/s or 1 Gb/s connection to/from OSC port Note: 100 Mb/s SFP port is used for interoperability connection to L130 Blade running with Release 19.1.1.
F-SC-E1	SFP cage populated with OC-3 SFP module	Front panel	155 Mb/s connection to/from OSC port Note: SFP port is used for interoperability connection to FLASHWAVE 9500 MH ROADM.
LCN1	SFP cage	Front panel	100Base-T/1000Base-T electrical or 1000Base-SX/LX10/EX/ZX optical connection to DCN Cloud Note: The LCN1 port cannot be used simultaneously with the UTLP port.
LCN2	SFP cage	Front panel	100Base-T/1000Base-T electrical or 1000Base-SX/LX10/EX/ZX optical connection to DCN Cloud Note: The LCN2 port cannot be used simultaneously with the UTLP port.

Table 18 (Cont.)
Connector and Function

Connector	Connector Type	Location	Function
LCN3	SFP cage	Front panel	100Base-T/1000Base-T electrical or 1000Base-SX/LX10/EX/ZX optical interblade communication connection for blades within one ROADM system Note: The LCN3 port cannot be used as a communication connection to another blade or device outside the virtual NE.
LCN4	SFP cage	Front panel	100Base-T/1000Base-T electrical or 1000Base-SX/LX10/EX/ZX optical interblade communication connection for blades within one ROADM system Note: The LCN4 port cannot be used as a communication connection to another blade or device outside the virtual NE.
LMP	RJ-45	Front panel	100Base-T/1000Base-T LMP connection for craft interface to WebGUI
MAINT	RJ-45	Rear panel	100/1000Base-T connection of maintenance port (for Internal use only)
MON	LC with shutter	Front panel	MON-T and MON-R have 5% (13 dB) splitter
OSC	LC with shutter	Front panel	Optical supervisory channel connection to/from E-SC E1 port or F-SC E1 port
PoE Note: Supported only on L100 Issue 13, FC95453R11-I13, and earlier.	RJ-45	Front panel	10Base-T/100Base-T power over Ethernet (PoE) connection (for future use) Note: The PoE application and cabling is suitable for indoor use only, and is not suitable for outside applications.
CONSOLE Note: Supported only on L100 Issue 14, FC95453R11-I14, and later.	RJ-45	Front panel	RS-232C Serial Interface for CRAFT IF
USB Flash Drive	USB 2.0 Standard A	Front panel	Secondary database
USER	RJ-45	Front panel	User channel supports maximum 10 Mb/s for both OC3 OSC and Ethernet OSC (100M/1GE).
UTLP	RJ-45	Front panel	100Base-T/1000Base-T Note: Connection to L160 Raman blade.

Note: The RST push-switch is used to activate a CPU blade software WARM reset.

2.2.2 LED Descriptions

Table 19
LED and Condition

LED	Color	Condition
ALM	Yellow	Minor alarm
	Red	Critical or major alarm
	Off	No alarm
FIND ME Note: Supported only on L100 Issue 14, FC95453R11-I14, and later.	Off	No power
	Blue	A CLI command has been used to activate this LED
	Off	Default state when powered on
C1...C9	Green	Incoming optical signal operating normally
	Red	Input port down One of the following alarms is present: ■ powerOutOfSpecificationHigh ■ powerOutOfSpecificationLow
	Off	Power off
E1	Green	Incoming optical signal operating normally
	Yellow	Payload Missing Indication (PMI) High Reflection (HREF) Backward Detect Indication-Optical Layer (BDI-O) One of the following alarms is present: ■ postBlockSpanAdjustmentInProgress ■ preBlockSpanAdjustmentInProgress
	Red	WDM Loss of Signal (LOS) WDM power out of specification One of the following alarms is present: ■ omsLossOfSignal ■ omsPowerOutOfSpecificationHigh ■ omsPowerOutOfSpecificationLow ■ oscPowerOutOfSpecificationHigh ■ oscPowerOutOfSpecificationLow ■ otsLossOfSignal ■ otsSpanlossOutOfSpecificationHigh ■ otsSpanlossOutOfSpecificationLow
	Off	Power off

Table 19 (Cont.)
LED and Condition

LED	Color	Condition
E-SC-E1 (on SFP module)	Green	Link up (with or without active data communication)
	Red	OSC LOS OSC fault OSC link down One of the following alarms is present: ■ equipmentRemoved ■ powerOutOfSpecificationHigh ■ powerOutOfSpecificationLow
	Blinking red	Mismatch of equipment and attributes (MEA)
FAN1, FAN2	Green	No error condition (normal operation)
	Red	Error condition (fan failed, fan removed)
	Blinking red	Fan mismatch
	Off	Power off
F-SC-E1	Green	Link up (without data communication)
	Blinking green	Link up under data communication
	Red	OSC LOS OSC fault OSC link down
	Blinking red	MEA
Entity Type	Green	Link up (with or without active data communication)
	Red	Error condition (link down, module failure, module removal) One of the following alarms is present: ■ LOS (eth) ■ POS (port)
	Blinking red	MEA
OSC	Green	Incoming optical signal operating normally
	Red	Input port down One of the following alarms is present: ■ powerOutOfSpecificationHigh ■ powerOutOfSpecificationLow
	Off	Power off

Table 19 (Cont.)
LED and Condition

LED	Color	Condition
SYS	Green	Normal operation
	Yellow	Not ready (warm-up or Software Download (SWDL) in progress) One of the following alarms is present: ■ databaseCorruption ■ databaseLockedDbAlarmPresent ■ databaseLockedDbRestoreInProgress ■ databaseLockedIfViolation ■ databaseLockedShelfProvModePresent ■ databaseLockedSoftwareUpgradeInProgress ■ databaseVersionMismatch ■ equipmentWarmup ■ firmwareBackwardCompatibleAll ■ firmwareDownloadOrActivationFailure ■ firmwareInitInProgress ■ firmwareVersionMismatch ■ shelfProvisioningMode ■ softwareStageInProgress ■ softwareVersionMismatch
	Red	Equipment fault Equipment in power-up state
	Blinking red	MEA
	Off	Power off

2.3 L110 Blade

In this section:

2.3.1 4x16 Splitter/Coupler PIU

The L110 (BDL1-3B11) Optical Mux Base blade houses up to two 4x16 splitter/coupler (PIL1-3PS2) PIUs. Each PIL1-3PS2 provides client ports for up to 16 add/drop 100 Gb/s or higher (including 150 and 200 Gb/s) channels. A filler panel is available for any unfilled PIU slots.

The number of BDL1-3B11 blades that can be installed depends upon the ROADM configuration.

The following table lists the number of BDL1-3B11 blades can be installed for 1- to 4-degree ROADM configurations.

Table 20
Number of Blades for up to Four Degrees of ROADM

Degrees of ROADM	Maximum Number of PIL1-3PS2 PIUs	Add/Drop Channels
1	6	96
2	8	128
3	7	112
4	6	96

The following figures show front and rear views of the Optical Mux Base blade.

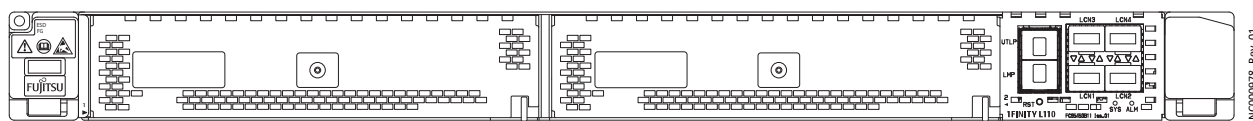


Figure 17
Front View of L110 Optical Mux Base Blade



Figure 18
Rear View of L110 Optical Mux Base Blade

Connector Descriptions

Table 21
Connector and Function

Connector	Connector Type	Location	Function
LCN1	SFP cage	Front panel	100Base-T/1000Base-T electrical or 1000Base-SX/LX10/EX/ZX optical connection to DCN Cloud Note: The LCN1 port cannot be used simultaneously with the UTLP port.
LCN2	SFP cage	Front panel	100Base-T/1000Base-T electrical or 1000Base-SX/LX10/EX/ZX optical connection to DCN Cloud Note: The LCN2 port cannot be used simultaneously with the UTLP port.
LCN3	SFP cage	Front panel	100Base-T/1000Base-T electrical or 1000Base-SX/LX10/EX/ZX optical interblade communication connection for blades within one ROADM system Note: The LCN3 port cannot be used as a communication connection to another blade or device outside the virtual NE.
LCN4	SFP cage	Front panel	100Base-T/1000Base-T electrical or 1000Base-SX/LX10/EX/ZX optical interblade communication connection for blades within one ROADM system Note: The LCN4 port cannot be used as a communication connection to another blade or device outside the virtual NE.
LMP	RJ-45	Front panel	100Base-T/1000Base-T LMP connection for craft interface to WebGUI
MAINT	RJ-45	Rear panel	100/1000Base-T connection of maintenance port
UTLP	RJ-45	Front panel	Future use

Note: The RST push-switch is used to activate a CPU WARM reset.

LED Descriptions

Table 22
LED and Condition

LED	Color	Condition
SYS	Green	Normal operation
	Yellow	Not ready (warm-up or SWDL in progress) One of the following alarms is present: ■ softwareVersionMismatch ■ databaseLockedIfViolation ■ databaseLockedShelfProvModePresent ■ databaseVersionMismatch ■ firmwareBackwardCompatibleAll ■ firmwareDownloadOrActivationFailure ■ firmwareInitInProgress ■ firmwareVersionMismatch ■ shelfProvisioningMode ■ softwareStageInProgress
	Red	Equipment fault Power-up state
	Blinking red	MEA
	Off	Power off
ALM	Yellow	Minor alarm
	Red	Critical or major alarm
	Off	No alarm
Entity Type	Green	Link up (with or without active data communication)
	Red	Error condition (link down, module failure, module removal) One of the following alarms is present: ■ LOS (eth) ■ POS (port)
	Blinking red	MEA
FAN1, FAN2	Green	No error condition (normal operation)
	Red	Error condition (fan failed, fan removed)
	Blinking red	Fan mismatch
	Off	Power off

Related Links

[Blade and PIU](#)
[L110 Specifications](#)
[L110 Blade LED Indicators](#)
[SFP Module](#)
[Install Blade in Housing](#)
[Remove Blade from Rack](#)
[Remove Blade from Housing](#)
[C-Band Frequencies and Wavelengths](#)

2.3.1

4x16 Splitter/Coupler PIU

Up to two 3PS2 splitter/coupler PIUs [PIL1-3PS2] can be installed in each L110 (BDL1-3B11) Optical Mux Base blade. Each PIL1-3PS2 PIU provides client ports for up to 16 add/drop narrowband channels at 100 Gb/s or higher (including 150 and 200 Gb/s). The following figure shows the front view of the PIL1-3PS2.

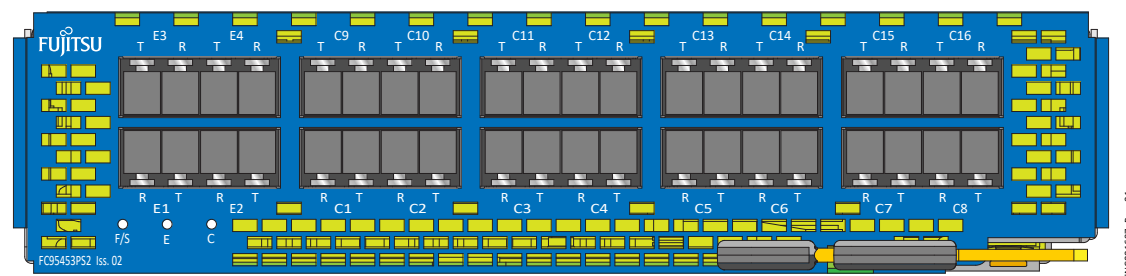


Figure 19
Front View of PIL1-3PS2

Note: Issue 04 and earlier units have optical port direction labels O/I instead of T/R.

Connector Descriptions

Table 23
Connector and Function

Connector	Connector Type	Function
C1...C16	LC with shutter	Client connection to transponder/muxponder
E1	LC with shutter	Add/drop connection to client port (C1...C9) of L100 Blade on ROADM degree 1

Table 23 (Cont.)
Connector and Function

Connector	Connector Type	Function
E2	LC with shutter	Add/drop connection to client port (C1...C9) of L100 Blade on ROADM degree 2
E3	LC with shutter	Add/drop connection to client port (C1...C9) of L100 Blade on ROADM degree 3
E4	LC with shutter	Add/drop connection to client port (C1...C9) of L100 Blade on ROADM degree 4

LED Descriptions

Table 24
LED and Condition

LED	Color	Condition
C	Red	C1...C16 port input down C1...C16 port power out of specification
	Green	Incoming optical signal operating normally
	Off	Power off
E	Red	E1...E4 port input down E1...E4 port power out of specification
	Green	Incoming optical signal operating normally
	Off	Power off
F/S	Red	Equipment fault or equipment in power-up state
	Blinking red	MEA
	Yellow	Not ready (warm-up or SWDL in progress)
	Green	Normal operation
	Off	Power off

Related Links

[Blade and PIU](#)

[4x16 Splitter/Coupler Specifications](#)

[PIL1-3PS2 PIU LED Indicators](#)

[Install Plug-In Unit](#)

[Remove Plug-In Unit](#)

[C-Band Frequencies and Wavelengths](#)

2.4

L160 Raman Amplifier

The L160 Blade is a carrier-grade (C + L) backward Raman optical amplifier blade designed for high OSNR performance and for the specific challenges of long-reach optical network spans. Currently, the L160 Blade can be deployed with the L100 Blade. The operation of the L160 Blade is fully integrated with the L100 Blade, and after it is installed, the L160 Blade functions as part of that NE. Hardware support for (C + L) amplification is built directly into the blade. Currently, C-Band pumps support C-Band applications and configurations. In-service L-Band amplification (C + L) can be added on the same blade in a future release.

The number of L160 Raman blades that can be installed depends upon the ROADM configuration.

Note: L100 Issue 14, FC95453R11-I14, and later blade must be used with L160 Raman blade.

Connector Descriptions

The following figures show front and rear view ports of the Raman blade.

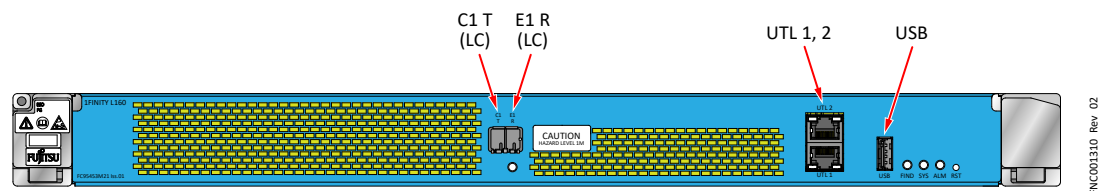


Figure 20
Front View of L160 Raman Blade

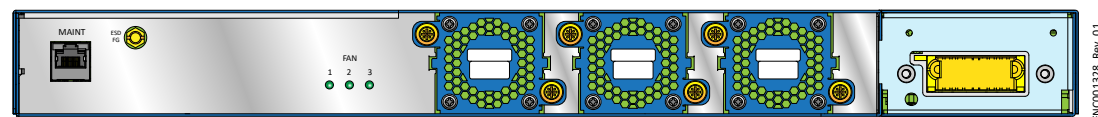


Figure 21
Rear View of L160 Raman Blade

Table 25
Connector and Function

Connector	Connector Type	Location	Function
E1 R	LC with shutter	Front panel	Connection from line Receive fiber
C1 T	LC with shutter	Front panel	Connection to L100 (BDL1-3R11) E1 R port
MAINT	RJ-45	Rear panel	100/1000Base-T connection of maintenance port

Table 25 (Cont.)
Connector and Function

Connector	Connector Type	Location	Function
USB	USB 2.0 Standard A	Front panel	Future
UTL1	RJ-45	Front panel	Connection to/from L100 (BDL1-3R11)
UTL2	RJ-45	Front panel	Future

Note: The RST push-switch is used to activate a CPU WARM reset.

LED Descriptions

The following figures show front and rear view LEDs of the Raman blade.

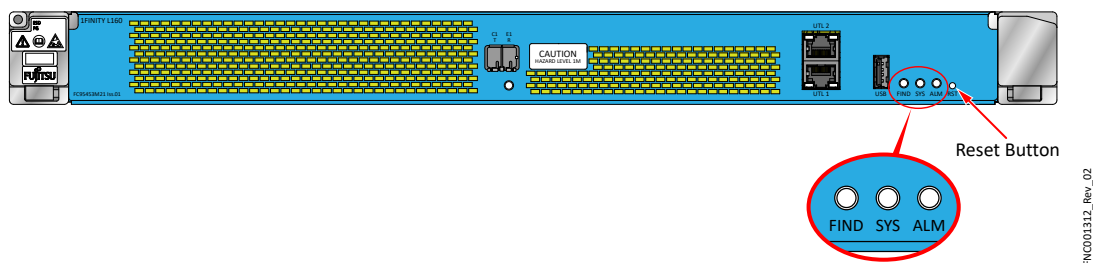


Figure 22
Front View LEDs of L160 Raman Blade

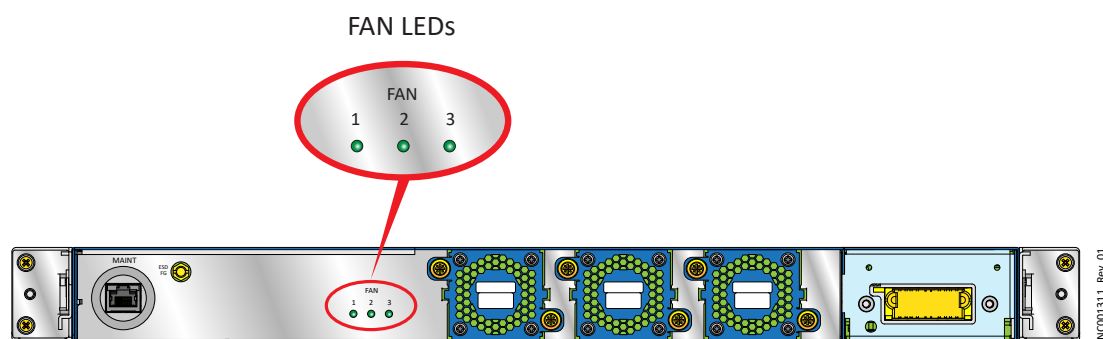


Figure 23
Rear View LEDs of L160 Raman Blade

Table 26
LED and Condition

LED		Color	Condition
SYS		Green	Normal operation
		Yellow	Not ready (warm-up, downloading, installing)
		Red	Equipment fault Power-up state
		Blinking red	MEA
		Blinking yellow	Running status of DHCP
		Off	Power off
ALM		Red	Critical or major alarm
		Yellow	Minor alarm
		Off	No alarm Power off
FIND ME		Blinking blue	CLI command to turn LED on
		Off	Normal operation, power off
E1 R		Green	Incoming optical signal operating normally
		Red	LOS
		Yellow	HREF
		Off	Power off
UTL1,2	ACT	Green	Link up (without data communication/under data communication)
		Off	No link up
	SPEED	Green	Linked 1000Base-T
		Off	Linked 100Base-T
			No link up
FAN 1, FAN 2, FAN 3		Green	No error condition (normal operation)
		Red	Error condition (fan failed, fan removed)
		Blinking red	Fan mismatch
		Off	Power off

L160 Raman Blade Operational Requirement and Considerations

- Minimum outside plant fiber distance is 40 km
- Minimum span loss is 15 dB
- Optical return loss is greater than 17.6 dB
- Total Loss (bulk loss) between L160 Raman blade and the outside plan transmission line is lesser than 1 dB
- Accumulated loss in the fiber (splices and connectors loss), particularly in the first 20 km, should be kept to a minimum as this could affect Raman gain. Performance is best with the cleanest possible transmission line.

2.5 Fan Unit

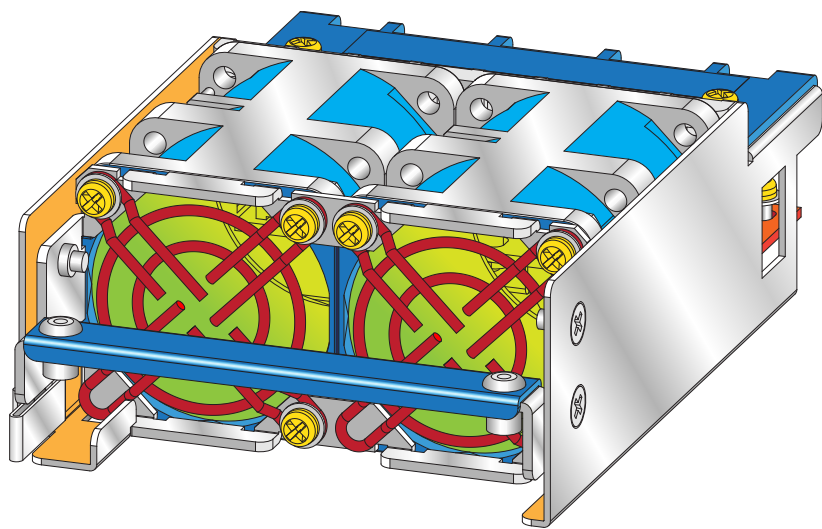
In this section:

- 2.5.1 L100 Fan Unit
- 2.5.2 L160 Fan Unit

2.5.1 L100 Fan Unit

The L100 Blades have field-replaceable Fan units installed at the rear of the blades to provide cooling. Each AC00-F011 Fan unit has two fans that pull cool air in from the front-side of the blade and exhaust the heated air out to the rear-side of the blade. The FAN LED indicators located at the rear-side of the blade are lit green when power (12 V DC) is applied to the blade and the Fan units are operating normally. The Fan units are hot swappable. The following figures show front and rear views of the Fan unit.

Note: Two Fan units are needed for each L100, L110 Blade.



s8102af_1

Figure 24
Front View of Fan Unit

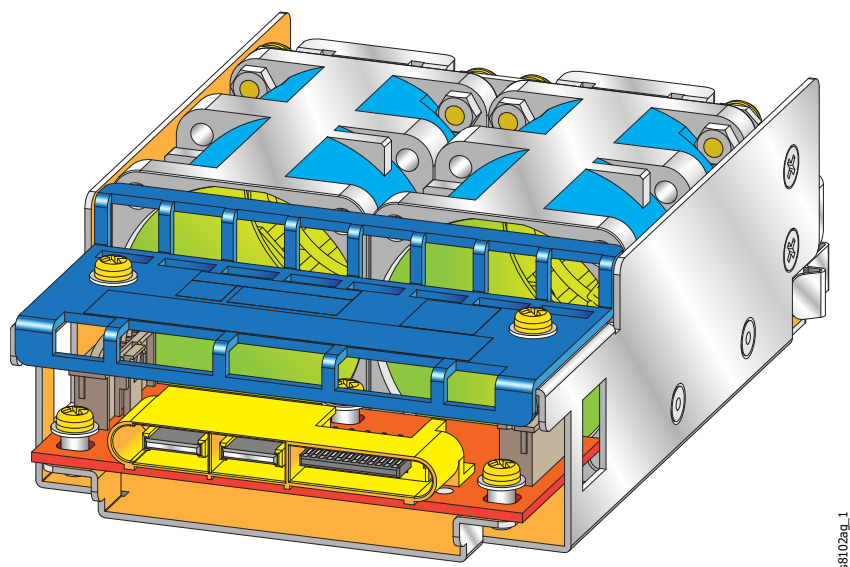


Figure 25
Rear View of Fan Unit

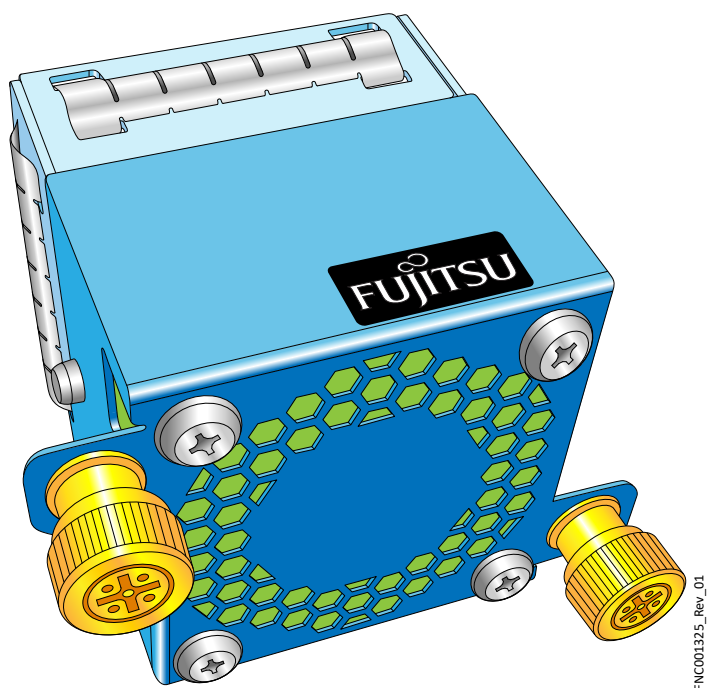
Related Links

[Install Fan Unit](#)
[Remove Fan Unit](#)

2.5.2 L160 Fan Unit

The L160 Blades have field-replaceable Fan units installed at the rear of the blade to provide cooling. Each ACL8-3PC1 Fan unit has one fan to perform the cool-and-exhaust function in the L160 Blade.

Note: L160 Blade comes prepopulated with Fan units. Three Fan units are needed for the L160 Blade.



FNC001325_Rev_01

Figure 26
Front View of Fan Unit

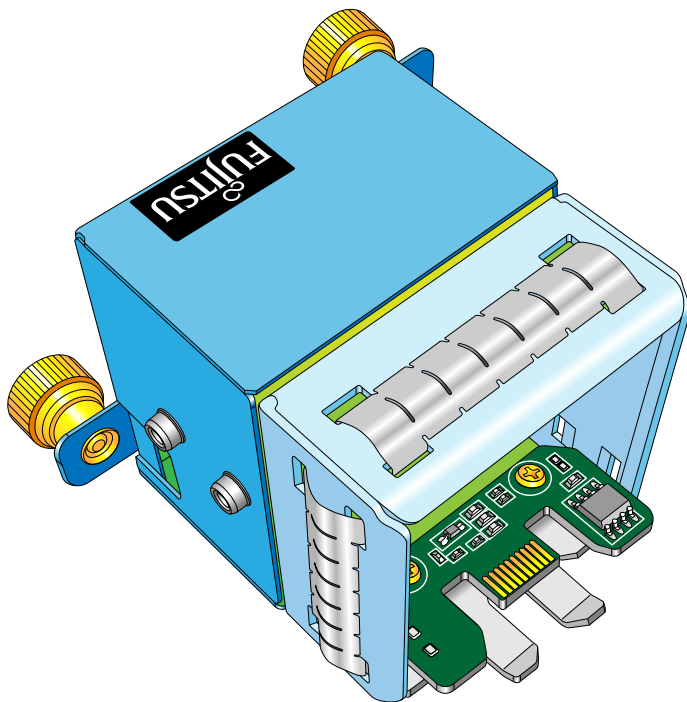


Figure 27
Rear View of Fan Unit

2.6

Power Terminal Adapter

The power terminal adapter is required when the L100 and L160 Blades are installed using the disaggregated option. The power terminal adapter provides dual –48 V DC (A and B) power feeds to the blade. One power terminal adapter is required for each blade. The following figures show front and rear views of the power terminal adapter.

Note: The power terminal adapter is not required if the 1FINITY six-slot Housing is used.

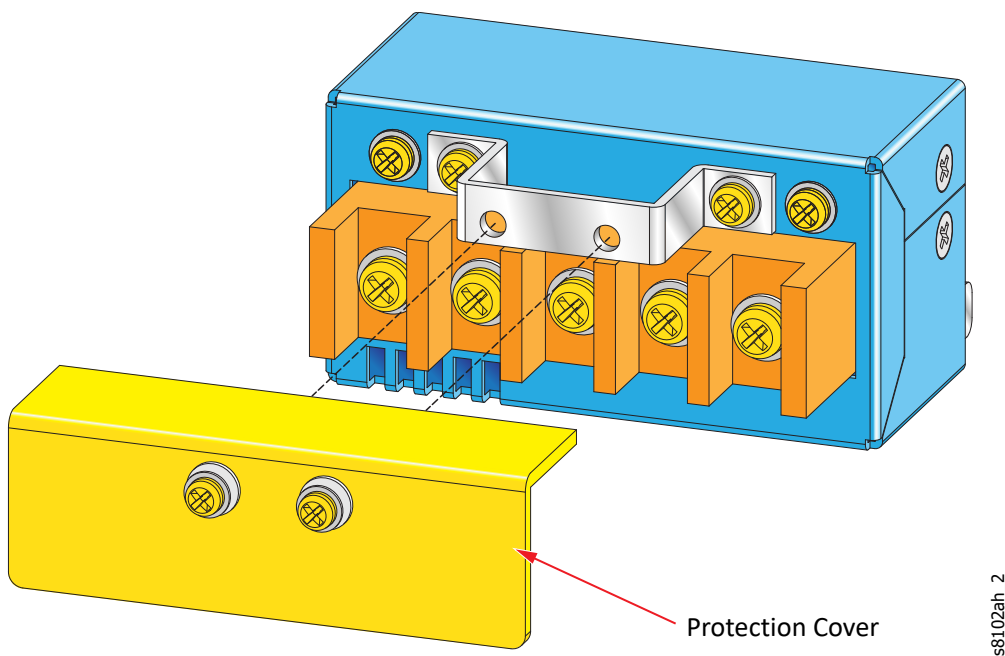


Figure 28
Front View of Power Terminal Adapter

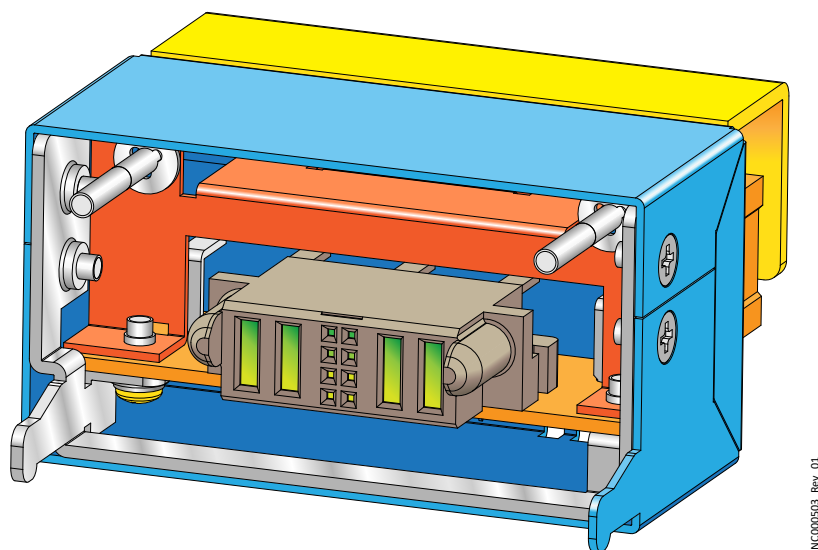


Figure 29
Rear View of Power Terminal Adapter

Related Links

[Install Power Terminal Adapter](#)
[Install Power Cables](#)

2.7

CD ROADM Cabling

1-Degree Small ROADM Cabling

Table 27

1-Degree Small ROADM Add/Drop Connections

From		To		
Degree	L100 Port	L110 Blade	Slot	Port
1	C9	1	1	E1
1	C8	1	2	E1
1	C7	2	1	E1
1	C6	2	2	E1
1	C5	3	1	E1
1	C4	3	2	E1

2-Degree Small ROADM Cabling

Table 28

2-Degree Small ROADM Cabling Pass-Through (Hub) Connections

From		To	
Degree	L100 Port	Degree	L100 Port
1	C1	2	C1

Table 29

2-Degree Small ROADM Add/Drop Connections

From		To		
Degree	L100 Port	L110 Blade	Slot	Port
1	C9	1	1	E1
1	C8	1	2	E1
1	C7	2	1	E1
1	C6	2	2	E1

Table 29 (Cont.)

2-Degree Small ROADM Add/Drop Connections

From		To		
Degree	L100 Port	L110 Blade	Slot	Port
1	C5	3	1	E1
1	C4	3	2	E1
1	C3	4	1	E1
1	C2	4	2	E1
2	C9	1	1	E2
2	C8	1	2	E2
2	C7	2	1	E2
2	C6	2	2	E2
2	C5	3	1	E2
2	C4	3	2	E2
2	C3	4	1	E2
2	C2	4	2	E2

3-Degree Small ROADM Cabling

Table 30

3-Degree Small ROADM Cabling Pass-Through (Hub) Connections

From		To	
Degree	L100 Port	Degree	L100 Port
1	C1	2	C1
1	C2	3	C1
2	C2	3	C2

Table 31
3-Degree Small ROADM Cabling Add/Drop Connections

From		To		
Degree	L100 Port	L110 Blade	Slot	Port
1	C9	1	1	E1
1	C8	1	2	E1
1	C7	2	1	E1
1	C6	2	2	E1
1	C5	3	1	E1
1	C4	3	2	E1
1	C3	4	1	E1
2	C9	1	1	E2
2	C8	1	2	E2
2	C7	2	1	E2
2	C6	2	2	E2
2	C5	3	1	E2
2	C4	3	2	E2
2	C3	4	1	E2
3	C9	1	1	E3
3	C8	1	2	E3
3	C7	2	1	E3
3	C6	2	2	E3
3	C5	3	1	E3
3	C4	3	2	E3
3	C3	4	1	E3

4-Degree Small ROADM Cabling

Table 32
4-Degree Small ROADM Cabling Pass-Through (Hub) Connections

From		To	
Degree	L100 Port	Degree	L100 Port
1	C1	2	C1
1	C2	3	C1
1	C3	4	C1
2	C2	3	C2
2	C3	4	C2
3	C3	4	C3

Table 33
4-Degree Small ROADM Cabling Add/Drop Connections

From		To		
Degree	L100 Port	L110 Blade	Slot	Port
1	C9	1	1	E1
1	C8	1	2	E1
1	C7	2	1	E1
1	C6	2	2	E1
1	C5	3	1	E1
1	C4	3	2	E1
2	C9	1	1	E2
2	C8	1	2	E2
2	C7	2	1	E2
2	C6	2	2	E2
2	C5	3	1	E2
2	C4	3	2	E2
3	C9	1	1	E3
3	C8	1	2	E3

Table 33 (Cont.)

4-Degree Small ROADM Cabling Add/Drop Connections

From		To		
Degree	L100 Port	L110 Blade	Slot	Port
3	C7	2	1	E3
3	C6	2	2	E3
3	C5	3	1	E3
3	C4	3	2	E3
4	C9	1	1	E4
4	C8	1	2	E4
4	C7	2	1	E4
4	C6	2	2	E4
4	C5	3	1	E4
4	C4	3	2	E4

3

Ordering Information

In this chapter:

- 3.1 Blade and PIU
- 3.2 Installation Hardware
- 3.3 SFP Module
- 3.4 LC/UPC Attenuator
- 3.5 Individual Licensed Feature
- 3.6 Software

This chapter describes the L100 Series Blades ordering information.

3.1

Blade and PIU

The following tables list the part numbers of the blades and PIUs.

Table 34
Pre-Load Part Numbers

Part Number	Description
PLL100F1910xA	L100 Blade preloaded with Release 19.1.x software
PLL100F1910xB	L100 Issue 14 and later blade preloaded with Release 19.1.x software
PLL110F1910xA	L110 Blade preloaded with Release 19.1.x software

Table 35
FC Part Numbers

Unit Name	Part Number	Description
BDL1-3M21	FC95453M21	L160 back-propagation Raman amplifier
ACL1-3PF1	FC95453PF1	Filler panel for empty slot in
PIL1-3PS2	FC95453PS2	4x16 splitter/coupler with AMP PIU
ACL8-3PC1	FC95643PC1	Replacement fan unit for L160 Blade

3.2 Installation Hardware

The following table lists accessories used with the L100 Blade.

Table 36
Accessories Equipment

Unit Name	Part Number	Description
AC00-F011	FC9549F011	12 V DC Fan unit
HS00-H011	FC9549H011	1FINITY six-slot 8RU Housing Note: The blade mounting bracket and power terminal module are not required if the 1FINITY six-slot Housing is used.
AC00-HA11	FC9549HA11	1FINITY six-slot Housing air filter
AC00-HC11	FC9549HC11	1FINITY six-slot Housing front cover
AC00-HF11	FC9549HF11	1FINITY six-slot Housing mounting flange for installation in 19-inch, 2-post rack Adjustable Range: 127...128.8 mm
AC00-HF21	FC9549HF21	1FINITY six-slot Housing mounting bracket for installation in 19-inch, 4-post rack Adjustable Range: 566.3...784.1 mm
AC00-HF31	FC9549HF31	1FINITY six-slot Housing mounting bracket for installation in 23-inch, 2-post rack Adjustable Range: 127...128.8 mm
AC00-HP11	FC9549HP11	1FINITY six-slot Housing filler panel
AC00-SD11	FC9549SD11	–48 V DC power terminal module Note: The power terminal module is not required if the 1FINITY six-slot Housing is used.
AC00-SR11	FC9549SR11	Blade mounting bracket for installation in 19-inch, 2-post rack Note: The blade mounting bracket is not required if the 1FINITY six-slot Housing is used. Adjustable Range: 127...180 mm
AC00-SR21	FC9549SR21	Blade mounting bracket for installation in 19-inch, 4-post rack Note: The blade mounting bracket is not required if the 1FINITY six-slot Housing is used. Adjustable Range: 367...460 mm

Table 36 (Cont.)
Accessories Equipment

Unit Name	Part Number	Description
AC00-SR31	FC9549SR31	Blade mounting bracket for installation in 23-inch, 2-post rack Note: The blade mounting bracket is not required if the 1FINITY six-slot Housing is used. Adjustable Range: 127...180 mm
AC00-SR41	FC9549SR41	Blade mounting bracket for installation in 23-inch, 4-post rack Note: The blade mounting bracket is not required if the 1FINITY six-slot Housing is used. Adjustable Range: 367...460 mm
USB Flash Drive	PC15B-0001-B162	USB flash drive formatted with 1FINITY encryption Note: The USB flash drive on the Main blade to remove the standing alarm against the secondary database. The standing alarm remains if USB flash drive is not installed. Note: USB flash drive is not supported for tributary blades.

The following table lists power harnesses used with the L100 Blade.

Table 37
Power Harnesses

Configuration	Part Number	Description	Length
Rack Mounted	PC660-0105-T043	1FINITY blade power harness (A and B feeds and grounds), 12 AWG, 1HL, #8, 90°	10 ft
	PC660-0105-T044	1FINITY blade power harness (A and B feeds and grounds), 12 AWG, 1HL, #8, 90°	25 ft
1FINITY Housing	PC660-0105-T041	1FINITY housing power harness (A and B feeds and grounds), 6 AWG, 2HL, 5/8 in. CD	12 ft
	PC660-0105-T042	1FINITY housing power harness (A and B feeds and grounds), 6 AWG, 2HL, 5/8 in. CD	25 ft
	PC660-0102-T022	1FINITY housing frame ground cable, 6 AWG, Green/Yellow (2) 2HL lug	2 ft

3.3 SFP Module

Table 38
SFP Modules

Unit Name	Part Number	Description	Rate	Target Distance	Connector	Compatible Blades
SFPF-B61F	FC9581B61F	1GE CWDM OSC long-haul 1511 nm SFP module	1000 Mb/s	Spans with loss up to 19 dB	SMF	BDL1-3R11
SFPF-B62F	FC9581B62F	OC-3/100 Mb/s CWDM OSC ultralong-haul 1511 nm SFP module	OC-3, 100Mb/s	Spans with loss up to 35 dB	SMF	BDL1-3R11
SFPF-B65F	FC9581B65F	1GE CWDM OSC ultralong-haul 1511 nm SFP module	1000 Mb/s	Spans with loss up to 27 dB	SMF	BDL1-3R11
SFPF-5210	FC95815210	10/100/1000Base-T SFP module	10 Mb/s, 100 Mb/s, or 1000 Mb/s	100 m	RJ-45	BDL1-3R11 BDL1-3B11
SFPF-5000	FC95815000	1000Base-SX SFP module	1000 Mb/s	300 m (62.5/125 μ m [OM1]) 550 m (50/125 μ m [OM2])	MMF	BDL1-3R11 BDL1-3B11
SFPF-5200	FC95815200	1000Base-LX SFP module	1000 Mb/s	10 km (9/125 μ m)	SMF	BDL1-3R11 BDL1-3B11

3.4

LC/UPC Attenuator

Table 39
LC/UPC Attenuator

Part Number	Description
37-442-612	1 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-461-612	2 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-441-612	3 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-462-612	4 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-440-612	5 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-463-612	6 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-443-612	7 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-464-612	8 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-444-612	9 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-465-612	10 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-424-612	11 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-425-612	12 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-426-612	13 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-427-612	14 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-445-612	15 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-428-612	16 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm

Table 39 (Cont.)
LC/UPC Attenuator

Part Number	Description
37-429-612	17 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-430-612	18 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-431-612	19 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm
37-466-612	20 dB fixed attenuator, LC/UPC, single mode, 55 dB return loss, length: 26.9 mm

3.5

Individual Licensed Feature

L100 Series Licensing Information

Table 40
Part Numbers

Unit Name	Part Number	Description
Fujitsu Lambda, preaudit	RTUL1xxLAM-FJTS-B	L1xx Lambda Fujitsu, 1 per drop, preaudit
Fujitsu Lambda, postaudit	RTUL1xxLAM-FJTS-E	L1xx Lambda Fujitsu, 1 per drop, postaudit
Alien Lambda, preaudit	RTUL1xxLAMINTRP-B	L1xx Lambda interoperability, 1 per drop port, preaudit
Alien Lambda, postaudit	RTUL1xxLAMINTRP-E	L1xx Lambda interoperability, 1 per drop port, postaudit

3.6 Software

L100 Series Software Licensing Information

Table 41
Part Numbers

Unit Name	Part Number	Description
Software	RTUL1xxAR191x	L100 software license Release 19.1.x, 1 per blade

4

Equipment Installation

In this chapter:

- | | | | |
|------|---------------------------------------|------|--|
| 4.1 | Precautions | 4.11 | Rectifier and Blade Wiring Connections |
| 4.2 | Site Selection | 4.12 | Install Power Cables |
| 4.3 | Unpack and Inspect Equipment | 4.13 | Install Fan Unit |
| 4.4 | Adjustable Range of Rack Brackets | 4.14 | Install Plug-In Unit |
| 4.5 | Install 19-Inch, 2-Post Rack Rail Kit | 4.15 | Install SFP Module |
| 4.6 | Install 23-Inch, 2-Post Rack Rail Kit | 4.16 | Install Filler Panel |
| 4.7 | Install 19-Inch, 4-Post Rack Rail Kit | 4.17 | Install Data Communication Network Cabling |
| 4.8 | Install 23-Inch, 4-Post Rack Rail Kit | 4.18 | Install Local Management Port Cabling |
| 4.9 | Install Power Terminal Adapter | 4.19 | Install CD ROADM Cabling |
| 4.10 | Install Blade in Housing | 4.20 | Install L160 Raman Cabling |

The procedures described in this section are used to install L100 Blades, PIUs, pluggable modules, and other required equipment in 19- or 23-inch, 2- or 4-post racks or the 1FINITY Housing.

The following tools and equipment are required:

- Common hand tools
- Digital Multimeter (DMM)
- Terminal lugs (for rack installation)

4.1

Precautions

Before beginning installation or removal, observe the following precautions:



Danger: Never look into the end of an optical cable. Permanent eye damage or blindness can occur if laser radiation is present.



Danger: Never handle exposed fiber with bare hands or touch it to bare skin. Fiber fragments can enter the skin and are difficult to detect and remove. Follow local safety precautions regarding fiber.



Warning: PIUs and pluggable optics can be damaged by ESD during installation, removal, storage, or shipment. Such equipment is stamped with antielectrostatic marks and requires special care in handling and storage. When handling PIUs and pluggable optics, keep in mind that they can be damaged by ESD during installation, removal, or whenever they are contacted so be sure to wear a wrist strap as described in the next Warning.



Warning: Wear an ESD wrist strap with a minimum resistance of 1 megohm that is properly secured to the ESD grounding wrist strap connection socket when handling PIUs and pluggable optics to prevent possible PIU damage. Before using the strap, check it for opens, shorts, and the minimum resistance value. If the strap does not pass these checks, do not use it.

4.2

Site Selection

Inspect the installation area, examine the racks to be used, and ensure that sufficient workspace is available to unpack and inspect the equipment.

Storing Equipment: When equipment is to be kept in storage, the equipment should be left in the shipping containers. The storage bags are coated with a conductive material to protect PIUs from damage by ESD. Inspect the shipping containers for signs of damage. Claims for shipping damage must be filed with the transportation company in accordance with their published procedure. Contact Fujitsu Sales Representative for any shipment questions.

4.3

Unpack and Inspect Equipment



Warning: The blade, PIU, or pluggable module (SFP/SFP+/CFP4) can be damaged by Electrostatic Discharge (ESD) during installation or removal. Ensure that ESD handling precautions are observed, when handling equipment to prevent possible ESD equipment damage.



Warning: Use two hands when handling the blade to avoid damaging the unit.

Step 1

Before unpacking the equipment, inspect the shipping containers for signs of damage.

Note: Claims for shipping damage must be filed with the transportation company in accordance with their published procedure. Contact Fujitsu Sales Representative for any shipment questions.

Step 2

Unpack and carefully remove the equipment from its shipping container and place it on a clean, level, and ESD-safe surface. Use two hands as shown in the following figure to remove a blade unit from container.



Attention: Leave the blade or PIU in the ESD-protective plastic bag until ready for installation.



Figure 30
Removing Blade from Packing Container

Step 3

Inspect the equipment to ensure that it is free of any defects, damage, or other debris. If the equipment is damaged, follow local practices/procedures for repair or replacement of the damaged unit, as applicable.

Note: Use the checklist accompanying each shipment (if applicable) to verify that all items are present with the shipment.

Step 4

Record the serial number and issue number of the equipment in accordance with local record-keeping procedures, if necessary.

✓ **This task is complete.**

Additional Instructions:

Perform the same procedure for each blade, PIU, or pluggable module (SFP/SFP+/CFP4) being installed.

4.4

Adjustable Range of Rack Brackets

The following table lists the adjustable range of the rack brackets.

Note: The adjustable range is the width between the front and rear flanges of the bracket.

Table 42
Adjustable Range of Rack Brackets

Configuration	Part Number	Description	Adjustable Range
Housing (aggregated)	FC9549HF11	Flange for 6-blade Housing, 19-inch, 2-post rack	127.0...128.8 mm
	FC9549HF21	Flange for 6-blade Housing, 19-inch, 4-post rack	566.3...784.1 mm
	FC9549HF31	Flange for 6-blade Housing, 23-inch, 2-post rack	127.0...128.8 mm
Standalone (disaggregated)	FC9549SR11	Mount rail for standalone blade, 19-inch, 2-post rack	127.0...180.0 mm
	FC9549SR21	Mount rail for standalone blade, 19-inch, 4-post rack	367.0...460.0 mm
	FC9549SR31	Mount rail for standalone blade, 23-inch, 2-post rack	127.0...180.0 mm
	FC9549SR41	Mount rail for standalone blade, 23-inch, 4-post rack	367.0...460.0 mm

4.5

Install 19-Inch, 2-Post Rack Rail Kit

Prerequisites:

- The 19-inch, 2-post EIA-310 compliant rack is installed according to local standards.
- Hex-nut driver set and socket wrench set
- Screwdriver set (including straight-slot and Phillips® head)
- Small level

Step 1

Determine the left and right bracket for installation.

Note: Each bracket is labeled left or right as shown in the following figure.

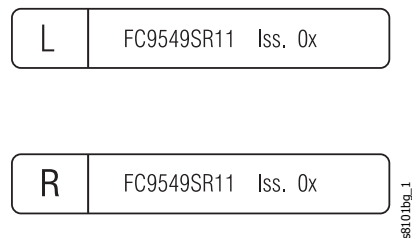


Figure 31

Left and Right Bracket Labels

Step 2

Loosen the flange nuts on both the left and right brackets, as shown in the following figure.

Note: Do not completely remove the nuts from the flange.

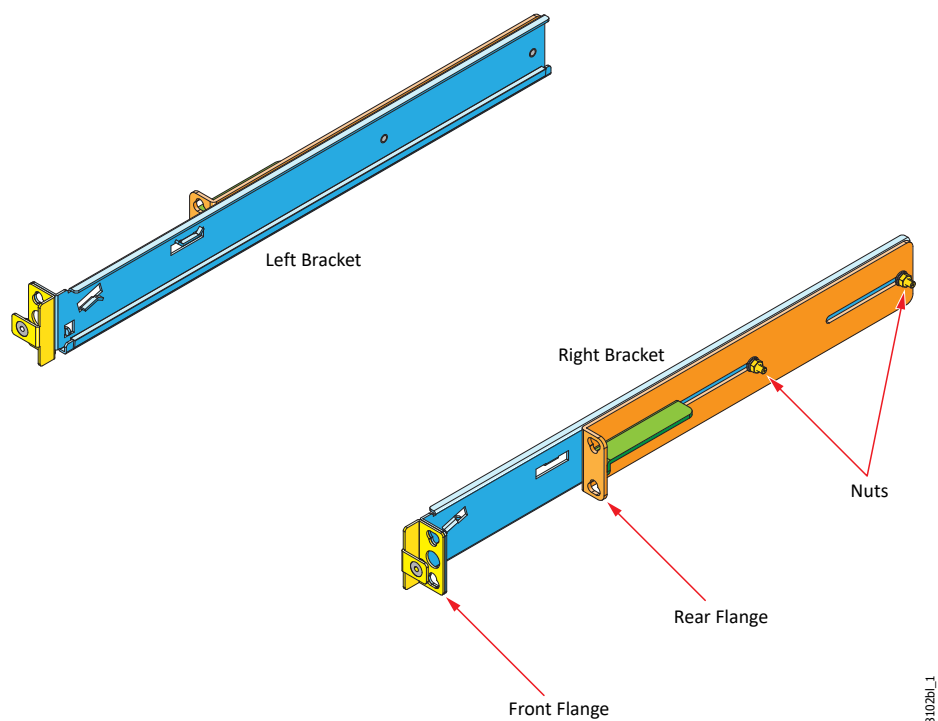


Figure 32
Loosening Flange Nuts 19-Inch, 2-Post Rack Bracket

Step 3

Adjust the width between the front flange and rear flange (on both brackets) to the width of the rack.

Step 4

Use the front flange on each bracket to attach the brackets to the front of the rack at the required position, as shown in the following figure.

Note: Do not completely tighten the bolts, the bolts are completely tightened later in the procedure.

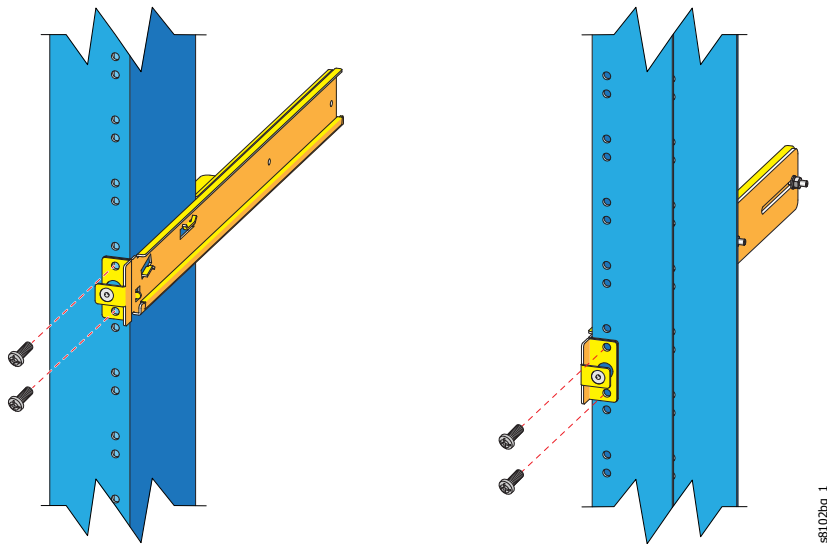


Figure 33
Attaching Front Flange 19-Inch, 2-Post Rack Bracket

Step 5

Use the rear flange on each bracket to attach the brackets to the rear of the rack at the required position, as shown in the following figure.

Note: Do not completely tighten the bolts, the bolts are completely tightened later in the procedure.

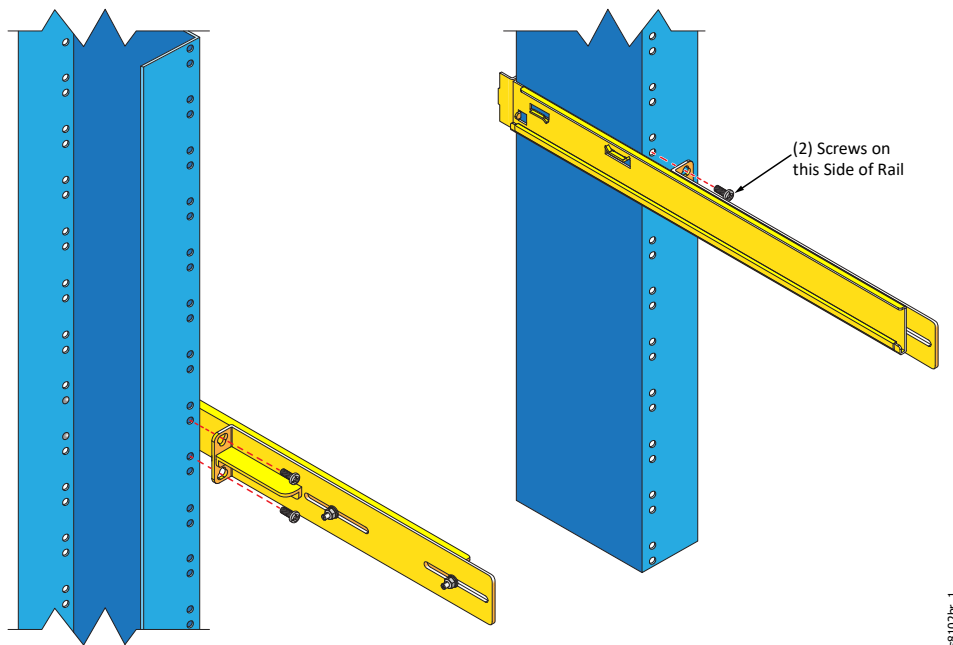


Figure 34
Attaching Rear Flange 19-Inch, 2-Post Rack Bracket

Step 6

Verify front and rear flanges on each bracket are installed in the correct rack position.

Step 7

Verify the installed brackets on the 19-inch, 2-post rack are spaced 440 mm apart and that brackets are level, as shown in the following figure.

Note: Confirm the bracket spacing is correct at the front and rear of the rack. Use the bolts on the flanges to adjust the spacing and tilt of the brackets.

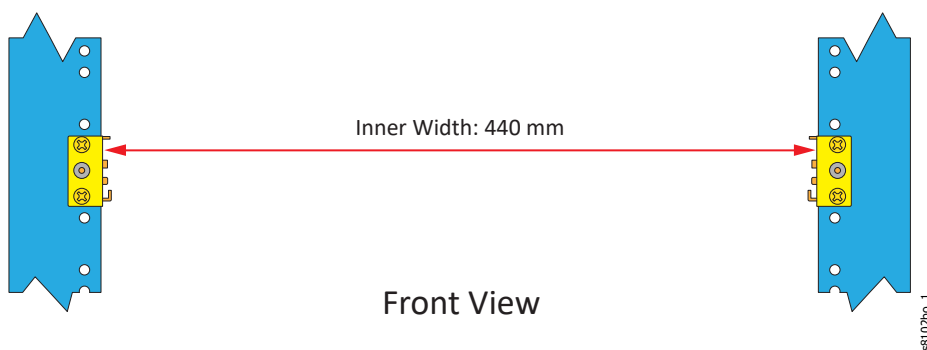


Figure 35

Bracket Spacing 19-Inch, 2-Post Rack

Step 8

Tighten all bolts completely on each bracket.

Note: Each bracket is mounted to the rack using four bolts, two in the front and two in the rear.

Step 9

Tighten the flange nuts on both the left and right brackets, as shown in the following figure.

Note: Do not use excessive force to tighten the nuts.

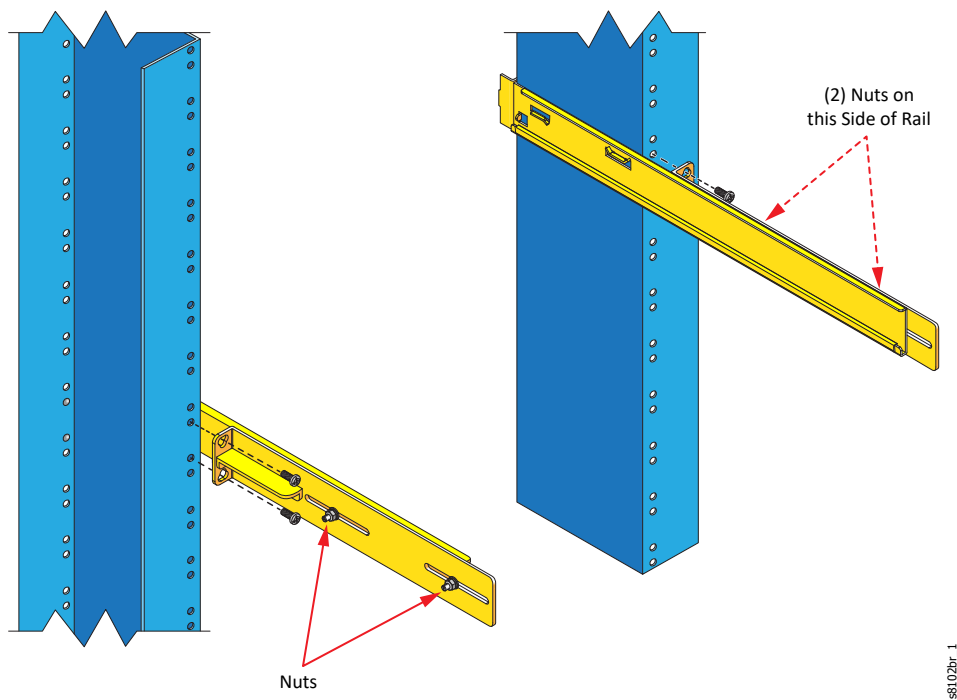


Figure 36
Tightening Flange Nuts 19-Inch, 2-Post Rack Bracket

The following figure shows the installed 19-inch, 2-post rack brackets.

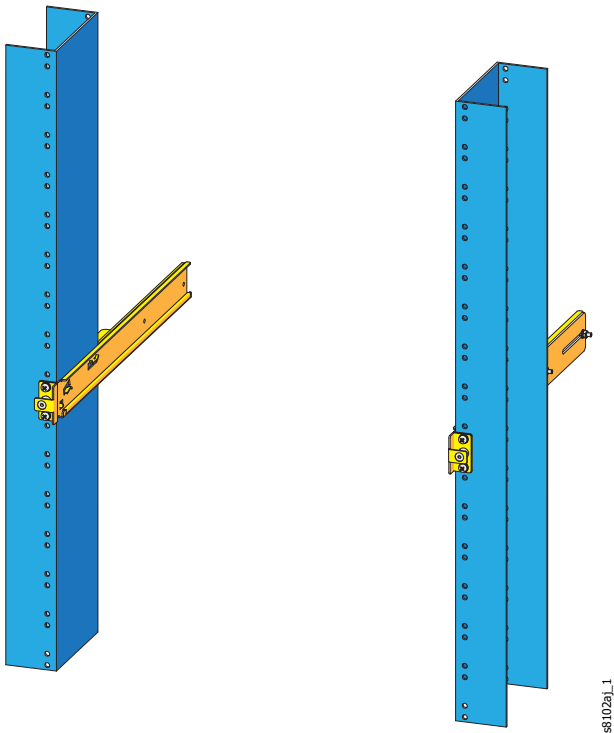


Figure 37
Installed 19-Inch, 2-Post Rack Brackets

Step Result:

The blade can be installed in the rack.

✓ **This task is complete.**

4.6

Install 23-Inch, 2-Post Rack Rail Kit

Prerequisites:

- The 23-inch, 2-post EIA-310 compliant rack is installed according to local standards.
- Hex-nut driver set and socket wrench set
- Screwdriver set (including straight-slot and Phillips head)
- Small level

Step 1

Determine the left and right bracket for installation.

Note: Each bracket is labeled left or right as shown in the following figure.

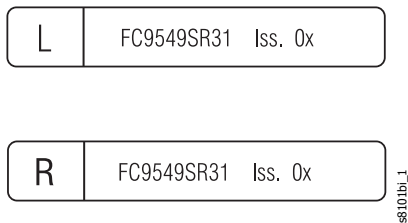


Figure 38

Left and Right Bracket Labels

Step 2

Loosen the flange nuts on both the left and right brackets, as shown in the following figure.

Note: Do not completely remove the nuts from the flange.

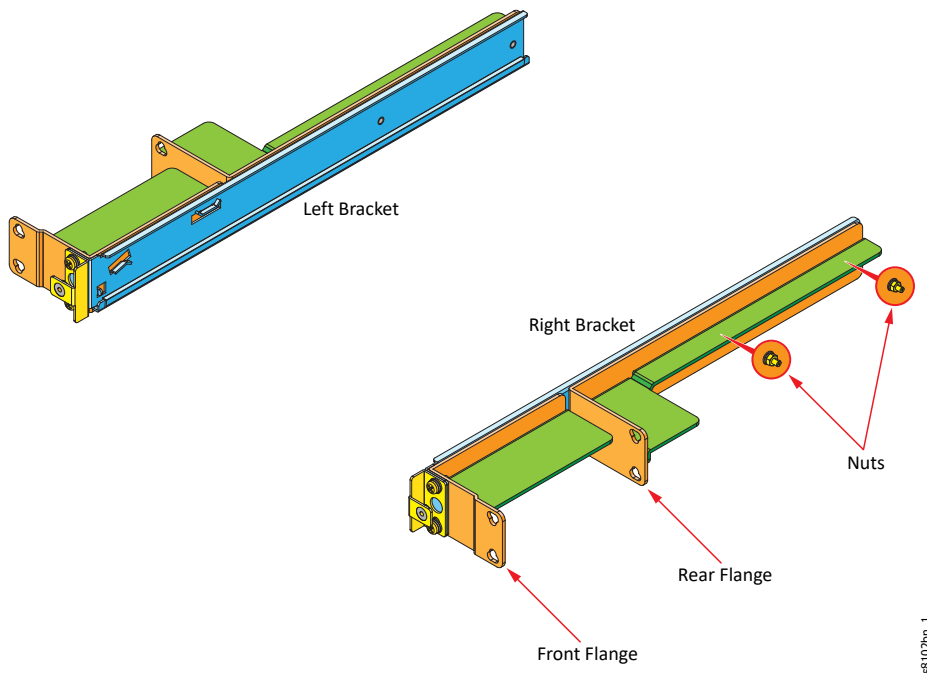


Figure 39
Loosening Flange Nuts 23-Inch, 2-Post Rack Bracket

Step 3

Adjust the width between the front flange and rear flange (on both brackets) to the width of the rack.

Step 4

Use the front flange on each bracket to attach the brackets to the front of the rack at the required position, as shown in the following figure.

Note: Do not completely tighten the bolts, the bolts are completely tightened later in the procedure.

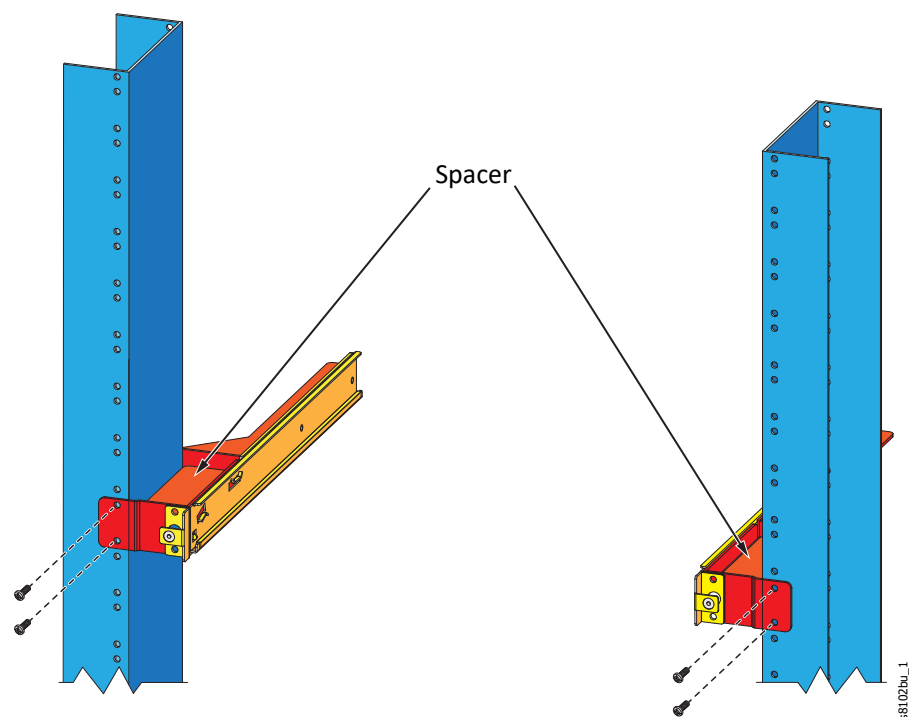


Figure 40
Attaching Front Flange 23-Inch, 2-Post Rack Bracket

Step 5

Use the rear flange on each bracket to attach the brackets to the rear of the rack at the required position, as shown in the following figure.

Note: Do not completely tighten the bolts, the bolts are completely tightened later in the procedure.

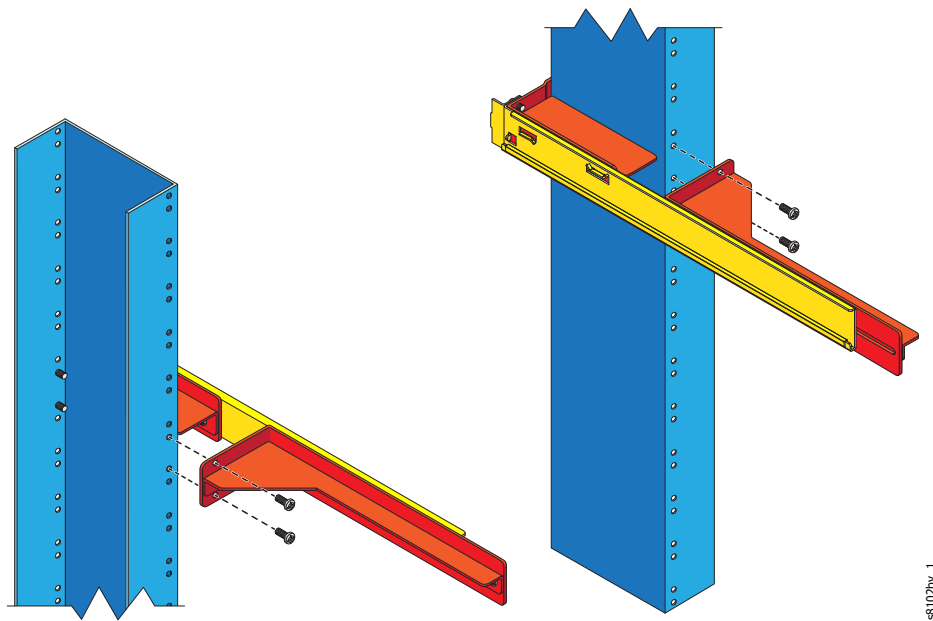


Figure 41
Attaching Rear Flange 23-Inch, 2-Post Rack Bracket

Step 6

Verify front and rear flanges on each bracket are installed in the correct rack position.

Step 7

Verify the installed brackets on the 23-inch, 2-post rack are spaced 440 mm apart and that brackets are level, as shown in the following figure.

Note: Confirm the bracket spacing is correct at the front and rear of the rack. Use the bolts on the flanges to adjust the spacing and tilt of the brackets.



Figure 42
Bracket Spacing 23-Inch, 2-Post Rack

Step 8

Tighten all bolts completely on each bracket.

Note: Each bracket is mounted to the rack using four bolts, two in the front and two in the rear.

Step 9

Tighten the flange nuts on both the left and right brackets, as shown in the following figure.

Note: Do not use excessive force to tighten the nuts.

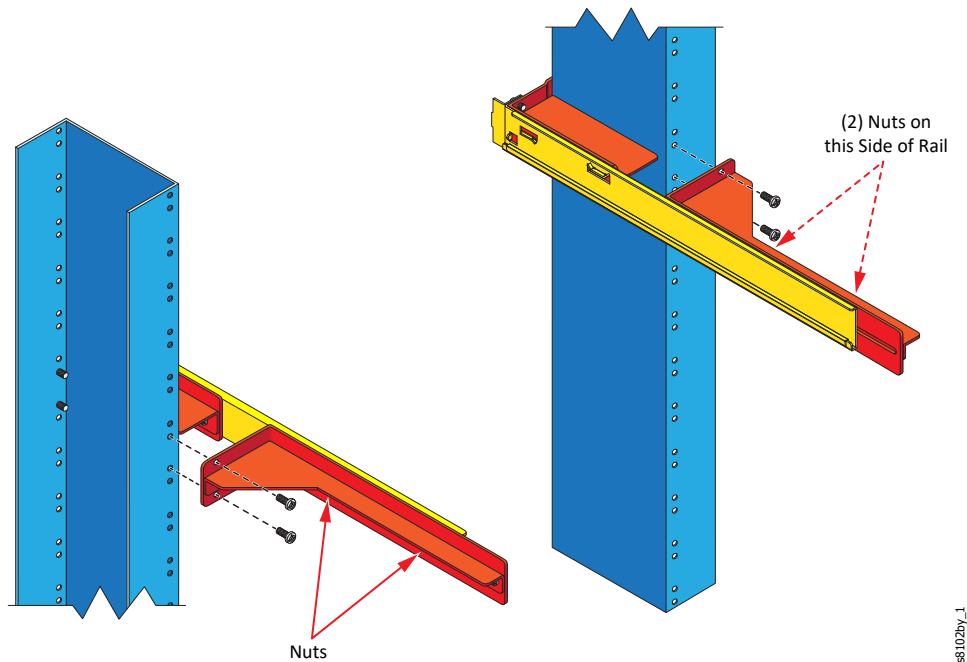


Figure 43
Tightening Flange Nuts 23-Inch, 2-Post Rack Bracket

The following figure shows the installed 23-inch, 2-post rack brackets.

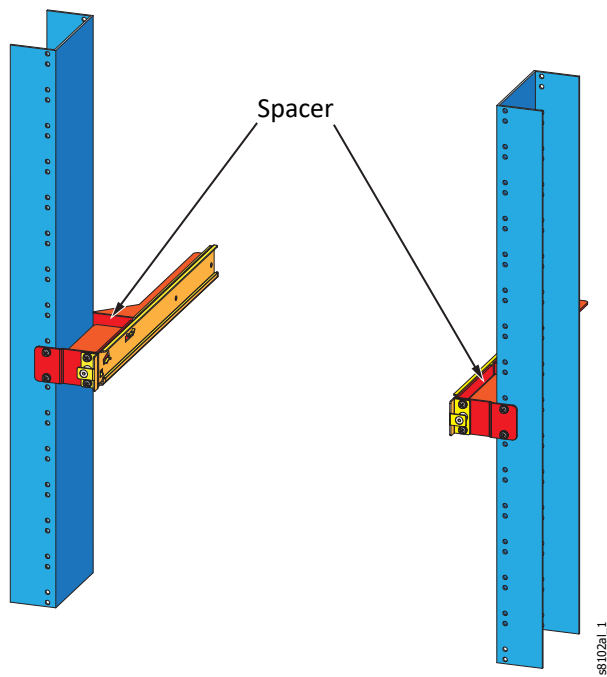


Figure 44
Installed 23-Inch, 2-Post Rack Brackets

Step Result:

The blade can be installed in the rack.

✓ **This task is complete.**

4.7

Install 19-Inch, 4-Post Rack Rail Kit

Prerequisites:

- The 19-inch, 4-post EIA-310 compliant rack is installed according to local standards.
- Hex-nut driver set and socket wrench set
- Screwdriver set (including straight-slot and Phillips head)
- Small level

Step 1

Determine the left and right bracket for installation.

Note: Each bracket is labeled left or right as shown in the following figure.

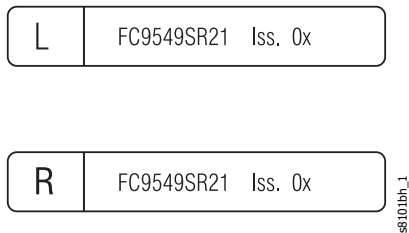


Figure 45
Left and Right Bracket Labels

Step 2

Loosen the flange screws on both the left and right brackets, as shown in the following figure.

Note: Do not completely remove the screws from the flange.

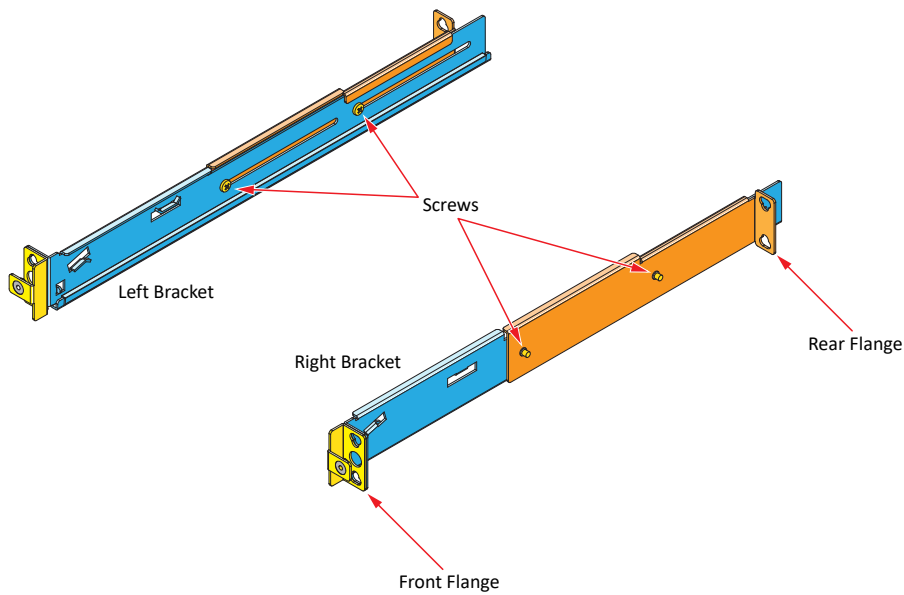


Figure 46
Loosening Flange Screws 19-Inch, 4-Post Rack Bracket

Step 3

Adjust the width between the front flange and rear flange (on both brackets) to the width of the rack.

Step 4

Use the front flange on each bracket to attach the brackets to the front of the rack at the required position, as shown in the following figure.

Note: Do not completely tighten the bolts, the bolts are completely tightened later in the procedure.

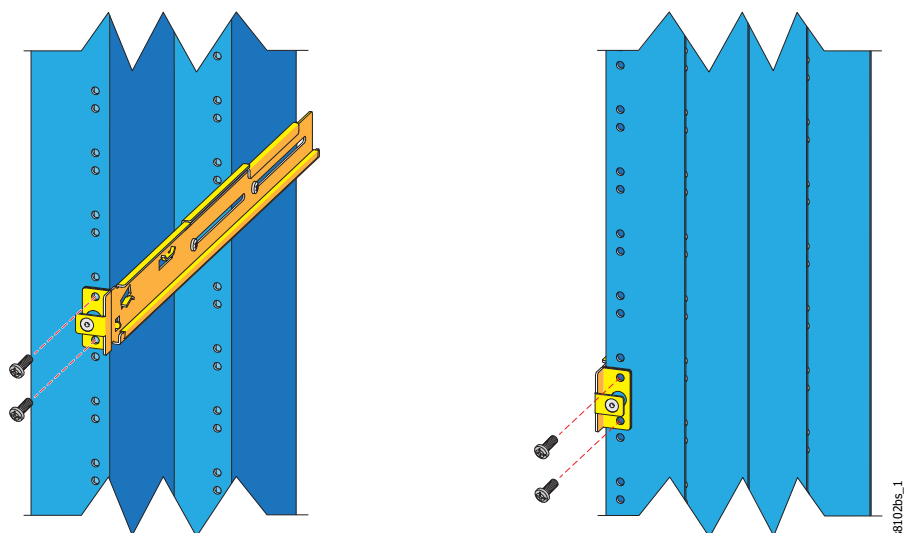


Figure 47
Attaching Front Flange 19-Inch, 4-Post Rack Bracket

Step 5

Use the rear flange on each bracket to attach the brackets to the rear of the rack at the required position, as shown in the following figure.

Note: Do not completely tighten the bolts, the bolts are completely tightened later in the procedure.

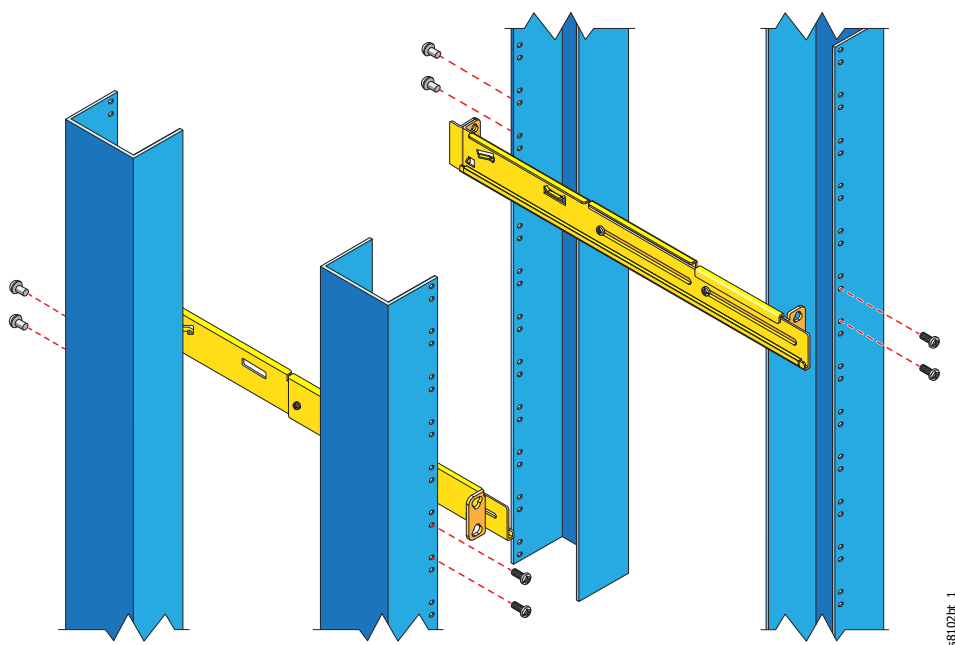


Figure 48
Attaching Rear Flange 19-Inch, 4-Post Rack Bracket

Step 6

Verify front and rear flanges on each bracket are installed in the correct rack position.

Step 7

Verify the installed brackets on the 19-inch, 4-post rack are spaced 440 mm apart and that brackets are level, as shown in the following figure.

Note: Confirm the bracket spacing is correct at the front and rear of the rack. Use the bolts on the flanges to adjust the spacing and tilt of the brackets.

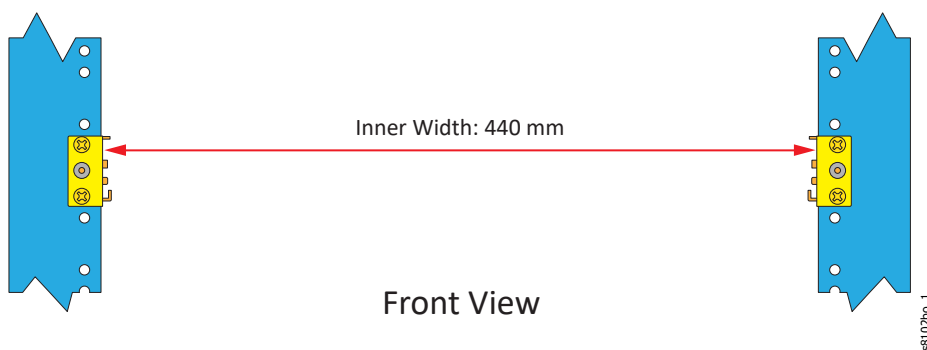


Figure 49

Bracket Spacing 19-Inch, 4-Post Rack

Step 8

Tighten all bolts completely on each bracket.

Note: Each bracket is mounted to the rack using four bolts, two in the front and two in the rear.

Step 9

Tighten the flange screws on both the left and right brackets, as shown in the following figure.

Note: Do not use excessive force to tighten the nuts.

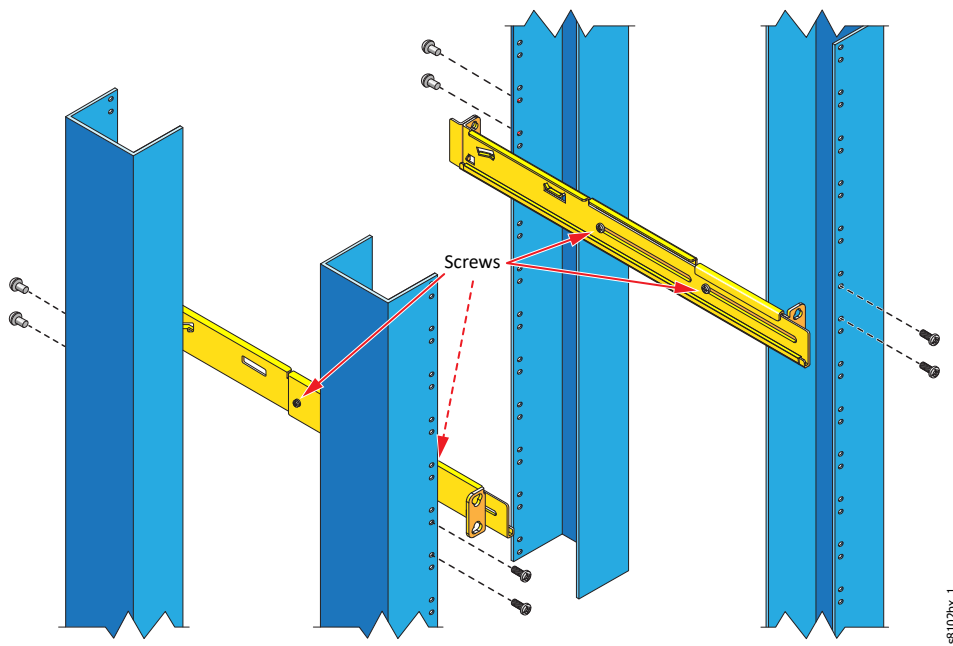


Figure 50
Tightening Flange Screws 19-Inch, 4-Post Rack Bracket

The following figure shows the installed 19-inch, 4-post rack brackets.

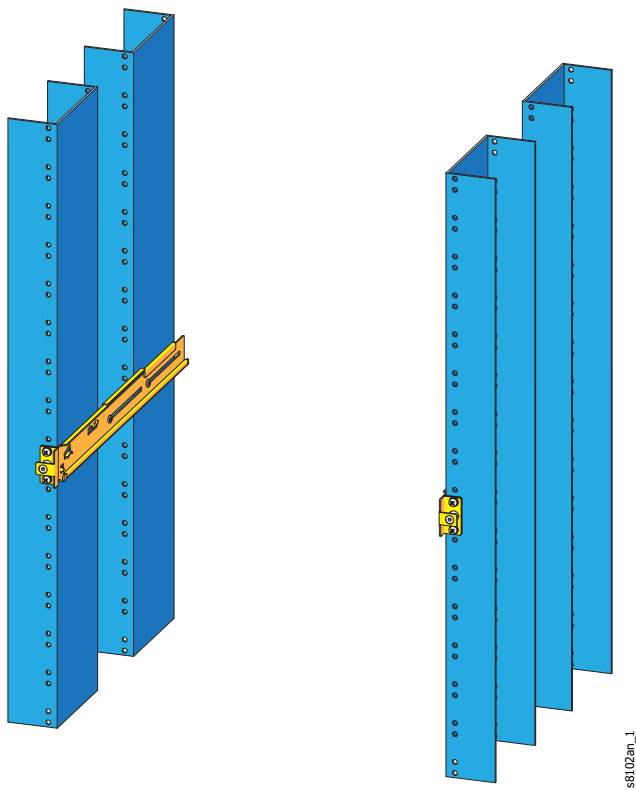


Figure 51
Installed 19-Inch, 4-Post Rack Brackets

Step Result:

The blade can be installed in the rack.

✓ **This task is complete.**

4.8

Install 23-Inch, 4-Post Rack Rail Kit

Prerequisites:

- The 23-inch, 4-post EIA-310 compliant rack is installed according to local standards.
- Hex-nut driver set and socket wrench set
- Screwdriver set (including straight-slot and Phillips head)
- Small level

Step 1

Determine the left and right bracket for installation.

Note: Each bracket is labeled left or right as shown in the following figure.

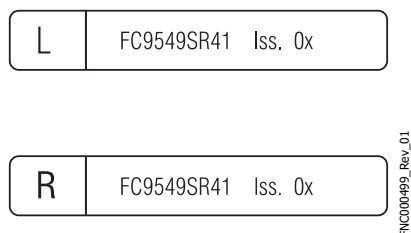


Figure 52

Left and Right Bracket Labels

Step 2

Loosen the flange screws on both the left and right brackets, as shown in the following figure.

Note: Do not completely remove the screws from the flange.

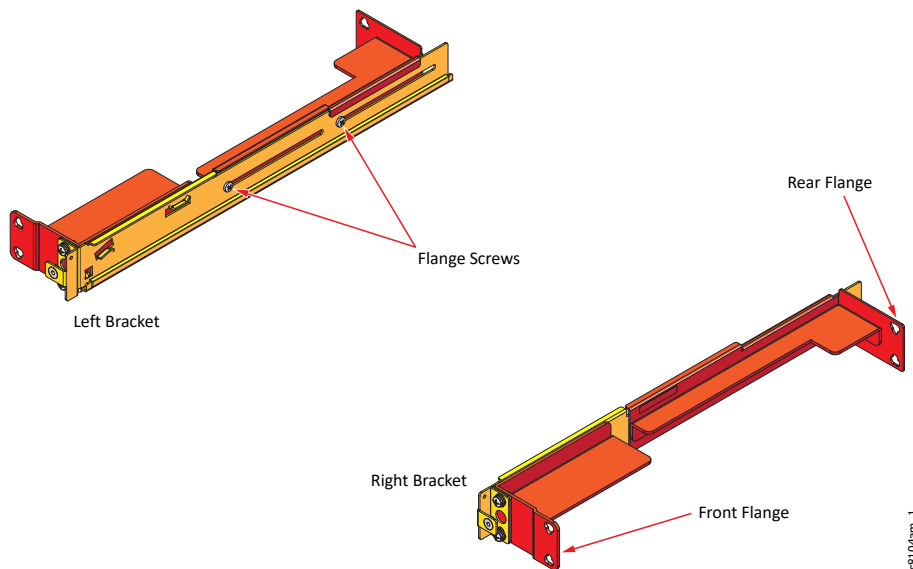


Figure 53
Loosening Flange Screws 23-Inch, 4-Post Rack Bracket

Step 3

Adjust the width between the front flange and rear flange (on both brackets) to the width of the rack.

Step 4

Use the front flange on each bracket to attach the brackets to the front of the rack at the required position, as shown in the following figure.

Note: Do not completely tighten the bolts, the bolts are completely tightened later in the procedure.

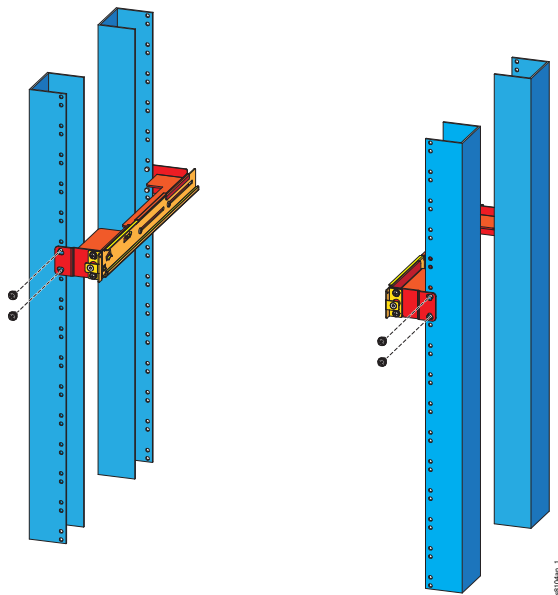


Figure 54
Attaching Front Flange 23-Inch, 4-Post Rack Bracket

Step 5

Use the rear flange on each bracket to attach the brackets to the rear of the rack at the required position, as shown in the following figure.

Note: Do not completely tighten the bolts, the bolts are completely tightened later in the procedure.

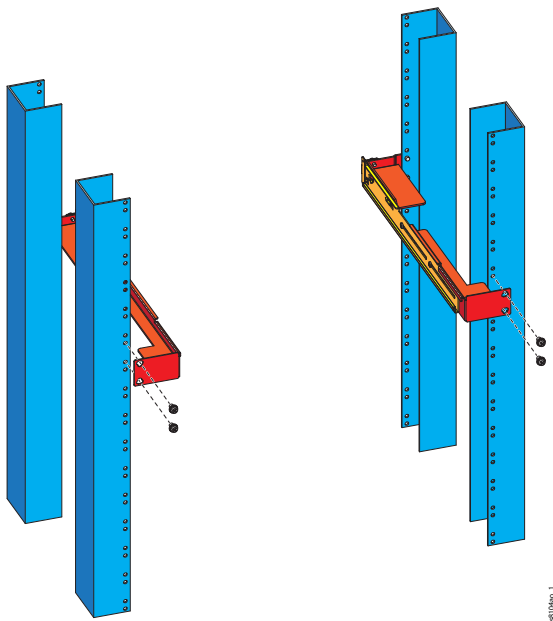


Figure 55
Attaching Rear Flange 23-Inch, 4-Post Rack Bracket

Step 6

Verify front and rear flanges on each bracket are installed in the correct rack position.

Step 7

Verify the installed brackets on the 23-inch, 4-post rack are spaced 440 mm apart and that brackets are level, as shown in the following figure.

Note: Confirm the bracket spacing is correct at the front and rear of the rack. Use the bolts on the flanges to adjust the spacing and tilt of the brackets.



Figure 56
Bracket Spacing 23-Inch, 4-Post Rack

Step 8

Tighten all bolts completely on each bracket.

Note: Each bracket is mounted to the rack using four bolts, two in the front and two in the rear.

Step 9

Tighten the flange screws on both the left and right brackets, as shown in the following figure.

Note: Do not use excessive force to tighten the nuts.

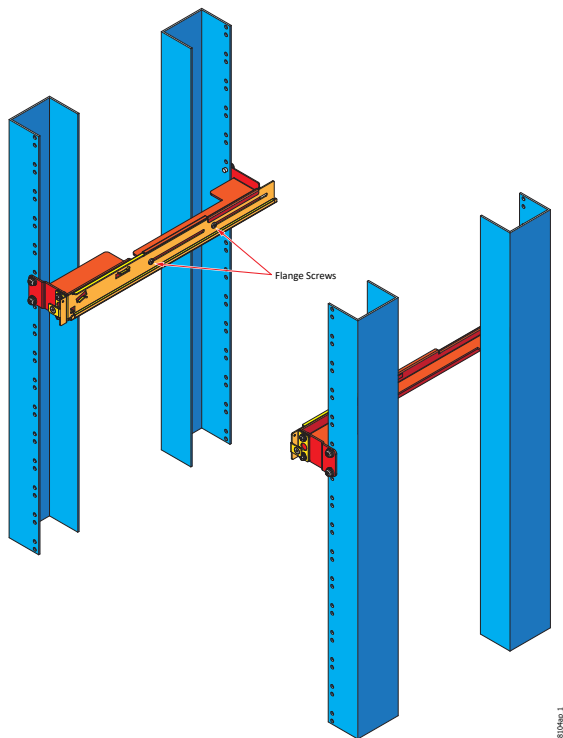


Figure 57
Tightening Flange Screws 23-Inch, 4-Post Rack Bracket

The following figure shows the installed 23-inch, 4-post rack brackets.

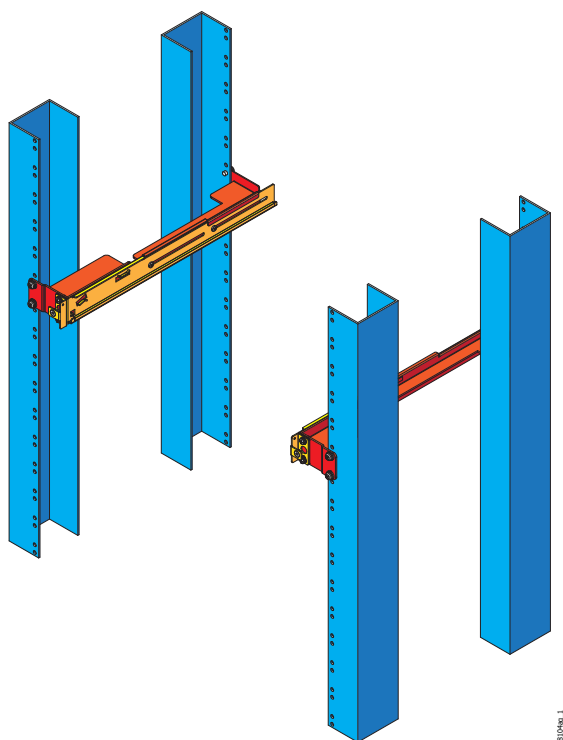


Figure 58
Installed 23-Inch, 4-Post Rack Brackets

Step Result:

The blade can be installed in the rack.

✓ **This task is complete.**

4.9

Install Power Terminal Adapter

Prerequisites:

- Screwdriver set (including straight-slot and Phillips head)
- The blade to be installed in a rack



Attention: The power terminal adapter is not used with the 1FINITY Housing option.

Note: Fujitsu recommends installing the power terminal adapter on the blade before installing the blade into the rack (2-post or 4-post).

Step 1

Remove the protection cover from the power terminal adapter.

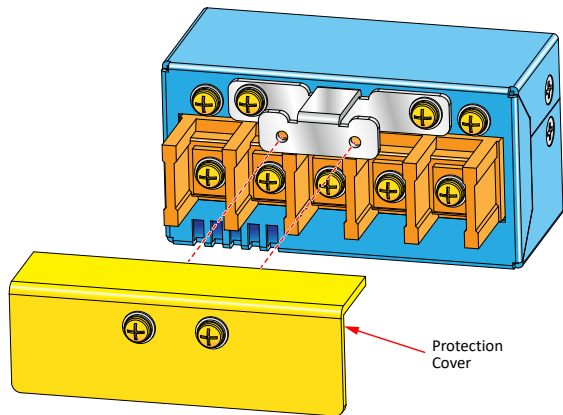


Figure 59
Removing Protection Cover

Step 2

Inspect the power terminal adapter for possible physical damage and note the connecting hooks on the rear of the adapter.

Note: If the power terminal adapter is observed to be damaged, set the adapter aside for repair and return paperwork processing. Obtain another power terminal adapter for installation.

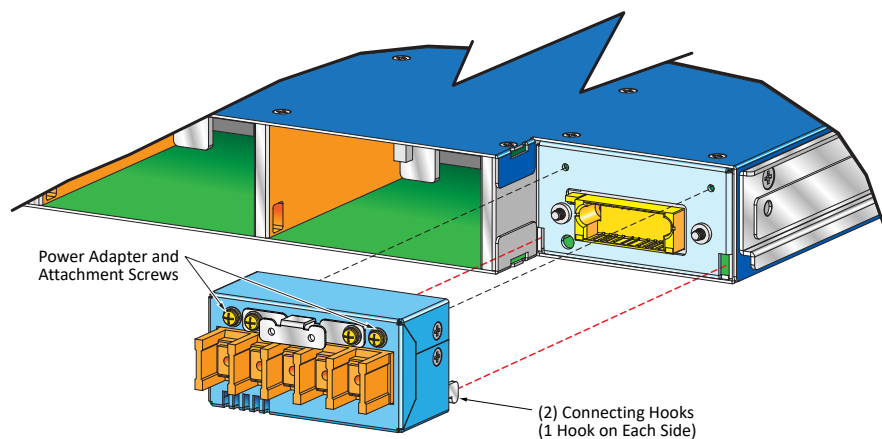


Figure 60
Power Terminal Adapter Description and Placement Location

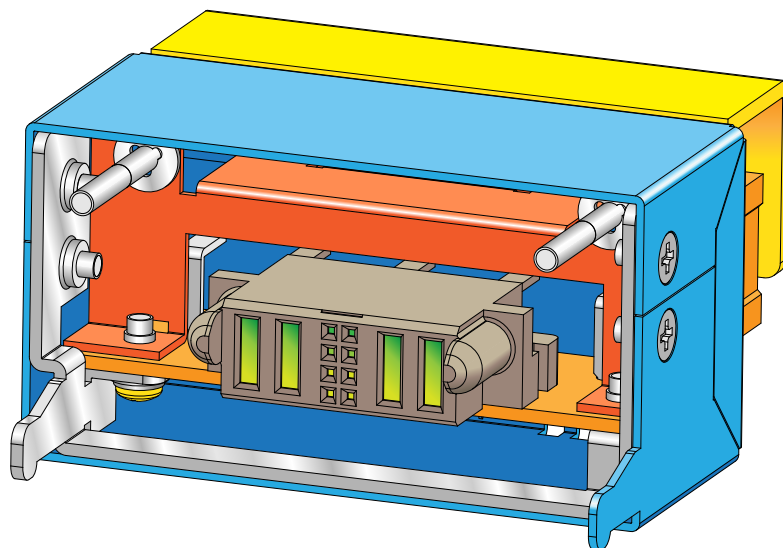


Figure 61
Rear View of Power Terminal Adapter

Step 3

Attach the power terminal adapter onto the rear of the blade on the right-hand side by performing the following substeps:

- a) Insert the hooks on the power terminal adapter into the holes on the rear of the blade.



Figure 62
Hooking Power Terminal Adapter

- b) Push the power terminal adapter until it meets the rear of the blade.

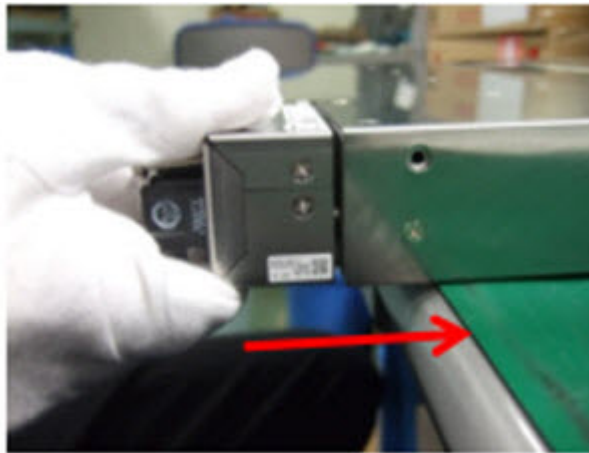


Figure 63
Pushing Power Terminal Adapter

- c) Press down slightly on the power terminal adapter to properly seat the adapter on the blade.

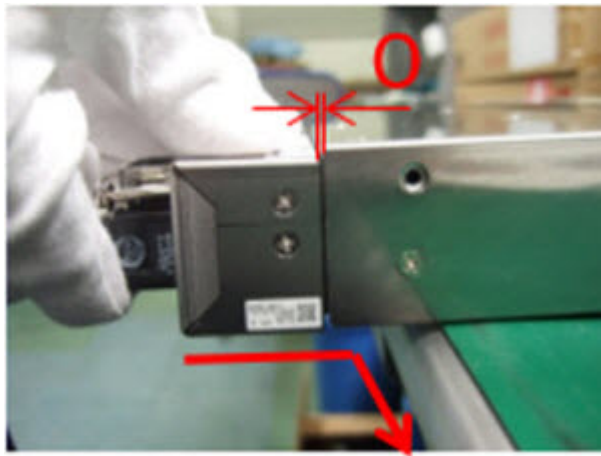


Figure 64
Seating Adapter

Step 4

Review the position of the power terminal adapter on the blade, verifying that no gap exists between the power terminal adapter and the blade and that the power terminal adapter is slightly below the top of the blade.



Figure 65
Correcting Power Terminal Adapter Placement

Step 5

On a flat ESD-safe surface, alternately tighten the attachment screws to secure the power terminal adapter onto the blade, careful not to overtighten them or tighten one screw completely before the other screw.



Caution: When tightening the two screws, alternate between them; do not overtighten either screw or tighten one completely before tightening the other.



Warning: The attachment screws mentioned in this step must be lined up straight in the holes and should easily turn, right up to the point of being tight. If any resistance occurs in the first turn, stop immediately, or risk damage to the blade (broken screws and screw holes on the blade).



Figure 66
Attachment Screws

Step 6

Replace the protection cover on the power terminal adapter.

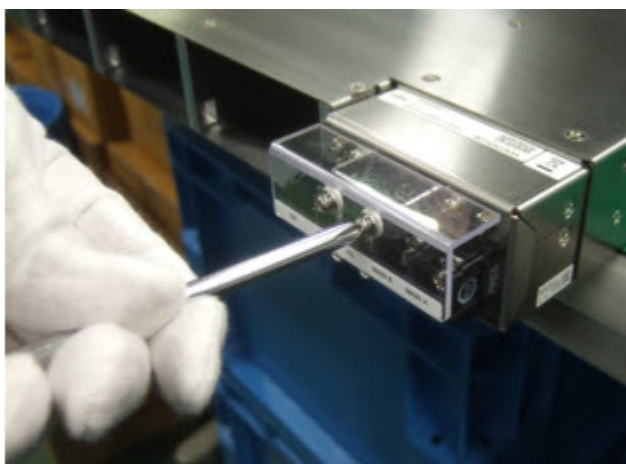


Figure 67
Attaching Protection Cover

✓ **This task is complete.**

Additional Instructions:

Perform the same procedure for each blade being installed.

4.10

Install Blade in Housing

Prerequisites:

- The 1FINITY Housing must be installed.
- Verify power terminal adapter (AC00-SD11) is not installed on the rear of the blade.
- The slot in the 1FINITY Housing where the blade is installed has power.
- Screwdriver set (including straight-slot and Phillips head)



Danger: The blade is heavy and should be handled carefully. Do not use the fiber management arms to lift and install the blade. Doing so may damage the blade, causing the blade to fall and personal injury. When lifting or installing, position two hands in the center of the blade to avoid dropping the blade.

Step 1

Insert the blade into the appropriate slot of the 1FINITY Housing and slide it back until it locks in place.

Note: Fujitsu recommends installing the blade without the PIUs installed.



Attention: When lifting or installing, position two hands in the center of the blade to avoid dropping the blade.

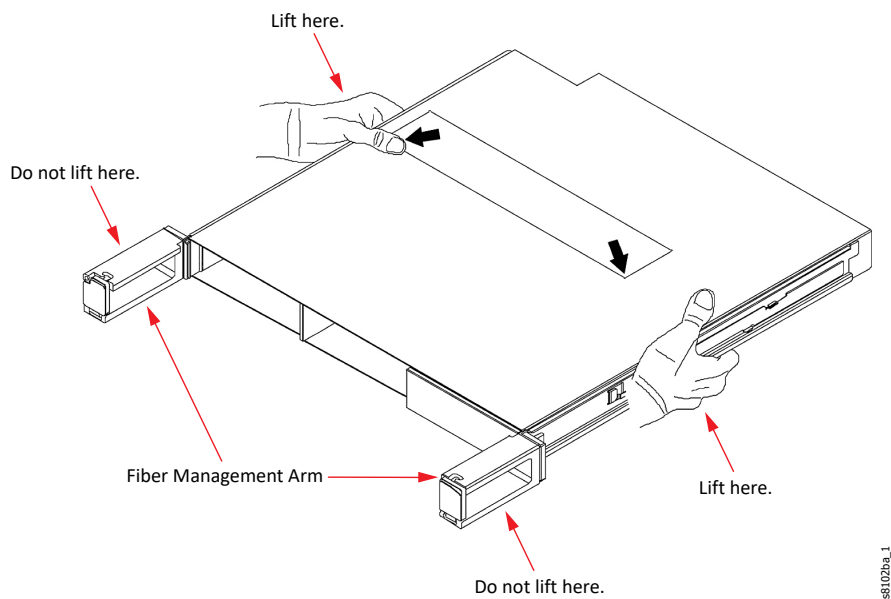


Figure 68
Proper Method to Lift Blade

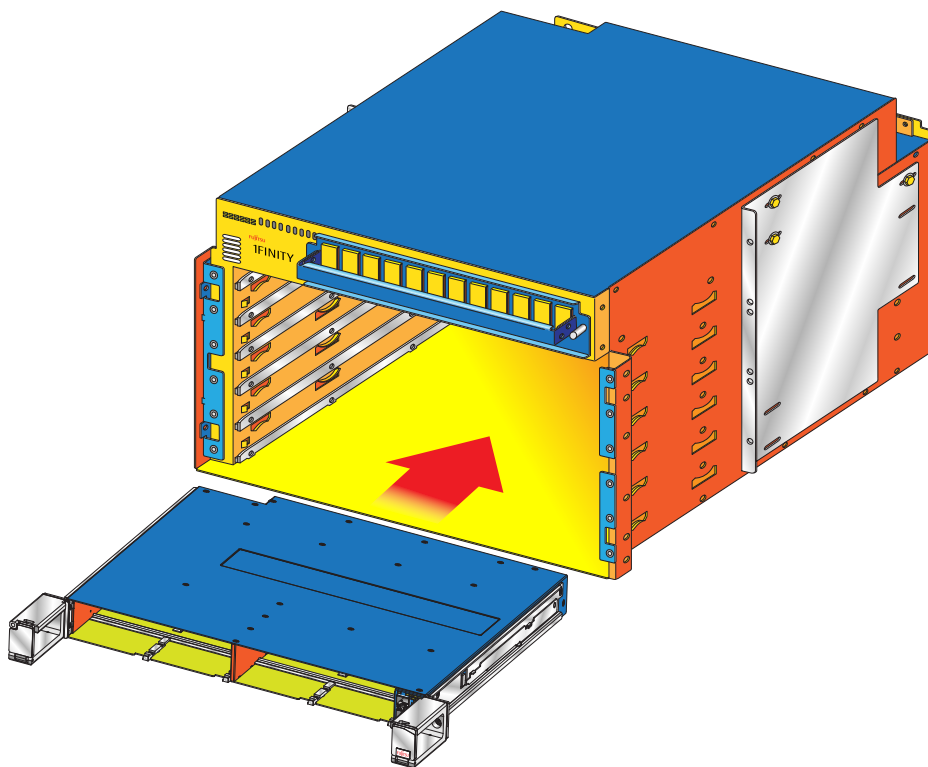


Figure 69
Installing Blade in 1FINITY Housing

Step 2

Unlatch the fiber restraint gates on each side of the blade.

Note: Slide the fiber restraint gate slightly inward to open the gate.

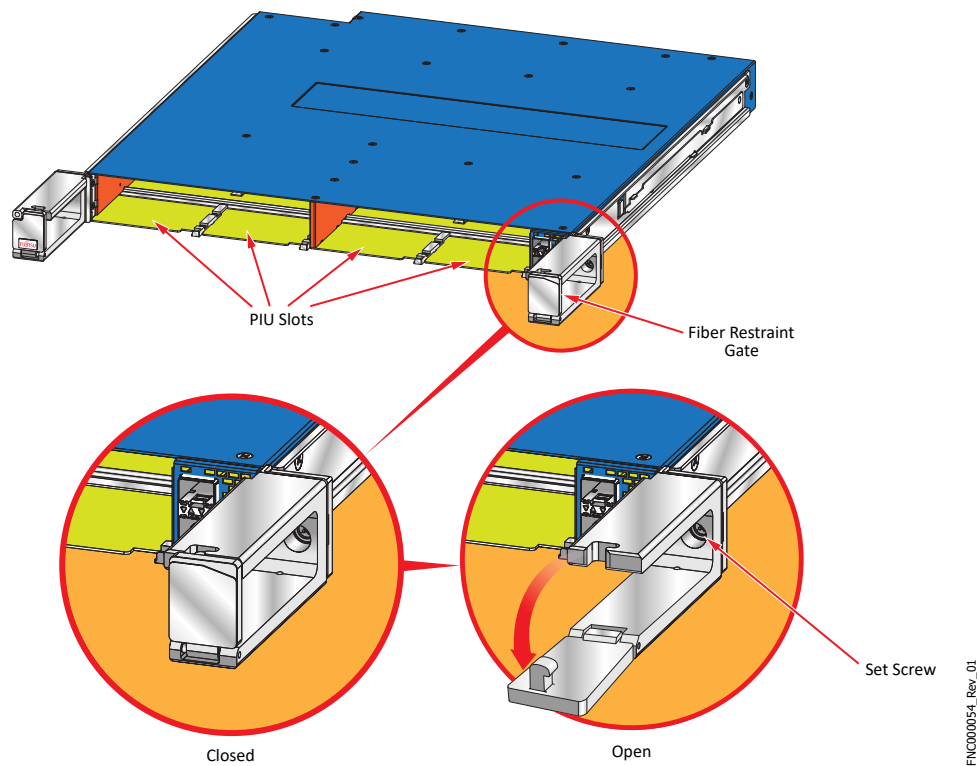


Figure 70
Unlatching Fiber Restraint Gate

Step 3

After opening the fiber restraint gates, secure the blade to the 1FINITY Housing using the set screws on the left- and right-hand sides of the blade.

Note: The set screws for securing the blade are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 4

Perform the same procedure for each blade to be installed.



Warning: After the blade is seated in the housing, it should not be removed, reseated, or power down for at least 15 minutes. Repetitive restart of the blade within 15 minutes causes database corruption, which leads to blade start-up failure and requires a return to factory for reprogramming.

✓ **This task is complete.**

4.11

Rectifier and Blade Wiring Connections

Previously, only the frame ground on the rectifier unit and on each L100 Blade was connected to the nearest common ground. The following figure shows an example of these connections on a rack-based rectifier and an L100 Blade.

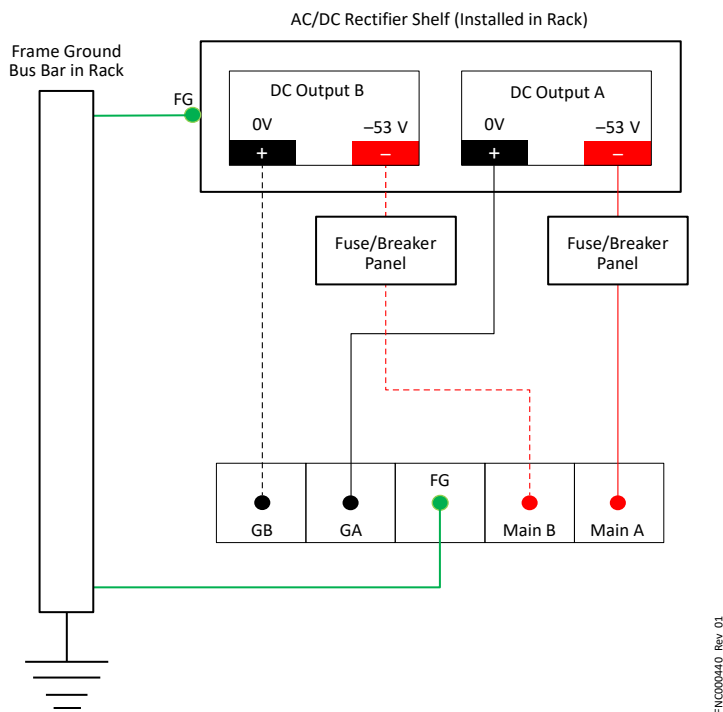
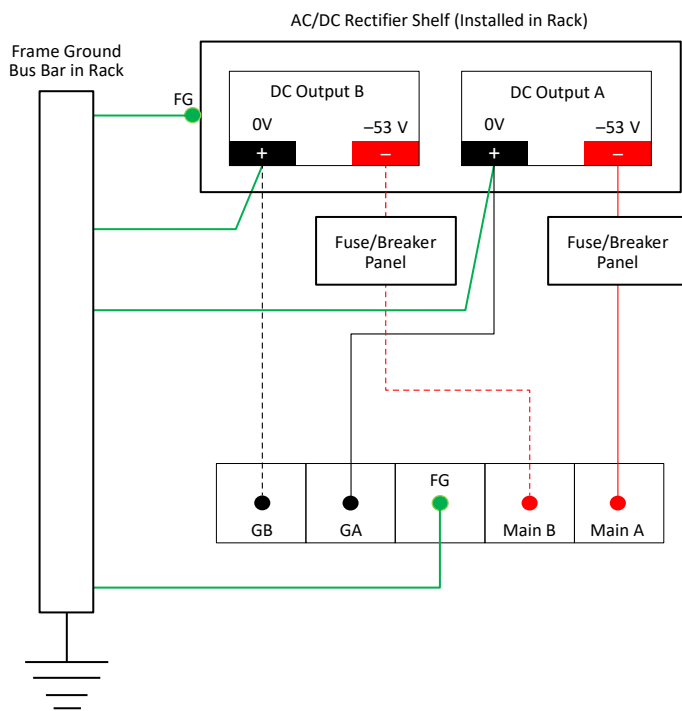


Figure 71
Existing Rectifier and Blade Wiring Connections

In addition to the frame ground connections shown in the preceding figure, Fujitsu recommends connecting the positive (0 V) connectors (A and B) on the rectifier to the nearest common ground. The following figure shows an example of these connections on a rack-based rectifier and an L100 Blade.



FNC000441_Rev_01

Figure 72
Recommended Rectifier and Blade Wiring Connections

4.12

Install Power Cables

The following links lead to related information:

[Power Specifications](#)

[Install Power Terminal Adapter](#)

[Remove Power Cables](#)

Tools required:

- Power terminal adapter attached to the blade
- Screwdriver set (including straight-slot and Phillips head)
- DMM
- Terminating cable lugs bent at 90 degrees



Danger: Make sure both A and B power feeds are turned off, at the source, before starting this procedure. Use properly insulated tools and perform all safety precautions in accordance with local safety practices when working on or around power terminals, power cables, and power connectors. Contacting power terminals, uninsulated conductors, cable wiring, or any connector that could be part of a live electrical circuit can result in death or serious personal injury.



Attention: This procedure is not applicable if using the 1FINITY Housing option.

Step 1

If not already done, remove the protection cover from the power terminal adapter.

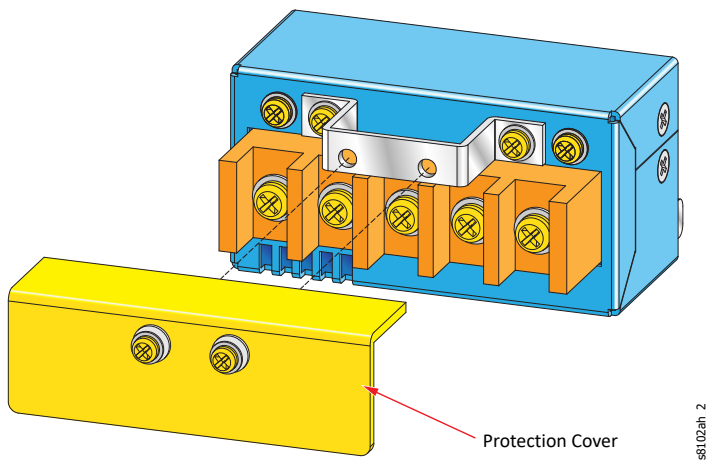


Figure 73
Removing Protection Cover

Step 2

Attach the power cables to the power terminal adapter by performing the following substeps:



Warning: Make sure to use terminating cable lugs bent at 90 degrees if stacking multiple blades in a rack. The cables make contact below the power terminal adapter if the lugs are not bent at 90 degrees.

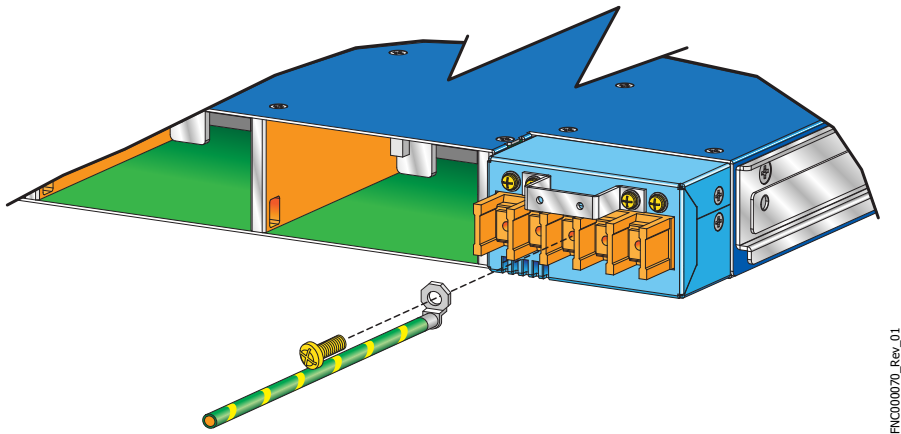


Figure 74
Installing Power Cables

- a) Attach the frame ground cable (green/yellow) to the terminal marked FG on the power terminal adapter.
- b) Attach the ground cable (black) for the A power feed to the terminal marked GA on the power terminal adapter.
- c) Attach the power feed A cable (red) to the terminal marked MAIN A on the power terminal adapter.

- d) Attach the ground cable (black/white) for the B power feed to the terminal marked GB on the power terminal adapter.
- e) Attach the power feed B cable (red/white) to the terminal marked MAIN B on the power terminal adapter.
- f) Route the power cables to the right of the blade as shown in the following figure.

Note: Routing the power cables to the left of the blade can obstruct access to the Fan unit slots.

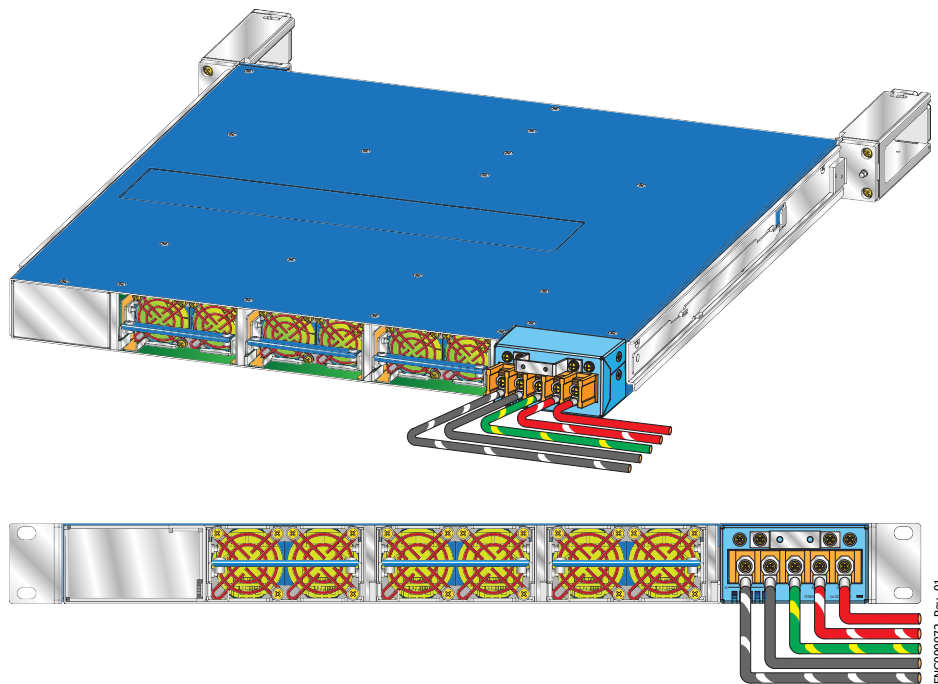


Figure 75
Power Cable Routing

Step 3

Turn on the power at the source.



Danger: Follow all local electrical safety procedures when working with a live electrical circuit to prevent possible serious injury or death.

Note: The SYS LED indicator on the front of the blade lights yellow initially and changes to green when the blade warm-up completes.

Step 4

Measure the voltage between the MAIN A and GA terminals on the power adapter as shown in the following figure.

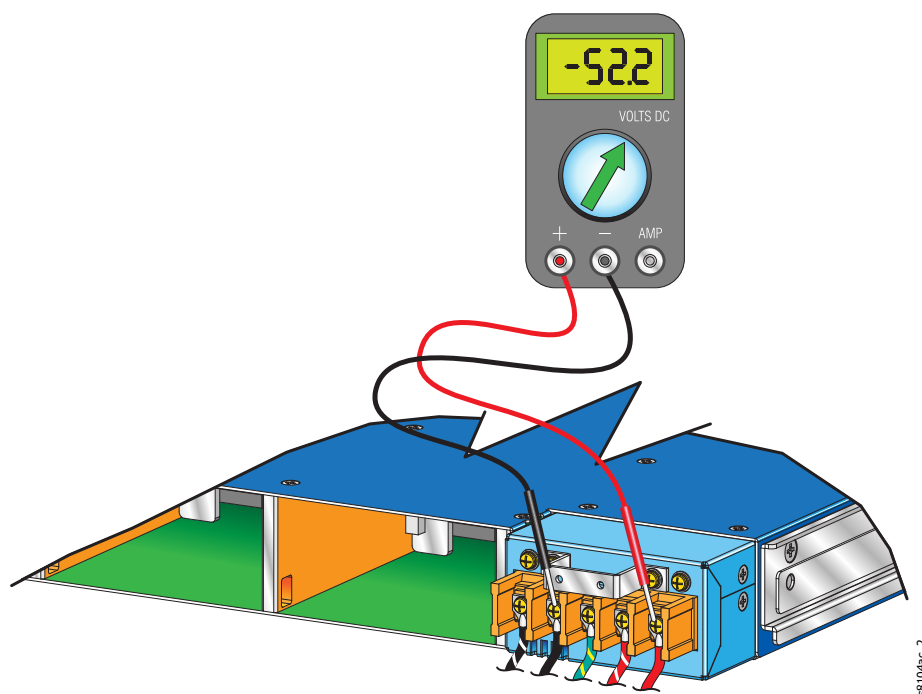


Figure 76
Measuring Voltage



Important: If the voltage is not between -40 and -57 V DC, contact the installation group or appropriate supervisor/manager, as applicable, before proceeding. After the trouble has been corrected, repeat [Step 4](#).

Step 5

Measure the voltage between the MAIN B and GB terminals on the power adapter.

Step 6

Replace the protection cover.

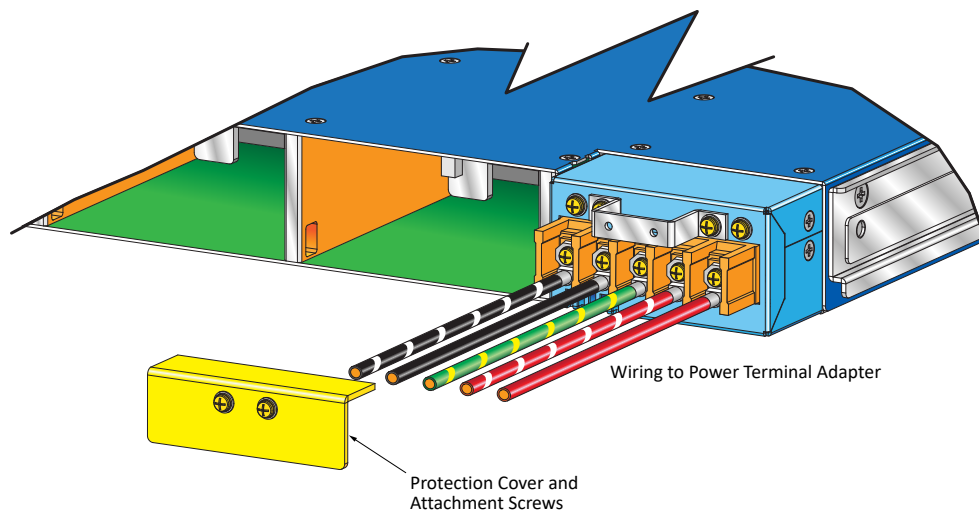


Figure 77
Replacing Protection Cover

✓ **This task is complete.**

Additional Instructions:

Perform the same procedure for each blade being installed.

4.13

Install Fan Unit

The following links lead to related information:

[Remove Fan Unit](#)

Note: A Fan unit must be installed in an active blade to maintain the safe operating temperature range (5...40 °C). Do not leave a Fan unit slot open for more than 3 minutes for an active blade.

Note: Fan filters are not required for individual L100 Blades (disaggregated option). For L100 Blades installed in the 1FINITY Housing (aggregated option), an air filter kit can be installed in the housing to protect all of the blades in the housing. The housing air filter should be replaced every 90 days.

Step 1

Insert the Fan unit into one of the slots located at the rear of the blade and press until Fan unit locks in place.

Note: This figure only shows one Fan unit being installed, but two or three Fan units must be installed, depending on the blade type.

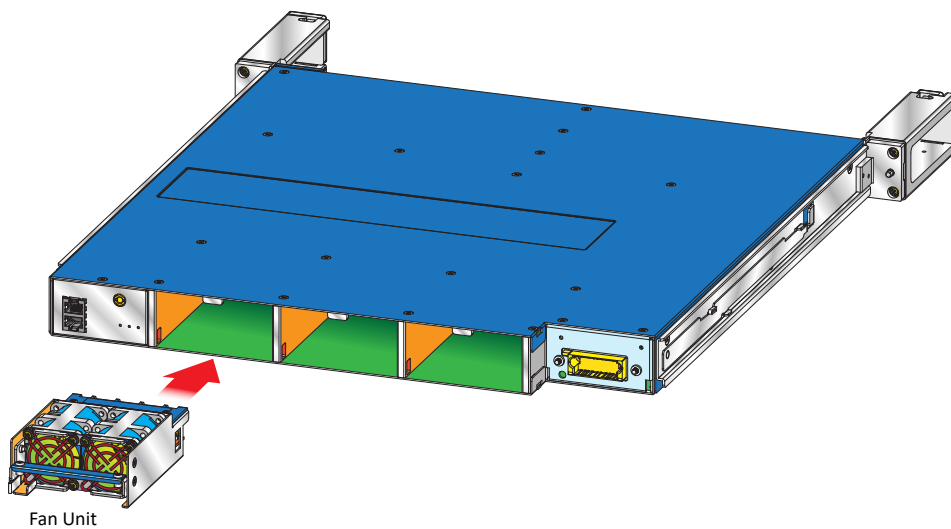


Figure 78
Installing Fan Unit

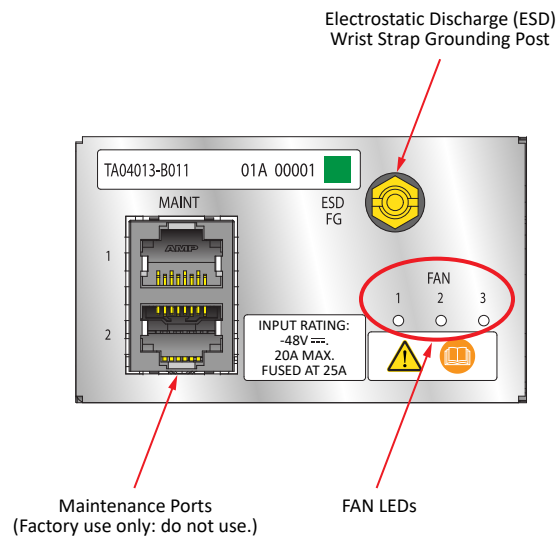


Figure 79
FAN LEDs

Step Result:

When the Fan unit is inserted, the fans should start running and the FAN LEDs on rear-left side of the blade lights green. The Fan unit has two or three FAN LEDs, depending on the blade type.

Note: The FAN LEDs change to green if power is applied to the blade.

✓ **This task is complete.**

Additional Instructions:

Perform the same procedure, as needed, to insert additional Fan units into the remaining Fan unit slots.

4.14

Install Plug-In Unit

The following links lead to related information:

[Remove Plug-In Unit](#)

Each PIU has a locking lever that has the following three functions:

- Acts as a stopper to prevent damage to the PIU or blade connector if the PIU is inserted using too much force

Note: This function only works when the locking lever is in the locked position.

- Helps to properly align the PIU and blade connectors during insertion
- Locks the PIU in place after complete insertion

Step 1

Ensure that ESD handling precautions are observed and remove the PIU from its ESD protective packaging.



Warning: Ensure that an appropriate ESD wrist strap is worn and is properly grounded.

Step 2

Inspect the PIU for any damage or debris. Carefully remove any debris. If the PIU is damaged, follow local practices/procedures for repair or replacement of a damaged PIU, as applicable. Do not proceed unless the PIU to be installed is of the correct type and is undamaged.



Warning: When installing the unit, use both hands to avoid dropping the unit.

Step 3

Verify the locking lever is in the locked position.

Step 4

Carefully align and insert the PIU into the appropriate slot until it stops, as shown in the following figure.

Note: The PIU should be slowly and carefully inserted into the blade to prevent damage to the PIU and/or blade. When the connector pins of the PIU engage the backplane, the PIU must be fully seated within 10 seconds. If it takes longer than 10 seconds to install the PIU, an equipment fault alarm is raised. Return to the first step and repeat the procedure to clear the alarm. Damage to the PIU and/or blade is more likely if the locking lever is in the released position.

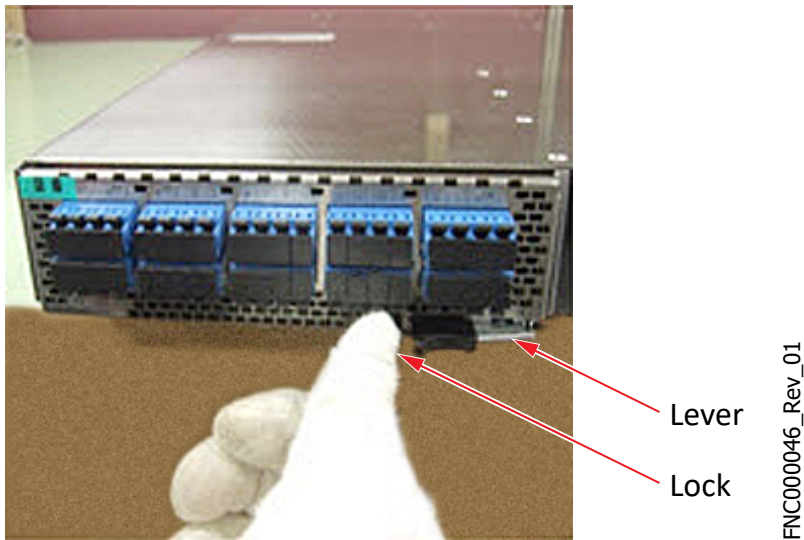


Figure 80
Inserting PIU

Step 5

Rotate the locking lever to the released position and gently press it in until the lever automatically starts to move, as shown in the following figure.

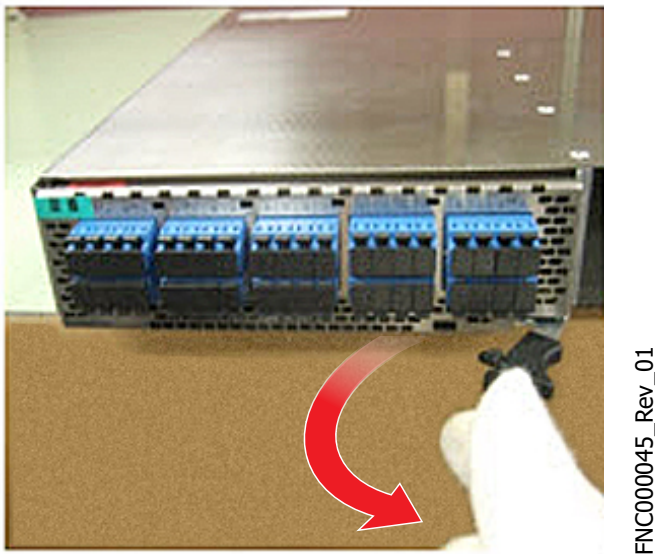


Figure 81
Releasing Locking Lever

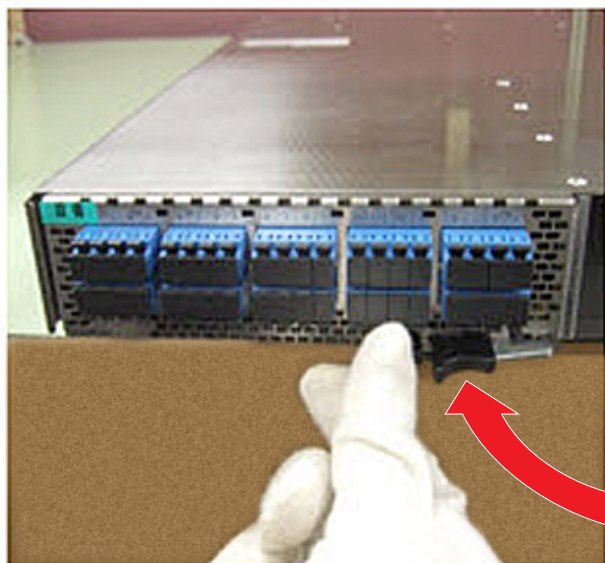
Step 6

Rotate the locking lever to the locked position to fully insert the PIU and lock it into place, as shown in the following figure. Wait 1 minute after installing PIU before removing the PIU.



Warning: To prevent damage to the PIU, never force a PIU into position.

Note: Make sure that the locking lever locks in place. If the locking lever is hard to lock into place, press the release button in the middle of the locking lever.



FNC000044_Rev_01

Figure 82
Locking PIU in Place

Step 7

Record the installation in accordance with local record-keeping procedures.

✓ **This task is complete.**

Additional Instructions:

Perform the same procedure to install all required PIUs.

4.15

Install SFP Module

The following links lead to related information:

[Remove SFP Module](#)



Danger: Never look into the end of a fiber-optic cable. Permanent eye damage or blindness can occur very quickly if the laser light is present. Follow local safety precautions regarding fiber.



Danger: Never handle exposed fiber with bare hands or touch it to bare skin. Fiber fragments can enter the skin and are difficult to detect and remove. Follow local safety precautions regarding fiber.

Step 1

Inspect the module for damage or debris, and remove the optical connector cover.

Step 2

Locate the PIU or blade into which the module is to be installed.

Step 3

If necessary, record the serial number and issue number of the module being installed.

Step 4

Install and secure the SFP module into the proper port on the front of the PIU or blade.



Warning: Do not push the module quickly into the slot (port) on the front of the PIU or blade. Seat the module with a slow but firm motion to avoid damage to the module and slot (port).

Step 5

Are fibers to be connected to the module at this time?

If YES:

Continue with next step.

If NO:

Attach a protective dust cap on the optical receptacle. This procedure is complete.

Step 6

Clean the optical connectors.

Step 7

Connect the fiber-optic cables to the module.



This task is complete.

4.16

Install Filler Panel

The following links lead to related information:

[Remove Filler Panel](#)

Step 1

Locate the PIU slot on the blade that requires the filler panel.

Step 2

Obtain the proper number of filler panels for installation into unoccupied PIU slots on the blade.

Step 3



Warning: Equipment can be damaged by ESD during installation, removal, or whenever handled. Ensure that an appropriate ESD wrist strap is worn and is properly grounded.

Install the filler panel by performing the following substeps:

- a) Ensure that the filler panel is oriented properly.
- b) Carefully and slowly press the filler panel into the empty PIU slot until the filler panel is completely inserted.

Note: Filler panels must be installed in any unoccupied slot on the blade.

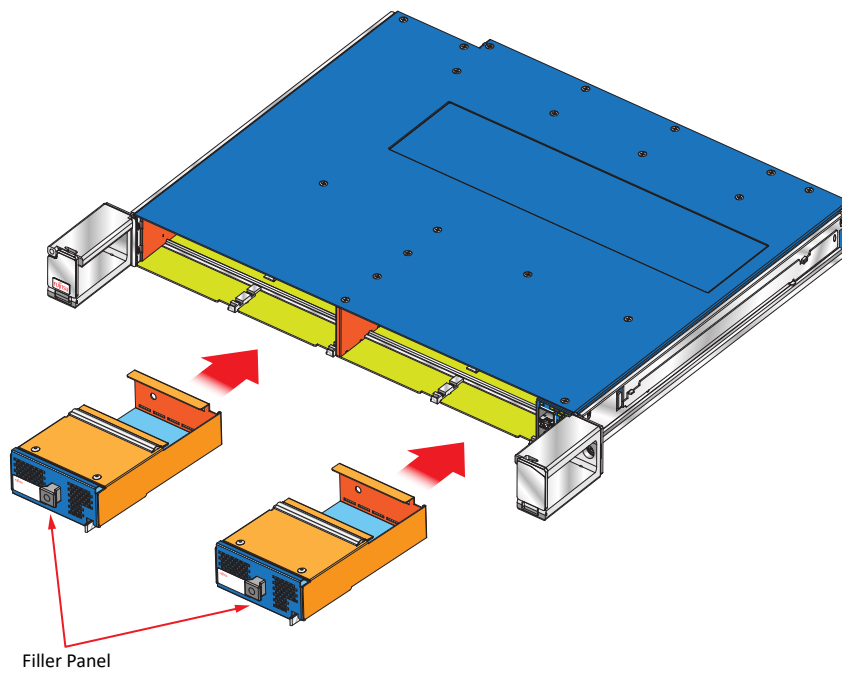


Figure 83
Installing Filler Panel

Step 4

Record the installation in accordance with local record-keeping procedures.

✓ **This task is complete.**

4.17

Install Data Communication Network Cabling

Prerequisite:

- The appropriate SFP modules must be installed in the LCN ports being cabled.

Step 1

Connect the fiber-optic cable from the DCN cloud to the LCN1 port on the Main L100 Blade.

Step 2

Connect a fiber-optic cable from the LCN3 port on the Main L100 Blade to the LCN4 port on the first tributary blade.

Step 3

Is another tributary blade to be installed?

If YES:

Continue with next step.

If NO:

Proceed to [Step 6](#).

Step 4

Connect a fiber-optic cable from the LCN3 port on the current tributary blade to the LCN4 port on the next tributary blade.

Step 5

Return to [Step 3](#).

Step 6

Connect a fiber-optic cable from the LCN3 port on the last tributary blade to the LCN4 port on the Main blade.

✓ **This task is complete.**

4.18

Install Local Management Port Cabling

Step 1

Connect one end of the RJ-45 cable to the Ethernet port on the PC.

Step 2

Connect the other end of the RJ-45 cable to the LMP on the front of the blade.

✓ **This task is complete.**

4.19

Install CD ROADM Cabling

Note: For each step in this procedure, verify that the dust cover has been removed from both the port and the fiber-optic cable. Verify that both the port and the fiber-optic cable connectors are free of dust and other contaminants before connecting the fiber-optic cable.

Step 1

Connect the fiber-optic cables from the transponder and/or the muxponder units to the appropriate client port (C1...C16) on a 4x16 splitter/coupler PIU.

Step 2

For add/drop connections, connect the fiber-optic cables from the appropriate edge ports (E1...E4) on a 4x16 splitter/coupler PIU to the appropriate client port (C1...C9) on the 1x9 WSS ROB.

Step 3

For pass-through (hub) connections, connect the fiber-optic cables to the appropriate client ports (C1...C9) on the 1x9 WSS ROB between the ROADM degrees.

✓ **This task is complete.**

4.20

Install L160 Raman Cabling

This procedure describes how to connect the fiber-optic cables between L100 ROB blade and L160 Raman blade.

Note: Follow the Raman cabling procedure when L160 Raman blade is used in the network.



Caution: Raman blade must be powered down when connecting the fiber-optic cables with L100 Blade. Turn on the power supply only after ensuring that the connections are complete in all aspects and no open-ended fiber exists.



Danger: L160 Raman blade emits very HIGH power light.



Danger: Never look into any ports of Raman blade directly. Never look into the end of a fiber, fiber cord, or fiber pigtail connected with any ports of Raman blade. Permanent eye damage or blindness can occur quickly when laser radiation is present.



Danger: Never expose bare skin to light from E1 port of Raman blade. Never bring bare hands near the surface of the connector of E1 port and near the fiber which is connected with E1 port of Raman blade.



Danger: All ports and the surface of a fiber connector should be cleaned thoroughly. If dust exists on the surface of a fiber connector or on the ports of Raman blade, it is burnt and melted due to the high power light. The burning or melting action causes damage to the fiber connector or to the blade.

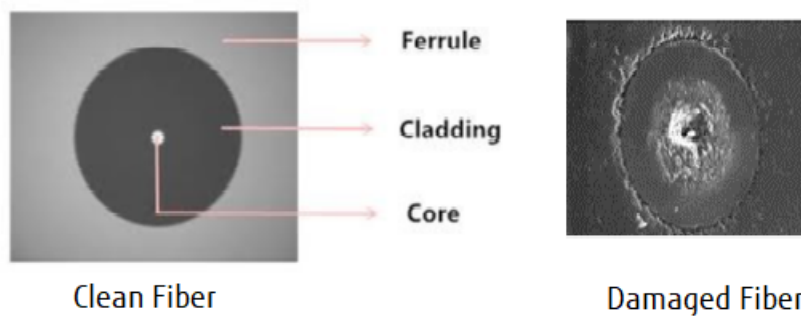


Figure 84
Clean and Damaged Fiber

Note: The fiber from the network should be connected to E1 (RCV) port of the ROB through the Raman blade, that is, the fiber from the network (RCV) should be connected to E1 (RCV) port of the Raman blade. The C1 (TRMT) port of the Raman blade should be connected to E1 (RCV) port of the ROB.

Note: The E1 (TRMT) port of ROB connected to the fiber, which is connected to the network, remains same for both Raman and non-Raman configurations.

Note: E1 and C1 ports of the Raman blade are unidirectional ports. E1 port only receives signal from the network and C1 port only transmits the signal to the L100 Blade.

Step 1

Connect the fiber-optic cables from the network side to the Edge port (E1) of the Raman blade.

Step 2

Connect the fiber-optic cables from the Client port (C1) of the Raman blade to the Edge port (E1) of the ROB.

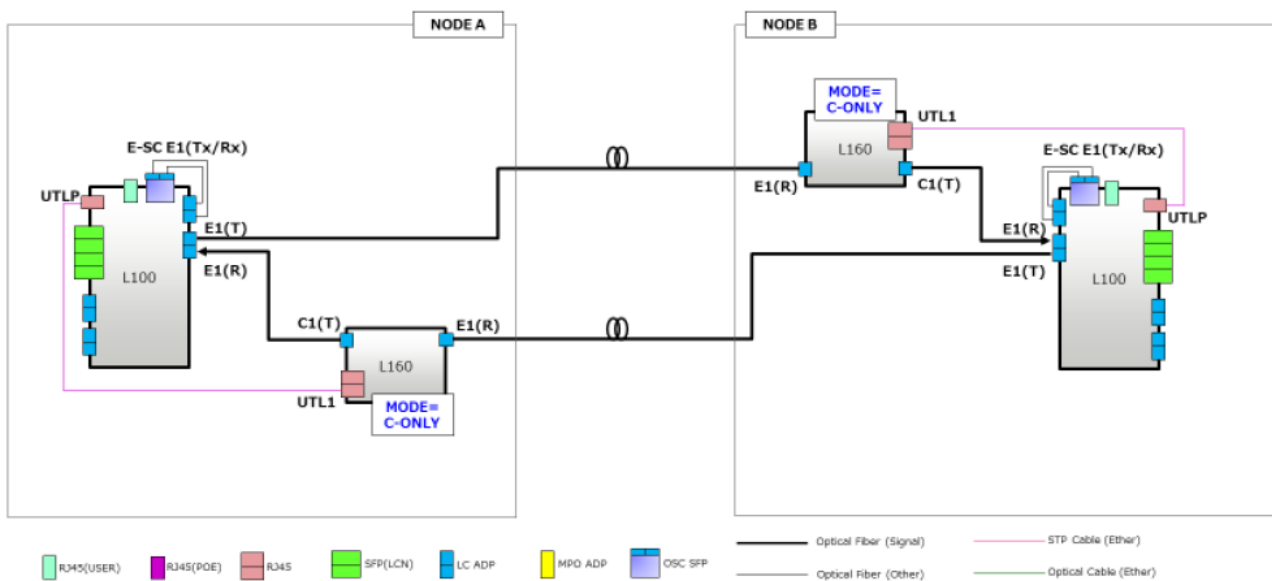


Figure 85
L100 and L160 Blade Cabling

Note: L100 Blade Issue 14, FC95453R11-I14, and later blade must be used with L160 Raman blade.

✓ **This task is complete.**

5

Initial System Turn Up

In this chapter:

- | | | | |
|-----|---|------|---|
| 5.1 | Log On | 5.10 | Set Up Tributary Blade with Auto Shelf ID Assignment Enabled |
| 5.2 | Set Up Main Blade (Initial) | 5.11 | Set Up Tributary Blade without Auto Shelf ID Assignment Enabled |
| 5.3 | Set WDM Configuration | 5.12 | Provision L100 Tributary Blade and Ports |
| 5.4 | CONSOLE Port (Serial Port) | 5.13 | Provision L110 Tributary Shelf, Ports, and Slots |
| 5.5 | Configure OSPF Routing | 5.14 | Provision L160 Tributary Blade |
| 5.6 | Configure IP Connection to DCN | 5.15 | Verify Blade Setup |
| 5.7 | Log On NE Through LMP | 5.16 | Set ROADM Constraints |
| 5.8 | Provision Main Blade Ports and Fan Units | | |
| 5.9 | Enable Auto Shelf ID Assignment for Tributary Blade Feature | | |

5.1

Log On

This procedure describes how to log on to the blade.

Note: The blade ships with a default username and password of fujitsu/1finity. The defaults are used in the procedures in this manual.

Note: Fujitsu recommends changing the default username after logging on for the first time.

Note: Wait 10 minutes after powered up to log on the system.

Step 1

Using a LAN cable, connect the PC to the blade connector labeled LMP. The IP address of the LMP is 192.168.1.1/30. The PC must have an IP address of 192.168.1.2/30.

Step 2

From the PC, open an SSH client session to the NE using the LMP IP address (192.168.1.1).

Note: Some SSH clients (for example, TeraTerm) provide an option to the user to choose an authentication method. In such a scenario, select the **Use keyboard-interactive to log in** option button and click **OK**.

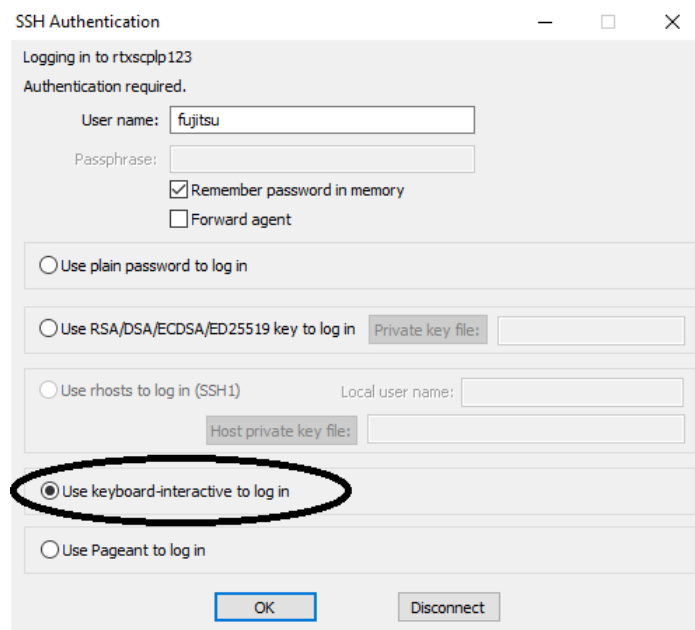


Figure 86
Use keyboard-interactive to log in Option Button

Step Result:

A command line window displays a logon command line.

Step 3

Log on to the NE using factory default user credentials.

factory default username:	fujitsu
factory default password:	1finity

Note: Fujitsu recommends changing the default username and password after initial login.

Step 4

If appropriate, change to CLI *configure* mode.

Command:

```
> configure
```

Output Example:

```
Entering configuration mode
[ok]
```

Step 5

Proceed to the procedure that required the login.

✓ **This task is complete.**

5.2

Set Up Main Blade (Initial)



Warning: After the blade is powered up, power down within 15 minutes causes database corruption, which leads to blade start-up failure and requires a return to factory for reprogramming.

Step 1

Apply power to the Main blade.

Step 2

Log on the blade using the LMP as described in [Log On](#).

Step 3

Retrieve the alarms.

Command:

```
> show alarms alarm-list
```

Step 4

Check if the *shelfProvisioningMode* alarm is present.

If YES:

Continue with next step.

If NO:

This procedure is complete.

Step 5

Execute the Main blade setup.

Command:

```
> request set-shelfmode shelfid <shelf-id> shelfrole <shelf-role>
```

Table 43

Input Parameter

Parameter	Value	Description
shelf-id	<1>	Shelf value for the blade
shelf-role	<MAIN>	Name of shelf role

Command Example:

```
> request set-shelfmode shelfid 1 shelfrole MAIN
```

Step 6

When the question "Really want to do this?" appears, enter "yes".

Note: When ZTP *auto-upgrade* flag is enabled, the ZTP-Boot starts the auto-upgrade process. If the ZTP server is not reachable, the *ztpBootFailure* alarm raises and ZTP-Boot retries auto-upgrade after 5 minutes. A user can set the *auto-upgrade* flag to false to clear the *ztpBootFailure* alarm. To enable/disable ZTP-Boot, refer to [ZTP-Boot](#). For more detailed ZTP information, refer to *1FINITY ZTP User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

Step Result:

The blade reboots.

Step 7

When the reboot completes, log on the blade using the LMP as described in [Log On](#).

Step 8

Troubleshoot external ZTP device, if after setting the shelf role to Main blade and ZTP alarms are on equipment.

Step 9

Set the system name.

Command:

```
% set system name <string, min: 7 chars, max: 20 chars>
```

Step 10

Perform the following command, if ZTP is not used.

Command:

```
% request restartsystem level warm
```



Attention: Perform a WARM restart for the system to turn up and for the tributary shelves to be recognized.

Step 11

Retrieve the alarms.

Command:

```
> show alarms alarm-list
```

Note: [Step 11](#) is optional.

Step 12

Verify that the *shelfProvisioningMode* alarm is not present.

✓ **This task is complete.**

5.3

Set WDM Configuration

Step 1

Set the WDM configuration.

Note: If autoprovisioning is enabled, the WDM configuration is automatically provisioned.

Command:

```
% set wdmcfg tidnum 1
```

✓ **This task is complete.**

5.4

CONSOLE Port (Serial Port)

In this section:

5.4.1 Enable/Disable CONSOLE Logon

5.4.2 CONSOLE Log On

Note: Only L100 Blade Issue 14 (FC95453R11-I14) and later supports CONSOLE port.

5.4.1

Enable/Disable CONSOLE Logon

Note: The console logon is enabled by default.

Step 1

Verify console logon status.

Command:

```
> show system console console-enabled
```

Output:

```
console-enabled true;  
[ok] [date-time]
```

Step 2

Configure console logon (if disabled).

Command:

```
% set system console console-enabled true
```

Output:

```
[ok] [date-time]
```

✓ **This task is complete.**

5.4.2

CONSOLE Log On

Prerequisites:

- Release 19.1.1 software or later is required to use the CONSOLE port.
- RJ-45 roll over cable is attached to the CONSOLE port.
- The user must use the CLI for management.
- The user must be logged on as *fujitsu*/1finity or must have Level-4 or Level-5 privileges assigned to perform this procedure.

Note: The CONSOLE port should only be used for debugging or recovery purposes.

Note: For remote access, connect the CONSOLE port to a terminal server.

Step 1

Connect to the blade using an RJ-45 roll over cable to the CONSOLE port of the blade.

Step 2

Log on as user *fujitsu* on the command line.

Note: The default username, *fujitsu*, is used for initial turn-up. After initial turn-up, a user must use an assigned username.

Step 3

At the *password* prompt, type *1finity*.

Note: The default password, *1finity*, is used for initial turn-up. After initial turn-up, a user must use a self-set password.

Step 4

Start a CLI session by entering *cli*.

Note: By default, the idle time-out period for CLI session is 30 minutes. Use `set idle-timeout` command to set the time-out period.

Output:

```
fujitsu login: fujitsu
Password:
```

```
fujitsu:~$ cli
```

```
NOTICE: THIS IS A PRIVATE COMPUTER SYSTEM. UNAUTHORIZED ACCESS OR USE MAY LEAD TO
PROSECUTION.
```

```
fujitsu connected from 1xx.xxx.xxx.xxx using console on fujitsu
fujitsu@fujitsu>conf
Entering configuration mode private
Current configuration users:
fujitsu ssh (cli from 1xx.xxx.xxx.xxx) on since 2017-07-17 01:13:55 private mode
autop-sys console (noaaa from 1xx.xxx.xxx.xxx) on since 2017-07-17 01:10:35
private mode
[ok][2017-07-17 01:21:05]
```

Step 5

If appropriate for the task, change to CLI *configure* mode.

Command:

```
> configure
```

Output:

```
Entering configuration mode
[ok][date-time]
```

✓ **This task is complete.**

5.5

Configure OSPF Routing

Step 1

Configure OSPF routing.

Command:

```
% set routing routing-instance <routing_instance_name> enabled routing-protocols
routing-protocol ospfv2 <protocol-instance-name> ospf instance
<routing_instance_name> ipv4 router-id <router-id> network <address> <mask> area
<area-id>
```

Parameter	Value	Description
routing_instance_name	<MCN>	Name of routing instance
protocol_instance_name	<OSC>	Name of routing protocol instance
router-id	<1xx.xxx.xxx.xxx>	OSPF router id per RFC 2328
address	<1xx.xxx.xxx.xxx>	IPv4 address, mask, and area ID defining the network on which OSPF routing is enabled
mask	<1xx.xxx.xxx.xxx>	
area-id	<x.x.x.x>	

Command Example:

```
% set routing routing-instance MCN enabled routing-protocols routing-protocol
ospfv2 OSC ospf instance MCN ipv4 router-id 1.x.x.xx network 1xx.x.x.x
2xx.xxx.xxx.x area 0.0.0.0
```

Step 2

Check the candidate configuration, if desired.

Command:

```
% show routing routing-instance
```

✓ This task is complete.

5.6

Configure IP Connection to DCN

In this section:

- 5.6.1 Configure LCN1 with Static IP Address
- 5.6.2 Configure LCN1 with DHCP
- 5.6.3 Configure IP Connection Through OSC

These procedures are alternate methods for connecting the NE to the DCN. Choose one method and perform that procedure.

After the connection is established, the NE can be managed from remote locations through the DCN.

5.6.1

Configure LCN1 with Static IP Address

Prerequisite:

- If port LCN1 requires a pluggable SFP module, the SFP is installed and its autoprovisioning is complete.

Overview

At the site where the NE is installed, the Local Area Network (LAN) connects to a router and the DCN is reachable through the router. This section describes the procedure to configure a static IP address on LCN1, configure the default next-hop IP address, and install a LAN cable connecting LCN1 to the LAN. The LAN cable can connect directly to a LAN switch port as shown in the following figure.

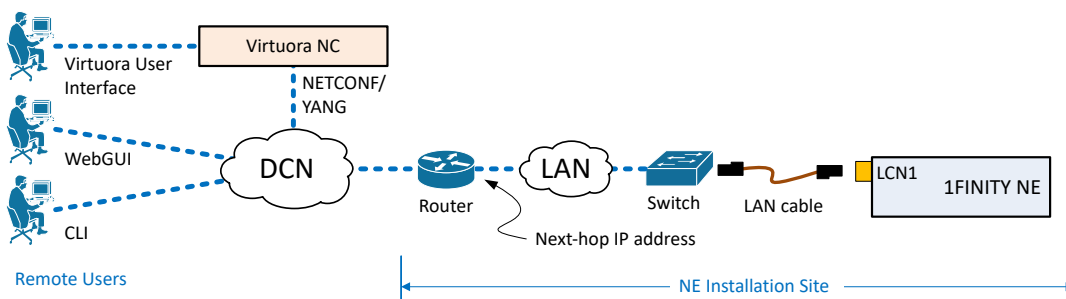


Figure 87

LAN Switch Port Connection

The following figure shows how to save LAN switch ports by connecting the LAN cable to a daisy chain of LAN cable connections between 1FINITY NEs.

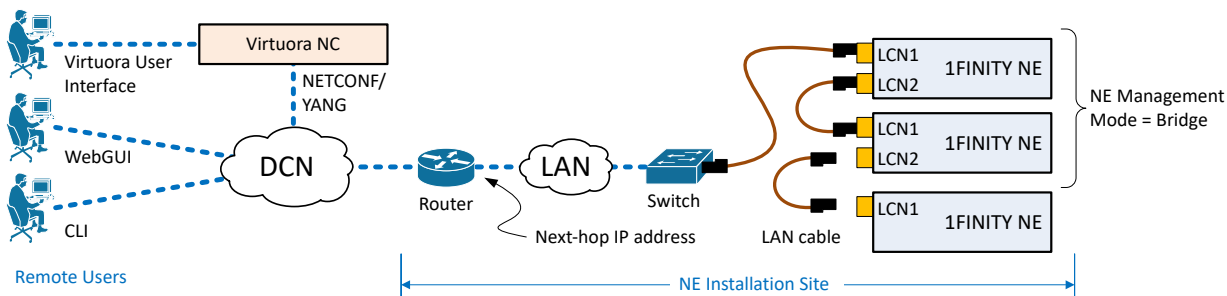


Figure 88

Daisy Chain LAN Connection

Note: The NE that is being turned up connects to the end of the daisy chain. The other NEs must be configured to bridge traffic between LCN1 and LCN2 (set [neMgmtMode](#) to *Bridge* or *MixMode*).

Step 1

Change to CLI configuration mode.

Command:

```
> configure
```

Set Static IP Address on LCN1

Step 2

Disable DHCP on LCN1 interface.

Command:

```
% set clientv4 client-if ip-1/0/0/LCN1 enable false
```

Step 3

Set IP address on the LCN1 interface.

Command:

```
% set interfaces interface ip-1/0/0/LCN1 type ip ipv4 enabled forwarding true  
address <ip_address> prefix-length <prefix-length>
```

Parameter	Value	Description
ip_address	<1xx.xxx.xxx.xxx>	IP address of the LCN interface
prefix-length	<0...32>	Length of the subnet prefix

Command Example:

```
% set interfaces interface ip-1/0/0/LCN1 type ip ipv4  
enabled forwarding true address 192.168.229.64 prefix-length 24
```

Step 4

Check the candidate configuration, if desired.

Command:

```
% show interfaces interface ip-1/0/0/LCN1
```

Configure Next-Hop

If the NE connects directly to the DCN, the next hop should be the address of the default router. Otherwise, the next hop should be the IP address of the neighboring NE that is closer to the DCN in the chain of GCC0-linked NEs.

Step 5

Configure the next hop. This step creates the routing instance if it does not already exist.

Command:

```
% set routing routing-instance <routing_instance_name> routing-protocols routing-protocol static <protocol_instance_name> static-routes ipv4 route 0.0.0.0/0 next-hop next-hop-address <ip_address>
```

Parameter	Value	Description
routing_instance_name	<MCN>	Name of routing instance
protocol_instance_name	<FNC>	Name of protocol instance
ip_address	<1xx.xxx.xxx.xxx>	IP address of next hop

Command Example:

```
% set routing routing-instance MCN routing-protocols routing-protocol static FNC static-routes ipv4 route 0.0.0.0/0 next-hop next-hop-address 192.168.229.254
```

Step 6

Check the candidate configuration, if desired.

Command:

```
% show routing routing-instance
```

Step 7

Save the changes.

Command:

```
% commit
```

Step 8

Exit configuration mode.

Command:

```
% exit
```

Connect LCN1 to DCN

Step 9

Connect the LCN1 port to the DCN.

✓ This task is complete.

5.6.2

Configure LCN1 with DHCP

Prerequisites:

- If port LCN1 requires a pluggable SFP module, the SFP is installed and its autoprovisioning is complete.
- A DHCP server is operating on the LAN.

Overview

At the site where the NE is installed, the LAN connects to a router and the DCN is reachable through the router. A DHCP server is operating on the LAN. LCN1 is configured with DHCP enabled by factory default. With this configuration, the DHCP server automatically assigns an IP address to LCN1 when it is connected to the LAN. To establish IP communication beyond the LAN, the user must configure the next-hop IP address on the NE routing instance.

This section describes the procedure to configure the default next-hop IP address, and install a LAN cable connecting LCN1 to the LAN. The LAN cable can connect directly to a LAN switch port as shown in the following figure.

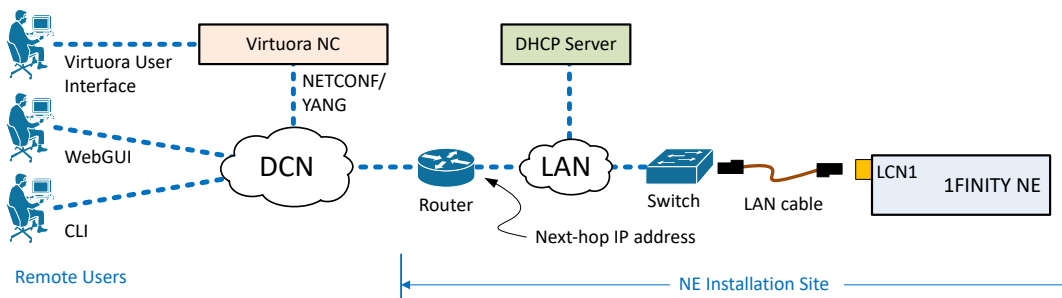


Figure 89
LAN Switch Port Connection

The following figure shows how to save LAN switch ports by connecting the LAN cable to a daisy chain of LAN cable connections between 1FINITY NEs.

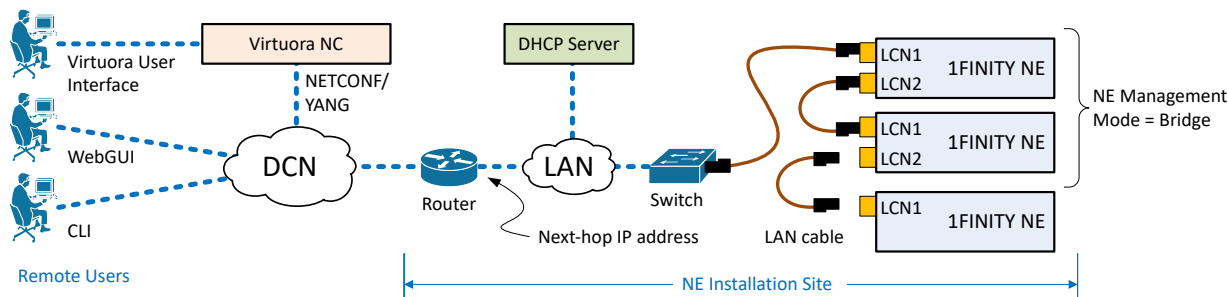


Figure 90
Daisy Chain LAN Connection

Note: The NE that is being turned up connects to the end of the daisy chain. The other NEs must be configured to bridge traffic between LCN1 and LCN2 (set `neMgmtMode` to *Bridge* or *MixMode*).

Step 1

Change to CLI configuration mode.

Command:

```
> configure
```

Configure Next-Hop

If the NE connects directly to the DCN, the next hop should be the address of the default router.

Step 2

Configure the next hop. This step creates the routing instance if it does not already exist.

Command:

```
% set routing routing-instance <routing_instance_name> routing-protocols routing-protocol static <protocol_instance_name> static-routes ipv4 route 0.0.0.0/0 next-hop next-hop-address <ip_address>
```

Parameter	Value	Description
<code>routing_instance_name</code>	<MCN>	Name of routing instance
<code>protocol_instance_name</code>	<FNC>	Name of protocol instance
<code>ip_address</code>	<1xx.xxx.xxx.xxx>	IP address of next hop

Command Example:

```
% set routing routing-instance MCN routing-protocols routing-protocol static FNC static-routes ipv4 route 0.0.0.0/0 next-hop next-hop-address 192.168.229.254
```


Step 3

Check the candidate configuration, if desired.

Command:

```
% show routing routing-instance
```

Step 4

Save the changes.

Command:

```
% commit
```

Step 5

Exit configuration mode.

Command:

```
% exit
```

Continue with the next task.

Connect LCN1 to DCN

Step 6

Connect the LCN1 port to the DCN.

Identify IP Address Assigned to LCN1 by DHCP

Step 7

Change to operational mode.

Command:

```
% exit
```

Step 8

Show IP address state on LCN1.

Command:

```
> show interfaces-state interface ip-1/0/0/LCN1 ipv4 address
```

Output Example:

IP	ORIGIN	PREFIX LENGTH
192.168.229.235	dhcp	24

✓ **This task is complete.**

5.6.3

Configure IP Connection Through OSC

Prerequisite:

- The NE is installed and powered up, OSC is configured, and DCN connectivity is available through the transport network using OSC.



Attention: To connect the NE to the DCN through the OSC, the user must first turn up the OSC, which requires significant OSC provisioning. OSC turn-up is not covered in this chapter because it fits in the chapter that covers turn-up of the NE transport functions.

The following figure shows a WDM network composed of 1FINITY ROADM and ILA NEs. The NEs at Site A and Site B connect to the DCN through the LCN1 interface. All other NEs connect to the DCN through OSC links.

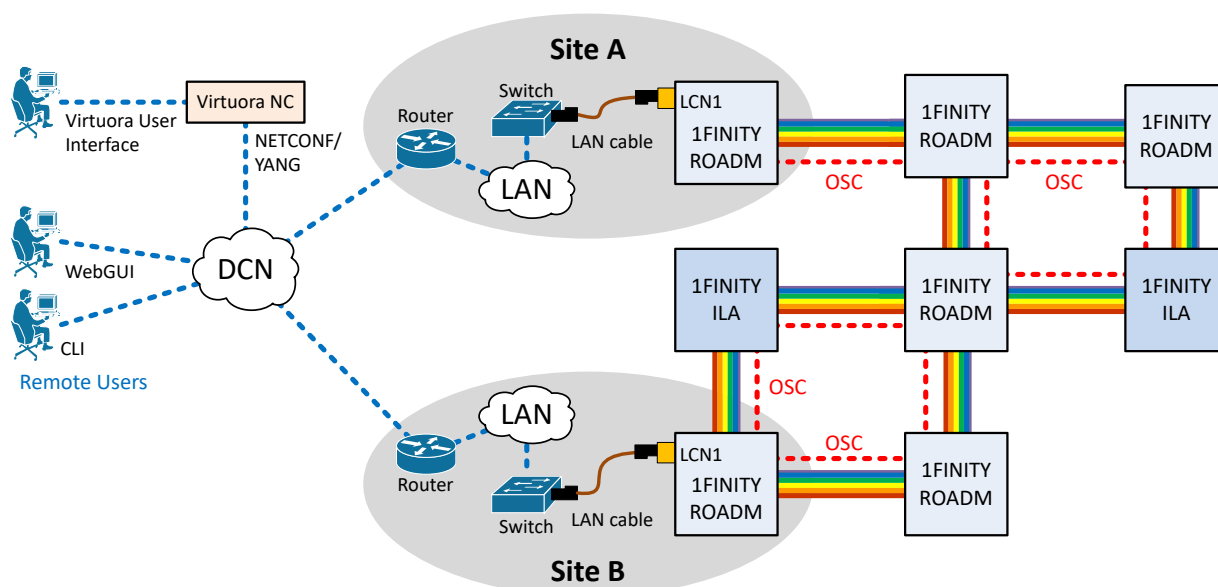


Figure 91
WDM Network

OSC is required (must be turned up) between every pair of adjacent NEs in the 1FINITY WDM network because it (OSC) supports various required functions, for example, optical power management and LLDP. If needed, OSC can carry DCN traffic.

Note: In this procedure, routing instances are configured with *OSPF routing protocol*. If desired, routing instances can be configured statically (using the *static routing protocol* and next-hops).

Conceptual Walk Through of Provisioning

Assume that the NEs at Site A and Site B already connect to the DCN through LCN1 and all OSC links are up and running. The objective is to use the OSC links to propagate DCN connectivity to the other NEs in the network.

The following figure shows provisioning for two representative NEs (Site B and Site C) with the understanding that similar provisioning is required at other NEs (to propagate DCN connectivity throughout the network).

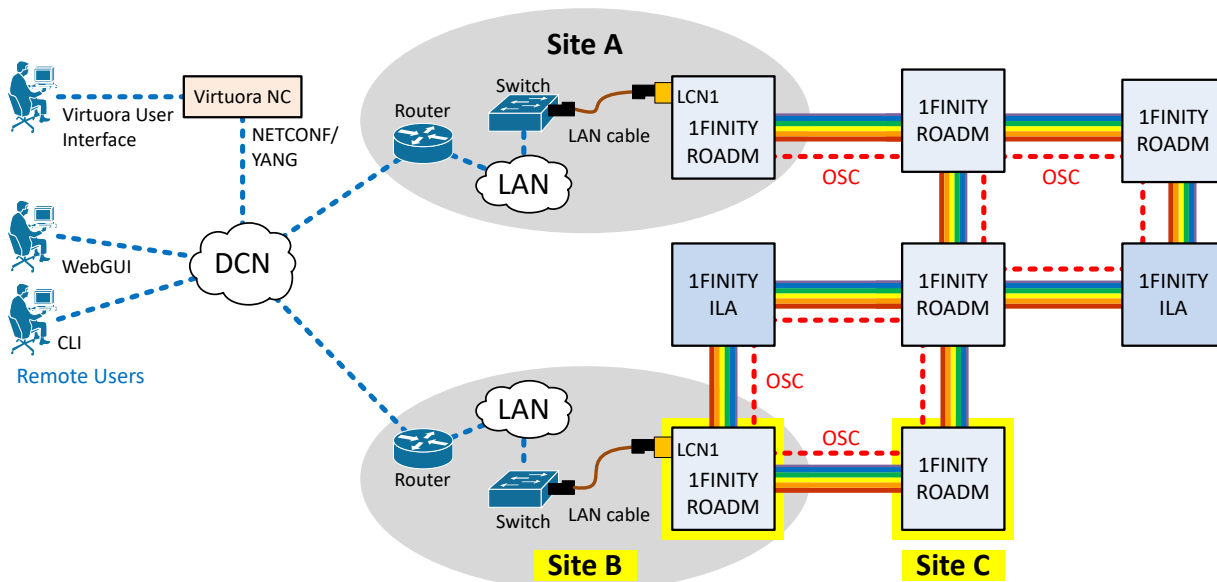


Figure 92
Provisioning Two Representative NEs

The following figure shows the initial configuration of NEs at Site B and Site C.

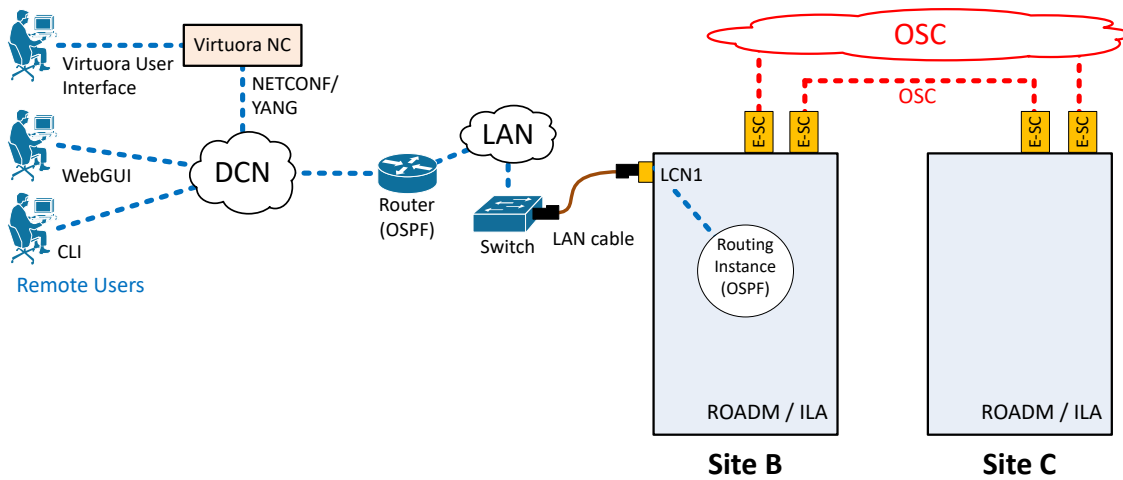


Figure 93
Initial Configuration of NEs

Subsequent provisioning can be conceptualized by performing the following steps:

1. Configure the NEM interface at each NE.

Note: NEM interfaces are needed to provide an IP address to share with unnumbered IP interfaces that are yet to be created.

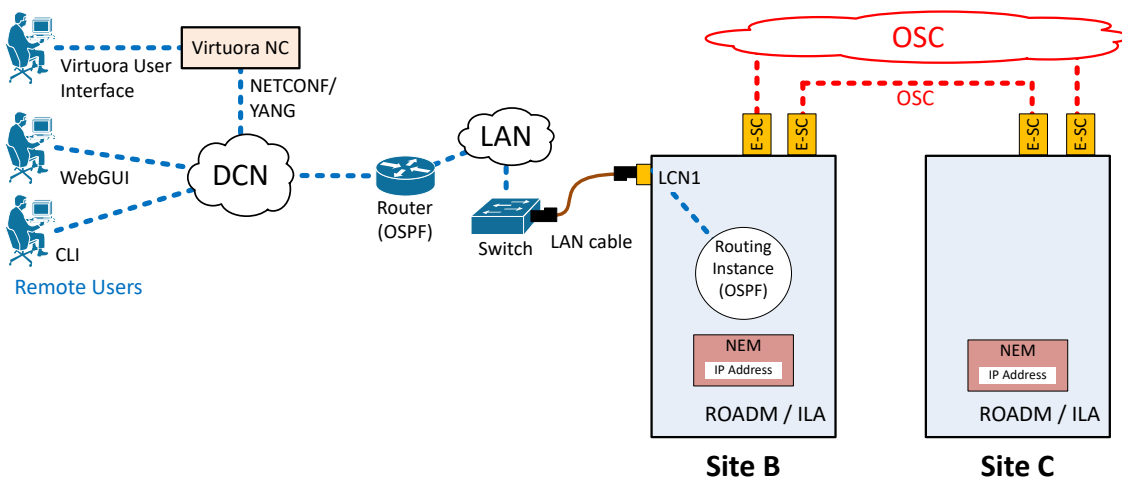


Figure 94
Configuring NEM Interface

2. Create an unnumbered IP interface to each OSC link and configure each of them to share the NE NEM IP address.

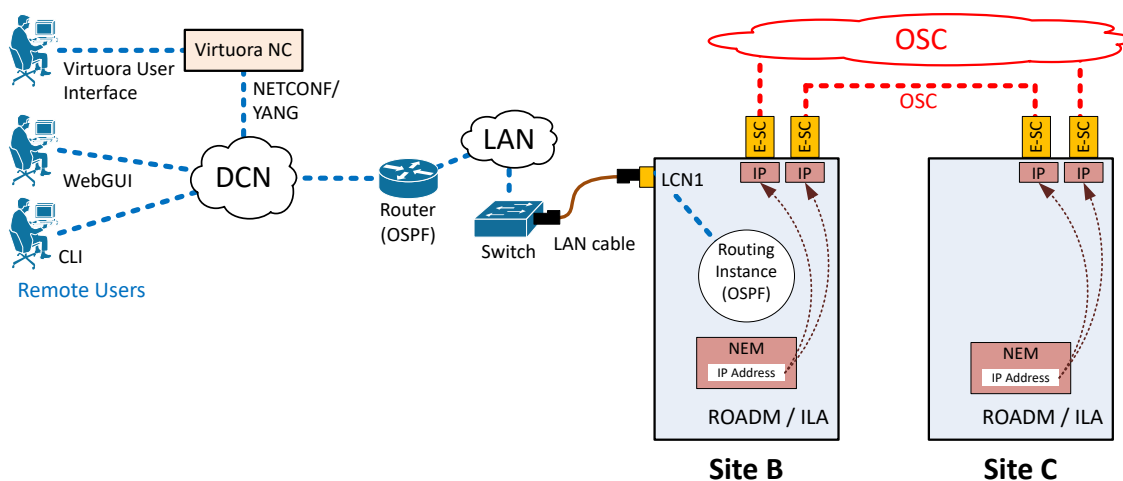


Figure 95

Creating Unnumbered IP Interface

- On each NE, create a routing instance (if it does not already exist) and assign the newly created IP interface(s) to the routing instance.

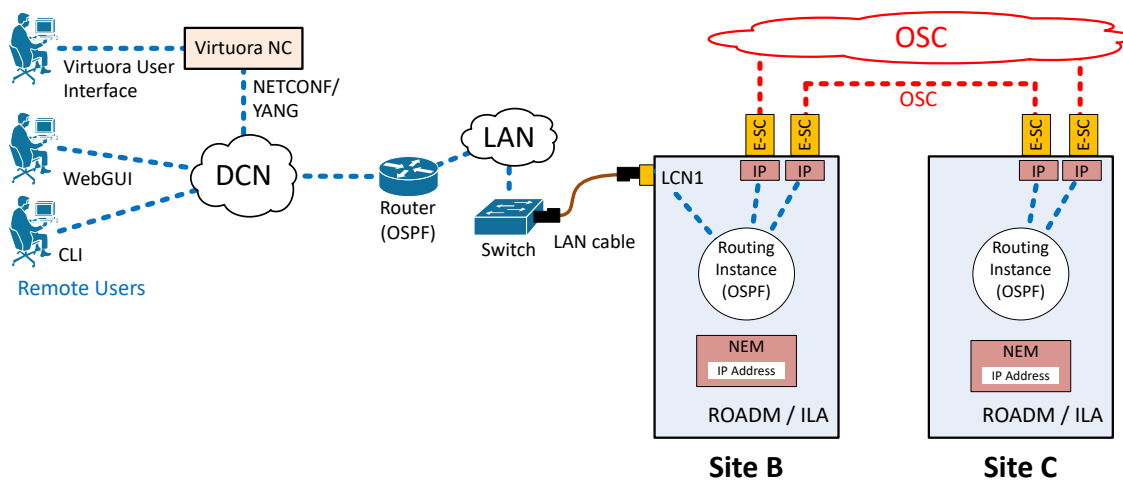


Figure 96 Creating Routing Instance

Remote users should be able to log on to the NE at Site C through the DCN.

The following steps describe provisioning for one NE in detail. Remember this provisioning is required for each NE involved in the network of OSC-linked NEs connecting the current NE (being turned up) with the DCN.

Step 1

Change to CLI configuration mode.

Command:

```
> configure
```

Configure NEM Interface

Step 2

Create the NEM interface and set its IP address.

Command:

```
% set interfaces interface ip-1/0/0/NEM type ip ipv4 forwarding true address  
<ip_address> prefix-length <prefix-length> preferred true
```

Parameter	Value	Description
ip_address	<1xx.xxx.xxx.xxx>	IP address of the NEM interface
prefix-length	<0...32>	Length of the subnet prefix

Command Example:

```
% set interfaces interface ip-1/0/0/NEM type ip ipv4 forwarding true address  
2.2.2.2 prefix-length 32 preferred true
```

Configure IP Interfaces for OSC Links

Prerequisite:

- All provisioning for each OSC link is complete, including provisioning of the ETH interface on the E-SC-E1 port.

Step 3

Create an IP interface on the E-SC-E1 port that supports the OSC link. Configure this IP interface to share the NEM IP address.

Command:

```
% set interfaces interface <interface> type ip ipv4 enabled forwarding true  
address-src ip-1/0/0/NEM mtu <mtu_size>
```

Parameter	Value	Description
interface	<ip-<shelf>/0/0/E-SC-E1>	IP interface on the E-SC-E1 port that supports the OSC link

Parameter	Value	Description
mtu_size	<68...1500>	Size, in octets, of the largest IPv4 packet that the interface can send and receive

Command Example:

```
% set interfaces interface ip-1/0/0/E-SC-E1 type ip ipv4 enabled
forwarding true address-src ip-1/0/0/NEM mtu 1500
```

Step 4

Check the candidate configuration, if desired.

Command:

```
% show interfaces interface <interface>
```

Step 5

Associate the newly created IP interface with the routing instance. This step creates the routing instance if it does not already exist.

Command:

```
% set routing routing-instance <routing_instance_name> interfaces interface
<interface>
```

Parameter	Value	Description
routing_instance_name	<MCN>	Name of routing instance
interface	<ip-<shelf>/0/0/E-SC-E1>	IP interface on the E-SC-E1 port that supports the OSC link

Command Example:

```
% set routing routing-instance MCN interfaces interface
ip-1/0/0/E-SC-E1
```

Step 6

Check the candidate configuration, if desired.

Command:

```
% show routing routing-instance
```

Step 7

If the NE has another OSC link, repeat the previous four steps for that OSC link.

Configure OSPF Routing

Step 8

Configure OSPF routing.

Command:

```
% set routing routing-instance <routing_instance_name> enabled routing-protocols
routing-protocol ospfv2 <protocol-instance-name> ospf instance
<routing_instance_name> ipv4 router-id <router-id> network <address> <mask> area
<area-id>
```

Parameter	Value	Description
routing_instance_name	<MCN>	Name of routing instance
protocol_instance_name	<OSC>	Name of routing protocol instance
router-id	<1xx.xxx.xxx.xxx>	OSPF router id per RFC 2328
address	<1xx.xxx.xxx.xxx>	IPv4 address, mask, and area ID defining the network on which OSPF routing is enabled
mask	<1xx.xxx.xxx.xxx>	
area-id	<x.x.x.x>	

Command Example:

```
% set routing routing-instance MCN enabled routing-protocols
routing-protocol ospfv2 OSC ospf instance MCN ipv4 router-id 1.1.1.32
network 172.1.1.0 255.255.255.0 area 0.0.0.0
```

Step 9

Check the candidate configuration, if desired.

Command:

```
% show routing routing-instance
```

Step 10

Save the changes.

Command:

```
% commit
```

Step 11

Exit configuration mode.

Command:

```
% exit
```

✓ This task is complete.

5.7

Log On NE Through LMP

Prerequisite:

- The NE is installed and powered up. Wait 10 minutes after powered up to log on the system.

Note: If this NE has multiple blades, log on through the Main blade LMP.

Step 1

Using a LAN cable, connect the PC to the blade connector labeled LMP. The IP address of the LMP is 192.168.1.1 /30. The PC must have an IP address of 192.168.1.2 /30.

Step 2

From the PC, open an SSH client session to the NE using the LMP IP address (192.168.1.1).

Step 3

Using an SSH client to connect, log on using factory default user credentials.

Note: Some SSH clients (for example, TeraTerm) provide an option to the user to choose an authentication method. In such a scenario, select the **Use keyboard-interactive to log in** option button and click **OK**.

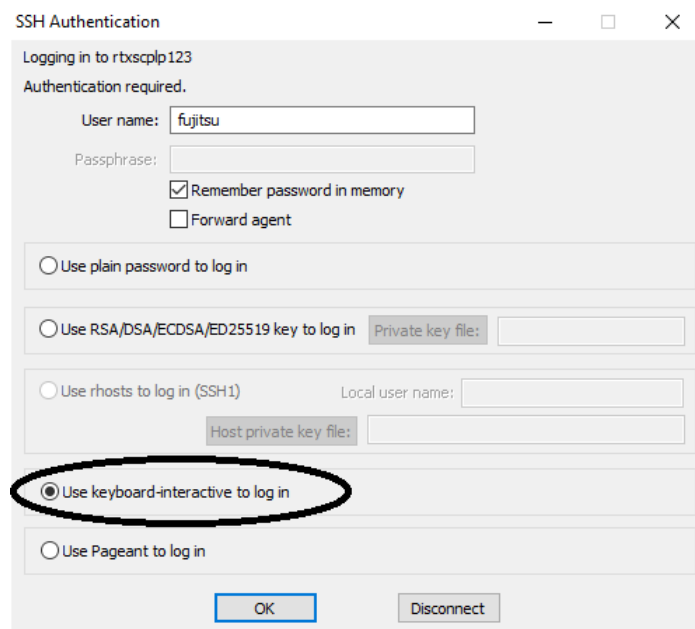


Figure 97

Use keyboard-interactive to log in Option Button

factory default username: fujitsu

factory default password: 1finity

Note: Fujitsu recommends changing the default username and password after initial login.

✓ **This task is complete.**

5.8

Provision Main Blade Ports and Fan Units

Step 1

Provision the Fan unit slots.

Note: If autoprovisioning is enabled, the Fan unit slots are automatically provisioned.

Commands:

```
% set eqpt shelf 1 slot FAN1 cardType AC00-F011 admin-status up
```

```
% set eqpt shelf 1 slot FAN2 cardType AC00-F011 admin-status up
```

Step 2

Provision the E1 port on the L100 Main blade.

Note: If autoprovisioning is enabled, the E1 port is automatically provisioned.

Command:

```
% set eqpt shelf 1 slot 0 cardType BDL1-3R11 subslot 0 port E1 admin-status up  
pluggableInterfaceType Edge num-lanes 128
```

Step 3

Provision the client ports (C1...C9) on the L100 Main blade.

Note: If autoprovisioning is enabled, the C1...C9 ports are automatically provisioned.

Command:

```
% set eqpt shelf 1 slot 0 cardType BDL1-3R11 subslot 0 port C1 admin-status up  
pluggableInterfaceType Client num-lanes 128
```

Note: Repeat this step for each client port (C1...C9) to be provisioned.

Step 4

Provision the logical port.

Note: If autoprovisioning is enabled, the logical port is automatically provisioned.

Command:

```
% set eqpt shelf 1 slot 0 subslot 0 port L1 pluggableInterfaceType Logical num-  
lanes 128 admin-status up
```

Step 5

Provision the OSC port.

Note: If autoprovisioning is enabled, the OSC port is automatically provisioned.

Command:

```
% set eqpt shelf 1 slot 0 subslot 0 port OSC pluggableInterfaceType OSC admin-  
status up
```

Step 6

Provision the E-SC-E1 port equipment.

Note: If autoprovisioning is enabled, the E-SC-E1 port is automatically provisioned.

Command:

```
% set eqpt shelf 1 slot 0 subslot 0 port E-SC-E1 pluggableInterfaceType 1GEX admin-status up
```

Note: If a 100ME SFP module is being used, specify 100ME for the pluggableInterfaceType instead of 1GEX.

Command:

```
% set eqpt shelf 1 slot 0 subslot 0 port E-SC-E1 pluggableInterfaceType 100ME admin-status up
```



Important: Fujitsu does not recommend using third-party OSC SFP module because the performance of third-party OSC module is not guaranteed.

Step 7

Provision the E-SC-E1 port interface.

Note: If autoprovisioning is enabled, the E-SC-E1 port interface is automatically provisioned.

Command:

```
% set interfaces interface eth-1/0/0/E-SC-E1 type ethernetCsmacd ethernet admin-status up mtu 1800 speed 1000
```

Step 8

Provision the UTLP equipment.

Command:

```
% set eqpt shelf 1 slot 0 subslot 0 port UTLP pluggableInterfaceType Fixed
```

Note: If a user provisions any UTLP equipment in an earlier release, use the following commands.

Commands:

```
% delete eqpt shelf 1 slot 0 subslot 0 port UTLP
```

```
% set eqpt shelf 1 slot 0 subslot 0 port UTLP pluggableInterfaceType Fixed
```

Note: If a user provisions any UTLP equipment and UTLP interface in an earlier release, use the following commands.

Commands:

```
% delete interfaces interface eth-1/0/0/UTLP
```

```
% delete eqpt shelf 1 slot 0 subslot 0 port UTLP
```

```
% set eqpt shelf 1 slot 0 subslot 0 port UTLP pluggableInterfaceType Fixed
```

Step 9

Provision the UTLP interface.

Command:

```
% set interfaces interface eth-1/0/0/UTLP type ethernetCsmacd ethernet duplex full
speed 1000 mtu 1522 auto-negotiation enabled admin-status up
```

✓ **This task is complete.**

5.9

Enable Auto Shelf ID Assignment for Tributary Blade Feature

Note: Auto Shelf ID Assignment feature is disabled by default.

Step 1

Connect to the Main blade.

Step 2

Enter configure mode.

Command:

```
> configure
```

Step 3

Enable tributary shelf ID assignment.

Command:

```
% set system trib-shelfid-assignment true
```

Step 4

Save the changes.

Command:

```
% commit
```

Step 5

Verify tributary shelf ID assignment.

Command:

```
% show system trib-shelfid-assignment
```

Output:

```
trib-shelfid-assignment true
```

✓ **This task is complete.**

5.10

Set Up Tributary Blade with Auto Shelf ID Assignment Enabled

Prerequisite:

- Verify that tributary blade is in shelf provisioning mode.

Step 1

Log on the Main blade.

Step 2

Verify whether tributary auto shelf ID assignment is enabled.

Command:

```
> show system trib-shelfid-assignment
```

Step 3

Clear the list of USIs of the tributary blade.

Command:

```
% request flush-shelf-prov-mode-blades
```

Step 4

Apply power to the tributary blade.

Step 5

Plug in all LCN SFP (to the Main and tributary blades) and connect the interconnect LAN cables.

Step 6

List the USI of the tributary blade.

Command:

```
> show shelf-prov-mode-blades
```

Output:

```
USI LBFJTU011725463R110100xxx shelf-id 0
```

Step 7

Set the desired shelf ID on the tributary blade manually.

Command:

```
> request set-shelf-id USI LBFJTU011725463R110100002 shelf-id 2
```

Step Result:

The tributary blade restarts and autoprovisions with the new shelf ID.

Step 8

After restart, retrieve the tributary blade information.

Command:

```
> show shelf-prov-mode-blades
```

Output:

```
USI LBFJTU011725463R110100xxx shelf-id 2
```

✓ **This task is complete.**

5.11

Set Up Tributary Blade without Auto Shelf ID Assignment Enabled

Prerequisite:

Note: The default name must be changed from *fujitsu* for the system to turn up as well as for the tributary shelves to be recognized.

- Verify that no SFP module is installed in the LCN ports of the blade that is being initialized as tributary.

Step 1

Apply power to the tributary blade.

Step 2

Log on the blade using the LMP as described in [Log On](#).

Step 3

Retrieve the alarms.

Command:

```
> show alarms alarm-list
```

Step 4

Check if the *shelfProvisioningMode* alarm is present.

If YES:

Continue with next step.

If NO:

Proceed to [Set Up Main Blade \(Initial\)](#).

Step 5

Connect LCN3 of the Main blade (already booted) to LCN4 port of the blade to be configured as tributary using either an optical fiber or a LAN cable (depending on the type of SFP module used).

Step 6

Execute the tributary blade setup.

Command:

```
> request set-shelfmode shelfid 2 shelfrole TRIB
```

Step 7

When the question "Really want to do this?" appears, enter "yes".

Step Result:

The blade reboots.

✓ **This task is complete.**

5.12

Provision L100 Tributary Blade and Ports

Note: The default name must be changed from *fujitsu* for the system to turn up as well as for the tributary shelves to be recognized.

Step 1

Provision the L100 tributary blade equipment.

Note: If autoprovisioning is enabled, the L100 tributary blade equipment is automatically provisioned.

Command:

```
% set eqpt shelf 2 type BDL1-3R11 shelf-mode NORMAL shelf-role TRIB admin-status up
```

Step 2

Provision the Fan unit slots.

Note: If autoprovisioning is enabled, the Fan unit slots are automatically provisioned.

Commands:

```
% set eqpt shelf 2 slot FAN1 cardType AC00-F011 admin-status up
```

```
% set eqpt shelf 2 slot FAN2 cardType AC00-F011 admin-status up
```

Step 3

Provision the E1 port on the L100 tributary blade.

Note: If autoprovisioning is enabled, the E1 port is automatically provisioned.

Command:

```
% set eqpt shelf 2 slot 0 cardType BDL1-3R11 subslot 0 port E1 admin-status up  
pluggableInterfaceType Edge num-lanes 128
```

Step 4

Provision the client ports (C1...C9) on the L100 tributary blade.

Note: If autoprovisioning is enabled, the client ports (C1...C9) are automatically provisioned.

Command:

```
% set eqpt shelf 2 slot 0 cardType BDL1-3R11 subslot 0 port C1 admin-status up  
pluggableInterfaceType Client num-lanes 128
```

Note: Repeat this step for each client port (C1...C9) to be provisioned.

Step 5

Provision the logical port.

Note: If autoprovisioning is enabled, the logical port is automatically provisioned.

Command:

```
% set eqpt shelf 2 slot 0 subslot 0 port L1 pluggableInterfaceType Logical num-  
lanes 128 admin-status up
```

Step 6

Provision the OSC port.

Note: If autoprovisioning is enabled, the OSC port is automatically provisioned.

Command:

```
% set eqpt shelf 2 slot 0 subslot 0 port OSC pluggableInterfaceType OSC admin-  
status up
```

Step 7

Provision the E-SC-E1 port equipment.

Note: If autoprovisioning is enabled, the E-SC-E1 port is automatically provisioned.

Note: If a 100ME SFP is being used, specify 100ME for pluggableInterfaceType.

Command:

```
% set eqpt shelf 2 slot 0 subslot 0 port E-SC-E1 pluggableInterfaceType 1GEX admin-  
status up
```



Important: Fujitsu does not recommend using third-party OSC SFP module because the performance of third-party OSC module is not guaranteed.

Step 8

Provision the E-SC-E1 port interface.

Note: If autoprovisioning is enabled, the E-SC-E1 port interface is automatically provisioned.

Command:

```
% set interfaces interface eth-2/0/0/E-SC-E1 type ethernetCsmacd ethernet admin-  
status up mtu 1800 speed 1000
```

Step 9

Provision the UTLP equipment.

Command:

```
% set eqpt shelf 2 slot 0 subslot 0 port UTLP pluggableInterfaceType Fixed
```

Note: If a user provisions any UTLP equipment in an earlier release, use the following commands.

Commands:

```
% delete eqpt shelf 2 slot 0 subslot 0 port UTLP
```

```
% set eqpt shelf 2 slot 0 subslot 0 port UTLP pluggableInterfaceType Fixed
```

Note: If a user provisions any UTLT equipment and UTLT interface in an earlier release, use the following commands.

Commands:

```
% delete interfaces interface eth-2/0/0/UTLT
% delete eqpt shelf 2 slot 0 subslot 0 port UTLT
% set eqpt shelf 2 slot 0 subslot 0 port UTLT pluggableInterfaceType Fixed
```

Step 10

Provision the UTLT port interface.

Command:

```
% set interfaces interface eth-2/0/0/UTLT type ethernetCsmacd ethernet duplex full
speed 1000 mtu 1522 auto-negotiation enabled admin-status up
```

✓ **This task is complete.**

5.13

Provision L110 Tributary Shelf, Ports, and Slots

Step 1

Provision the L110 tributary blade.

Note: If autoprovisioning is enabled, the L110 tributary blade is automatically provisioned.

Command:

```
% set eqpt shelf 3 type BDL1-3B11 shelf-mode NORMAL shelf-role TRIB admin-status up
```

Step 2

Provision the Fan unit slots.

Note: If autoprovisioning is enabled, the Fan unit slots are automatically provisioned.

Commands:

```
% set eqpt shelf 3 slot FAN1 cardType AC00-F011 admin-status up
```

```
% set eqpt shelf 3 slot FAN2 cardType AC00-F011 admin-status up
```

Step 3

Provision the PIL1-3PS2 PIU equipment.

Note: If autoprovisioning is enabled, the PIL1-3PS2 equipment is automatically provisioned.

Command:

```
% set eqpt shelf 3 slot 1 cardType PIL1-3PS2 admin-status up
```

Step 4

Provision a PIL1-3PS2 edge port (E1...E4).

Note: If autoprovisioning is enabled, the PIL1-3PS2 edge ports (E1...E4) are automatically provisioned.

Command:

```
% set eqpt shelf 3 slot 1 subslot 0 port E1 pluggableInterfaceType Edge admin-status up num-lanes 128
```

Note: Repeat this step for each edge port (E1...E4) to be provisioned.

Step 5

Provision a PIL1-3PS2 client port (C1...C16).

Note: If autoprovisioning is enabled, the PIL1-3PS2 PIU client ports (C1...C16) are automatically provisioned.

Command:

```
% set eqpt shelf 3 slot 1 subslot 0 port C1 pluggableInterfaceType Client admin-status up num-lanes 128
```

Note: Repeat this step for each client port (C1...C16) to be provisioned.

✓ **This task is complete.**

5.14

Provision L160 Tributary Blade

Note: The L160 Blade must be provisioned after provisioning the L100 Main blade ports and before provisioning the L100 Main blade OTS interface.



Danger: L160 Raman blade emits very HIGH power light.



Danger: Never look into any ports of Raman blade directly. Never look into the end of a fiber, fiber cord, or fiber pigtail connected with any ports of Raman blade. Permanent eye damage or blindness can occur quickly when laser radiation is present.



Danger: Never expose bare skin to light from E1 port of Raman blade. Never bring bare hands near the surface of the connector of E1 port and near the fiber which is connected with E1 port of Raman blade.



Danger: All ports and the surface of a fiber connector should be cleaned thoroughly. If dust exists on the surface of a fiber connector or on the ports of Raman blade, it is burnt and melted due to the high power light. The burning or melting action causes damage to the fiber connector or to the blade.

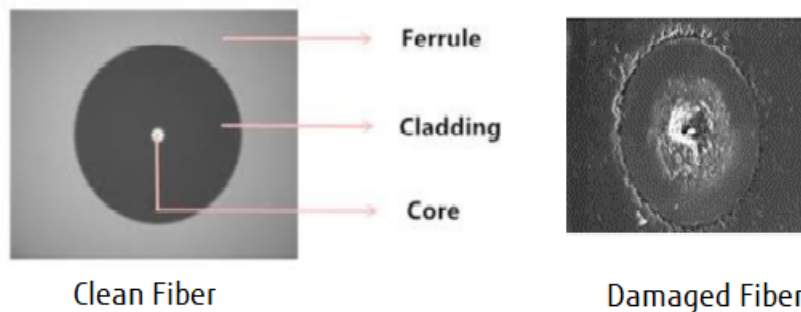


Figure 98
Clean and Damaged Fiber

Note: L100 Blade Issue 14, FC95453R11-I14, and later blade must be used with L160 Raman blade.

Step 1

Provision the L160 tributary blade.

Command:

```
% set eqpt shelf 101 type BDL1-3M21 shelf-mode NORMAL shelf-role TRIB admin-status up
```

Step 2

Provision the Fan unit slots.

Note: If autoprovisioning is enabled, the Fan unit slots are automatically provisioned.

Commands:

```
% set eqpt shelf 101 slot FAN1 cardType ACL8-3PC1 admin-status up
% set eqpt shelf 101 slot FAN2 cardType ACL8-3PC1 admin-status up
% set eqpt shelf 101 slot FAN3 cardType ACL8-3PC1 admin-status up
```

Step 3

Provision the E1 port on the L160 tributary blade.

Note: If autoprovisioning is enabled, the E1 port is automatically provisioned.

Command:

```
% set eqpt shelf 101 slot 0 cardType BDL1-3M21 subslot 0 port E1 admin-status up
pluggableInterfaceType Edge num-lanes 128
```

Step 4

Provision the client port C1 on the L160 tributary blade.

Note: If autoprovisioning is enabled, the client port C1 is automatically provisioned.

Command:

```
% set eqpt shelf 101 slot 0 cardType BDL1-3M21 subslot 0 port C1 admin-status up
pluggableInterfaceType Client num-lanes 128
```

Step 5

Save the changes.

Command:

```
% commit
```

✓ **This task is complete.**

Additional Instructions:

Perform the same procedure for each L160 additional blade to be provisioned.

5.15

Verify Blade Setup

Step 1

Log on the Main blade through the blade LMP interface.

Step 2

Verify that each blade is configured as intended.

Command:

```
> show table eqpt shelf <shelfid>
```

Parameter	Value	Description
shelf-id	<1...4>	Shelf value for L100
	<101,103,105,107>	Shelf value for L130
	<220>	Shelf value for L200
	<9>	Shelf value for L140

Command Example:

```
> show table eqpt shelf 1
```

Output:

```
fujitsu@fujitsu % show eqpt shelf 1 slot 0 subslot 0 port C1
pluggableInterfaceType Client;
num-lanes 128;
admin-status up;
ains disabled;
remoteSrcPeerPort-id "";
remoteDestPeerPort-id "";
[ok] [2021-05-12 09:44:03]
```

✓ This task is complete.

5.16

Set ROADM Constraints

Step 1

Set ROADM constraints.

Command:

```
% set roadm-constraints equipment-port-group contention-gr-1 ports 3/1/0/C1
3/1/0/C2 3/1/0/C3 3/1/0/C4 3/1/0/C5 3/1/0/C6 3/1/0/C7 3/1/0/C8 3/1/0/C9 3/1/0/C10
3/1/0/C11 3/1/0/C12 3/1/0/C13 3/1/0/C14 3/1/0/C15
3/1/0/C16
```

✓ **This task is complete.**

6

System Administration

In this chapter:

- | | | | |
|-----|--|------|--|
| 6.1 | Default Settings and IP Port Information | 6.10 | Syslog Administration |
| 6.2 | About CLI | 6.11 | Provision Proxy-ARP |
| 6.3 | System Name | 6.12 | LCN Port Commands |
| 6.4 | System Time and Date | 6.13 | Default Gateway Commands |
| 6.5 | ILF Administration | 6.14 | System Services |
| 6.6 | System Feature Administration | 6.15 | Web Services |
| 6.7 | User Administration | 6.16 | FTP/SFTP File Transfer, Client Session |
| 6.8 | CLI Session Administration | 6.17 | Logon from NE to Remote Host |
| 6.9 | Security Administration | | |

6.1

Default Settings and IP Port Information

The following table lists IP port ranges for 1FINITY product management interfaces, protocols, and services.

Note: The values shown in bold are default values.

Management Interface/ Protocol/Service	Port Range
SSH	22 , 2000...65535
Telnet	23
FTP	21
SFTP	2000... 2202 ...65535
Webserver-HTTP	80 , 2000...65535
Webserver-HTTPS	443 , 2000...65535
NETCONF/SSH	830 , 2000...65535
CLI	Uses SSH
WebGUI	Uses HTTP/HTTPS
REST	Uses HTTP/HTTPS
NTP	123
SNMP	161 , 2000...65535

The following table shows default values for CLI/SSH, NETCONF, and Web server parameters.

Management Interface/ Protocol/Service	Default value
cli-ssh-port	22
netconf-enabled	true
netconf-timeout	30m
netconf-ssh-port	830
webserver-http-port	80
webserver-timeout	30m
webserver-http-enabled	false

The following table shows maximum number of individual sessions supported.

Management Interface/ Protocol/Service	Number of Sessions
NETCONF	20
SNMP	5
WebView	5
CLI	9

Note: The total number of sessions supported concurrently is 20, where each WebView session is counted as 2 and each (netconf/snmp/CLI) session is counted as 1.

6.2 About CLI

In this section:

- | | | | |
|-------|--|-------|------------------------|
| 6.2.1 | CLI Operational Mode | 6.2.5 | CLI User Permissions |
| 6.2.2 | CLI Configuration Mode | 6.2.6 | CLI Command Completion |
| 6.2.3 | CLI Command Prompts | 6.2.7 | CLI Command Comments |
| 6.2.4 | User Guide CLI Command Prompt Convention | 6.2.8 | CLI Messages |

The CLI provides a mechanism for users to enter commands that configure and manage the NE and perform operations, administrative, and maintenance tasks on the NE. The CLI is rendered using data models defined in YANG modules. The CLI is hierarchical, which means that the configuration commands have nested layers.

CLI provides two command modes: *operational mode* and *configuration mode*.

Note: For more detailed CLI information, refer to *1FINITY CLI User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

6.2.1 CLI Operational Mode

Operational mode is the initial mode after successful logon to the CLI. It is primarily used for viewing the system status, controlling the CLI environment, monitoring and troubleshooting network connectivity, and initiating the configuration mode.

When the CLI session is in the operational mode, the command prompt terminates with the ">" character.

Command Example:
fujitsu@fujitsu>

Key operational mode commands are *show* and *request*.

Operational Mode Commands

Command	Description
clear	Clear parameter
commit	Confirm a pending command
compare	Compare running configuration to another configuration or a file
configure	Manipulate software configuration information
describe	Display transparent command information
exit	Exit the management session

Command	Description
file	Perform file operations
ftp	Open a ftp session to another host
help	Provide help information
id	Show user id information
monitor	Show real-time debugging information
quit	Exit the management session
request	Make system-level requests
set	Set CLI properties
set-path	Set relative show path
sftp	Open a sftp session to another host
show	Show information about the system
ssh	Start a secure shell to another host
ssh-key-remove	Remove host from know hosts file
start	Start specified program or command
telnet	Establish a Telnet session to another host
top	Exit to top level and optionally run command
up	Exit one level of configuration

6.2.2

CLI Configuration Mode

To change CLI Session from operational mode to configuration mode, enter the *configure* command.

Command:

```
> configure
```

When the CLI session is in the configuration mode, the command prompt terminates with the "%" character.

Command Example:

```
fujitsu@fujitsu%
```

In configuration mode, all changes to the device configuration are initially done to a copy of the active configuration called the *candidate configuration*. These changes do not take effect until a successful *commit* or *commit confirm* command is entered. Commit activates the candidate configuration on the device. Candidate configurations enable the user to alter the configuration without causing potential damage to the current network operations.

Configuration Mode Commands

Command	Description
annotate	Add a comment to a statement
commit	Commit current set of changes
compare	Show configuration differences
copy	Copy a list entry
delete	Delete a data element
describe	Display transparent command information
edit	Edit a sub-element
exit	Exit from this level
help	Provide help information
insert	Insert a parameter
load	Load configuration from an ASCII file
move	Move a parameter
quit	Exit from this level
rename	Rename an identifier
request	Make system-level requests
resolved	Conflicts resolved
revert	Copy configuration from running
rollback	Roll back database to last committed version
run	Run an operational-mode command
save	Save configuration to an ASCII file
set	Set a parameter
show	Show a parameter
status	Display users currently editing the configuration
tag	Manipulate statement tags
top	Exit to top level and optionally run command
up	Exit one level of configuration
validate	Validate current configuration

6.2.3

CLI Command Prompts

Name	Default Value	Screen Display
prompt1 (The operational mode prompt)	"\u@\h> "	fujitsu@fujitsu>
prompt2 (The configuration mode prompt)	"\u@\h% "	fujitsu@fujitsu%



Caution: Fujitsu does not recommend changing the values of prompt1 or prompt2.

If the default settings are kept, as recommended, the CLI command prompt changes with user, host, and CLI command mode as shown in the following table.

Username	Hostname	Operational Mode Prompt	Configuration Mode Prompt
fujitsu	fujitsu	fujitsu@fujitsu>	fujitsu@fujitsu%
fujitsu	AcmeNet-T100-31	fujitsu@AcmeNet-T100-31>	fujitsu@AcmeNet-T100-31%
fred	AcmeNet-T100-31	fred@AcmeNet-T100-31>	fred@AcmeNet-T100-31%

Note: For 1FINITY NEs, the hostname is the value of *system name*. (Refer to [System Name](#).)

6.2.4

User Guide CLI Command Prompt Convention

In this user guide, CLI commands are typically illustrated using only a single-character, either (>) or (%) for the command prompt. The single-character prompt indicates the applicable CLI command mode:

- (>) indicates that the command is entered in *operational* mode
- (%) indicates that the command is entered in *configuration* mode

The single-character prompt should be interpreted as an abbreviated representation of the full CLI command prompt, which includes `fujitsu@fujitsu` preceding the single character.

The following example shows a CLI command entered in configuration mode.

Command:

```
% show system services telnet
```

If a user has the username fred and is logged onto a system with hostname AcmeNet-T100-31, the screen displays:

Command Example:

```
fred@AcmeNet-T100-31% show system services telnet
```

Note: For 1FINITY NEs, the hostname is the value of *system name*. (Refer to [System Name](#).)

6.2.5

CLI User Permissions

Certain commands require Level-4 (root/superuser) permissions, for example, configuration and provisioning. The default username and password (fujitsu/1finity) provide Level-4 permissions that allow access to all functions on the system.

6.2.6

CLI Command Completion

Typing the full command or option name is not required for the CLI to recognize it. To display possible completions, type the partial command followed immediately by <tab> or <space> (the latter requires the *complete-on-space* option to be enabled).

If the partially typed command uniquely identifies a command, the full command name appears. Otherwise, a list of possible completions is displayed.

Completion is disabled inside quotes. To enter an argument containing spaces, enclose it with either the backslash (\) or double quotation mark (") character.

Command completion applies to filenames and directories. Space completion is disabled when entering a filename.

The following examples illustrate how command completion works.

Example 1

The user enters *com* followed by <tab> or <space>.

Command:

```
> com
```

The system displays two possible completions.

Example:

```
> com
Possible completions:
  commit - Confirm a pending commit
  compare - Compare running configuration to another configuration or a file
```

Example 2

The user enters `co` followed by <tab> or <space>.

Command:

```
> co
```

The system displays three possible completions.

Example:

```
> co
Possible completions:
  commit      - Confirm a pending commit
  compare     - Compare running configuration to another configuration or a file
  configure   - Manipulate software configuration information
```

6.2.7

CLI Command Comments

The `#` symbol indicates that the rest of the line is a comment. In the following example, the CLI ignores `#enable_complete_on_space`.

Command Example:

```
> set complete-on-space true #enable_complete_on_space
```

If `#` character is to be entered as an argument, it has to be prefixed with a backslash (`\`) or used inside quotes (`"`).

6.2.8

CLI Messages

CLI messages appear when a user enters or exits configure mode, commits to a configuration, and enters an invalid command or value.

Example:

```
> show c
-----^ syntax error:
Possible commands starting with c:
  cli - Display cli information
  commit - Display configuration changes
[error]
```

Example:

```
> show configuration
-----^
syntax error: element does not exist
[error]
```

When the user commits to a configuration, the CLI first validates the configuration and, if a problem exists, indicates what the problem is. For example, if an identifier is missing or a value is out of range, a message indicates where the errors are.

Example:

```
% set eqpt shelf 1 slot 0 cardType BDT3-T300 subslot 0 port C11
pluggableInterfaceType QSFP28 admin-status up
[ok]

[edit]
% commit
Aborted: 'eqpt shelf 1 slot 0 subslot 0 port C11' : pluggable/port/card
mismatch
[error]

[edit]
%
```

6.3

System Name

The factory default system name is *fujitsu*.

Note: The default name must be changed from *fujitsu* for the system to turn up as well as for the tributary shelves to be recognized.

Note: The NE hostname is the value of system name.

Show System Name

Command:

```
% show system name
```

Set System Name

Command:

```
% set system name <string, min: 7 chars, max: 20 chars>
```



Remember: In configuration mode (command prompt ends with %), changes are made to the candidate configuration and do not affect in the running configuration until a successful *commit* or *commit confirm* command is entered.

Perform a WARM reboot for the system to turn up as well as for the tributary shelves to be recognized.

Command:

```
> request restartSystem level warm
```

Note: Changing the system name generally changes the CLI command prompt. (Refer to [CLI Command Prompts](#).)

6.4

System Time and Date

In this section:

- 6.4.1 Show System Time and Date
- 6.4.2 Set System Time and Date Manually
- 6.4.3 Configure NTP
- 6.4.4 Configure System Time Zone

User can set system time and date in two ways: manually or through the network using NTP. Regardless of how the user chooses to set system time and date, the user still needs to set the system time zone.

6.4.1

Show System Time and Date

Command:

```
> show system-state clock datetime
```

Output Example:

```
system-state clock datetime 2018-04-26T10:23:44
[ok]
```

6.4.2

Set System Time and Date Manually

Command:

```
> request set-current-datetime current-datetime <YYYY-MM-DDTHH:MM:SS>
```

Command Example:

```
> request set-current-datetime current-datetime 2018-04-26T10:23:44
```

Note: If the NTP is enabled, this operation fails with message `status set-current-datetime is unavailable` when NTP is enabled.

6.4.3

Configure NTP

Show System NTP Configuration

Command:

```
% show system ntp
```

Enable/Disable NTP

Command:

```
% set system ntp <enabled|disabled>
```

Add NTP Server

Command:

```
% set system ntp servers <server_name> address <ip_address> prefer <prefer>
```

Parameter	Value	Description
server_name	<ntp>	Name of server
ip_address	<1xx.xxx.xxx.xxx>	IP address of server
prefer	<True, False>	True—Enable server preference False—Disable server preference

Command Example:

```
% set system ntp servers ntp-b.nist.gov address 132.163.96.5
[ok]
```

Setting prefer is optional. If omitted, prefer is set to false. Repeat, as required, for multiple NTP servers.

Remove NTP Server

Command:

```
% delete system ntp servers <server_name>
```

6.4.4

Configure System Time Zone

Set System Time Zone

Command:

```
% set system clock timezone-name <timezone>
```

Parameter	Value	Description
timezone	<CST6CDT>	Name of time zone (TZ) database, for example, America/Chicago, Europe/Stockholm, or Asia/Tokyo

Show System Time Zone

Command:

```
% show system clock timezone-name
```

Output Example:

```
timezone-name CST6CDT;  
[ok]
```

Note: The system displays the time zone abbreviation, not the TZ database name.

Note: The system shall allow customer to set the date and time to both backward and forward date/time. If customer wants to set date and time backward then the limitation is that the customer can go to the date when OSL was built which can be retrieved via show sw-version.

6.5

ILF Administration

Enable Temporary License Agreement

Command:

```
% set licenses license <license-id> license-temp-agree yes
```

Install License Key

Command:

```
% set licenses license <license-id> license-key <license-key>
```

Show License Status

Command:

```
> show licenses-status license-status
```

6.6

System Feature Administration

In this section:

6.6.1	AutoP	6.6.5	ZTP
6.6.2	RASIS	6.6.6	ZTP-Boot
6.6.3	neMgmtMode	6.6.7	Zeroize System
6.6.4	Telemetry, Streaming	6.6.8	Configure AINS

6.6.1

AutoP

Show AutoP Configuration

Command:

```
% show system autoP
```

Enable/Disable Autoprovisioning

Autoprovisioning is enabled by factory default.

Command:

```
% set system autoP <true|false>
```

6.6.2

RASIS

Note: For more detailed RASIS information, refer to *1FINITY RASIS User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

Provision to trigger-tech-info On-Demand Logging

Note: For L-series blades, on-demand logging supports MicroBlade Log (MBLOG) and Firmware Log (FWLOG).

Trigger-tech-info uses the CLI command to write log messages to the hardware.

Command Examples:

```
> request trigger-tech-info shelf-id <shelf num> trigger <log type>
> request trigger-tech-info shelf-id 1 trigger MBLOG
[ok]
> request trigger-tech-info shelf-id 1 trigger FWLOG
[ok]
```

An alarm raises when a specific log (MBLOG, FWLOG) write is in progress. When write is completed, the alarm clears.

Provision to Create-tech-info

The log (MBLOG, FWLOG) files are stored in `/var/ftp/rasis_log` directory. By default, create-tech-info uses ftp path (`/var/ftp`).

- To collect all log files in a single file.

Command:

```
> request create-tech-info shelf-id <x> slot-id <y> single-file <true/false>
```

Table 44
Input Parameter

Parameter	Value	Description
shelf-id	<0,1>	Shelf number
slot-id	<0,1>	Slot number
single-file	<True, False>	True—Single tech-info file is generated. False—Multiple small tech-info files are generated.

Command Example:

```
> request create-tech-info shelf-id 1 slot-id 1 single-file true
```

- To transfer the log files from hardware to a remote path.

Command:

```
> request transfer sftp action upload local-file-path remote-file-path
```

Table 45
Input Parameter

Parameter	Value	Description
Local file path	<code>/var/ftp/rasis_log/</code>	Path of the file should be shared
Remote file path	<code>//user:password@host_ip_address/home/user/</code>	Remote path of the file

Command Example:

```
> request transfer sftp action upload local-file-path /var/ftp/rasis_log/
TECHINFO_L130_id1_1_20170131201637.tar.gz
remote-file-path /user:password@host_ip_address/home/user/
TECHINFO_L130_id1_1_20170131201637.tar.gz
```

6.6.3 neMgmtMode

Set NE Management Mode

Command:

```
% set system neMgmtMode <mgmtmode>
```

Parameter	Value	Description
mgmtmode	<Bridge, MixMode, Router>	Bridge—Layer 2 bridging runs between management interfaces.
		MixMode—Routing runs between all management interfaces, except LCN1 and LCN2 which are bridged.
		Router—Routing runs between management interfaces.

Show NE Management Mode

Command:

```
% show system neMgmtMode
```

6.6.4

Telemetry, Streaming

Note: For more detailed streaming telemetry information, refer to *1FINITY Telemetry User Guide*, available on the Fujitsu *Partners* website, <https://partners.fnc.fujitsu.com/documentation>.

Enable Streaming Telemetry

Streaming telemetry must be configured by the user via an XML configuration file loaded to the NE. The telemetry feature checks the configuration file on system start-up and if the file is not found, telemetry service shuts down.

By default, streaming telemetry is disabled.

Command:

```
> request telemetry action <telemetry-action> config-file <telemetry-filename>
```

Command Example:

```
> request telemetry action configure config-file /var/ftp/  
telemetry.tar.gz
```

Disable Streaming Telemetry

Command:

```
> request telemetry action delete
```

6.6.5

ZTP

Note: For more detailed ZTP information, refer to *1FINITY ZTP User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

Show ZTP Configuration

Command:

```
> show system ztp
```

Output Example:

```
ztp-enabled true;
[ok]
```

Enable/Disable ZTP

Command:

```
% set system ztp ztp-enabled <false|true>
```

Show ZTP Operational Data

Command:

```
> show system ztp ztp-oper-data
```

Output Example:

```
system ztp ztp-oper-data controller-reg-status not-registered
system ztp ztp-oper-data controller-ip-addr 0.0.0.0
[ok]
```

6.6.6

ZTP-Boot

ZTP-Boot is provisioned by default on L100 Blade.

Both system ztp auto-upgrade and system ztp ztp-enabled flags control the enabling and disabling of ZTP-Boot. The ztp-enabled flag controls the enabling and disabling of ZTP-announcement. If either flag is set to false, ZTP-Boot is disabled. By default, both flags are enabled.

After device turn-up, if auto-upgrade or ztp-enabled is false, the auto-upgrade process is not started.

Enable/Disable ZTP-Boot

Command:

```
% set system ztp ztp-enabled <false|true> auto-upgrade <false|true>
```

6.6.7

Zeroize System

Prerequisites:

- User must have Level-4 or Level-5 privileges assigned to perform this procedure
- System backup has been performed (if required)

Use this procedure if a blade provisioned for encryption is decommissioned or repurposed, and used in nonencryption mode.

Note: The factory default image is stored in a separate partition in the blade. On every software upgrade, the factory default image stored in the partition gets upgraded. Hence zeroization at any point returns the blade to the image which is actively running.

Note: System zeroization is denied when SW validation is in progress. Blade sends a *sysZeroizationFailed* alarm if the zeroization process fails.

Table 46
Subsets of Zeroized Data Overwritten on Partition File

Zeroized Data Subsets	
Critical security parameters	■ User passwords ■ PSK used for encryption ■ Web certificates ■ RADIUS files
User data files	■ \$HOME folders
Log files	■ Encryption-related log files ■ <i>/var/rasis/*</i>
User configuration files	■ <i>/var/shared/confd/bankX/*.cdb</i> ■ <i>/var/shared/confd/default/*.cdb</i> ■ <i>/var/confd/*.cdb</i>



Notice: This step is for zeroizing the tributary blade. For zeroizing the Main blade, proceed to [Step 2](#).

Step 1

Place the blade into shelf provisioning mode.

Step 2

Perform zeroization on the system (blade).



Caution: The system zeroization process can take up to 2 hours to complete. Do not power down the system during this 2-hour time frame and until communication can be reestablished with the blade after the zeroization process.

Command:

```
> request zeroize-system
```

Output:

The system will be reset to the factory status and cannot be recovered. It will take a couple of hours to complete. Really want to do this? [no,yes]

Step 3

Enter `yes` to start zeroization of system.

Output:

```
status Successful
status-message Zeroization of system has started.
```

Step Result:

The blade is reset to the factory default software status. The blade reboots after 3...6 minutes.

Note: `sysZeroizationCompleted` alarm is raised before the system is going down for reboot.

Step 4

After the system zeroization process is completed and the system reboots, reestablish communication with the blade.

✓ **This task is complete.**

6.6.8

Configure AINS

User can specify the target port for configuring AINS.

Step 1

Retrieve systemwide provisionable timeout.

Command:

```
> show system sys-vstimer
```

Step 2

Provision system valid signal timer (VSTIMER).

Note: Supported values for VSTIMER are 00-01 to 48-00.

Command Example:

```
% set system sys-vstimer 08-00
```

Step 3

Set admin-status on the target port to DOWN.

Command Example:

```
% set eqpt shelf 1 slot 0 subslot 0 port E1 admin-status down
```

Step 4

Provision AINS on the target port.

Note: AINS can be enabled only when admin status of the target port is set to DOWN.

Command Example:

```
% set eqpt shelf 1 slot 0 subslot 0 port E1 ains enabled
```

Step Result:

Alarm free time (ACTVST) increases every alarm-free minute for the entity under test.

Note: When VSTIMER expires, AINS setting is disabled and admin-status automatically transitions to UP. After transition, the alarm is raised as normal severity.

Step 5

Save the changes.

Command:

```
% commit
```

Step 6

Retrieve current timer count value as ACTVST for the target port.

Command Example:

```
> show eqpt shelf 1 slot 0 subslot 0 port E1 ACTVST
```

✓ **This task is complete.**

6.7

User Administration

In this section:

- 6.7.1 Password Rules
- 6.7.2 Username Rules
- 6.7.3 User Group Permissions
- 6.7.4 Create/Edit/Delete User
- 6.7.5 Show Users
- 6.7.6 Change User Password

6.7.1

Password Rules

Password Modes

The two modes for password content are BASIC and ENHANCED. The default mode is BASIC. ENHANCED mode has more restrictive content and length rules. The password mode can be changed using CLI command:

Command:

```
% set security systemwide pcontent <BASIC|ENHANCED>
```

In the YANG model, the password mode is set using keyword *pcontent* in node *security:systemwide*.

Password Length

The minimum password length is set by PMIN. The maximum password length is 128. PMIN can be changed using CLI command:

Command:

```
% set security systemwide pmin <6...20>
```

In the YANG model, PMIN is value of keyword *pmin* in node *security:systemwide*.

General Content Rules

The password character set includes the following characters:

- Lowercase alphabetic: a...z
- Uppercase alphabetic: A...Z
- Numeric: 0...9
- Special: ! @ \$ % ^ () _ + ~ { } [] . - & *

Password must not contain the same username (case insensitive).

Passwords are stored in hashed SHA-256 format on NE.

BASIC Mode Content Rules

Minimum Length = 6 or PMIN, whichever is greater.

The password must contain at least one alphabetic and one non-alphabetic character.

ENHANCED Mode Content Rules

Minimum Length = 8 or PMIN, whichever is greater.

The password must contain at least two characters from each of the following categories:

- Lowercase alphabetic: a...z
- Uppercase alphabetic: A...Z
- Numeric: 0...9
- Special: ! @ \$ % ^ () _ + ~ { } [] . - & *

Password Scenarios

If a password value is entered on a command line, characters following # are ignored (interpreted as a comment). For example, in the following command the password is set to *abc12* (ignoring *#34*).

Command Example:

```
% set users user user1 password abc12#34
```

The following command fails because \ causes # to be interpreted literally and # is not a valid username character.

Command Example:

```
% set users user user1 password abc12\#34
```

6.7.2

Username Rules

If a username is already in the `/etc/passwd` file or in the current database, it cannot be used to create a new user.

The following rules are used to assign a username:

- The first character must be a lowercase alphabetic character: a...z
- Remaining characters can be any of the following characters:
 - Lowercase alphabetic: a...z
 - Numeric: 0...9
 - Hyphen: -
 - Underscore: _

Username Length

UMIN determines the minimum length of usernames. The default value of UMIN is 3. The range of UMIN is 3...32. The value of UMIN can be changed using CLI command.

Command:

```
% set security systemwide umin <3...32>
```

In the YANG model, UMIN is set by the value of keyword *umin* in node *security:systemwide*.

Username Scenarios

If a username value is entered on a command line, characters following # are ignored (interpreted as a comment). For example, in the following command the username is set to *user1* (ignoring #345).

Command Example:

```
% set users user user1#345
```

6.7.3

User Group Permissions

The *user group* defines the user level of access to the system. Configuration, turn-up, and certain other commands require Level-4 (root) user group permissions. The default username and password (fujitsu/1finity) provide Level-4 permissions that allow access to all functions on the system.

Table 47
Groups and Permission Levels

Group	Role	Access Privilege Description
Level-1	Read-Only User	Read-only access privileges, including retrieving alarm and PM information Note: Level-1 cannot access security- or encryption-related information.
Level-2	PM Operator	Level-1 access privileges, plus can reset PMs
Level-3	Equipment Operator	Level-2 access privileges, plus access to provision equipment and interfaces
Level-4	Super User	Full access to system, provisioning, security, and crypto-encryption functions
Level-5	System Operator	Level-3 access privileges, plus access to system functions Note: Level-5 cannot access security and crypto-encryption functions. Note: Level-5 cannot add, change, or delete users.
Level-6	Security Operator	Security access privileges only, plus can add, change, or delete users Note: Level-6 cannot access system, provisioning, and crypto-encryption functions.
Level-7	Crypto-User	Provision data-encryption interfaces and read the encryption status on the encryption-enabled interfaces Note: Level-7 only applies to products that support data encryption.
Level-8	Crypto-Officer	Level-7 access privileges, plus can create crypto-user, crypto-officer, and can set global encryption policy Note: Level-8 only applies to products that support data encryption. Note: A Level-4 user must create the first Level-8 user account. Additional Level-8 user accounts can be created by a Level-4 or Level-8 user.

6.7.4

Create/Edit/Delete User

Users can be created and edited with the following CLI command, where `<username>` conforms to username rules, `<password>` conforms to password rules, and `<level-1...level-6>` declares the user group.

Command:

```
% set users user <username> password <password> group <level-1...level-6>
```

The maximum number of users that can be created is 100.

In the YANG model, each *user* is a child node of the *users* node.



Remember: In configuration mode (command prompt ends with %), changes are made to the candidate configuration and do not affect in the running configuration until a successful *commit* or *commit confirm* command is entered.

Users can be deleted with the following CLI command.

Command:

```
% delete users user <username>
```

6.7.5

Show Users

The following CLI command displays all users in the candidate configuration.

Command:

```
% show users
```

The following CLI command compares users in the running configuration to users in the candidate configuration.

Command:

```
% compare running users
```

6.7.6

Change User Password

Any user can change their own password using the following CLI command, where <newPassword> conforms to password rules and does not match <currentPassword>.

Command:

```
> request change-password currentPassword <currentPassword> newPassword  
<newPassword> newPasswordConfirm <newPassword>
```

6.8

CLI Session Administration

In this section:

6.8.1 Enable/Disable CLI Complete-on-Space

6.8.2 Change Idle Timeout

6.8.3 Force Logoff

6.8.1

Enable/Disable CLI Complete-on-Space



Hint: Setting `complete-on-space` to `true` can speed up the CLI session.

Any user can enable and disable the `complete-on-space` setting for their current CLI session. At the start of every CLI session, `complete-on-space` is set to `false`, which means CLI command completion is only triggered by the `<tab>` key. If the user wants the `<space>` key to trigger CLI command completion, set `complete-on-space` to `true` using the following CLI command.

Command:

```
% set complete-on-space <true|false>
```

6.8.2

Change Idle Timeout

Users are automatically logged off the system if the session is idle longer than *idle-timeout* seconds. Users can change the value of *idle-timeout* for the current session using the following CLI command.

Command:

```
% set idle-timeout <0...8192>
```

6.8.3

Force Logoff

Level-4 or Level-6 privileges are required to perform this command.

The following CLI command terminates all CLI sessions for user `<username>`.

Command:

```
> request system logout user <username>
```

6.9

Security Administration

In this section:

- 6.9.1 SSH Server Service Host Keys
- 6.9.2 Systemwide Security Settings
- 6.9.3 Default User Security Settings
- 6.9.4 User Login and Authentication Log
- 6.9.5 RADIUS Authentication
- 6.9.6 TACACS+ Authentication

6.9.1

SSH Server Service Host Keys

The SSH host key is used by clients of the system SSH server service.

The following CLI command generates a new SSH host key, replacing the current SSH host key.

Command:

```
> request generate-ssh-host-key
```

The following CLI command shows the current SSH host key status and fingerprint.

Command:

```
> request show-ssh-host-key
```

6.9.2

Systemwide Security Settings

The following CLI command displays the system security settings.

Command:

```
% show security systemwide
```

Output Example:

```
umin          3;
pmin          6;
pcontent      BASIC;
prot          OFF;
smt           Y;
dural         1;
maxinv        3;
pre-login-banner "Welcome to the FUJITSU 1FINITY\nCopyright Fujitsu Network
Communications";
post-login-banner "NOTICE: THIS IS A PRIVATE COMPUTER SYSTEM. UNAUTHORIZED
ACCESS OR USE MAY LEAD TO PROSECUTION.";
lastlogin     n;
warn          Not-Settable;
debug         n;
authentication-order [ LOCAL ];
[ok]
```

6.9.3

Default User Security Settings

The following CLI command displays default user security settings.

Command:

```
% show security defaults
```

Output Example:

```
uage          OFF;
page          OFF;
minit         OFF;
reauth        n;
idle-timeout  1800;
[ok]
```

6.9.4

User Login and Authentication Log

The following CLI command displays entries in the security log file: /var/log/auth.log.

Command:

```
> request security-log-get num-of-lines <num-of-lines> read-from-linenumner <read-
from-linenumner>
```

Parameter	Value	Description
num-of-lines	<1, 2, ...>	Number of lines
read-from-linenumner	<2>	Line number to read from

6.9.5

RADIUS Authentication

Note: For more detailed RADIUS information, refer to *1FINITY RADIUS User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

Enable/Disable RADIUS Authentication

Enter the following command.

Command:

```
% set security systemwide authentication-order
```

The user is prompted to enter the order in which the 1FINITY device authenticates users.

```
[list] ():
```

To enable RADIUS authentication, include RADIUS in the list. To disable RADIUS authentication, omit RADIUS from the list. Value LOCAL must appear last in the list.

Command Example (RADIUS enable):

```
% set security systemwide authentication-order
[list] (LOCAL): RADIUS LOCAL
[ok]
```

```
[edit]
% commit
```

Command Example (RADIUS disable):

```
% set security systemwide authentication-order
[list] (RADIUS LOCAL): LOCAL
[ok]
```

```
[edit]
% commit
```

Show Authentication Order

Command:

```
% show security systemwide authentication-order
```

6.9.6

TACACS+ Authentication

Note: For more detailed TACACS+ protocol information, refer to *1FINITY TACACS+ Protocol User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

Enable/Disable TACACS+ Authentication

Enter the following command.

Command:

```
% set security systemwide authentication-order
```

The user is prompted to enter the order in which the 1FINITY device authenticates users.

```
[list] ():
```

To enable TACACS+ authentication, include `TACACS` in the list. To disable TACACS+ authentication, omit `TACACS` from the list. Value `LOCAL` must appear last in the list.

Command Example (TACACS+ enable):

```
% set security systemwide authentication-order
[list] (LOCAL): TACACS LOCAL
[ok]
```

```
[edit]
% commit
```

Command Example (TACACS+ disable):

```
% set security systemwide authentication-order
[list] (TACACS LOCAL): LOCAL
[ok]
```

```
[edit]
% commit
```

Show Authentication Order

Command:

```
% show security systemwide authentication-order
```

6.10

Syslog Administration

The 1FINITY NEs support syslog filtering, saving syslogs to local syslog files, and sending syslogs to the remote syslog server. The filtering can be done based on facility and severity. When the filtered syslogs are saved locally, a maximum of four such files can be created.

Note: For more detailed syslog information, refer to *1FINITY Syslog User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

Display Syslog

Command:

```
> request syslog-get num-of-lines <#lines> read-from-linenumner <starting-line-number>
```

This command displays the contents of the syslog log `/var/log/syslog`.

Clear/Reset Syslog

Command:

```
> request clear-syslog
```

This command deletes the contents of the syslog log `/var/log/syslog`.

Check Local Syslog Filtering

Command:

```
% show syslog actions file log-file
```

Configure Local Syslog Filtering

Command:

```
% set syslog actions file log-file <log-file-name> log-selector log-facility <log-facility-selection> severity <severity-selection>
```

This command captures user logon/logoff events to a local syslog file on disk.

Delete Local Syslog Filtering

Command:

```
% delete syslog actions file log-file <log-file-name>
```

Check Remote Syslog Filtering

Command:

```
% show syslog actions remote
```

Configure Remote Syslog Filtering

Command:

```
% set syslog actions remote destination <arbitrary-endpoint-name> log-selector  
<log-facility-selection> severity <severity-selection>
```

This command captures user logon/logoff events to a remote syslog server.

Delete Remote Syslog Filtering

Command:

```
% delete syslog actions remote destination <endpoint-name>
```

6.11

Provision Proxy-ARP

Prerequisite:

- The user must be logged on as fujitsu/1finity or have Level-4 privileges assigned to perform this procedure.

This procedure describes how to provision Proxy-ARP for the L100 Blade.

Note: The network management system accesses the RNE through a DCN router that in turn interfaces with a GNE on which an embedded communication channel, for example, OSC, with an IP subnet and has been configured. This GNE has an OSPF or static route provisioned on its OSC port, and has Proxy-ARP provisioned on the specific interfaces used to communicate with the DCN router, thus, enabling the network management system to connect to the RNE.

Note: The user can run OSPF on the embedded side or can provision static routes.

Step 1

Enter configuration mode, if necessary.

Command:

```
> configure
```

Step 2

Is the LCN1 port provisioned?

If YES:

Proceed to [Step 4](#).

If NO:

Continue with next step.

Step 3

Enable Ethernet and line rate on the LCN1 port.

Command:

```
% set interface interfaces eth-1/0/0/LCN1 type ethernetCsmacd enabled ethernet  
speed 100 admin-status up
```

Note: By default, Ethernet interface and IP interface are autoprovisioned.

Step 4

Set IP address on the LCN1 interface.

Command:

```
% set interfaces interface <interface name> type <type> ip ipv4 <enabled/disabled>  
forwarding <true/false> address <ip_address> netmask <netmask address>
```

Parameter	Value	Description
interface name	<ip-1/0/0/LCN1>	Name of IP interface instance
type	<ipv4>	Type of IP interface
enabled/disabled	<Enabled, Disabled>	Enabled—Enable IPv4 Disabled—Disable IPv4
true/false	<True, False>	True—Enable forwarding False—Disable forwarding
ip_address	<1xx.xxx.xxx.xxx>	IP address of the LCN interface
netmask address	<1xx.xxx.xxx.xxx>	Subnet mask address

Command Example:

```
% set interfaces interface ip-1/0/0/LCN1 type ip ipv4 enabled forwarding true
address 10.25.25.100 netmask 255.255.255.0
```

Note: Disable client IPv4 over LCN1 IP interface.

Step 5

Set NEM IP address for the GNE.

Command:

```
% set interfaces interface ip-1/0/0/NEM type ip ipv4 forwarding true address
10.25.25.100 prefix-length 32 preferred true
```

Step 6

Set NEM IP address for the RNE.

Command:

```
% set interfaces interface ip-1/0/0/NEM type ip ipv4 forwarding true address
10.25.25.100 prefix-length 32 preferred true
```

Step 7

Provision the E-SC-E1 port equipment.

Command:

```
% set eqpt shelf 1 slot 0 subslot 0 port E-SC-E1 pluggableInterfaceType 1GEX admin-
status up
```

Step 8

Enable Ethernet and line rate on the E-SC-E1 port.

Command:

```
% set interfaces interface eth-1/0/0/E-SC-E1 type ethernetCsmacd ethernet speed
1000 admin-status up
```

Step 9

Provision an IP interface on the E-SC-E1 port that supports the OSC link.

Command:

```
% set interfaces interface ip-1/0/0/E-SC-E1 type ip ipv4 enabled forwarding true
address-src ip-1/0/0/NEM mtu 1500
```

Step 10

Configure an onlink static route over OSC interface.

Command:

```
% set routing routing-instance MCN routing-protocols routing-protocol static FNC
static-routes ipv4 route 192.168.200.1/32 next-hop onlink-next-hop-addr
192.168.200.1 onlink-outgoing-intf ip-1/0/0/E-SC-E1
```

Step 11

Commit the changes.

Command:

```
% commit
```

Step 12

Enable Proxy-ARP for IPv4 connection.

Command:

```
% set interfaces interface ip-1/0/0/LCN1 ipv4 proxy-arp enabled
```

Step 13

Check the configuration.

Command:

```
% show routing routing-instance
```

✓ **This task is complete.**

6.12

LCN Port Commands

Note: The LCN port can have only one address at a time.

Note: When a new LCN IP address is entered, the address that is already on the LCN must be deleted before the changes are committed.

Set LCN

This command provisions IPv4 or IPv6 address interface for LCN port.

Command:

```
% set interfaces interface <interface> <ipv4 | ipv6> <enabled> forwarding <packet-  
forwarding> <true | false> address <ipv4-address | ipv6-address> prefix-length  
<prefix-length>
```

Delete LCN

This command deletes the IP address interface for LCN port.

Command:

```
% delete interfaces <interface> <ipv4-address | ipv6-address>
```


6.13

Default Gateway Commands

Default Gateway Commands

This command creates routing instance for network gateway.

Command:

```
% set routing routing-instance MCN routing-protocols routing- protocol static FNC
static-routes ipv4 route 0.0.0.0/0 next-hop next-hop-address 1xx.xxx.xxx.xxx
```

6.14

System Services

In this section:

- 6.14.1 Show System Services
- 6.14.2 FTP Server Service
- 6.14.3 NETCONF Service
- 6.14.4 SFTP Server Service

- 6.14.5 SNMP Agent Service
- 6.14.6 SSH Server Service
- 6.14.7 Telnet Server Service
- 6.14.8 Web Server Services

6.14.1

Show System Services

The following command displays the current configuration of all system services.

Command:

```
% show system services
```

Output Example:

```
ssh-server {
    ssh-server-enabled true;
    ssh-server-port      22;
    algorithms {
        mac          [ hmac-md5 hmac-sha1 hmac-sha2-256 hmac-sha2-512 hmac-sha1-96
hmac-md5-96 ];
        encryption [ aes128-ctr aes192-ctr aes256-ctr aes128-cbc aes256-cbc 3des-
cbc ];
    }
}
web-server {
    webgui-enabled true;
    webgui-timeout 30m;
    rest {
        rest-enabled false;
    }
    http {
        http-port 80;
    }
}
ftp {
    ftp-server {
        ftp-server-enabled false;
        ftp-server-port      21;
    }
}
sftp {
    sftp-server {
        sftp-server-enabled false;
        sftp-server-port      2202;
    }
}
telnet {
    telnet-enabled false;
    telnet-port      23;
}
netconf {
    netconf-enabled true;
    netconf-port      830;
    netconf-timeout 30m;
}
snmp {
    snmp-enabled      false;
    snmp-port          161;
    authFailureTrap    enabled;
    alarm-trap          enabled;
    event-trap          enabled;
    tca-trap            enabled;
}
[ok]
```

6.14.2

FTP Server Service

The FTP server service, if enabled, listens on port 21.

Show FTP Server Configuration

Command:

```
% show system services ftp ftp-server
```

Enable/Disable FTP Server

By default, the FTP server service is disabled.

Command:

```
% set system services ftp ftp-server ftp-server-enabled <true|false>
```

6.14.3

NETCONF Service

Note: For more detailed NETCONF information, refer to *1FINITY NETCONF Protocol User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

Show NETCONF Service Configuration

Command:

```
% show system services netconf
```

Enable/Disable NETCONF

By default, the NETCONF service is enabled.

Command:

```
% set system services netconf netconf-enabled <true|false>
```

Set NETCONF Port

By default, the NETCONF service port is set to 830.

Command:

```
% set system services netconf netconf-port <830|2000...6021|6023...32767|61001...65535>
```

Set NETCONF Timeout

By default, the NETCONF timeout is set to 30 minutes.

Command:

```
% set system services netconf netconf-timeout <nYnMnDnHnMnS>
```

6.14.4

SFTP Server Service

The SFTP server service, if enabled, listens on port 2202 by default.

Show SFTP Server Configuration

Command:

```
% show system services sftp
```

Enable/Disable SFTP Server

By default, the SFTP server service is disabled.

Command:

```
% set system services sftp sftp-server sftp-server-enabled <true|false>
```

Set SFTP Server Port

By default, the SFTP server port is set to 2202.

Command:

```
% set system services netconf netconf-port <2000...6021|6023...32767|61001...65535>
```

6.14.5

SNMP Agent Service

Note: For more detailed SNMP information, refer to *1FINITY SNMP User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

Show SNMP Agent Configuration

Command:

```
% show system services snmp
```

Enable/Disable SNMP Agent

By default, the SNMP agent service is disabled.

Command:

```
% set system services snmp snmp-enabled <true|false>
```

Set SNMP Agent Port

By default, the SNMP agent port is set to 161.

Command:

```
% set system services snmp snmp-port <161|2000...6021|6023...32767|61001...65535>
```

Enable/Disable SNMP Alarm Trap Notification

By default, the SNMP alarm trap notification is enabled.

Command:

```
% set system services snmp alarm-trap <enabled|disabled>
```

Enable/Disable SNMP Authentication Failure Trap Notification

By default, the SNMP authentication failure trap notification is enabled.

Command:

```
% set system services snmp authFailureTrap <enabled|disabled>
```

Enable/Disable SNMP Event Trap Notification

By default, the SNMP common event trap notification is enabled.

Command:

```
% set system services snmp event-trap <enabled|disabled>
```

Enable/Disable SNMP TCA Trap Notification

By default, the SNMP TCA trap notification is enabled.

Command:

```
% set system services snmp tca-trap <enabled|disabled>
```

6.14.6

SSH Server Service

Note: CLI uses the SSH server service.

Show SSH Server Configuration

Command:

```
% show system services ssh-server
```

Enable/Disable SSH Server

By default, the SSH server service is enabled.

Command:

```
% set system services ssh-server ssh-server-enabled <true|false>
```

Set SSH Server Port

By default, the SSH server port is set to 22.

Command:

```
% set system services ssh-server ssh-server-port <22|2000...6021|6023...32767|61001...65535>
```

6.14.7

Telnet Server Service

The Telnet server service, if enabled, listens on port 23.

Show Telnet Server Configuration

Command:

```
% show system services telnet
```

Enable/Disable Telnet Server

By default, the Telnet server service is disabled.

Command:

```
% set system services telnet telnet-enabled <true|false>
```

6.14.8

Web Server Services

Web server services include HTTP, HTTPS, WebGUI, and REST. HTTP and HTTPS are standard web server services. WebGUI and REST are 1FINITY-specific web services that make use of the HTTP/HTTPS server services:

- **HTTP service**—Provides an insecure web server service with HTTP protocol
- **HTTPS service**—Provides a secure web server service with HTTPS protocol
- **WebGUI service**—Provides a graphical user interface (through a web browser) for users to interact with and configure the 1FINITY system
- **REST service**—Provides a web service that allows users or software systems to interact with and configure the 1FINITY system through a YANG-data-model-based REST API

6.15

Web Services

In this section:

- 6.15.1 HTTP Server Service
- 6.15.2 HTTPS Server Service
- 6.15.3 REST API Service
- 6.15.4 WebGUI Service
- 6.15.5 Log On NE with WebGUI

Web server services include HTTP, HTTPS, WebGUI, and REST. HTTP and HTTPS are standard web server services. WebGUI and REST are 1FINITY-specific web services that make use of the HTTP/HTTPS server services:

- **HTTP service**—Provides an insecure web server service with HTTP protocol
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- **WebGUI service**—Provides a graphical user interface (through a web browser) for users to interact with and configure the 1FINITY system
- **REST service**—Provides a web service that allows users or software systems to interact with and configure the 1FINITY system through a YANG-data-model-based REST API

The following CLI command displays the configuration of all web-server services.

Command:

```
% show system services web-server
```

Output Example:

```
webgui-enabled true;
webgui-timeout 30m;
rest {
    rest-enabled false;
}
http {
    http-port 80;
}
https {
    system-generated-certificate;
    https-port 443;
}
[ok]
```

6.15.1

HTTP Server Service

Show HTTP Server Configuration

Command:

```
% show system services web-server
```

Note: This command displays the configuration of all web-server services.

Enable HTTP Server

By default, the HTTP server service is disabled.

Command:

```
% set system services web-server http
```

Set HTTP Server Port

By default, the HTTP server port is set to 80.

Command:

```
% set system services web-server http http-port <80|2000...6021|6023...32767|61001...65535>
```

6.15.2

HTTPS Server Service

Show HTTPS Server Configuration

Command:

```
% show system services web-server
```

Note: This command displays the configuration of all web-server services.

Enable HTTPS Server

By default, the HTTPS server service is disabled.

Command:

```
% set system services web-server https system-generated-certificate
```

Set HTTPS Server Port

By default, the HTTPS server port is set to 443.

Command:

```
% set system services web-server https https-port <443|2000...6021|6023...32767|61001...65535>
```

6.15.3

REST API Service

The REST API web service uses the HTTP/HTTPS server service.

Note: For more detailed REST API information, refer to *1FINITY REST API User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

Show REST API Configuration

Command:

```
% show system services web-server
```

Note: This command displays the configuration of all web-server services.

Enable REST API Service

By default, the REST API service is disabled.

Command:

```
% set system services web-server rest rest-enabled <false|true>
```

6.15.4

WebGUI Service

The WebGUI service uses the HTTP/HTTPS server service.

Note: For more detailed WebGUI information, refer to *1FINITY WebGUI User Guide*, available on the Fujitsu Partners website, <https://partners.fnc.fujitsu.com/documentation>.

Show WebGUI Configuration

Command:

```
% show system services web-server
```

Note: This command displays the configuration of all web-server services.

Enable WebGUI Service

By default, the WebGUI service is disabled.

Command:

```
% set system services web-server webgui-enabled <false|true>
```

Set WebGUI Service Timeout

By default, the WebGUI service timeout is set to 30 minutes.

Command:

```
% set system services web-server webgui-timeout <nYnMnDnHnMnS>
```

6.15.5

Log On NE with WebGUI

Before a user can log on the NE using the WebGUI, the NE HTTP/HTTPS server service must be enabled, the NE WebGUI service must be enabled, the NE must be available on the network, and the user must know the NE IP address and have login credentials (username and password).

1. Open a web browser.
2. In the address box of the web browser, enter one of the following IP addresses:
`http://<ip_address>`
`https://<ip_address>`

The WebGUI login screen appears.

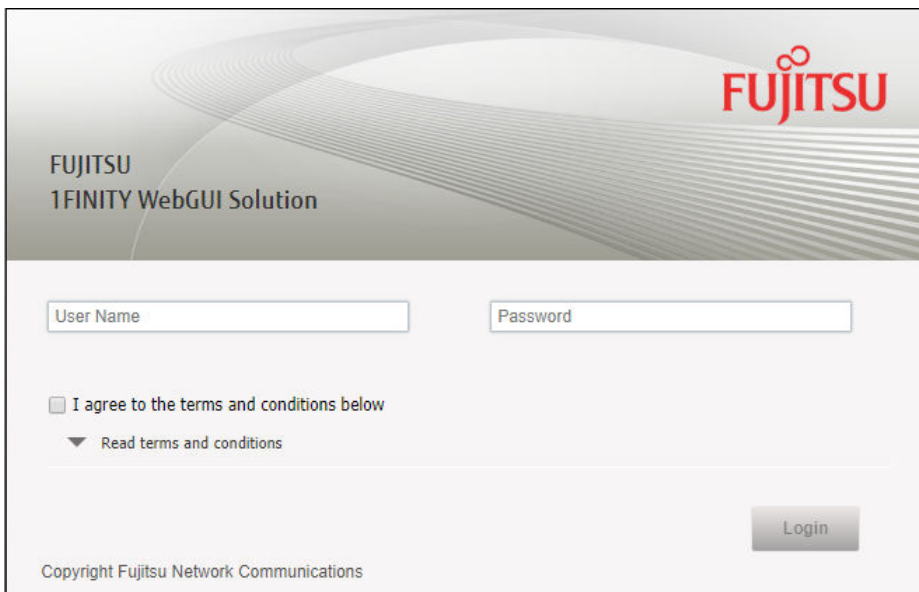
The image shows the Fujitsu 1FINITY WebGUI Solution login screen. At the top, there is a header with the Fujitsu logo and the text "FUJITSU 1FINITY WebGUI Solution". Below the header, there are two input fields: "User Name" and "Password". Underneath these fields, there is a checkbox labeled "I agree to the terms and conditions below" and a link "Read terms and conditions". At the bottom right, there is a "Login" button. At the bottom left, there is a copyright notice: "Copyright Fujitsu Network Communications".

Figure 99
Log On Screen

3. Enter username and password.

6.16

FTP/SFTP File Transfer, Client Session

The FTP/SFTP file transfer directory is `/var/ftp`.

Show Files in Transfer Directory

Command:

```
> request show-file filename <path>
```

Parameter	Value	Description
path	<code></var/ftp></code>	Full path of file transfer directory

Request FTP/SFTP File Transfer

Use this command to open an FTP/SFTP client session with a remote FTP/SFTP server and either download a file from the server or upload a file to the server.

Command:

```
> request transfer <ftp|sftp> action [upload|download] local-file-path
<local-file-path> remote-file-path <remote-file-path>
```

Parameter	Value	Description
ftp sftp	<code><ftp, sftp></code>	FTP—Use FTP file transfer protocol SFTP—Use SFTP file transfer protocol
upload download	<code><upload, download></code>	Upload—Upload to server Download—Download from server
local-file-path	<code>/var/ftp/filename</code>	Full path filename of local file
remote-file-path	<code>ftp://user:password@host/filename</code> <code>sftp://user:password@host/filename</code>	Uniform Resource Identifier (URI) path to remote file

Command Example (FTP Upload):

```
> request transfer ftp action upload local-file-path /var/ftp/1FINITY-071716.dbs
remote-file-path ftp://anonymous:anonymous@192.168.1.2/1FINITY-071716.dbs
```

Command Example (FTP Download):

```
> request transfer ftp action download local-file-path /var/ftp/1FINITY-071716.PGM
remote-file-path ftp://anonymous:anonymous@192.168.1.2/1FINITY-071716.PGM
```

Command Example (SFTP Upload):

```
> request transfer sftp action upload local-file-path /var/ftp/1FINITY-071716.dbs
remote-file-path sftp://anonymous:anonymous@192.168.1.2/1FINITY-071716.dbs
```

Command Example (SFTP Download):

```
> request transfer sftp action download local-file-path /var/ftp/
1FINITY-071716.PGM remote-file-path sftp://
anonymous:anonymous@192.168.1.2/1FINITY-071716.PGM
```

Delete File from Transfer Directory

Command:

```
> request delete-file filename <filename>
```

Parameter	Value	Description
filename	/var/ftp/filename	Full path filename of local file

6.17

Logon from NE to Remote Host

In this section:

6.17.1 Telnet Client Session

6.17.2 SSH Client Session

6.17.1

Telnet Client Session

This command creates a Telnet session from the 1FINITY node (the Telnet client) to a Telnet server host.

Command:

```
> telnet host <host-address> port <port number> user <username>
```

Parameter	Value	Description
host-address	<1xx.xxx.xxx.xxx>	Hostname or IP address
port number	23	Port number
username	<user 1, user 2>	Username for logging on to the remote host

6.17.2

SSH Client Session

Open SSH Client Session

Command:

```
> ssh host <host-address> port <port number> user <username>
```

Parameter	Value	Description
host-address	<1xx.xxx.xxx.xxx>	Hostname or IP address
port number	22	Port number
username	<user 1, user 2>	Username for logging on to the remote host

Remove SSH Server Key Entry

This command removes the SSH key from the 1FINITY node (the SSH client) that was given to it by an SSH server host.

Command:

```
> ssh-key-remove -host <hostname or IP address>
```

7

ROADM Turn-Up and Operations

In this chapter:

- | | | | |
|-----|---|------|--|
| 7.1 | Enable Flexible Grid | 7.10 | Provision Cross-Connects |
| 7.2 | Enable OSNR for Main Blade | 7.11 | Adding Degree to ROADM Node |
| 7.3 | Provision Raman Port Connections | 7.12 | Adding L110 Blade to ROADM |
| 7.4 | Provision L100 Main Blade OTS/OMS/OCH Interfaces | 7.13 | Adding Raman Blade to ROADM Node |
| 7.5 | Set 100 Mb/s OSC Power Pad | 7.14 | Provision L100 Blade for FLASHWAVE 9500 Interoperability |
| 7.6 | Provision Direct Access (Colorless) Channels | 7.15 | Reprovision L100 Blade for FLASHWAVE 9500 Interoperability |
| 7.7 | Provision L110 Tributary Blade OCH Interfaces | 7.16 | Interoperability with L130 Blade |
| 7.8 | Link PIL1-3PS2 Client Port to Transponder/Muxponder Edge Port | | |
| 7.9 | Provision Add/Drop Port Connections | | |

7.1

Enable Flexible Grid

Step 1

If required, activate the flexible grid capability ILF.

Note: The key can be regular or golden.

Command:

```
% set licenses license FLEXGRID license-key <key string>
```

Step 2

Set the WDM configuration.

Command:

```
% set wdmcfg grid-type flex
```

Step 3

For equipment port configuration, the number of lanes should be 128. Provision the E1 port on the L100 Main blade.

Command:

```
% set eqpt shelf 1 slot 0 cardType BDL1-3R11 subslot 0 port E1 admin-status up  
pluggableInterfaceType Edge num-lanes 128
```

Note: This command needs to be performed on the L100 tributary shelves as well.

Step 4

Provision OTS, OMS, and OCH interfaces. Refer to [Provision L100 Main Blade OTS/OMS/OCH Interfaces](#).

✓ **This task is complete.**

7.2

Enable OSNR for Main Blade

Note: By default, the *OSNR* is disabled for a Main blade.

Step 1

Enter configure mode.

Command:

```
> configure
```

Step 2

Enable OSNR for all blades in NE.

Note: OSNR setting must be enabled in L100 Blade to support interoperability with L130 Blade.

Command:

```
% set wdmcfg osnr enable
```

Step 3

Save the changes.

Command:

```
% commit
```

Step 4

Verify OSNR settings.

Command:

```
% show wdmcfg osnr
```

Output:

```
osnr enable
```

✓ **This task is complete.**

7.3

Provision Raman Port Connections

Prerequisites:

- Appropriate Raman fiber-optic cable connection must be completed. For more information, refer to [Install L160 Raman Cabling](#).
- L100 and L160 Blades must be provisioned

This procedure describes how to provision port-connection between Raman blade and L100 Blade.

Note: L100 Blade Issue 14, FC95453R11-I14, and later blade must be used with L160 Raman blade.

Step 1

Provision the edge port E1 of L100 Blade.

Note: If autoprovisioning is enabled, the E1 port is automatically provisioned.

Command:

```
% set eqpt shelf 1 slot 0 cardType BDL1-3R11 subslot 0 port E1 admin-status up
pluggableInterfaceType Edge num-lanes 128
```

Step 2

Provision the client port C1 on the Raman tributary blade.

Note: If autoprovisioning is enabled, the client port C1 is automatically provisioned.

Command:

```
% set eqpt shelf 101 slot 0 cardType BDL1-3M21 subslot 0 port C1 admin-status up
pluggableInterfaceType Client num-lanes 128
```

Step 3

Provision port-connection between Raman client port and L100 edge port.

Command:

```
% set port-connection D1-RAMAN src-shelf 101 src-slot 0 src-subslot 0 src-port C1
dst-shelf 1 dst-slot 0 dst-subslot 0 dst-port E1
```

✓ **This task is complete.**

7.4

Provision L100 Main Blade OTS/OMS/OCH Interfaces

Step 1

Enable the Optical Transmission Section (OTS) for the Main L100 Blade.

Command:

```
% set interfaces interface ots-1/0/0/E1 type opticalTransport ots degree-number 1
operation-mode ope-mode-full-spec admin-status up
```

Step 2

Provision the Optical Multiplex Section (OMS) and postoutadj value for the Main L100 Blade.

Command:

```
% set interfaces interface ots-1/0/0/E1 ots oms admin-status up postoutadj <value>
```

Note: The postoutadj design value for the network is calculated during network design and planning process. Postoutadj value range is within -15.0...0.0 dB. The default value is 0.0 dB. Setting should be implemented in initial system turn-up.

Contact Fujitsu Support Center for assistance on the postoutadj value.

Step 3

Enable the OTS for a tributary L100 Blade.

Command:

```
% set interfaces interface ots-2/0/0/E1 type opticalTransport ots degree-number 2
operation-mode ope-mode-full-spec admin-status up
```

Note: Repeat this step to enable the OTS for each tributary L100 Blade.

Step 4

Provision the OMS and a postoutadj value for a tributary L100 Blade.

Command:

```
% set interfaces interface ots-2/0/0/E1 ots oms admin-status up postoutadj <value>
```

Note: Repeat this step to enable the OMS for each tributary L100 Blade.

Step 5

Provision an OCH interface.

Command:

```
% set interfaces interface ots-2/0/0/E1 ots oms och 1 admin-status up center-
frequency 196.1
```

Note: Repeat this step for each OCH interface to be provisioned.

Note: With flexible grid enabled, the slot width can be provisioned for a value between 37.5 to 100 GHz (in 12.5 GHz increments) with a maximum of 128 channels.

Step 6

Verify the interfaces are provisioned correctly.

Command:

```
% show interfaces
```

✓ **This task is complete.**

7.5

Set 100 Mb/s OSC Power Pad

The *OSC power pad* value (FC9581B62F) is calculated during network design and planning process. Pad value is 5, 10, or 15 dB. The default value is 0.0 dB.

Contact Fujitsu Technical Assistance Center for assistance on the *OSC power pad* value.



Important: Fujitsu does not recommend using third-party OSC SFP module because the power of third-party OSC module is not guaranteed.

Step 1

Place the blade in CLI configuration mode.

Command:

```
> configure
```

Step 2

Set the *OSC power pad* value on the L100 Main blade.

Command:

```
% set interfaces interface ots-1/0/0/E1 type opticalTransport ots operation-mode  
ope-mode-full-spec degree-number 1 fiber-type nz-dsf osc-attenuation 10 admin-  
status up
```

Note: The value must be set with an incremental value of 0.1 within the 0.0...15.0 range. For example, the values being set must be 0.1, 0.2, 0.3...1.0, 1.1, 1.2...14.9, 15.0.

Note: An LC/UPC attenuator with the same value must be attached to the corresponding *OSC (R)* port.

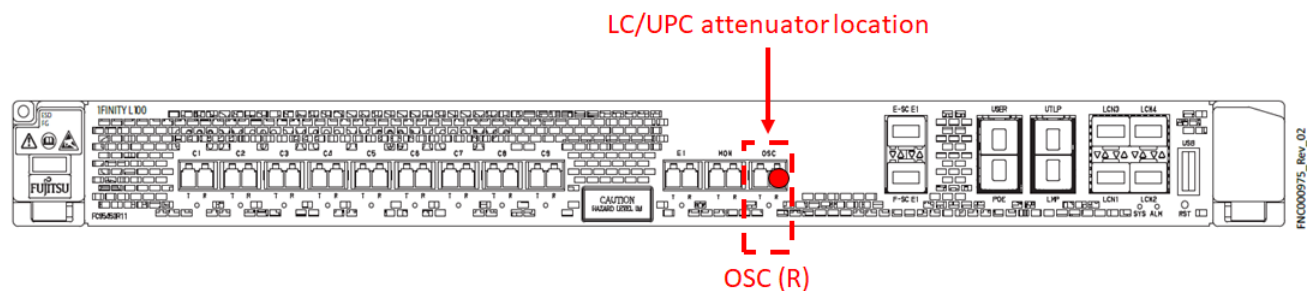


Figure 100
LC/UPC Attenuator

Step 3

Save the changes.

Command:

```
% commit
```

Step 4

Repeat the preceding steps for each L100 tributary blade.

✓ **This task is complete.**

7.6

Provision Direct Access (Colorless) Channels

This procedure describes how to provision colorless connections in the L100 ROADM blade.

Note: The following procedure uses client port C9 as an example. Any port, C1...C9, can be provisioned for direct access channels.

Step 1

Provision colorless port-connection on a client port.

Command:

```
% set port-connection FCON-MAIN-C9 mode colorless shelf 1 slot 0 subslot 0 port C9
```

Step 2

Enable the OCH interface on the colorless client port.

Command:

```
% set interfaces interface och-1/0/0/C9 type opticalChannel enabled och
```

Step 3

Provision OTS OCH interface with desired center-frequency.

Command:

```
% set interfaces interface ots-1/0/0/E1 ots oms och 1 admin-status up center-  
frequency 196.00 slot-width 50
```

Note: In this example, center-frequency is set as 196.00 with slot-width 50 GHz.

Step 4

Provision a colorless cross-connect through edge and client port of the Main blade.

Commands:

```
% set fabric och-connection XCON-M-C9E1-1 src-if och-1/0/0/C9 dst-if ots-1/0/0/E1  
dst-och 1
```

```
% set fabric och-connection XCON-M-E1C9-1 src-if ots-1/0/0/E1 src-och 1 dst-if  
och-1/0/0/C9
```

✓ **This task is complete.**

7.7

Provision L110 Tributary Blade OCH Interfaces

Step 1

Provision an OCH interface on a PIL1-3PS2 PIU.

Command:

```
% set interfaces interface och-3/1/0/C1 type opticalChannel enabled och
```

Step 2

Repeat [Step 1](#) for each OCH interface (1...16) to be provisioned.



This task is complete.

7.8

Link PIL1-3PS2 Client Port to Transponder/Muxponder Edge Port

Step 1

Link a PIL1-3PS2 client port to a remote transponder/muxponder edge port.

Commands:

```
% set eqpt shelf 3 slot 1 subslot 0 port C1 remoteDestPeerPort-id "FWCDS-01 1-1-2"
```

```
% set eqpt shelf 3 slot 1 subslot 0 port C1 remoteSrcPeerPort-id "FWCDS-01 1-1-2"
```

Note: The preceding example shows a link to a FLASHWAVE CDS port.

```
% set eqpt shelf 3 slot 1 subslot 0 port C3 remoteDestPeerPort-id "T200-NE1 1/1/0/E1"
```

```
% set eqpt shelf 3 slot 1 subslot 0 port C3 remoteSrcPeerPort-id "T200-NE1 1/1/0/E1"
```

Note: The preceding example shows a link to a 1FINITY T200 port.

✓ **This task is complete.**

7.9

Provision Add/Drop Port Connections

Step 1

Provision an ADD connection between a PIL1-3PS2 edge port and the L100 client port.

Command:

```
% set port-connection PORT-ADD src-shelf 3 src-slot 1 src-subslot 0 src-port E1
dst-shelf 1 dst-slot 0 dst-subslot 0 dst-port C1
```

Step 2

Provision a DROP connection between the L100 client port and a PIL1-3PS2 edge port.

Command:

```
% set port-connection PORT-DROP src-shelf 1 src-slot 0 src-subslot 0 src-port C1
dst-shelf 3 dst-slot 1 dst-subslot 0 dst-port E1
```

Step 3

Repeat the preceding steps for each ADD/DROP connection to be provisioned.

Step 4

Provision pass-through connections between two CD ROADM degrees.

Commands:

```
% set port-connection PASS-THRH-1 src-shelf 2 src-slot 0 src-subslot 0 src-port C2
dst-shelf 1 dst-slot 0 dst-subslot 0 dst-port C2
```

```
% set port-connection PASS-THRH-2 src-shelf 1 src-slot 0 src-subslot 0 src-port C2
dst-shelf 2 dst-slot 0 dst-subslot 0 dst-port C2
```

✓ **This task is complete.**

Additional Instructions:

Perform the same procedure for each pass-through connection to be provisioned.

7.10

Provision Cross-Connects

Step 1

Provision a DROP cross-connect from an OCH on an edge port of an L100 Blade to a client port on a PIL1-3PS2 PIU.

Command:

```
% set fabric och-connection XCON-DROP src-if ots-2/0/0/E1 src-och 1 dst-if
och-3/1/0/C1
```

Step 2

Provision a ADD cross-connect from a client port on a PIL1-3PS2 to an OCH on an edge port of an L100 Blade.

Command:

```
% set fabric och-connection XCON-ADD src-if och-3/1/0/C1 dst-if ots-2/0/0/E1 dst-
och 1
```

Step 3

Provision pass-through cross-connects between two CD ROADM degrees.

Commands:

```
% set fabric och-connection XCON-PASS-1 src-if ots-1/0/0/E1 src-och 2 dst-if
ots-2/0/0/E1 dst-och 2
```

```
% set fabric och-connection XCON-PASS-2 src-if ots-2/0/0/E1 src-och 2 dst-if
ots-1/0/0/E1 dst-och 2
```

✓ **This task is complete.**

7.11

Adding Degree to ROADM Node

In this section:

- 7.11.1 Provision L100 Tributary Blade and Ports
- 7.11.2 Provision L100 Tributary Blade OTS/OMS/OCH Interfaces
- 7.11.3 Provision Add/Drop Port Connections
- 7.11.4 Provision Cross-Connects

Perform initial setup for the tributary L100 Blade to be added as described in [Set Up Tributary Blade without Auto Shelf ID Assignment Enabled](#).

Note: The default name must be changed from *fujitsu* for the system to turn up as well as for the tributary shelves to be recognized.

7.11.1

Provision L100 Tributary Blade and Ports

Note: The default name must be changed from *fujitsu* for the system to turn up as well as for the tributary shelves to be recognized.

Step 1

Provision the L100 tributary blade equipment.

Note: If autoprovisioning is enabled, the L100 tributary blade equipment is automatically provisioned.

Command:

```
% set eqpt shelf 2 type BDL1-3R11 shelf-mode NORMAL shelf-role TRIB admin-status up
```

Step 2

Provision the Fan unit slots.

Note: If autoprovisioning is enabled, the Fan unit slots are automatically provisioned.

Commands:

```
% set eqpt shelf 2 slot FAN1 cardType AC00-F011 admin-status up
```

```
% set eqpt shelf 2 slot FAN2 cardType AC00-F011 admin-status up
```

Step 3

Provision the E1 port on the L100 tributary blade.

Note: If autoprovisioning is enabled, the E1 port is automatically provisioned.

Command:

```
% set eqpt shelf 2 slot 0 cardType BDL1-3R11 subslot 0 port E1 admin-status up  
pluggableInterfaceType Edge num-lanes 128
```

Step 4

Provision the client ports (C1...C9) on the L100 tributary blade.

Note: If autoprovisioning is enabled, the client ports (C1...C9) are automatically provisioned.

Command:

```
% set eqpt shelf 2 slot 0 cardType BDL1-3R11 subslot 0 port C1 admin-status up
pluggableInterfaceType Client num-lanes 128
```

Note: Repeat this step for each client port (C1...C9) to be provisioned.

Step 5

Provision the logical port.

Note: If autoprovisioning is enabled, the logical port is automatically provisioned.

Command:

```
% set eqpt shelf 2 slot 0 subslot 0 port L1 pluggableInterfaceType Logical num-
lanes 128 admin-status up
```

Step 6

Provision the OSC port.

Note: If autoprovisioning is enabled, the OSC port is automatically provisioned.

Command:

```
% set eqpt shelf 2 slot 0 subslot 0 port OSC pluggableInterfaceType OSC admin-
status up
```

Step 7

Provision the E-SC-E1 port equipment.

Note: If autoprovisioning is enabled, the E-SC-E1 port is automatically provisioned.

Note: If a 100ME SFP is being used, specify 100ME for pluggableInterfaceType.

Command:

```
% set eqpt shelf 2 slot 0 subslot 0 port E-SC-E1 pluggableInterfaceType 1GEX admin-
status up
```



Important: Fujitsu does not recommend using third-party OSC SFP module because the performance of third-party OSC module is not guaranteed.

Step 8

Provision the E-SC-E1 port interface.

Note: If autoprovisioning is enabled, the E-SC-E1 port interface is automatically provisioned.

Command:

```
% set interfaces interface eth-2/0/0/E-SC-E1 type ethernetCsmacd ethernet admin-  
status up mtu 1800 speed 1000
```

Step 9

Provision the UTLP equipment.

Command:

```
% set eqpt shelf 2 slot 0 subslot 0 port UTLP pluggableInterfaceType Fixed
```

Note: If a user provisions any UTLP equipment in an earlier release, use the following commands.

Commands:

```
% delete eqpt shelf 2 slot 0 subslot 0 port UTLP
```

```
% set eqpt shelf 2 slot 0 subslot 0 port UTLP pluggableInterfaceType Fixed
```

Note: If a user provisions any UTLP equipment and UTLP interface in an earlier release, use the following commands.

Commands:

```
% delete interfaces interface eth-2/0/0/UTLP
```

```
% delete eqpt shelf 2 slot 0 subslot 0 port UTLP
```

```
% set eqpt shelf 2 slot 0 subslot 0 port UTLP pluggableInterfaceType Fixed
```

Step 10

Provision the UTLP port interface.

Command:

```
% set interfaces interface eth-2/0/0/UTLP type ethernetCsmacd ethernet duplex full  
speed 1000 mtu 1522 auto-negotiation enabled admin-status up
```

Continue with the next task.

7.11.2

Provision L100 Tributary Blade OTS/OMS/OCH Interfaces

Step 1

Enable optical transport.

Command:

```
% set interfaces interface ots-2/0/0/E1 type opticalTransport enabled
```

Step 2

Provision the OTS interface.

Command:

```
% set interfaces interface ots-2/0/0/E1 type opticalTransport ots circuit-id OTS-2
fiber-type smf degree-number 2 operation-mode owb-fj-base admin-status up
```

Step 3

Provision the OMS interface.

Command:

```
% set interfaces interface ots-2/0/0/E1 type opticalTransport ots oms circuit-id
OMS-2
```

Step 4

Provision the OCH interface.

Command:

```
% set interfaces interface ots-2/0/0/E1 type opticalTransport ots oms och 1 center-
frequency 196.10 slot-width 50
```

Step 5

Repeat the preceding step for each OCH interface (1...96) to be provisioned.

Step 6

Verify the interfaces are provisioned correctly.

Command:

```
% show interfaces
```

Continue with the next task.

7.11.3

Provision Add/Drop Port Connections

Step 1

Provision an ADD connection between a PIL1-3PS2 edge port and the L100 client port.

Command:

```
% set port-connection PORT-ADD src-shelf 3 src-slot 1 src-subslot 0 src-port E1
dst-shelf 1 dst-slot 0 dst-subslot 0 dst-port C1
```

Step 2

Provision a DROP connection between the L100 client port and a PIL1-3PS2 edge port.

Command:

```
% set port-connection PORT-DROP src-shelf 1 src-slot 0 src-subslot 0 src-port C1
dst-shelf 3 dst-slot 1 dst-subslot 0 dst-port E1
```

Step 3

Repeat the preceding steps for each ADD/DROP connection to be provisioned.

Step 4

Provision pass-through connections between two CD ROADM degrees.

Commands:

```
% set port-connection PASS-THRH-1 src-shelf 2 src-slot 0 src-subslot 0 src-port C2
dst-shelf 1 dst-slot 0 dst-subslot 0 dst-port C2
```

```
% set port-connection PASS-THRH-2 src-shelf 1 src-slot 0 src-subslot 0 src-port C2
dst-shelf 2 dst-slot 0 dst-subslot 0 dst-port C2
```

Continue with the next task.

Additional Instructions:

Perform the same procedure for each pass-through connection to be provisioned.

7.11.4

Provision Cross-Connects

Step 1

Provision a DROP cross-connect from an OCH on an edge port of an L100 Blade to a client port on a PIL1-3PS2 PIU.

Command:

```
% set fabric och-connection XCON-DROP src-if ots-2/0/0/E1 src-och 1 dst-if
och-3/1/0/C1
```

Step 2

Provision a ADD cross-connect from a client port on a PIL1-3PS2 to an OCH on an edge port of an L100 Blade.

Command:

```
% set fabric och-connection XCON-ADD src-if och-3/1/0/C1 dst-if ots-2/0/0/E1 dst-
och 1
```

Step 3

Provision pass-through cross-connects between two CD ROADM degrees.

Commands:

```
% set fabric och-connection XCON-PASS-1 src-if ots-1/0/0/E1 src-och 2 dst-if
ots-2/0/0/E1 dst-och 2
```

```
% set fabric och-connection XCON-PASS-2 src-if ots-2/0/0/E1 src-och 2 dst-if
ots-1/0/0/E1 dst-och 2
```



This task is complete.

7.12

Adding L110 Blade to ROADM

In this section:

- 7.12.1 Provision L110 Tributary Shelf, Slots, and Ports Using WebGUI
- 7.12.2 Provision L110 Tributary Blade OCH Interfaces
- 7.12.3 Provision Add/Drop Port Connections
- 7.12.4 Provision Cross-Connects

Perform initial setup for the tributary L110 Blade to be added as described in [Set Up Tributary Blade without Auto Shelf ID Assignment Enabled](#).

Note: The default name must be changed from *fujitsu* for the system to turn up as well as for the tributary shelves to be recognized.

7.12.1

Provision L110 Tributary Shelf, Slots, and Ports Using WebGUI

Step 1

Press the arrow icon that appears when hovering over an existing blade in the Equipment tree.

Step Result:

The `/equipment:eqpt/shelf` tab appears.

Step 2

Select the checkbox next to the BDL1-3B11 shelf.

Step Result:

The `/equipment:eqpt/shelf("n")` tab appears.

Note: The value "n" is the number of the shelf being provisioned.

Step 3

Select the *slot* tab.

Step 4

Press the add button (+ sign) in the *slot* tab.

Step Result:

The **Add List Item** dialog box opens.

Step 5

Select the slot number (1 or 2) from the *slotID* dropdown list and click **Ok**.

Step Result:

The slot is added to slot table.

Step 6

Select the checkbox next to the slot (1 or 2) being provisioned.

Step Result:

The `/equipment:eqpt/shelf(n)/slot("x")` tab appears.

Note: The value "x" is the number of the slot being provisioned.

Step 7

In the `/equipment:eqpt/shelf(n)/slot("x")` tab, select **up** from the **admin-status** dropdown list.

Step 8

Select **PIL1-3PS2** from the **cardType** dropdown list.

Step 9

To provision another slot, select the **slot** tab and repeat [Step 4](#) through [Step 8](#).

Step 10

In the `/equipment:eqpt/shelf(n)/slot("x")` tab, select the **subslot** tab.

Step 11

Press the add button (+ sign) in the **subslot** tab.

Step Result:

The **Add List Item** dialog box opens.

Step 12

Select the subslot number **0** from the **subslotID** dropdown list and click **Ok**.

Step Result:

The subslot is added to subslot table.

Step 13

Select the checkbox next to subslot 0.

Step Result:

The `/equipment:eqpt/shelf(n)/slot(x)/subslot("0")` tab appears.

Step 14

Select the **port** tab.

Step 15

Press the add button (+ sign) in the **port** tab.

Step Result:

The **Add List Item** dialog box opens.

Step 16

Select the edge port (E1 or E2) from the **portID** dropdown list and click **Ok**.

Step Result:

The port is added to port table.

Step 17

Select the checkbox next to edge port.

Step Result:

The `/equipment:eqpt/shelf(n)/slot(x)/subslot(0)/port("Ey")` tab appears.

Note: The value "y" is the number of the edge port being provisioned.

Step 18

In the `/equipment:eqpt/shelf(n)/slot(x)/subslot(0)/port("Ey")` tab, select **up** from the **admin-status** dropdown list.

Step 19

Select **Edge** from the **pluggableInterfaceType** dropdown list.

Step 20

Enter **96** in the **num-lanes** field.

Step 21

Select **Commit** to commit the changes.

Step 22

Close the `/equipment:eqpt/shelf(n)/slot(x)/subslot(0)/port("Ey")` tab.

Step 23

Repeat [Step 15](#) through [Step 22](#) for each edge port to be provisioned.

Step 24

Select the add button (+ sign) in the **port** tab.

Step Result:

The **Add List Item** dialog box opens.

Step 25

Select the client port (C1...C16) from the **portID** dropdown list and click **Ok**.

Step Result:

The port is added to port table.

Step 26

Select the checkbox next to client port.

Step Result:

The `/equipment:eqpt/shelf(n)/slot(x)/subslot(0)/port("Cz")` tab appears.

Note: The value "z" is the number of the client port being provisioned.

Step 27

In the `/equipment:eqpt/shelf(n)/slot(x)/subslot(0)/port("Cz")` tab, select **up** from the **admin-status** dropdown list.

Step 28

Select **Client** from the **pluggableInterfaceType** dropdown list.

Step 29

Press **Commit** to commit the changes.

Step 30

Close the `/equipment:eqpt/shelf(n)/slot(x)/subslot(0)/port("Cz")` tab.

Step 31

Repeat [Step 24](#) through [Step 30](#) for each client port to be provisioned.

Step 32

If a second slot is provisioned, repeat [Step 10](#) through [Step 31](#) to provision the edge and client ports for the second slot.

Step 33

Close the following tabs:

- `/equipment:eqpt/shelf, /equipment:eqpt/shelf(n)`
- `/equipment:eqpt/shelf(n)/slot("x")`
- `/equipment:eqpt/shelf(n)/slot("x")/subslot("0")`

Continue with the next task.

7.12.2

Provision L110 Tributary Blade OCH Interfaces

Step 1

Provision an OCH interface on a PIL1-3PS2 PIU.

Command:

```
% set interfaces interface och-3/1/0/C1 type opticalChannel enabled och
```

Step 2

Repeat [Step 1](#) for each OCH interface (1...16) to be provisioned.

✓ **This task is complete.**

7.12.3

Provision Add/Drop Port Connections

Step 1

Provision an ADD connection between a PIL1-3PS2 edge port and the L100 client port.

Command:

```
% set port-connection PORT-ADD src-shelf 3 src-slot 1 src-subslot 0 src-port E1
dst-shelf 1 dst-slot 0 dst-subslot 0 dst-port C1
```

Step 2

Provision a DROP connection between the L100 client port and a PIL1-3PS2 edge port.

Command:

```
% set port-connection PORT-DROP src-shelf 1 src-slot 0 src-subslot 0 src-port C1
dst-shelf 3 dst-slot 1 dst-subslot 0 dst-port E1
```

Step 3

Repeat the preceding steps for each ADD/DROP connection to be provisioned.

Step 4

Provision pass-through connections between two CD ROADM degrees.

Commands:

```
% set port-connection PASS-THRH-1 src-shelf 2 src-slot 0 src-subslot 0 src-port C2
dst-shelf 1 dst-slot 0 dst-subslot 0 dst-port C2
```

```
% set port-connection PASS-THRH-2 src-shelf 1 src-slot 0 src-subslot 0 src-port C2
dst-shelf 2 dst-slot 0 dst-subslot 0 dst-port C2
```

Continue with the next task.

Additional Instructions:

Perform the same procedure for each pass-through connection to be provisioned.

7.12.4

Provision Cross-Connects

Step 1

Provision a DROP cross-connect from an OCH on an edge port of an L100 Blade to a client port on a PIL1-3PS2 PIU.

Command:

```
% set fabric och-connection XCON-DROP src-if ots-2/0/0/E1 src-och 1 dst-if
och-3/1/0/C1
```

Step 2

Provision a ADD cross-connect from a client port on a PIL1-3PS2 to an OCH on an edge port of an L100 Blade.

Command:

```
% set fabric och-connection XCON-ADD src-if och-3/1/0/C1 dst-if ots-2/0/0/E1 dst-och 1
```

Step 3

Provision pass-through cross-connects between two CD ROADM degrees.

Commands:

```
% set fabric och-connection XCON-PASS-1 src-if ots-1/0/0/E1 src-och 2 dst-if ots-2/0/0/E1 dst-och 2
```

```
% set fabric och-connection XCON-PASS-2 src-if ots-2/0/0/E1 src-och 2 dst-if ots-1/0/0/E1 dst-och 2
```

✓ **This task is complete.**

7.13

Adding Raman Blade to ROADM Node

In this section:

7.13.1 Provision Raman Port Connections

Perform the correct Raman fiber cabling connection as described in [Install L160 Raman Cabling](#).

Note: If L160 blade is not provisioned already, in order to provision refer to [Provision L160 Tributary Blade](#)

7.13.1

Provision Raman Port Connections

Prerequisites:

- Appropriate Raman fiber-optic cable connection must be completed. For more information, refer to [Install L160 Raman Cabling](#).
- L100 and L160 Blades must be provisioned

This procedure describes how to provision port-connection between Raman blade and L100 Blade.

Note: L100 Blade Issue 14, FC95453R11-I14, and later blade must be used with L160 Raman blade.

Step 1

Provision the edge port E1 of L100 Blade.

Note: If autoprovisioning is enabled, the E1 port is automatically provisioned.

Command:

```
% set eqpt shelf 1 slot 0 cardType BDL1-3R11 subslot 0 port E1 admin-status up
pluggableInterfaceType Edge num-lanes 128
```

Step 2

Provision the client port C1 on the Raman tributary blade.

Note: If autoprovisioning is enabled, the client port C1 is automatically provisioned.

Command:

```
% set eqpt shelf 101 slot 0 cardType BDL1-3M21 subslot 0 port C1 admin-status up
pluggableInterfaceType Client num-lanes 128
```

Step 3

Provision port-connection between Raman client port and L100 edge port.

Command:

```
% set port-connection D1-RAMAN src-shelf 101 src-slot 0 src-subslot 0 src-port C1  
dst-shelf 1 dst-slot 0 dst-subslot 0 dst-port E1
```

✓ **This task is complete.**

7.14

Provision L100 Blade for FLASHWAVE 9500 Interoperability

If the user is using the FLASHWAVE 9500 MH interoperability (ope-mode-mh-interop) mode, the user needs to use the SONET OSC (F-SC-E1) port.

Step 1

Activate the FLASHWAVE 9500 INTEROP ILF.

Command:

```
% set licenses COMPAT_9500 license-key <key string>
```

Step 2

Insert the OC3-rate SFP in the F-SC-E1 SFP slot.

Step 3

Provision the F-SC-E1 port equipment.

Command:

```
% set eqpt shelf 1 slot 0 subslot 0 port F-SC-E1 pluggableInterfaceType 155M admin-status up
```

Step 4

Provision the F-SC-E1 port interface.

Note: If autoprovisioning is enabled, the F-SC-E1 port is automatically provisioned.

Command:

```
% set interfaces interface fsc-1/0/0/F-SC-E1 type sonet enabled oc rate OC3 admin-status up
```

Step 5

Provision OTS interface for FLASHWAVE 9500 interoperability mode.

Command:

```
% set interfaces interface ots-1/0/0/E1 type opticalTransport ots degree-number 1 operation-mode ope-mode-mh-interop admin-status up
```

Step Result:

Blade reboots.

Note: If a user is using the Base (ope-mode-base) or Full-Spec (ope-mode-full-spec) mode, the user needs to use the Ethernet OSC (E-SC-E1) port.

Step 6

Provision the OMS and postoutadj value for the L100 Blade.

Command:

```
% set interfaces interface ots-1/0/0/E1 ots oms admin-status up postoutadj <value>
```

Step 7

Provision an OCH interface.

Command:

```
% set interfaces interface ots-2/0/0/E1 ots oms och 1 admin-status up center-  
frequency 196.1
```

Note: Repeat [Step 7](#) for each OCH interface to be provisioned.

✓ **This task is complete.**

7.15

Reprovision L100 Blade for FLASHWAVE 9500 Interoperability

Note: Reprovisioning of the L100 Blade is necessary if the E-SC-E1 SFP slot is already in use.

Step 1

Remove the SFP in the E-SC-E1 SFP slot.

Step 2

Delete the E-SC-E1 port equipment.

Command:

```
% delete eqpt shelf 1 slot 0 subslot 0 port E-SC-E1
```

Step 3

Delete the E-SC-E1 port interface.

Commands:

```
% delete interfaces interface ots-1/0/0/E1
```

```
% delete interfaces interface eth-1/0/0/E-SC-E1
```

Step 4

Proceed to [Provision L100 Blade for FLASHWAVE 9500 Interoperability](#).



This task is complete.

7.16

Interoperability with L130 Blade

The interoperability between L100 and L130 Blades is automatic. Turn up both the L100 and L130 Blades normally and connect the fibers between the L100 and L130 Blades. The connection between the blades is autoprovisioned.

Note: When L100 Blade supports interoperability with L130 Release 19.1.1, the span supported between L130 and L100 Blades must be Raman amplifiers (with L160 Blade).

Note: OSNR setting must be enabled in L100 Blade to support interoperability with L130 Blade.

Note: When L100 Blade supports interoperability with L130 Blade, the OSNR setting on the L100 must be set to enable. Otherwise, an outstanding *osnrModeMismatch* alarm is raised on the L100 Blade.

8

System Maintenance

In this chapter:

- 8.1 Trouble Clearing
- 8.2 Performance Monitoring Counters
- 8.3 Equipment Replacement
- 8.4 Upgrades
- 8.5 Backup and Restore

8.1 Trouble Clearing

In this section:

- | | |
|------------------------------------|--|
| 8.1.1 L100 Blade LED Indicators | 8.1.5 Retrieve Alarms and PMs |
| 8.1.2 L110 Blade LED Indicators | 8.1.6 Alarms and Conditions |
| 8.1.3 L160 Blade LED Indicators | 8.1.7 Secondary Database Alarms and Events |
| 8.1.4 PIL1-3PS2 PIU LED Indicators | 8.1.8 LED Control |

8.1.1 L100 Blade LED Indicators

The following figures show front and rear views of the L100 Blade.

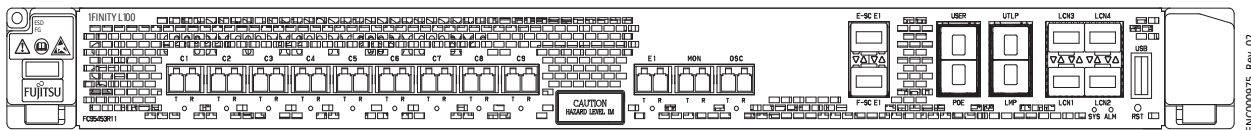


Figure 101
Front View of L100 1x9 WSS ROB (Issue 13 and earlier)

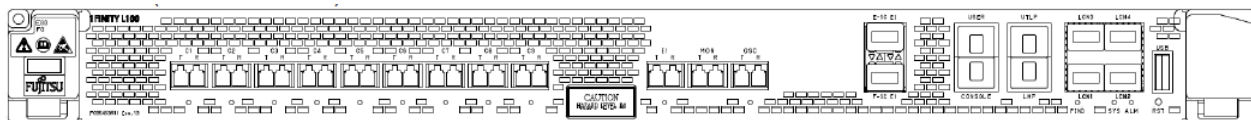


Figure 102
Front View of L100 1x9 WSS ROB (Issue 14 and later)

Note: Issue 05 and earlier units have optical port direction labels O/I instead of T/R.

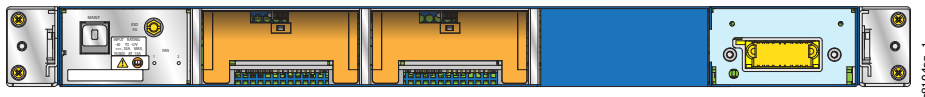


Figure 103
Rear View of L100 1x9 WSS ROB

Note: The Entity Type, E-SC-E1, and F-SC-E1 LED indicators are on the front of the SFP module installed in the corresponding SFP cage. The FAN LED indicators are on the rear of the blade. All other LED indicators are on the front of the blade.

Note: In a multiblade configuration, consisting of a Main blade and one or more tributary blades, only the ALM LED on the Main blade lights when an alarm occurs on any of the blades.

Table 48
LED and Condition

LED	Color	Condition
ALM	Yellow	Minor alarm
	Red	Critical or major alarm
	Off	No alarm
C1...C9	Green	Incoming optical signal operating normally
	Red	Input port down One of the following alarms is present: ■ powerOutOfSpecificationHigh ■ powerOutOfSpecificationLow
	Off	Power off
E1	Green	Incoming optical signal operating normally
	Yellow	PMI HREF BDI-O One of the following alarms is present: ■ postBlockSpanAdjustmentInProgress ■ preBlockSpanAdjustmentInProgress
	Red	WDM LOS WDM power out of specification One of the following alarms is present: ■ omsLossOfSignal ■ omsPowerOutOfSpecificationHigh ■ omsPowerOutOfSpecificationLow ■ oscPowerOutOfSpecificationHigh ■ oscPowerOutOfSpecificationLow ■ otsLossOfSignal ■ otsSpanlossOutOfSpecificationHigh ■ otsSpanlossOutOfSpecificationLow
	Off	Power off

Table 48 (Cont.)
LED and Condition

LED	Color	Condition
E-SC-E1 (on SFP module)	Green	Link up (with or without active data communication)
	Red	OSC LOS OSC fault OSC link down One of the following alarms is present: ■ equipmentRemoved ■ powerOutOfSpecificationHigh ■ powerOutOfSpecificationLow
	Blinking red	MEA
FAN1, FAN2	Green	No error condition (normal operation)
	Red	Error condition (fan failed, fan removed)
	Blinking red	Fan mismatch
	Off	Power off
F-SC-E1	Green	Link up (without data communication)
	Blinking green	Link up under data communication
	Red	OSC LOS OSC fault OSC link down
	Blinking red	MEA
Entity Type	Green	Link up (with or without active data communication)
	Red	Error condition (link down, module failure, module removal) One of the following alarms is present: ■ LOS (eth) ■ POS (port)
	Blinking red	MEA
OSC	Green	Incoming optical signal operating normally
	Red	Input port down One of the following alarms is present: ■ powerOutOfSpecificationHigh ■ powerOutOfSpecificationLow
	Off	Power off

Table 48 (Cont.)
LED and Condition

LED	Color	Condition
SYS	Green	Normal operation
	Yellow	Not ready (warm-up or SWDL in progress) One of the following alarms is present: ■ databaseCorruption ■ databaseLockedDbAlarmPresent ■ databaseLockedDbRestoreInProgress ■ databaseLockedIlfViolation ■ databaseLockedShelfProvModePresent ■ databaseLockedSoftwareUpgradeInProgress ■ databaseVersionMismatch ■ equipmentWarmup ■ firmwareBackwardCompatibleAll ■ firmwareDownloadOrActivationFailure ■ firmwareInitInProgress ■ firmwareVersionMismatch ■ shelfProvisioningMode ■ softwareStagelInProgress ■ softwareVersionMismatch
	Red	Equipment fault Equipment in power-up state
	Blinking red	MEA
	Off	Power off

8.1.2 L110 Blade LED Indicators

The following figures show front and rear views of the Optical Mux Base blade.

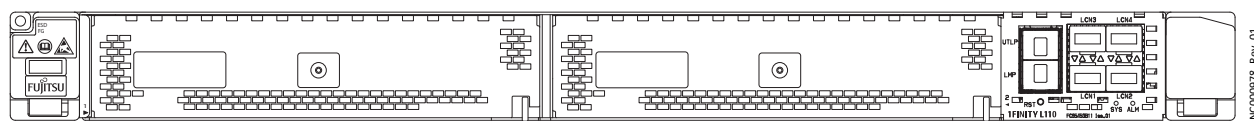


Figure 104
Front View of L110 Optical Mux Base Blade



Figure 105
Rear View of L110 Optical Mux Base Blade

Note: The Entity Type LED indicators are on the front of the SFP module installed in the corresponding SFP cage. The FAN LED indicators are on the rear of the blade. All other LED indicators are on the front of the blade.

Note: In a multiblade configuration, consisting of a Main blade and one or more tributary blades, only the ALM LED on the Main blade lights when an alarm occurs on any of the blades.

Table 49
LED and Condition

LED	Color	Condition
SYS	Green	Normal operation
	Yellow	Not ready (warm-up or SWDL in progress) One of the following alarms is present: ■ databaseCorruption ■ databaseLockedDbRestoreInProgress ■ databaseLockedIfViolation ■ databaseLockedShelfProvModePresent ■ databaseLockedSoftwareUpgradeInProgress ■ equipmentWarmup ■ firmwareDownloadOrActivationFailure ■ shelfProvisioningMode ■ softwareVersionMismatch ■ databaseVersionMismatch ■ firmwareBackwardCompatibleAll ■ firmwareInitInProgress ■ firmwareVersionMismatch ■ softwareStageInProgress
	Red	Equipment fault Power-up state
	Blinking red	MEA
	Off	Power off
ALM	Yellow	Minor alarm
	Red	Critical or major alarm
	Off	No alarm

Table 49 (Cont.)
LED and Condition

LED	Color	Condition
Entity Type	Green	Link up (with or without active data communication)
	Red	Error condition (link down, module failure, module removal) One of the following alarms is present: ■ LOS (eth) ■ POS (port)
	Blinking red	MEA
FAN1, FAN2	Green	No error condition (normal operation)
	Red	Error condition (fan failed, fan removed)
	Blinking red	Fan mismatch
	Off	Power off

8.1.3 L160 Blade LED Indicators

The following figures show front and rear views of the Raman blade.

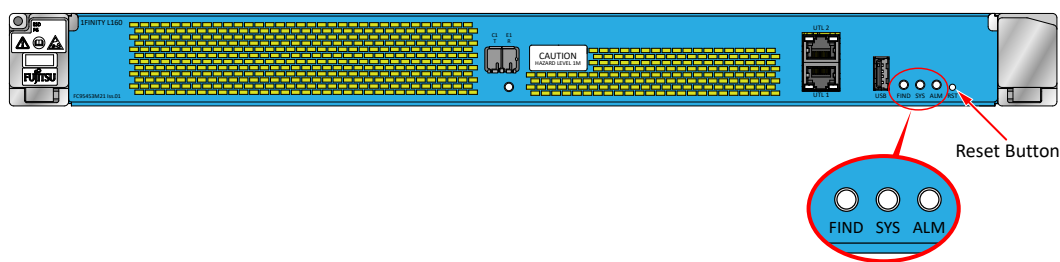


Figure 106
Front View LEDs of L160 Raman Blade

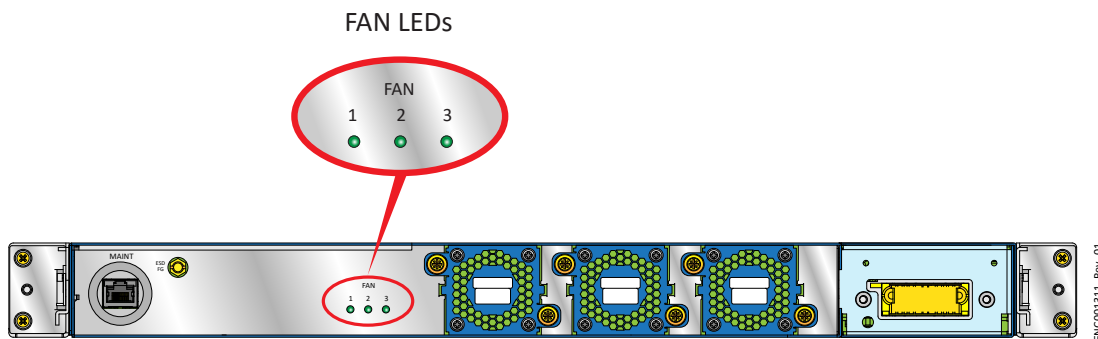


Figure 107
Rear View LEDs of L160 Raman Blade

Note: The FAN LED indicators are on the rear of the blade. All other LED indicators are on the front of the blade.

Note: In a multiblade configuration, consisting of a Main blade and one or more tributary blades, only the ALM LED on the Main blade lights when an alarm occurs on any of the blades.



Danger: L160 Raman blade emits very HIGH power light.



Danger: Never look into any ports of Raman blade directly. Never look into the end of a fiber, fiber cord, or fiber pigtail connected with any ports of Raman blade. Permanent eye damage or blindness can occur quickly when laser radiation is present.



Danger: Never expose bare skin to light from E1 port of Raman blade. Never bring bare hands near the surface of the connector of E1 port and near the fiber which is connected with E1 port of Raman blade.



Danger: All ports and the surface of a fiber connector should be cleaned thoroughly. If dust exists on the surface of a fiber connector or on the ports of Raman blade, it is burnt and melted due to the high power light. The burning or melting action causes damage to the fiber connector or to the blade.

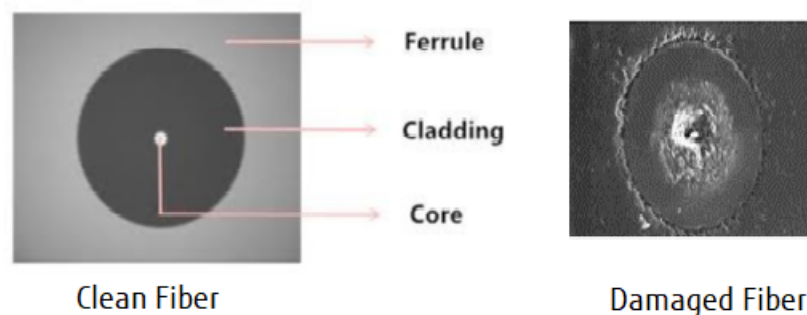


Figure 108
Clean and Damaged Fiber

Table 50
LED and Condition

LED	Color	Condition
SYS	Green	Normal operation
	Yellow	Not ready (warm-up, downloading, installing) One of the following alarms is present: ■ equipmentWarmup ■ firmwareBackwardCompatibleAll ■ firmwareDownloadOrActivationFailure ■ firmwareInitInProgress ■ firmwareVersionMismatch
	Red	Equipment fault Power-up state
	Blinking red	MEA (equipmentMismatchAttributes)
	Blinking yellow	Running status of DHCP
	Off	Power off
ALM	Red	Critical or major alarm
	Yellow	Minor alarm
	Off	No alarm Power off
FIND ME	Blinking blue	CLI command turned LED on
	Off	No power or default state after power up

Table 50 (Cont.)
LED and Condition

LED		Color	Condition
E1 R		Green	Incoming optical signal operating normally
		Red	LOS (lossOfSignal)
		Yellow	HREF (reflectionTooHigh)
		Off	Power off
UTL1,2	ACT	Green	Link up (without data communication/under data communication)
		Off	No link up
	SPEED	Green	Linked 1000Base-T
		Off	Linked 100Base-T
			No link up
			FAN 1, FAN 2, FAN 3
Red	Error condition (fan failed, fan removed) One of the following alarms is present: ■ equipmentFault ■ equipmentRemoved		
Blinking red	Fan mismatch (equipmentMismatchAttributes)		
Off	Power off		

8.1.4 PIL1-3PS2 PIU LED Indicators

The following figure shows the front view of the PIL1-3PS2 PIU.

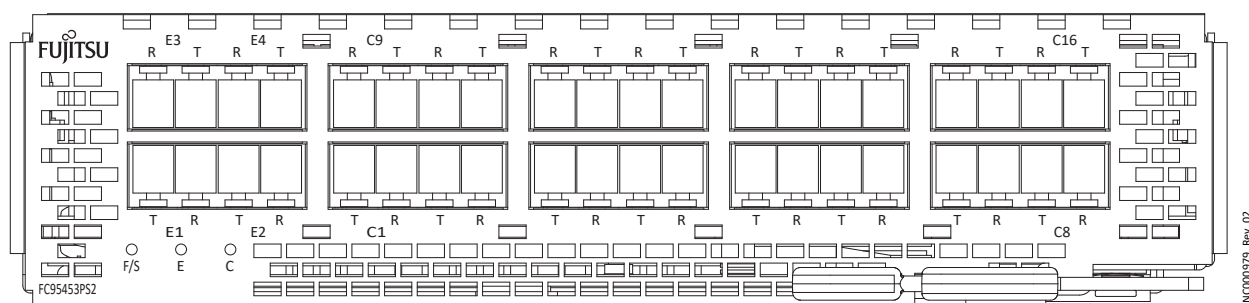


Figure 109
Front View of PIL1-3PS2

Note: Issue 04 and earlier units have optical port direction labels O/I instead of T/R.

Table 51
LED and Condition

LED	Color	Condition
C	Red	C1...C16 port input down C1...C16 port power out of specification
	Green	Incoming optical signal operating normally
	Off	Power off
E	Red	E1...E4 port input down E1...E4 port power out of specification
	Green	Incoming optical signal operating normally
	Off	Power off
F/S	Red	Equipment fault or equipment in power-up state
	Blinking red	MEA
	Yellow	Not ready (warm-up or SWDL in progress)
	Green	Normal operation
	Off	Power off

8.1.5

Retrieve Alarms and PMs

Prerequisite:

- User must be logged on to the Main blade.

The following CLI command examples may be useful in locating or narrowing the scope of a problem.

Step 1

Retrieve the alarms.

Command:

```
> show alarms alarm-list
```

Step 2

Retrieve the PM for the network port.

Command:

```
> show pm-equipment shelf 1 slot 1 subslot 0 port C1 pm-record opticalPowerReceive  
time-period-index 15-min 0
```

Step 3

Retrieve the PM for the client port.

Command:

```
> show pm-equipment shelf 1 slot 0 subslot 0 port C1 pm-oper-data  
opticalPowerReceive pm-time-period-index 15-min
```

Step 4

Retrieve the interface PM.

Command:

```
> show pm-interfaces interface eth-1/0/0/E-SC-E1 pm-oper-data codingViolation pm-  
time-period-index 1-day
```

✓ **This task is complete.**

8.1.6

Alarms and Conditions

This section describes alarms and condition codes and includes the following types of information:

- Entity type
- Severity (warning, minor, major, or critical)
- Service affected (service-affecting or non-service-affecting)

- Direction (transmit, receive, or not applicable [n/a])
- Location (near end, far end, or n/a)
- Description
- Probable causes
- Suggested corrective actions

The following abbreviations are used in the alarms and condition code tables in this section.

NSA	Non-service-affecting
SA	Service-affecting
RX	Receive
TX	Transmit
FEND	Far end
NEND	Near end
n/a	Not applicable

administrativeDown

Entity Type	Severity	Service Affected	Direction	Location	Description
PIL1-3PS2, PIL1-3PS3, AC00-F011	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the PIU or Fan unit is set to DOWN.		Change the administrative state to UP.		
BDL1-3B11, BDL1-3J11, BDL1-3R11	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the blade is set to DOWN.		Change the administrative state to UP.		
1GEX	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the SFP module is set to DOWN.		Change the administrative state to UP.		

Entity Type	Severity	Service Affected	Direction	Location	Description
100ME	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the SFP module is set to DOWN.		Change the administrative state to UP.		
1GEBT	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the SFP module is set to DOWN.		Change the administrative state to UP.		
155M	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the SFP module is set to DOWN.		Change the administrative state to UP.		
ots	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the OTS entity is set to DOWN.		Change the administrative state to UP.		
oms	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the OMS entity is set to DOWN.		Change the administrative state to UP.		
och	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the OCH entity is set to DOWN.		Change the administrative state to UP.		

Entity Type	Severity	Service Affected	Direction	Location	Description
eth	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the ETH entity is set to DOWN.		Change the administrative state to UP.		
Port (Edge)	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the edge port is set to DOWN.		Change the administrative state to UP.		
Port (Client)	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the client port is set to DOWN.		Change the administrative state to UP.		
Port (OSC)	warning	non-service-affecting	n/a	NEND	Administrative State Down: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The administrative state of the OSC port is set to DOWN.		Change the administrative state to UP.		

backwardDefectIndication

Entity Type	Severity	Service Affected	Direction	Location	Description
oms	warning	non-service-affecting	receive	FEND	Backward Defect Indication
	Probable Cause		Suggested Corrective Action		
	An OMS or OTS layer optical signal failure, for example, a WDM LOS, has occurred in the downstream direction.		Clear the downstream WDM receive signal failure. Caution: The OSC must be active for traffic to be restored. A failure of the OSC may keep traffic from recovering on a span.		

bitErrorRateSignalDegrade

Entity Type	Severity	Service Affected	Direction	Location	Description
OC3 OSC (BDL1-3R11)	major	non-service-affecting	receive	NEND	The bit error rate signal degrade count has exceeded the threshold value.
		Probable Cause	Suggested Corrective Action		
		■ The OSC receive power has fallen below or exceeded the specified limit. ■ A connector in the OSC optical link is dirty. ■ The opposite-end optical transmitter is creating bit errors. Points of large optical reflection exist along the OSC link. ■ A hardware problem in the blade could be causing line BER problems. ■ The BERSD-L threshold setting may not be appropriate for the application in use.	1. Verify the local receive and upstream transmit level. 2. Clean the optical connectors. 3. Verify good optical connections to the local and far-end blades and at the Lightwave Distribution Frame (LDF) panels along the link. 4. Replace the far-end blade. 5. Retrieve the facility and check the BERSD-L threshold setting in accordance with local practices.		

bitErrorRateSignalFail

Entity Type	Severity	Service Affected	Direction	Location	Description
OC3 OSC (BDL1-3R11)	major	non-service-affecting	receive	NEND	The bit error rate signal failure count has exceed the threshold value.
		Probable Cause	Suggested Corrective Action		
		■ The OSC receive power has fallen below or exceeded the specified limit. ■ A connector in the OSC optical link is dirty. ■ The opposite-end optical transmitter is creating bit errors. Points of large optical reflection exist along the OSC link. ■ A hardware problem in the blade could be causing line BER problems. ■ The BERSF-L threshold setting may not be appropriate for the application in use.	1. Verify the local receive and upstream transmit level. 2. Clean the optical connectors. 3. Verify good optical connections to the local and far-end blades and at the LDF panels along the link. 4. Replace the far-end blade. 5. Retrieve the facility and check the BERSF-L threshold setting in accordance with local practices.		

certificateNotInstalled

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	non-alarmed	non-service-affecting	n/a	NEND	Missing certificate file
Probable Cause			Suggested Corrective Action		
A request db-activate command is issued with system certificate enabled and the appropriate files are not present in the /var/confd/webui/cert path.			1. SFTP or DFP the .pem file to the appropriate path. 2. Execute the following command to install the certificate. <pre>request re-install-certificate <certid></pre>		

databaseCorruption

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	major	non-service-affecting	n/a	NEND	Database Corruption
Probable Cause			Suggested Corrective Action		
The database has become corrupted.			Call Fujitsu for technical assistance at 1-800-USE-FTAC (1-800-873-3822).		

databaseLockedDbAlarmPresent

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	Database System Locked; Database Alarm Present
Probable Cause			Suggested Corrective Action		
The database is locked due to a database alarm.			Clear the database alarm to unlock the database.		

databaseLockedDbRestoreInProgress

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	Database System Locked; Database Restore In Progress
Probable Cause			Suggested Corrective Action		
The database is locked due to a database restore in progress.			No action is required. The condition clears when the database restore completes.		

databaseLockedILFViolation

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	Database System Locked; ILF Violation
Probable Cause			Suggested Corrective Action		
The temporary license has expired.			Purchase a valid key to enable the license and unlock the database.		

databaseLockedShelfProvModePresent

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	Database System Locked; Shelf Provisioning Mode Present
Probable Cause			Suggested Corrective Action		
Blade is in shelf provisioning mode.			Provision the shelf ID and shelf role.		

databaseLockedSoftwareUpgradeInProgress

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	Database System Locked; Software Upgrade In Progress
Probable Cause			Suggested Corrective Action		
The database is locked due to a software upgrade in progress.			No action is required. The condition clears when the software upgrade completes.		

databaseVersionMismatch

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	major	non-service-affecting	n/a	NEND	Database Version Mismatch
Probable Cause			Suggested Corrective Action		
The database system of a tributary blade does not match the database system read from the Main blade.			1. If executing a new turn-up, perform a WARM reset on the tributary blade. 2. If the condition does not clear or if the tributary blade is carrying traffic, call Fujitsu for technical assistance at 1-800-USE-FTAC (1-800-873-3822).		

equipmentCurrExceededFeed

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	critical	non-service-affecting	n/a	NEND	The current drawn is greater than 80 percent of the fuse rating.
Probable Cause			Suggested Corrective Action		
Current drawn is greater than 80 percent of unit fuse feed rating.			Increase the rating for the fuse feed.		

equipmentFault

Entity Type	Severity	Service Affected	Direction	Location	Description
AC00-F011	minor	non-service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	Either FAN1 or FAN2 has failed.		Replace the failed Fan unit as soon as possible.		
AC00-F011	critical	service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	Both FAN1 and FAN2 have failed.		Replace the failed Fan units as soon as possible.		
BDL1-3B11, BDL1-3R11	minor	non-service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A BDL1-3B11 or BDL1-3R11 blade has failed.		Replace the failed blade as soon as possible.		
BDL1-3B11, BDL1-3R11	critical	service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A BDL1-3B11 or BDL1-3R11 blade has failed.		Replace the failed blade as soon as possible.		
PIL1-3PS2	minor	non-service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A PIL1-3PS2 PIU has failed.		Replace the failed PIU as soon as possible.		
PIL1-3PS2	critical	service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A PIL1-3PS2 has failed.		Replace the failed PIU as soon as possible.		
1GEX SFP	minor	non-service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A 1GEX SFP module has failed.		Replace the failed SFP module as soon as possible.		

Entity Type	Severity	Service Affected	Direction	Location	Description
1GEX SFP	critical	service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A 1GEX SFP module has failed.		Replace the failed SFP module as soon as possible.		
100ME SFP	minor	non-service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A 100ME SFP module has failed.		Replace the failed SFP module as soon as possible.		
100ME SFP	critical	service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A 100ME SFP module has failed.		Replace the failed SFP module as soon as possible.		
1GEBT SFP	minor	non-service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A 1GEBT SFP module has failed.		Replace the failed SFP module as soon as possible.		
1GEBT SFP	critical	service-affecting	n/a	NEND	Equipment Fault: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A 1GEBT SFP module has failed.		Replace the failed SFP module as soon as possible.		

equipmentInterConnectFailure

Entity Type	Severity	Service Affected	Direction	Location	Description
	major	non-service-affecting	n/a	NEND	Interblade communication Link Failure: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	A failure has occurred in the interblade communication links that connect the blades.		1. Check for loose connectors on each interblade communication link. 2. Check for and replace any damaged interblade communication cables.		

equipmentLedOn

Entity Type	Severity	Service Affected	Direction	Location	Description
PIL1-3PS2	warning	non-service-affecting	n/a	NEND	Equipment Common Visual LED Activated: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	An LED on the PIU has been intentionally activated.		Deactivate the LED.		
BDL1-3M21, BDL1-3R11, BDL1-3B11	warning	non-service-affecting	n/a	NEND	Equipment Common Visual LED Activated: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	An LED on the blade has been intentionally activated.		Deactivate the LED.		

equipmentMiscabledConnection

Entity Type	Severity	Service Affected	Direction	Location	Description
1GEX, 1GEBT	major	non-service-affecting	n/a	NEND	Interblade communication Miscabled Connection: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The interblade communication cables are not installed correctly.		Perform eqpt-reset with reset ADOPTSHELF. > request eqpt-reset shelf-id 1 reset ADOPTSHELF		

equipmentMismatchAttributes

Entity Type	Severity	Service Affected	Direction	Location	Description
AC00-F011	warning	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type> and the admin-status is down
	Probable Cause		Suggested Corrective Action		
	■ A Fan unit is mismounted. ■ A Fan unit slot is not provisioned correctly.		1. Remove and remount the Fan unit. 2. Correct the Fan unit slot provisioning.		

Entity Type	Severity	Service Affected	Direction	Location	Description
AC00-F011	minor	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	■ A Fan unit is mismounted. ■ A Fan unit slot is not provisioned correctly.		1. Remove and remount the Fan unit. 2. Correct the Fan unit slot provisioning.		
BDL1-3B11, BDL1-3R11, 1GEX, 1GEBT	major	non-service-affecting	n/a	NEND	Interblade communication Miscabled Connection: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The interblade communication cables are not installed correctly.		1. Remove and remount the SFP module. 2. Replace the SFP module with the correct SFP module.		
PIL1-3PS2	warning	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type> and the admin-status is down
	Probable Cause		Suggested Corrective Action		
	■ The PIU installed in the specified slot is mismounted. ■ The PIU does not match the slot for which it is provisioned.		1. Remove and remount the PIU module. 2. Replace the PIU module with the correct PIU module.		
PIL1-3PS2	minor	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	■ The PIU installed in the specified slot is mismounted. ■ The PIU does not match the slot for which it is provisioned.		1. Remove and remount the SFP module. 2. Replace the SFP module with the correct SFP module.		
BDL1-3B11, BDL1-3R11, 1GEX, 1GEBT	major	non-service-affecting	n/a	NEND	Interblade communication Miscabled Connection: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The interblade communication cables are not installed correctly.		1. Remove and remount the PIU. 2. Replace the PIU with the correct PIU.		

Entity Type	Severity	Service Affected	Direction	Location	Description
1GEX	warning	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type> and the admin-status is down
	Probable Cause		Suggested Corrective Action		
	■ The SFP module installed in the specified port is mismounted. ■ The SFP module does not match the port for which it is provisioned.		1. Remove and remount the SFP module. 2. Replace the SFP module with the correct SFP module.		
1GEX	minor	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	■ The SFP module installed in the specified port is mismounted. ■ The SFP module does not match the port for which it is provisioned.		1. Remove and remount the SFP module. 2. Replace the SFP module with the correct SFP module.		
1GEX	critical	service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	■ The SFP module installed in the specified port is mismounted and has an interface provisioned on it. ■ The SFP module does not match the port for which it is provisioned and has an interface provisioned on it.		1. Remove and remount the SFP module. 2. Replace the SFP module with the correct SFP module.		
100ME	warning	non-service-affecting	n/a	NEND	Interblade communication Miscabled Connection: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	■ The SFP module installed in the specified port is mismounted. ■ The SFP module does not match the port for which it is provisioned.		1. Remove and remount the SFP module. 2. Replace the SFP module with the correct SFP module.		

Entity Type	Severity	Service Affected	Direction	Location	Description
100ME	minor	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	- The SFP module installed in the specified port is mismounted. - The SFP module does not match the port for which it is provisioned.		- Remove and remount the SFP module. - Replace the SFP module with the correct SFP module.		
100ME	critical	service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	- The SFP module installed in the specified port is mismounted and has an interface provisioned on it. - The SFP module does not match the port for which it is provisioned and has an interface provisioned on it.		- Remove and remount the SFP module. - Replace the SFP module with the correct SFP module.		
1GEBT	warning	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type> and the admin-status is down
	Probable Cause		Suggested Corrective Action		
	- The SFP module installed in the specified port is mismounted. - The SFP module does not match the port for which it is provisioned.		- Remove and remount the SFP module. - Replace the SFP module with the correct SFP module.		
1GEBT	minor	non-service-affecting	n/a	NEND	Mismatch of Equipment and Attributes: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	- The SFP module installed in the specified port is mismounted. - The SFP module does not match the port for which it is provisioned.		- Remove and remount the SFP module. - Replace the SFP module with the correct SFP module.		
BDL1-3B11, BDL1-3R11, 1GEX, 1GEBT	major	non-service-affecting	n/a	NEND	Interblade communication Miscabled Connection: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The interblade communication cables are not installed correctly.		- Remove and remount the SFP module. - Replace the SFP module with the correct SFP module.		

equipmentRemoved

Entity Type	Severity	Service Affected	Direction	Location	Description
AC00-F011	minor	non-service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	A Fan unit has been removed.		Replace the Fan unit.		
AC00-F011	critical	service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	Both Fan units have been removed.		Replace the Fan units.		
PIL1-3PS2	minor	non-service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The PIU is not properly seated or has been removed.		Reseat or replace the PIU. If a PIU is not required in the slot generating the alarm, the alarm can be cleared by deleting the equipment slot.		
PIL1-3PS2	critical	service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The PIU is not properly seated or has been removed and has at least one interface provisioned on it.		Reseat or replace the PIU. If a PIU is not required in the slot generating the alarm, the alarm can be cleared by deleting the equipment slot.		
1GEX	minor	non-service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The SFP module is not properly seated or has been removed.		Reseat or replace the SFP module. If a SFP module is not required in the port generating the alarm, the alarm can be cleared by deleting the equipment port.		
1GEX	critical	service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The SFP module is not properly seated or has been removed and has an interface provisioned on it.		Reseat or replace the SFP module. If a SFP module is not required in the port generating the alarm, the alarm can be cleared by deleting the equipment port.		

Entity Type	Severity	Service Affected	Direction	Location	Description
100ME	minor	non-service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The SFP module is not properly seated or has been removed.		Reseat or replace the SFP module. If a SFP module is not required in the port generating the alarm, the alarm can be cleared by deleting the equipment port.		
100ME	critical	service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The SFP module is not properly seated or has been removed and has an interface provisioned on it.		Reseat or replace the SFP module. If a SFP module is not required in the port generating the alarm, the alarm can be cleared by deleting the equipment port.		
155M	minor	non-service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The SFP module is not properly seated or has been removed.		Reseat or replace the SFP module. If a SFP module is not required in the port generating the alarm, the alarm can be cleared by deleting the equipment.		
155M	critical	service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The SFP module is not properly seated or has been removed and has an interface provisioned on it.		Reseat or replace the SFP module. If a SFP module is not required in the port generating the alarm, the alarm can be cleared by deleting the equipment port.		
1GE BT	minor	non-service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The SFP module is not properly seated or has been removed.		Reseat or replace the SFP module. If a SFP module is not required in the port generating the alarm, the alarm can be cleared by deleting the equipment port.		
1GE BT	critical	service-affecting	n/a	NEND	Equipment Removed: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	The SFP module is not properly seated or has been removed and has an interface provisioned on it.		Reseat or replace the SFP module. If a SFP module is not required in the port generating the alarm, the alarm can be cleared by deleting the equipment port.		

equipmentShelfProvisioningMode

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	The shelf provisioning mode is set to an incorrect value.
		Probable Cause		Suggested Corrective Action	
		The shelf provisioning mode is incorrect.		1. If the blade is set to factory defaults in a new turn-up, initiate a WARM reset to make this blade the Main blade. 2. For a tributary blade in a new turn-up, change the shelf type to TRIB. 3. If the NE is carrying traffic and the shelf provision command is sent by accident, cancel the shelf provisioning command to clear the condition.	

equipmentWarmup

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3R11	warning	non-service-affecting	n/a	NEND	Equipment Warm-Up: TYPE = <unit type>
		Probable Cause		Suggested Corrective Action	
		The blade is in a warm-up condition following power-up.		No action is required. The condition appears after the facility is provisioned, and the condition clears when the unit achieves the desired output wavelength.	
PIL1-3PS2	warning	non-service-affecting	n/a	NEND	Equipment Warm-Up: TYPE = <unit type>
		Probable Cause		Suggested Corrective Action	
		The PIU is in a warm-up condition following insertion.		No action is required. The condition appears after the facility is provisioned, and the condition clears when the unit achieves the desired output wavelength.	

fanCoolingFail

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	minor	non-service-affecting	n/a	NEND	Equipment Temperature Exceeds Operating Spec: TYPE=<unit type>
Probable Cause			Suggested Corrective Action		
A Fan unit has failed.			If blade is installed in a 1FINITY Housing equipped with an air filter, check the air filter and replace if it is clogged. Note: Fujitsu recommends replacement of the Fan filter every 3 months. Check for the following conditions and correct as needed: ■ Foreign material blocking or in the vicinity of the air intakes ■ A structure or object impeding normal air intake or exhaust ■ An external airflow from another device opposing the normal blade exhaust Note: The blade front panel should be cleaned if dust accumulates on it to the point where the dust is visible. The cooling capacity of the blades is reduced if the intake holes are clogged or obstructed. If this trend continues unchecked, the fan rotation speed increases to maintain the cooling performance, up to the point where the fanCoolingFail alarm is raised. Note: Replace the Fan unit.		

firmwareBackwardCompatibleAll

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	warning	non-service-affecting	n/a	NEND	Firmware Version Limited Compatibility: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
The FW on the interface unit specified in the alarm is not compatible with the system software. All features are supported.			Load the appropriate FW.		

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	minor	non-service-affecting	n/a	NEND	Firmware Version Limited Compatibility: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The FW on the interface unit specified in the alarm is not compatible with the system software. All features are supported.		Load the appropriate FW.		

firmwareDownloadOrActivationFailure

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	minor	service-affecting	n/a	NEND	Firmware Download (FWDL)/Activation Failure: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The FWDL or activation process failed.		Retry the FWDL or activation process.		
BDL1-3B11, BDL1-3R11	critical	service-affecting	n/a	NEND	FWDL/Activation Failure: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The FWDL or activation process failed.		Retry the FWDL or activation process.		

firmwareInitInProgress

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	warning	non-service-affecting	n/a	NEND	Firmware INIT-EQPT in progress: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	A FW initialization is in progress.		No action is required. This condition clears when the FW initialization completes.		

firmwareVersionMismatch

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	minor	service-affecting	n/a	NEND	Firmware Version Mismatch: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The FW on the tributary blade specified in the alarm is not compatible with the system software.		Perform the FW upgrade procedure on the indicated blade.		
BDL1-3B11, BDL1-3R11	major	service-affecting	n/a	NEND	Firmware Version Mismatch: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The FW on the tributary blade specified in the alarm is not compatible with the system software.		Perform the FW upgrade procedure on the indicated blade.		

forwardDefectIndication

Entity Type	Severity	Service Affected	Direction	Location	Description
och	warning	service-affecting	receive	NEND	Forward Defect Indication
	Probable Cause		Suggested Corrective Action		
	An OCH signal failure, for example, a WDM LOS, has occurred in the upstream direction.		Clear the upstream WDM transmit signal failure. Caution: The OSC must be active for traffic to be restored. A failure of the OSC may keep traffic from recovering on a span.		

ilfViolationCritical

Entity Type	Severity	Service Affected	Direction	Location	Description
ILF	critical	non-service-affecting	n/a	NEND	ILF Violation
		Probable Cause	Suggested Corrective Action		
		The 30-day trial period for using an ILF, without obtaining a license key, has expired.	This alarm is issued daily and contains a countdown counter. One of the following actions must be performed before the countdown timer expires: ■ Purchase and load a license key for the ILF. ■ Delete the ILF. Important: If the ILF is not deleted and a license key is not purchased and loaded within 60 days, the database is locked. Traffic is not affected, but provisioning of the TID is blocked.		

ilfViolationMajor

Entity Type	Severity	Service Affected	Direction	Location	Description
ILF	major	non-service-affecting	n/a	NEND	ILF Violation
		Probable Cause	Suggested Corrective Action		
		An ILF is activated without a license key.	1. Purchase and load a license key for the ILF before the 30-day trial period expires. 2. Delete the ILF before the 30-day trial period expires. Important: If the ILF is not deleted and a license is not purchased and loaded, this alarm is upgraded to the critical alarm, ilfViolationCritical.		

incompatibleFirmware

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	major	non-service-affecting	n/a	NEND	Incompatible Firmware: TYPE = <unit type>
		Probable Cause		Suggested Corrective Action	
		Firmware on the blade is incompatible with the running software.		1. Upgrade the firmware on the blade by executing <code>request fw-update shelf-id x force true overwrite true</code> command. 2. After the firmware upgrade on all affected blades is completed, restart the system by executing <code>request restartSystem level cold</code> command.	

incompatibleHardware

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3R11	critical	service-affecting	n/a	NEND	Incompatible Hardware: TYPE = <unit type>
		Probable Cause		Suggested Corrective Action	
		An operation is performed on a blade that its hardware version (issue number) does not support.		Replace the blade with a new blade that its hardware version (issue number) is compatible with the operation.	

ipNotReachable

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	The IP address, specified in a ping request, is not reachable.
		Probable Cause		Suggested Corrective Action	
		The IP address is not reachable.		Correct the TCP/IP connectivity problem, and repeat the ping request.	

ipPingRequestTimedOut

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	A ping request to an IP address has timed out.
					Probable Cause
					Suggested Corrective Action
					The ping request timed out.
					Correct the TCP/IP connectivity problem, and repeat the ping request.

ipReachable

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	The IP address, specified in a ping request, is reachable.
					Probable Cause
					Suggested Corrective Action
					The IP address is reachable.
					No action is required.

lampTest

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	warning	non-service-affecting	n/a	NEND	Lamp Test
					Probable Cause
					Suggested Corrective Action
					A lamp test is in progress.
					None

linkDown

Entity Type	Severity	Service Affected	Direction	Location	Description
eth	major	non-service-affecting	receive	NEND	Link Down
					Probable Cause
					Suggested Corrective Action
					The local device has detected link errors.
					None

Entity Type	Severity	Service Affected	Direction	Location	Description
eth	major	service-affecting	receive	NEND	Link Down
	Probable Cause		Suggested Corrective Action		
	The local device has detected link errors.		None		

IldpFail

Entity Type	Severity	Service Affected	Direction	Location	Description
eth	non-alarmed	non-service-affecting	n/a	NEND	Neighbor discovery fail
	Probable Cause		Suggested Corrective Action		
	Communication between the local and remote amplifier cannot be established through the OSC (E-SC-E1) link.		Check the condition of the OSC (E-SC-E1) link at both the near and far ends.		

lossOfSignal

Entity Type	Severity	Service Affected	Direction	Location	Description
ots	critical	service-affecting	receive	NEND	Loss of Signal
	Probable Cause		Suggested Corrective Action		
	A loss of incoming signal exists on the specified OMS entity.		1. Check the local and upstream SFP modules for removal or failure. 2. Check the fiber link and perform trouble-clearing procedures, as necessary. 3. Verify the local receive and remote transmit optical levels. 4. Verify good optical connections to the local and far-end SFP modules.		

Entity Type	Severity	Service Affected	Direction	Location	Description
oms	critical	service-affecting	receive	NEND	Loss of Signal
	Probable Cause		Suggested Corrective Action		
	A loss of incoming signal exists on the specified OMS entity.		1. Check the local and upstream SFP modules for removal or failure. 2. Check the fiber link and perform trouble-clearing procedures, as necessary. 3. Verify the local receive and remote transmit optical levels. 4. Verify good optical connections to the local and far-end SFP modules.		
eth	major	non-service-affecting	receive	NEND	Loss of Signal
	Probable Cause		Suggested Corrective Action		
	A loss of incoming signal exists on the specified Ethernet entity.		1. Check the local and upstream SFP modules for removal or failure. 2. Check the fiber link and perform trouble-clearing procedures, as necessary. 3. Verify the local receive and remote transmit optical levels. 4. Verify good optical connections to the local and far-end SFP modules.		
eth	major	service-affecting	receive	NEND	Loss of Signal
	Probable Cause		Suggested Corrective Action		
	A loss of incoming signal exists on the specified Ethernet entity.		1. Check the local and upstream SFP modules for removal or failure. 2. Check the fiber link and perform trouble-clearing procedures, as necessary. 3. Verify the local receive and remote transmit optical levels. 4. Verify good optical connections to the local and far-end SFP modules.		

lossOfSynchronization

Entity Type	Severity	Service Affected	Direction	Location	Description
eth	major	non-service-affecting	receive	NEND	Loss of Synchronization
Probable Cause			Suggested Corrective Action		
A codeword synchronization problem exists on the 100GE signal.			Verify that the cable length and signal meet the specified requirements.		

modeMismatch

Entity Type	Severity	Service Affected	Direction	Location	Description
OTS interface	major	service-affecting	receive	NEND	OTS command mode mismatch
Probable Cause			Suggested Corrective Action		
An OTS configuration mismatch condition on interoperating with FLASHWAVE 9500.			Configure a span operation interoperate with FLASHWAVE 9500 full spec mode, class interoperability mode, multi-haul interoperability mode, or Open ROADM mode. Provision the mode through the OTS operation mode command under OTS configuration.		

openConnectionIndication

Entity Type	Severity	Service Affected	Direction	Location	Description
och	critical	service-affecting	receive	NEND	Open Connection Indication
Probable Cause			Suggested Corrective Action		
The Wavelength Channel (WCH) is not provisioned or equipped.			Verify that the correct PIU type is installed. Caution: The OSC must be active for traffic to be restored. A failure of the OSC may keep traffic from recovering on a span. Check the channel provisioning and correct, if necessary.		

oscControlFailure

Entity Type	Severity	Service Affect	Direction	Location	Description
OTS	non-alarmed	non-service-affecting	receive	NEND	OSC Control Fail
	Probable Cause		Suggested Corrective Action		
	The OSC communication is incorrect.		Check the local receive and upstream transmit condition.		

oscModeMismatch

Entity Type	Severity	Service Affect	Direction	Location	Description
100ME	major	service-affecting	receive	NEND	Different OSC Mode is automatically set by the system on the span with L160 Raman blade provision or not.
	Probable Cause		Suggested Corrective Action		
	L160 Raman blade is not provisioned on both NEs on the span.		Verify that the L160 Raman blade provisioning on both NEs on the span is correct.		

oscPowerOutOfSpecificationHigh

Entity Type	Severity	Service Affected	Direction	Location	Description
Port (Edge)	critical	service-affecting	receive	NEND	OPROSC Power Out of Specification–High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	Optical Receive Power OSC is too high.		1. Verify that the appropriate transmitter is installed at the far end. 2. Verify that the transmit output level is within specifications. 3. Install an appropriate optical attenuator in the line at the receiver.		

oscPowerOutOfSpecificationLow

Entity Type	Severity	Service Affected	Direction	Location	Description
Port (Edge)	critical	service-affecting	receive	NEND	OPROSC Power Out of Specification–Low: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
Optical Receive Power OSC is too low.			1. Verify that the appropriate transmitter is installed at the far end. 2. Verify that the transmit output level is within specifications. 3. Install an appropriate optical attenuator in the line at the receiver.		

osnrModeMismatch

Entity Type	Severity	Service Affect	Direction	Location	Description
OTS	major	service-affecting	receive	NEND	OSNR Mode Mismatch
Probable Cause			Suggested Corrective Action		
A mismatch exists between local NE and remote NE values for the OSNR enhancement setting.			Verify the provisioning of the OSNR enhancement setting between the two NEs.		

otsSpanlossPowerOutOfSpecificationHigh

Entity Type	Severity	Service Affected	Direction	Location	Description
ots	critical	service-affecting	receive	NEND	SPANLOSS Out of Specification–High
Probable Cause			Suggested Corrective Action		
■ The threshold level has been improperly set. ■ Attenuation on existing span has increased.			1. Verify the proper threshold levels with local standards. 2. If needed, reset the PM threshold to correct. 3. Follow local practices to check the physical characteristics of the existing span.		

otsSpanlossPowerOutOfSpecificationLow

Entity Type	Severity	Service Affected	Direction	Location	Description
ots	critical	service-affecting	receive	NEND	SPANLOSS Out of Specification–Low
Probable Cause			Suggested Corrective Action		
■ The threshold level has been improperly set. ■ Attenuation on existing span has decreased.			1. Verify the proper threshold levels with local standards. 2. If needed, reset the PM threshold to correct. 3. Follow local practices to check the physical characteristics of the existing span.		

payloadMissingIndication

Entity Type	Severity	Service Affected	Direction	Location	Description
oms	warning	service-affecting	receive	NEND	Payload Missing Indication
Probable Cause			Suggested Corrective Action		
■ No optical signal is detected in any of the OCH carriers. This function is implemented through the OSC. ■ No optical signal is detected in the OTS payload. PMI suppresses the WDM lossOfSignal alarm at the far end.			Clear the trouble upstream that is preventing the node from receiving PMI reports from its provisioned channels.		

portLossOfLight

Entity Type	Severity	Service Affected	Direction	Location	Description
Port (Edge)/PIL1-3PS2	critical	service-affecting	transmit	NEND	Port Loss of Light: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
A Loss of Light (LOL) has occurred on the receive side of the edge port and the admin-status is down.			Verify the local receive and upstream transmit level.		

Entity Type	Severity	Service Affected	Direction	Location	Description
Port (Edge)/PIL1-3PS2	critical	service-affecting	receive	NEND	Port Loss of Light: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	A LOL has occurred on the receive side of the edge port.		Verify the local receive and upstream transmit level.		
Port (OSC)/BDL1-3R11	major	service-affecting	receive	NEND	Port Loss of Light: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	A LOL has occurred on the receive side of the OSC port.		Verify the local receive and upstream transmit level.		
Subport (Edge)/BDL1-3R11	critical	service-affecting	transmit	NEND	Port Loss of Light: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	A LOL has occurred on the receive side of the edge subport and the admin-status is down.		Verify the local receive and upstream transmit level.		
Subport (Edge)/BDL1-3R11	critical	service-affecting	receive	NEND	Port Loss of Light: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	A LOL has occurred on the receive side of the edge subport.		Verify the local receive and upstream transmit level.		
Subport (Client)/BDL1-3R11	critical	service-affecting	transmit	NEND	Port Loss of Light: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	A LOL has occurred on the receive side of the client subport and the admin-status is down.		Verify the local receive and upstream transmit level.		

postBlockAutoPowerReduction

Entity Type	Severity	Service Affected	Direction	Location	Description
oms	warning	non-service-affecting	n/a	NEND	Post Block Automatic Power Reduction: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
APR has been triggered for the amplifier.			Check the fiber connections to/from the PIU.		

postBlockAutoShutoffDisabled

Entity Type	Severity	Service Affected	Direction	Location	Description
oms	warning	non-service-affecting	n/a	NEND	Post Block Laser Safety Override Active: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
A command is issued to forcefully turn on the laser.			None		

postBlockManualLaserShutdown

Entity Type	Severity	Service Affected	Direction	Location	Description
oms	warning	non-service-affecting	n/a	NEND	Post Block Manual Laser Shutdown: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
The user has changed laser shutdown to manual.			Change laser shutdown to automatic.		

postBlockSpanAdjustmentInProgress

Entity Type	Severity	Service Affected	Direction	Location	Description
oms	warning	non-service-affecting	n/a	NEND	Post Block Span Adjustment in Progress: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
Post block span adjustment is in progress.			No action is required. This condition automatically clears when the post block span adjustment completes.		

powerOutOfSpecificationHigh

Entity Type	Severity	Service Affected	Direction	Location	Description
Subport (Edge)	critical	service-affecting	transmit	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause			Suggested Corrective Action	
	The optical power on the indicated transmit edge subport is too high.			Check the optical transmit power at the indicated subport.	
Subport (Edge)	critical	service-affecting	receive	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause			Suggested Corrective Action	
	The optical power on the indicated receive edge subport is too high.			Check the optical receive power at the indicated subport.	
Subport (Client)	critical	service-affecting	transmit	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause			Suggested Corrective Action	
	The optical power on the indicated transmit client subport is too high.			Check the optical receive power at the indicated subport.	
Port (Edge)	critical	service-affecting	transmit	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause			Suggested Corrective Action	
	The optical power on the indicated transmit edge port is too high.			Check the optical transmit power at the indicated port.	

Entity Type	Severity	Service Affected	Direction	Location	Description
Port (Edge)	critical	service-affecting	receive	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive edge port is too high.			Check the optical receive power at the indicated port.	
Port (Client)	critical	service-affecting	transmit	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit client port is too high.			Check the optical transmit power at the indicated port.	
Port (Client)	critical	service-affecting	receive	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive client port is too high.			Check the optical receive power at the indicated port.	
Port (OSC)	major	non–service-affecting	receive	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive OSC port is too high.			Check the optical receive power at the indicated port.	
1GEX	minor	non–service-affecting	transmit	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEX interface is too high.			None	
1GEX	minor	non–service-affecting	receive	NEND	Power Out of Specification–High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive 1GEX interface is too high.			Check the optical receive power at the indicated port.	

Entity Type	Severity	Service Affected	Direction	Location	Description
1GEX	critical	non-service-affecting	transmit	NEND	Power Out of Specification-High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEX interface is too high.			None	
1GEX	critical	non-service-affecting	receive	NEND	Power Out of Specification-High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive 1GEX interface is too high.			Check the optical receive power at the indicated port.	
1GEBT	minor	non-service-affecting	transmit	NEND	Power Out of Specification-High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEBT interface is too high.			Check the optical transmit power at the indicated port.	
1GEBT	minor	non-service-affecting	receive	NEND	Power Out of Specification-High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive 1GEBT interface is too high.			Check the optical receive power at the indicated port.	
1GEBT	critical	non-service-affecting	transmit	NEND	Power Out of Specification-High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEBT interface is too high.			Check the optical transmit power at the indicated port.	
1GEBT	critical	non-service-affecting	receive	NEND	Power Out of Specification-High: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive 1GEBT interface is too high.			Check the optical receive power at the indicated port.	

powerOutOfSpecificationLow

Entity Type	Severity	Service Affected	Direction	Location	Description
Subport (Edge)	critical	service-affecting	transmit	NEND	Power Out of Specification–Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit edge subport is too low.		Check the optical transmit power at the indicated subport.		
Subport (Edge)	critical	service-affecting	receive	NEND	Power Out of Specification–Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit edge subport is too low.		Check the optical transmit power at the indicated subport.		
Subport (Edge)	critical	service-affecting	transmit	NEND	Power Out of Specification–Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit edge subport is too low.		Check the optical transmit power at the indicated subport.		
Port (Edge)	critical	service-affecting	transmit	NEND	Power Out of Specification–Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit edge subport is too low.		Check the optical transmit power at the indicated port.		
Port (Edge)	critical	service-affecting	receive	NEND	Power Out of Specification–Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit edge subport is too low.		Check the optical transmit power at the indicated port.		
Port (Client)	critical	service-affecting	transmit	NEND	Power Out of Specification–Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit edge subport is too low.		Check the optical transmit power at the indicated port.		

Entity Type	Severity	Service Affected	Direction	Location	Description
Port (Client)	critical	service-affecting	receive	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit edge subport is too low.			Check the optical transmit power at the indicated port.	
Port (OSC)	major	non-service-affecting	receive	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit edge port is too low.			Check the optical transmit power at the indicated port.	
1GEX	minor	non-service-affecting	transmit	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEX interface is too low.			None	
1GEX	minor	non-service-affecting	receive	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEX interface is too low.			Check the optical transmit power at the indicated interface.	
1GEX	critical	service-affecting	transmit	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEX interface is too low.			Check the optical transmit power at the indicated interface.	
1GEX	critical	service-affecting	receive	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive 1GEX interface is too low.			Check the optical receive power at the indicated interface.	

Entity Type	Severity	Service Affected	Direction	Location	Description
1GEBT	minor	non-service-affecting	transmit	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEX interface is too low.		Check the optical transmit power at the indicated interface.		
1GEBT	minor	non-service-affecting	receive	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive 1GEX interface is too low.		Check the optical receive power at the indicated interface.		
1GEBT	critical	service-affecting	transmit	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated transmit 1GEX interface is too low.		Check the optical transmit power at the indicated interface.		
1GEBT	critical	service-affecting	receive	NEND	Power Out of Specification—Low: TYPE = <unit type>
	Probable Cause		Suggested Corrective Action		
	The optical power on the indicated receive 1GEX interface is too low.		Check the optical receive power at the indicated interface.		

powerProblemA

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	major	non-service-affecting	n/a	NEND	Power Input Fail: TYPE=<unit type>
	Probable Cause		Suggested Corrective Action		
	■ The A power feed on the –48 V DC power adapter has failed. ■ The –48 V DC A power feed has failed at the source.		1. Check the power connections on the A power feed on the –48 V DC power adapter and/or the –48 V DC power source. 2. Check the input voltage at the A power feed on the –48 V DC power adapter.		

Entity Type	Severity	Service Affected	Direction	Location	Description
	critical	service-affecting	n/a	NEND	Power Input Fail: TYPE=<unit type>
Probable Cause			Suggested Corrective Action		
■ The A power feed on the –48 V DC power adapter has failed. ■ The –48 V DC A power feed has failed at the source.			1. Check the power connections on the A power feed on the –48 V DC power adapter and/or the –48 V DC power source. 2. Check the input voltage at the A power feed on the –48 V DC power adapter.		

powerProblemB

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	major	non-service-affecting	n/a	NEND	Power Input Fail: TYPE=<unit type>
Probable Cause			Suggested Corrective Action		
■ The A power feed on the –48 V DC power adapter has failed. ■ The –48 V DC B power feed has failed at the source.			1. Check the power connections on the B power feed on the –48 V DC power adapter and/or the –48 V DC power source. 2. Check the input voltage at the B power feed on the –48 V DC power adapter.		
BDL1-3B11, BDL1-3R11	critical	service-affecting	n/a	NEND	Power Input Fail: TYPE=<unit type>
Probable Cause			Suggested Corrective Action		
■ The A power feed on the –48 V DC power adapter has failed. ■ The –48 V DC B power feed has failed at the source.			1. Check the power connections on the B power feed on the –48 V DC power adapter and/or the –48 V DC power source. 2. Check the input voltage at the B power feed on the –48 V DC power adapter.		

preBlockSpanAdjustmentInProgress

Entity Type	Severity	Service Affected	Direction	Location	Description
oms	warning	non-service-affecting	n/a	NEND	Pre Block Span Adjustment in Progress: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
Pre-block span adjustment is in progress.			No action is required. The condition automatically clears when the pre-block span adjustment completes.		

reflectionTooHigh

Entity Type	Severity	Service Affected	Direction	Location	Description
Port (Edge)	major	non-service-affecting	n/a	NEND	Reflection Too High: TYPE = <unit type>
Probable Cause			Suggested Corrective Action		
Optical reflection is too high on the edge port. Attention: This alarm is reported as it approaches the point to trigger Automatic Power Reduction (APR). Although the alarmed condition is NSA, the required trouble-clearing actions to clear the alarm can disrupt traffic.			1. Check the transmit output fiber. 2. Clean the connectors. 3. Replace the fiber. 4. Replace the patch cord. 5. Replace the PIU.		

shelfProvisioningMode

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	Shelf Provisioning Mode
Probable Cause			Suggested Corrective Action		
The blade provisioning mode is incorrect. This alarm is caused by the blade being set at the factory defaults or by the user sending an incorrect shelf provisioning command.			1. If the blade is set to factory defaults in a new turn-up, initiate a WARM reset to make this blade the Main blade. 2. For a tributary blade in a new turn-up, change the shelf type to TRIB.		

softwareReset

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	minor	non-service-affecting	n/a	NEND	Software Reset: TYPE=<unit type>
Probable Cause			Suggested Corrective Action		
A system software reset is in progress.			No action is required. This alarm clears when the system software reset completes.		

softwareStageInProgress

Entity Type	Severity	Service Affected	Direction	Location	Description
BDL1-3B11, BDL1-3R11	major, non-alarmed	non-service-affecting	n/a	NEND	Software Stage In Progress
Probable Cause			Suggested Corrective Action		
Software staging is in progress.			No action is required. The alarm automatically clears when software staging completes.		

softwareVersionMismatch

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	major	non-service-affecting	n/a	NEND	Software Version Mismatch
Probable Cause			Suggested Corrective Action		
A mismatch failure of the file configuration among the active banks has occurred.			1. In a new turn-up, perform a WARM reset. 2. If the installed blade contains the desired software and database, call Fujitsu for technical assistance at 1-800-USE-FTAC (1-800-873-3822). 3. Download the latest release of software to all blades, and switch to the new load. If any problems exist, call Fujitsu for technical assistance at 1-800-USE-FTAC (1-800-873-3822).		

sysNameChanged

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	warning, minor	non-service-affecting	n/a	NEND	The system name is changed.
Probable Cause			Suggested Corrective Action		
The system name is changed.			Perform a WARM reboot to clear the condition.		

sysNtpNotSynchronized

Entity Type	Severity	Service Affected	Direction	Location	Description
COM	minor	non-service-affecting	n/a	NEND	NTP is not synchronized with the server.
Probable Cause			Suggested Corrective Action		
The time is not synced with the NTP server.			No action is required. The alarm automatically clears when the time synchronizes with the NTP server.		

thresholdCrossingAlertCodingViolation

Entity Type	Severity	Service Affected	Direction	Location	Description
Ether OSC (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The threshold for coding violations is exceeded for the section.
Probable Cause			Suggested Corrective Action		
■ Optical path problems exist between the line-termination nodes. ■ Low or high light levels can be caused by dirty fiber. ■ The blade has failed. ■ Bit interleaved parity (BIP) errors exceed a threshold.			1. Check the optical output level at the upstream transmitter. 2. Check the optical receive level at the input to the local blade. 3. Replace the blade. 4. Clean the optical connectors. 5. Check the optical receive level at the input of the local blade. 6. Clear outstanding lossOfSignal alarms to clear the PM counts.		

thresholdCrossingAlertErroredSeconds

Entity Type	Severity	Service Affected	Direction	Location	Description
Ether OSC (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The threshold for errored seconds has been crossed.
	Probable Cause			Suggested Corrective Action	
	■ Optical path problems exist between the upstream node and the near end. ■ The blade has failed.			1. Clean the optical connectors. 2. Check the optical receive level at the input to the local blade. 3. Clear outstanding lossOfSignal alarms to clear the PM counts. 4. Replace the blade.	

thresholdCrossingAlertOpticalReceivePowerTooHigh

Entity Type	Severity	Service Affected	Direction	Location	Description			
EQPT	warning	non-service-affecting	receive	NEND	The optical receive power level is above the threshold on one of the following ports: ■ For BDL1-3R11: - E-SC-E1 - F-SC-E1 - LCN1...4 ■ For BDL1-3B11: LCN1...4			
					Probable Cause		Suggested Corrective Action	
					Optical receive power is too high.		1. Verify that the appropriate transmitter is installed at the far end. 2. Verify that the transmit output level at the far end is within specifications. 3. Install an appropriate optical attenuator in the line at the receiver.	

thresholdCrossingAlertOpticalReceivePowerTooLow

Entity Type	Severity	Service Affected	Direction	Location	Description
EQPT	warning	non-service-affecting	receive	NEND	The optical receive power level is above the threshold on one of the following ports: ■ For BDL1-3R11: • E-SC-E1 • F-SC-E1 • LCN1...4 ■ For BDL1-3B11: LCN1...4
					Probable Cause
					Suggested Corrective Action
					Optical receive power is too low. 1. Verify that the appropriate transmitter is installed at the far end. 2. Verify that the transmit output level is within specifications at the far end. 3. Replace the transmitter at the far end if the level is too low. 4. Clean the fiber connections in the path.

thresholdCrossingAlertOpticalTransmitPowerTooHigh

Entity Type	Severity	Service Affected	Direction	Location	Description
EQPT	warning	non-service-affecting	receive	NEND	The optical receive power level is above the threshold on one of the following ports: ■ For BDL1-3R11: • E-SC-E1 • F-SC-E1 • LCN1...4 ■ For BDL1-3B11: LCN1...4
					Probable Cause
					Suggested Corrective Action
					Optical receive power is too high. 1. Verify that the transmitter output level is within specifications. 2. Replace the blade if necessary.

thresholdCrossingAlertOpticalTransmitPowerTooLow

Entity Type	Severity	Service Affected	Direction	Location	Description
EQPT	warning	non-service-affecting	receive	NEND	The optical receive power level is above the threshold on one of the following ports: ■ For BDL1-3R11: • E-SC-E1 • F-SC-E1 • LCN1...4 ■ For BDL1-3B11: LCN1...4
					Probable Cause
					Suggested Corrective Action
					Optical receive power is too low. 1. Verify that the transmitter output level is within specifications. 2. Clean the port and the fiber attached to it. 3. Replace the blade if necessary.

thresholdCrossingAlertOscOpticalReceivePowerTooHigh

Entity Type	Severity	Service Affected	Direction	Location	Description
PORT=E1 (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The optical receive power at the OSC port is above the threshold level.
					Probable Cause
					Suggested Corrective Action
					Optical OSC receive power is too high. 1. Verify that the appropriate transmitter is installed at the far end. 2. Verify that the transmit output level at the far end is within specifications. 3. Install an appropriate optical attenuator at the receiver.

thresholdCrossingAlertOscOpticalReceivePowerTooLow

Entity Type	Severity	Service Affected	Direction	Location	Description
PORT=E1 (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The optical receive power at the OSC port is below the threshold level.
					Probable Cause
					Suggested Corrective Action
					Optical OSC receive power is too low.
					1. Verify that the appropriate transmitter is installed at the far end. 2. Verify that the transmit output level at the far end is within specifications. Replace the transmitter if the level is too low. 3. Clean fiber connections in the path.

thresholdCrossingAlertOscOpticalTransmitPowerTooHigh

Entity Type	Severity	Service Affected	Direction	Location	Description
PORT=E1 (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The optical transmit power at the OSC port is above the threshold level.
					Probable Cause
					Suggested Corrective Action
					Optical OSC transmitter power is too high.
					Replace the blade if necessary.

thresholdCrossingAlertOscOpticalTransmitPowerTooLow

Entity Type	Severity	Service Affected	Direction	Location	Description
PORT=E1 (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The optical transmit power at the OSC port is below the threshold level.
					Probable Cause
					Suggested Corrective Action
					Optical OSC transmitter power is too low.
					1. Verify that the transmit output level is within specifications. 2. Replace the blade if necessary.

thresholdCrossingAlertSeverelyErroredSeconds

Entity Type	Severity	Service Affected	Direction	Location	Description
Ether OSC (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The threshold for severely errored seconds has been crossed.
	Probable Cause			Suggested Corrective Action	
	- Optical path problems exist between the upstream node and the near end. - The blade has failed.			- Check the optical output level at the upstream transmitter. - Check the optical receive level at the input to the local BDL1-3R11 blade. - Replace the blade.	

thresholdCrossingAlertSpanlossVariationTooHigh

Entity Type	Severity	Service Affected	Direction	Location	Description
PORT=E1 (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The span loss variation is above the threshold level.
	Probable Cause			Suggested Corrective Action	
	■ The threshold level is improperly set. ■ Attenuation on the existing span has increased.			1. Verify the proper threshold levels with local standards. If needed, reset the PM threshold to correct. 2. Follow local practices to check the physical characteristics of the existing span.	

thresholdCrossingAlertSpanlossVariationTooLow

Entity Type	Severity	Service Affected	Direction	Location	Description
PORT=E1 (BDL1-3R11)	warning	non-service-affecting	receive	NEND	The span loss variation is below the threshold level.
	Probable Cause			Suggested Corrective Action	
	■ The threshold level is improperly set. ■ Attenuation on the existing span has decreased.			1. Verify the proper threshold levels with local standards. If needed, reset the PM threshold to correct. 2. Follow local practices to check the physical characteristics of the existing span.	

trailTraceMismatch

Entity Type	Severity	Service Affected	Direction	Location	Description
OC3 OSC (BDL1-3R11)	major	non-service-affecting	receive	NEND	A mismatch exists on the trail trace identifier.
	Probable Cause			Suggested Corrective Action	
	The Receive Source Access Point Identifier (RCVSAPI) is different from Expected Source Access Point Identifier (EXPSAPI), the Receive Destination Access Point Identifier (RCVDAPI) is different from Expected Destination Access Point Identifier (EXPDAPI), or both conditions exist.			1. Ensure that the local EXPSAPI, EXPDAPI, or both are set correctly to match the ones transmitted from the far end. 2. Ensure that the far-end transmitting Source Access Point Identifier (SAPI) and Destination Access Point Identifier (DAPI) match the local EXPSAPI and/or EXPDAPI. 3. If the problem persists, verify the fiber connections for add and drop connections.	

unequippedIndication

Entity Type	Severity	Service Affected	Direction	Location	Description
OC3 OSC (BDL1-3R11)	major	service-affecting	receive	NEND	An unequipped indication is received.
	Probable Cause			Suggested Corrective Action	
	A WCH is not provisioned or equipped.			1. Verify that the correct blade type is installed. 2. Check the channel provisioning and correct if necessary. Caution: The OSC must be active for traffic to be restored. A failure of the OSC may keep traffic from recovering on a span.	

ztpBootFailure

Entity Type	Severity	Service Affect	Direction	Location	Description
COM	major	non-service-affecting	n/a	NEND	ZTP Boot failure.
	Probable Cause			Suggested Corrective Action	
	ZTP-Boot retries auto-upgrade after 5 minutes.			No action is required. ZTP-Boot retries auto-upgrade after 5 minutes. Note: This alarms clears at the end of 5 minutes.	

ztpBootInProgress

Entity Type	Severity	Service Affect	Direction	Location	Description
COM	minor	non-service-affecting	n/a	NEND	ZTP Boot in progress.
Probable Cause			Suggested Corrective Action		
A ZTP Boot is in progress.			No action is required. Note: This alarms clears when auto-upgrade completes or aborts and ZTP Boot closes.		

8.1.7

Secondary Database Alarms and Events

Entity Type	Severity	Service affected	Direction	Location	Description
EQPT (shelf AID)	warning	non-service-affecting	n/a	NEND	Secondary storage is not present or available.
Probable Cause			Suggested Corrective Action		
Raised when the secondary storage media is faulty or is not present on a given blade.			Disable secondary database if USB flash drive is not used. Note: Clears when a USB flash drive is present in the USB slot.		
COM	minor	non-service-affecting	n/a	NEND	No secondary database
Probable Cause			Suggested Corrective Action		
If secondary storage is not reachable, the NE raises a <i>noSecondaryDatabase</i> alarm.			1. Connect secondary storage. 2. After NE can successfully back up database in secondary storage, this alarm clears.		
COM	minor	non-service-affecting	n/a	NEND	Software synchronization is in progress.
Probable Cause			Suggested Corrective Action		
Raised when copying OSL to secondary storage (USB flash drive).			Clears when copying is complete.		

Entity Type	Severity	Service affected	Direction	Location	Description
COM	minor	non-service-affecting	n/a	NEND	Secondary storage OSL is corrupted.
	Probable Cause		Suggested Corrective Action		
	Main blade is default. Secondary software is corrupted.		1. Disconnect secondary storage (USB flash drive). 2. Upgrade load of the Main blade to desired load which supports secondary database feature (Release 2.4 and later releases). 3. Run <code>ss-overwrite</code> command. 4. Perform WARM restart on Main blade. 5. Clear the secondary OSL corruption alarm.		
COM	warning	non-service-affecting	n/a	NEND	Database locked: Secondary software corruption
	Probable Cause		Suggested Corrective Action		
	Secondary software is corrupted.		After <i>secondarySoftwareCorruption</i> alarm is cleared, this database lock condition is cleared.		
COM	warning	non-service-affecting	n/a	NEND	Database locked: Secondary software corruption
	Probable Cause		Suggested Corrective Action		
	Secondary software is corrupted.		After <i>secondarySoftwareCorruption</i> alarm is cleared, this database lock condition is cleared.		
BDL1-3R11	minor	non-service-affecting	n/a	NEND	USB flash drive encryption check failed
	Probable Cause		Suggested Corrective Action		
	Secondary storage drive failed; Encryption password check		1. Check whether the USB flash drive is inserted in the tributary blade. If it is, remove the USB flash drive from the tributary blade and insert in the Main blade. 2. If the USB flash drive is in the Main blade and an alarm is present, run <code>request format-usb</code> command to format the USB flash drive.		
COM	major	non-service-affecting	n/a	NEND	Software not ready
	Probable Cause		Suggested Corrective Action		
	Main blade software not ready		1. Perform WARM restart. 2. If WARM restart did not clear the alarm, try COLD restart. 3. If COLD restart did not clear the alarm, replace the Main blade.		

Entity Type	Severity	Service affected	Direction	Location	Description
COM	warning	non-service-affecting	n/a	NEND	Database system locked: Software not ready
	Probable Cause		Suggested Corrective Action		
	Main blade software not ready		After <i>softwareNotReady</i> alarm is cleared, the database locked alarm is cleared.		

Events

UsbSafeToRemove:

- Indication that USB flash drive can be safely removed
- Alarm raised when user does graceful USB flash drive remove

Alarms and LED Behavior

Alarm	Service Affected	Severity	Alarm LED	System LED
noSecondaryStorage	non-service-affecting	n/a	none	none
noSecondaryDatabase	non-service-affecting	minor	yellow	none
softwareSyncInProgress	non-service-affecting	minor	yellow	yellow
secondarySoftwareCorruption	non-service-affecting	minor	yellow	none
invalidUSB	non-service-affecting	minor	yellow	none
softwareNotReady	non-service-affecting	major	red	none
databaseLockedSecondarySoftwareCorruption	non-service-affecting	warning	none	yellow
databaseLockedSoftwareNotReady	non-service-affecting	warning	none	yellow

8.1.8 LED Control

The LED control is a mechanism by which the user can control turning on or off the FIND ME LED.

LED Control Behavior

Enter the following command.

Command:

```
> request led-control shelf-id <shelf-id> enable <ON | OFF>
```

8.2

Performance Monitoring Counters

The automatic Performance Monitoring (PM) function aids in the maintenance, operations, and trouble clearing of the network-side facility. The PM functions refer to in-service monitoring of equipment conditions and transmission quality using PM parameter data registers. The PM data is collected for critical equipment as well as for facility path, section, and line parameters.

The PM functions accumulate data for specific performance parameters. These parameters enable service providers to offer the following features:

- Detection of gradual increase in error rate and correction of certain maintenance problems before service is affected
- Guaranteed service to end users by using PM data as evidence of high signal-path quality

The PM data is stored in the following registers using two different protocols:

- Cumulative registers, which have no timing component. Generally, these registers hold up to $2^{64} - 1$ counts and roll back to a value of 1 when the maximum count is exceeded.
- Timed registers, sometimes referred to as binned registers, of which two types exist:
 - 15-minute registers that are automatically initialized on the quarter hour. Thirty-two 15-minute registers provide history storage for the previous 8 hours of PM counts.
 - 1-day registers that are automatically initialized every 24 hours (typically at midnight). Thirty-two 1-day registers provide history storage for the previous 32 days activity.

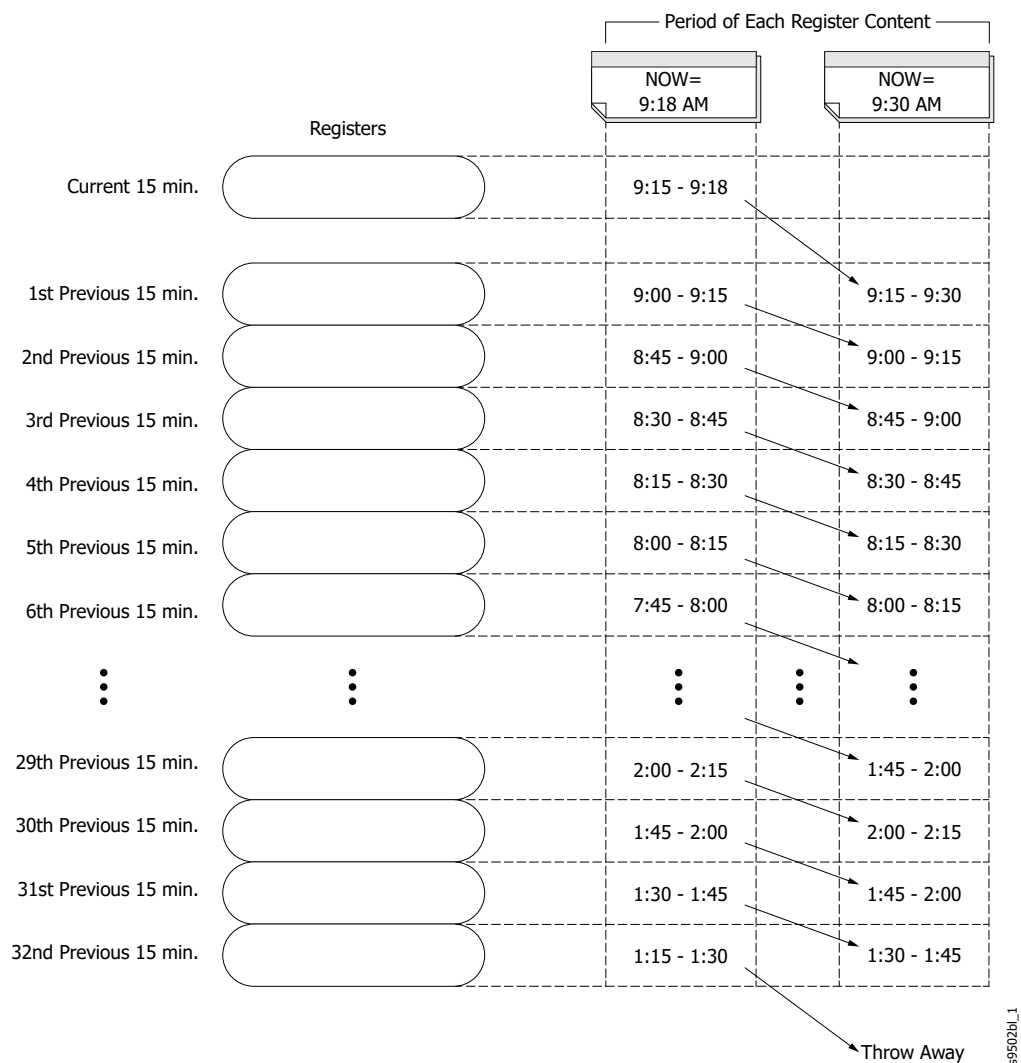


Figure 110
Performance Registers—Current and Previous 15 Minutes

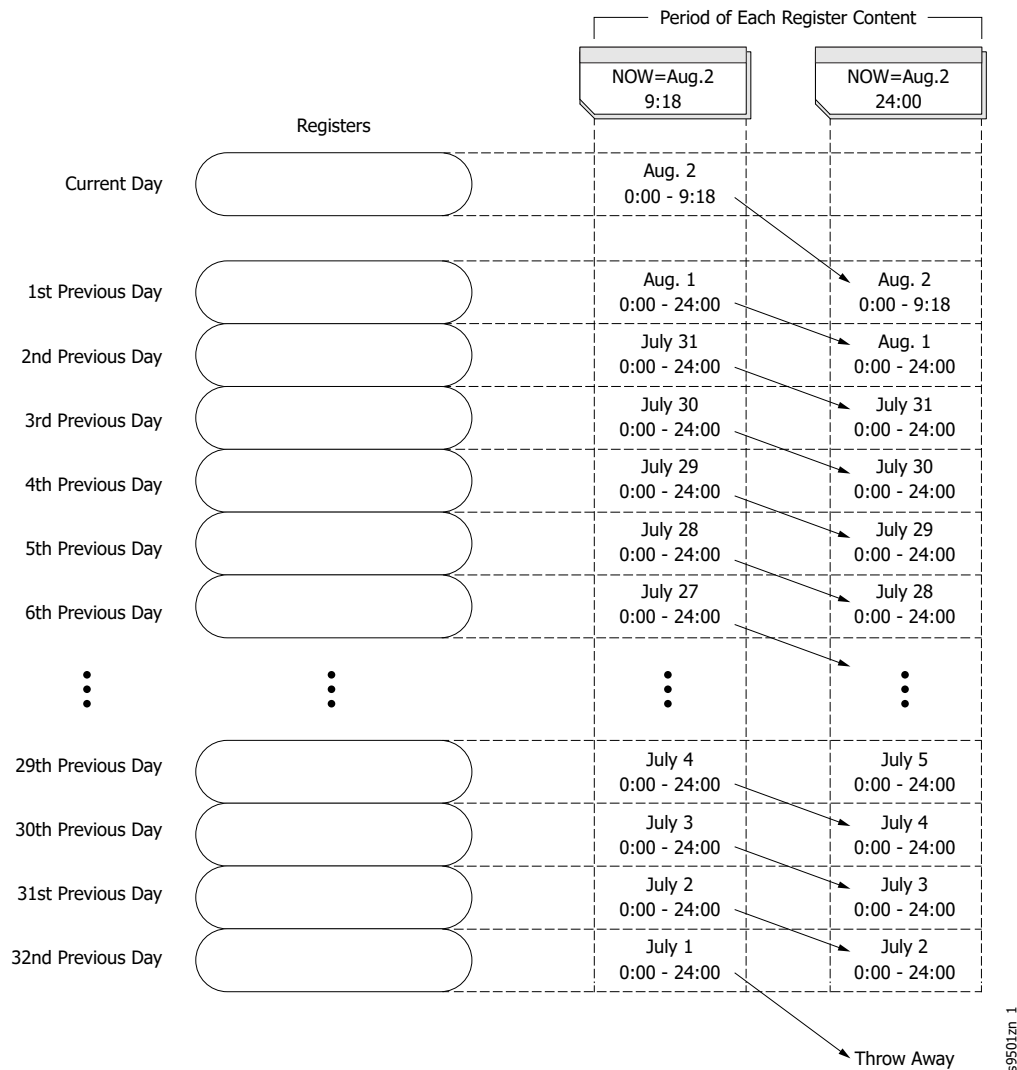


Figure 111
Performance Registers—Current Day and Previous 32 Days

For some timed parameters, Threshold Crossing Alerts (TCAs) are generated when the PM count exceeds a preset threshold. For other timed parameters, thresholds are not supported, and these registers can hold up to $2^{63} - 1$ counts before automatically rolling back to a value of 1 when the maximum count is exceeded.

The current timed registers store the number of events within their specified time limits and forward that information to the first previous register before they are initialized, for example, information from the first previous register is passed on to the second previous register and information from the second previous register is passed on to the third previous register. Information from the last previous register is discarded.

The following abbreviations are used in the performance monitoring counter tables in this section:

RX	Receive
TX	Transmit

NEND	Near end
n/a	Not applicable

Table 52
Equipment PM Counters (BDL1-3B11)

Counter	Entity Type	PM Type	Location	Direction	Valid Operating Range
temperature	Equipment	Metered	NEND	n/a	−99.0...99.0
temperatureAvg	Equipment	Metered	NEND	n/a	−99.0...99.0
temperatureMax	Equipment	Metered	NEND	n/a	−99.0...99.0
temperatureMin	Equipment	Metered	NEND	n/a	−99.0...99.0

Table 53
Equipment PM Counters (BDL1-3R11)

Counter	Entity Type	PM Type	Location	Direction	Valid Operating Range
temperature	Equipment	Metered	NEND	n/a	−99.0...99.0
temperatureAvg	Equipment	Metered	NEND	n/a	−99.0...99.0
temperatureMax	Equipment	Metered	NEND	n/a	−99.0...99.0
temperatureMin	Equipment	Metered	NEND	n/a	−99.0...99.0

Table 54
Shelf PM Counters

Counter	Time Period	PM Type	Location	Direction	Index	Valid Operating Range	Validity
memoryUsage	15-Min	Metered	NEND	NA	0...32	−99.0...99.0	False/True
	1-Day	Metered	NEND	NA	0...32	−99.0...99.0	False/True
memoryUsageMin	15-Min	Metered	NEND	NA	0...32	−99.0...99.0	False/True
	1-Day	Metered	NEND	NA	0...32	−99.0...99.0	False/True
memoryUsageMax	15-Min	Metered	NEND	NA	0...32	−99.0...99.0	False/True
	1-Day	Metered	NEND	NA	0...32	−99.0...99.0	False/True
memoryUsageAvg	15-Min	Metered	NEND	NA	0...32	−99.0...99.0	False/True
	1-Day	Metered	NEND	NA	0...32	−99.0...99.0	False/True

Table 54 (Cont.)
Shelf PM Counters

Counter	Time Period	PM Type	Location	Direction	Index	Valid Operating Range	Validity
cpuUsage	15-Min	Metered	NEND	NA	0...32	−99.0...99.0	False/True
	1-Day	Metered	NEND	NA	0...32	−99.0...99.0	False/True
cpuUsageMin	15-Min	Metered	NEND	NA	0...32	−99.0...99.0	False/True
	1-Day	Metered	NEND	NA	0...32	−99.0...99.0	False/True
cpuUsageMax	15-Min	Metered	NEND	NA	0...32	−99.0...99.0	False/True
	1-Day	Metered	NEND	NA	0...32	−99.0...99.0	False/True
cpuUsageAvg	15-Min	Metered	NEND	NA	0...32	−99.0...99.0	False/True
	1-Day	Metered	NEND	NA	0...32	−99.0...99.0	False/True

Table 55
Facility PM Counters (BDL1-3R11)

Counter	Time Period	Entity Type	PM Type	Location	Direction	Valid Operating Range
codingViolation	15-Min	EtherOsc	Binned	NEND	RX	1...90...2 ⁶³ -1
	1-Day	EtherOsc	Binned	NEND	RX	1...8640...2 ⁶³ -1
erroredSeconds	15-Min	EtherOsc	Binned	NEND	RX	1...90...900
	1-Day	EtherOsc	Binned	NEND	RX	1...8640...86400
severelyErroredSeconds	15-Min	EtherOsc	Binned	NEND	RX	1...3...900
	1-Day	EtherOsc	Binned	NEND	RX	1...7...86400

Table 56
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
opticalPowerReceive	Port (C1...C9)	n/a	Metered	NEND	RX	−10.6...15.0 (pass through)
	Subport (E1)	n/a	Metered	NEND	RX	−40.9...−3.4
	Port (OSC)	n/a	Metered	NEND	RX	−1.1...7.5
	Subport (L1)	n/a	Metered	NEND	RX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
opticalPowerReceiveAvg	Port (C1...C9)	n/a	Metered	NEND	RX	n/a
	Subport (E1)	n/a	Metered	NEND	RX	n/a
	Port (OSC)	n/a	Metered	NEND	RX	n/a
	Subport (L1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveMin	Port (C1...C9)	n/a	Metered	NEND	RX	n/a
	Subport (E1)	n/a	Metered	NEND	RX	n/a
	Port (OSC)	n/a	Metered	NEND	RX	n/a
	Subport (L1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveMax	Port (C1...C9)	n/a	Metered	NEND	RX	n/a
	Subport (E1)	n/a	Metered	NEND	RX	n/a
	Port (OSC)	n/a	Metered	NEND	RX	n/a
	Subport (L1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveOms	Port (E1)	n/a	Metered	NEND	RX	–35.8...13.2 (warm-up) –26.8...13.2 (low gain) –35.8...1.2 (high gain)
opticalPowerReceiveOmsAvg	Port (E1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveOmsMax	Port (E1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveOmsMin	Port (E1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveOsc	Port (E1)	n/a	Metered	NEND	RX	–39.9...7.0
opticalPowerReceiveOscAvg	Port (E1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveOscMax	Port (E1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveOscMin	Port (E1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveOts	Port (E1)	n/a	Metered	NEND	RX	–34.4...14.1 (warm-up) –26.6...14.1 (low gain) –34.4...8.0 (high gain)
opticalPowerReceiveOtsAvg	Port (E1)	n/a	Metered	NEND	RX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
opticalPowerReceiveOtsMax	Port (E1)	n/a	Metered	NEND	RX	n/a
opticalPowerReceiveOtsMin	Port (E1)	n/a	Metered	NEND	RX	n/a
opticalPowerTransmit	Port (C1...C9)	n/a	Metered	NEND	TX	−10.4...15.7 (pass through)
	Subport (E1)	n/a	Metered	NEND	TX	−21.7...11.3
	Subport (L1)	n/a	Metered	NEND	TX	n/a
	Subport (C1...C9)	n/a	Metered	NEND	TX	−12.4...−7.9 (add/drop) −10.4...−5.4 (pass through)
opticalPowerTransmitAvg	Port (C1...C9)	n/a	Metered	NEND	TX	n/a
	Subport (E1)	n/a	Metered	NEND	TX	n/a
	Subport (L1)	n/a	Metered	NEND	TX	n/a
	Subport (C1...C9)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitMax	Port (C1...C9)	n/a	Metered	NEND	TX	n/a
	Subport (E1)	n/a	Metered	NEND	TX	n/a
	Subport (L1)	n/a	Metered	NEND	TX	n/a
	Subport (C1...C9)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitMin	Port (C1...C9)	n/a	Metered	NEND	TX	n/a
	Subport (E1)	n/a	Metered	NEND	TX	n/a
	Subport (L1)	n/a	Metered	NEND	TX	n/a
	Subport (C1...C9)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitOms	Port (E1)	n/a	Metered	NEND	TX	−15.0...23.4
opticalPowerTransmitOmsAvg	Port (E1)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitOmsMax	Port (E1)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitOmsMin	Port (E1)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitOsc	Port (E1)	n/a	Metered	NEND	TX	−2.8...6.8
opticalPowerTransmitOscAvg	Port (E1)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitOscMax	Port (E1)	n/a	Metered	NEND	TX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
opticalPowerTransmitOscMin	Port (E1)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitOts	Port (E1)	n/a	Metered	NEND	TX	-2.5...23.5
opticalPowerTransmitOtsAvg	Port (E1)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitOtsMax	Port (E1)	n/a	Metered	NEND	TX	n/a
opticalPowerTransmitOtsMin	Port (E1)	n/a	Metered	NEND	TX	n/a
reflectionPower	Port (E1)	n/a	Metered	NEND	RX	17.6...99.9
reflectionPowerAvg	Port (E1)	n/a	Metered	NEND	RX	n/a
reflectionPowerMax	Port (E1)	n/a	Metered	NEND	RX	n/a
reflectionPowerMin	Port (E1)	n/a	Metered	NEND	RX	n/a
spanloss	Port (E1)	SMF (Full) ULL (Full)	Metered	NEND	RX	0.0...37.0 (warm-up) 0.0...25.0 (low gain) 8.0...37.0 (high gain)
		SMF (Limited) ULL (Limited)	Metered	NEND	RX	1.0...37.0 (warm-up) 1.0...28.0 (low gain) 11.0...37.0 (high gain)
spanloss	Port (E1)	ELEAF (Full) OLEAF (Full)	Metered	NEND	RX	0.0...34.0 (warm-up) 0.0...22 (low gain) 8.0...34 (high gain)
		ELEAF (Limited) OLEAF (Limited)	Metered	NEND	RX	1.0...37.0 (warm-up) 1.0...25.0 (low gain) 11.0...37.0 (high gain)

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
spanloss	Port (E1)	TrueWave® (Full) TrueWave-C (Full)	Metered	NEND	RX	0.0...33.0 (warm-up) 0.0...21.0 (low gain) 8.0...33.0 (high gain)
		TrueWave (Limited) TrueWave-C (Limited)	Metered	NEND	RX	1...36 (warm-up) 1.0...24.0 (low gain) 11.0...36 (high gain)
spanloss	Port (E1)	DSF (Full)	Metered	NEND	RX	0.0...29.0 (warm-up) 0.0...17.0 (low gain) 8.0...29.0 (high gain)
		DSF (Limited)	Metered	NEND	RX	1.0...32.0 (warm-up) 1.0...20.0 (low gain) 11.0...32.0... (high gain)
spanloss	Port (E1)	NZ-DSF (Full)	Metered	NEND	RX	0.0...33.0 (warm-up) 0.0...21.0 (low gain) 8.0...33.0 (high gain)
		NZ-DSF (Limited)	Metered	NEND	RX	1.0...36.0 (warm-up) 1.0...24.0 (low gain) 11.0...36.0 (high gain)
spanLossAvg	Port (E1)	SMF (Full) ULL (Full)	Metered	NEND	RX	n/a
		SMF (Limited) ULL (Limited)	Metered	NEND	RX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
spanLossAvg	Port (E1)	ELEAF (Full) OLEAF (Full)	Metered	NEND	RX	n/a
		ELEAF (Limited) OLEAF (Limited)	Metered	NEND	RX	n/a
spanLossAvg	Port (E1)	TrueWave (Full) TrueWave-C (Full)	Metered	NEND	RX	n/a
		TrueWave (Limited) TrueWave-C (Limited)	Metered	NEND	RX	n/a
spanLossAvg	Port (E1)	DSF (Full)	Metered	NEND	RX	n/a
		DSF (Limited)	Metered	NEND	RX	n/a
spanLossAvg	Port (E1)	NZ-DSF (Full)	Metered	NEND	RX	n/a
		NZ-DSF (Limited)	Metered	NEND	RX	n/a
spanLossMax	Port (E1)	SMF (Full) ULL (Full)	Metered	NEND	RX	n/a
		SMF (Limited) ULL (Limited)	Metered	NEND	RX	n/a
spanLossMax	Port (E1)	ELEAF (Full) OLEAF (Full)	Metered	NEND	RX	n/a
		ELEAF (Limited) OLEAF (Limited)	Metered	NEND	RX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
spanLossMax	Port (E1)	TrueWave (Full) TrueWave-C (Full)	Metered	NEND	RX	n/a
		TrueWave (Limited) TrueWave-C (Limited)	Metered	NEND	RX	n/a
spanLossMax	Port (E1)	DSF (Full)	Metered	NEND	RX	n/a
		DSF (Limited)	Metered	NEND	RX	n/a
spanLossMax	Port (E1)	NZ-DSF (Full)	Metered	NEND	RX	n/a
		NZ-DSF (Limited)	Metered	NEND	RX	n/a
spanLossMin	Port (E1)	SMF (Full) ULL (Full)	Metered	NEND	RX	n/a
		SMF (Limited) ULL (Limited)	Metered	NEND	RX	n/a
spanLossMin	Port (E1)	ELEAF (Full) OLEAF (Full)	Metered	NEND	RX	n/a
		ELEAF (Limited) OLEAF (Limited)	Metered	NEND	RX	n/a
spanLossMin	Port (E1)	TrueWave (Full) TrueWave-C (Full)	Metered	NEND	RX	n/a
		TrueWave (Limited) TrueWave-C (Limited)	Metered	NEND	RX	n/a
spanLossMin	Port (E1)	DSF (Full)	Metered	NEND	RX	n/a
		DSF (Limited)	Metered	NEND	RX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
spanLossMin	Port (E1)	NZ-DSF (Full)	Metered	NEND	RX	n/a
		NZ-DSF (Limited)	Metered	NEND	RX	n/a
spanLossVariation	Port (E1)	SMF (Full) ULL (Full)	Metered	NEND	RX	n/a
		SMF (Limited) ULL (Limited)	Metered	NEND	RX	n/a
spanLossVariation	Port (E1)	ELEAF (Full) OLEAF (Full)	Metered	NEND	RX	n/a
		ELEAF (Limited) OLEAF (Limited)	Metered	NEND	RX	n/a
spanLossVariation	Port (E1)	TrueWave (Full) TrueWave-C (Full)	Metered	NEND	RX	n/a
		TrueWave (Limited) TrueWave-C (Limited)	Metered	NEND	RX	n/a
spanLossVariation	Port (E1)	DSF (Full)	Metered	NEND	RX	n/a
		DSF (Limited)	Metered	NEND	RX	n/a
spanLossVariation	Port (E1)	NZ-DSF (Full)	Metered	NEND	RX	n/a
		NZ-DSF (Limited)	Metered	NEND	RX	n/a
spanLossVariationAvg	Port (E1)	SMF (Full) ULL (Full)	Metered	NEND	RX	n/a
		SMF (Limited) ULL (Limited)	Metered	NEND	RX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
spanLossVariationAvg	Port (E1)	ELEAF (Full) OLEAF (Full)	Metered	NEND	RX	n/a
		ELEAF (Limited) OLEAF (Limited)	Metered	NEND	RX	n/a
spanLossVariationAvg	Port (E1)	TrueWave (Full) TrueWave-C (Full)	Metered	NEND	RX	n/a
		TrueWave (Limited) TrueWave-C (Limited)	Metered	NEND	RX	n/a
spanLossVariationAvg	Port (E1)	DSF (Full)	Metered	NEND	RX	n/a
		DSF (Limited)	Metered	NEND	RX	n/a
spanLossVariationAvg	Port (E1)	NZ-DSF (Full)	Metered	NEND	RX	n/a
		NZ-DSF (Limited)	Metered	NEND	RX	n/a
spanLossVariationMax	Port (E1)	SMF (Full) ULL (Full)	Metered	NEND	RX	n/a
		SMF (Limited) ULL (Limited)	Metered	NEND	RX	n/a
spanLossVariationMax	Port (E1)	ELEAF (Full) OLEAF (Full)	Metered	NEND	RX	n/a
		ELEAF (Limited) OLEAF (Limited)	Metered	NEND	RX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
spanLossVariationMax	Port (E1)	TrueWave (Full) TrueWave-C (Full)	Metered	NEND	RX	n/a
		TrueWave (Limited) TrueWave-C (Limited)	Metered	NEND	RX	n/a
spanLossVariationMax	Port (E1)	DSF (Full)	Metered	NEND	RX	n/a
		DSF (Limited)	Metered	NEND	RX	n/a
spanLossVariationMax	Port (E1)	NZ-DSF (Full)	Metered	NEND	RX	n/a
		NZ-DSF (Limited)	Metered	NEND	RX	n/a
spanLossVariationMin	Port (E1)	SMF (Full) ULL (Full)	Metered	NEND	RX	n/a
		SMF (Limited) ULL (Limited)	Metered	NEND	RX	n/a
spanLossVariationMin	Port (E1)	ELEAF (Full) OLEAF (Full)	Metered	NEND	RX	n/a
		ELEAF (Limited) OLEAF (Limited)	Metered	NEND	RX	n/a
spanLossVariationMin	Port (E1)	TrueWave (Full) TrueWave-C (Full)	Metered	NEND	RX	n/a
		TrueWave (Limited) TrueWave-C (Limited)	Metered	NEND	RX	n/a
spanLossVariationMin	Port (E1)	DSF (Full)	Metered	NEND	RX	n/a
		DSF (Limited)	Metered	NEND	RX	n/a

Table 56 (Cont.)
Port PM Counters (BDL1-3R11)

Counter	Entity Type	Fiber Type (Capacity)	PM Type	Location	Direction	Valid Operating Range
spanLossVariationMin	Port (E1)	NZ-DSF (Full)	Metered	NEND	RX	n/a
		NZ-DSF (Limited)	Metered	NEND	RX	n/a

Table 57
Port PM Counters (PIL1-3PS2)

Counter	Entity Type	PM Type	Location	Direction	Valid Operating Range
opticalPowerReceive	Port (E1...E4)	Metered	NEND	RX	-15.0...4.0
	Port (C1...C16)	Metered	NEND	RX	-7.5...9.1
opticalPowerReceiveAvg	Port (E1...E4)	Metered	NEND	RX	n/a
	Port (C1...C16)	Metered	NEND	RX	n/a
opticalPowerReceiveMax	Port (E1...E4)	Metered	NEND	RX	n/a
	Port (C1...C16)	Metered	NEND	RX	n/a
opticalPowerReceiveMin	Port (E1...E4)	Metered	NEND	RX	n/a
	Port (C1...C16)	Metered	NEND	RX	n/a
opticalPowerTransmit	Port (E1...E4)	Metered	NEND	TX	-9.0...12.0
	Port (C1...C16)	Metered	NEND	TX	-14.2...9.9
opticalPowerTransmitAvg	Port (E1...E4)	Metered	NEND	TX	n/a
	Port (C1...C16)	Metered	NEND	TX	n/a
opticalPowerTransmitMax	Port (E1...E4)	Metered	NEND	TX	n/a
	Port (C1...C16)	Metered	NEND	TX	n/a
opticalPowerTransmitMin	Port (E1...E4)	Metered	NEND	TX	n/a
	Port (C1...C16)	Metered	NEND	TX	n/a

Table 58
Port PM Counters (SFP Modules)

Counter	Entity Type	Time Period	PM Type	Location	Direction	Valid Operating Range
laserBiasCurrent	Port (SFP Modules)	15-Min	Metered	NEND	TX	n/a
		1-Day	Metered	NEND	TX	n/a
opticalPowerReceive	Port (SFP Modules)	15-Min	Metered	NEND	RX	n/a
		1-Day	Metered	NEND	RX	n/a
opticalPowerReceiveAvg	Port (SFP Modules)	15-Min	Metered	NEND	RX	n/a
		1-Day	Metered	NEND	RX	n/a
opticalPowerReceiveMax	Port (SFP Modules)	15-Min	Metered	NEND	RX	n/a
		1-Day	Metered	NEND	RX	n/a
opticalPowerReceiveMin	Port (SFP Modules)	15-Min	Metered	NEND	RX	n/a
		1-Day	Metered	NEND	RX	n/a
opticalPowerTransmit	Port (SFP Modules)	15-Min	Metered	NEND	TX	n/a
		1-Day	Metered	NEND	TX	n/a
opticalPowerTransmitAvg	Port (SFP Modules)	15-Min	Metered	NEND	TX	n/a
		1-Day	Metered	NEND	TX	n/a
opticalPowerTransmitMax	Port (SFP Modules)	15-Min	Metered	NEND	TX	n/a
		1-Day	Metered	NEND	TX	n/a
opticalPowerTransmitMin	Port (SFP Modules)	15-Min	Metered	NEND	TX	n/a
		1-Day	Metered	NEND	TX	n/a

8.3 Equipment Replacement

In this section:

- 8.3.1 Replace Fan Unit
- 8.3.2 Replace Plug-In Unit
- 8.3.3 Replace SFP Module
- 8.3.4 Replace Blade in Rack
- 8.3.5 Replace Tributary Blade with Tributary Auto Shelf ID Assignment Enabled
- 8.3.6 Replace Blade in Housing

8.3.1 Replace Fan Unit

Step 1

Obtain a replacement Fan unit from a supply of spares.

Note: If two units are available, bring both units to the blade with the failed unit.

Step 2

Inspect and verify the replacement Fan unit to be installed is free from defects, debris, and physical damage.

Note: If defects exist, set the Fan unit aside for repair paperwork processing and return to [Step 1](#).

Step 3

Record the serial number and issue number of the replacement Fan unit to be installed, if necessary.

Step 4

Remove the Fan unit being replaced as described in [Remove Fan Unit](#), and within 3 minutes of removing the failed unit to avoid exceeding the operating temperature range (5...40 °C), install the replacement Fan unit in the proper Fan unit slot (as described in [Install Fan Unit](#)).



Caution: Remove only one Fan unit at a time if power is applied to the blade.



Remember: Set aside the failed Fan unit for repair paperwork processing.

Step 5

Verify that the FAN LED for the replacement Fan unit is lit green.

Note: If the light is not green, reseal the replacement Fan unit. If that does not clear the problem and a second replacement unit is available, install it immediately. Record the serial number of the replacement unit. If necessary, call Fujitsu at 1-800-USE-FTAC (1-800-873-3822) for technical assistance.

Step 6

Log on the NE, if necessary.

Step 7

Retrieve all alarms and standing conditions currently being reported on the NE.

Command:

```
> show alarms alarm-list
```

Step 8

Verify that no alarms or standing conditions are currently being reported on the NE related to the replacement of the Fan unit.

Note: Clear any alarms or standing conditions being reported for the replacement unit. If unable to clear the alarms, call Fujitsu at 1-800-USE-FTAC (1-800-873-3822) for technical assistance.

✓ **This task is complete.**

Additional Instructions:

Follow local practice/procedures for repair or replacement of equipment, as applicable.

8.3.2

Replace Plug-In Unit

Prerequisites:

- Traffic has been removed from the PIU (if the unit is handling traffic).
- Replacement PIU is available.
- If the operating temperature is not in the range of 5...40 °C, do not perform this procedure until the operating temperature is in this range.
- User must be logged on as fujitsu/1finity or have Level-4 or Level-5 privileges assigned to perform this procedure.

This procedure lists the general steps required for replacing the PIU in the blade. This procedure is limited to replacing a PIU with the exact same PIU type.



Danger: This equipment contains high-power optical laser devices that transmit light at a dangerous level. Never look into the end of a fiber-optic cable because permanent eye damage or blindness could occur if the laser light is present. Follow local safety precautions regarding fiber-optic cables.



Danger: Never handle exposed fiber with bare hands or touch it to bare skin. Fiber fragments can enter the skin and are difficult to detect and remove. Follow local safety precautions regarding fiber.



Warning: Replace only one plug-in at a time, if power is applied to blade, to maintain proper air flow and prevent damage due to potential overheating. The first PIU must be replaced before proceeding to replace the next PIU.



Attention: Do not replace the PIU if replacing a contained SFP, SFP+, or CFP4 module can clear the alarm.

Step 1

Retrieve alarms.

Command:

```
> show alarms alarm-list
```

Step 2

Are any alarms and standing conditions currently being reported?

If YES:

Clear all active alarms and standing conditions, as necessary, before continuing this procedure. After these alarms and standing conditions are cleared, continue with the next step.

If NO:

Continue with next step.

Step 3

Is it possible that replacing a single pluggable optics module clears the current alarms?

If YES:

Do not continue this procedure. Replace the suspected pluggable optics module. Continue with next step only if replacing the modules does not clear the problem.

If NO:

Continue with next step.

Step 4

Inspect the fiber connections to the PIU. Verify that each fiber is clearly marked or labeled to identify the correct connection location.

Step 5

Is each fiber clearly marked to show its corresponding connector?

If YES:

Continue with next step.

If NO:

Mark or label fibers according to local practice and continue with the next step.

Step 6

Ensure that an appropriate ESD wrist strap is worn and is properly grounded onto the ESD wrist strap grounding post on the chassis.

Step 7

Disconnect the fiber-optic cables from the PIU to be removed.



Hint: Carefully and slowly press the LC connector tab on the fiber-optic cable of an SFP, SFP+, or CFP4 module to release and remove the fiber connectors from the modules.

Step 8

Obtain a replacement PIU.

Step 9

Inspect the replacement PIU to be installed for possible physical damage or debris.

Step 10

Does the replacement PIU appear to be free from defects?

If YES:

Continue with next step.

If NO:

Set the PIU aside for repair paperwork processing and obtain another replacement PIU.

Step 11

Record the serial number and issue number of the replacement PIU to be installed, if necessary.

Step 12

Prepare slot for PIU removal.

Command:

```
> request prepare-removal shelf-id 1 slot-id x
```

Note: In this example, x = 1...4.

Step Result:

The LED turns red for any port with an installed pluggable module, but traffic remains UP.

Step 13

Carefully release the latch securing the SFP, SFP+, or CFP4 module to the PIU.

Step 14

Carefully grasp the latch, and slowly remove the SFP, SFP+, or CFP4 module from the PIU being replaced and place them on an ESD-protective surface.

Step 15

Install temporary optical connector covers into the optical connectors on the modules.

Step 16

Carefully install the removed modules into the same ports in the replacement PIU.

Step 17

After all modules have been transferred to the replacement unit, remove the PIU being replaced from the blade, and set it aside for repair paperwork processing.

Step 18

Carefully align and install the replacement PIU into the appropriate slot.

Note: The PIU should be slowly and carefully inserted into the blade to prevent damage to the PIU and/or blade. When the connector pins of the PIU engage the backplane, the PIU must be fully seated within 10 seconds. If it takes longer than 10 seconds to install the PIU, an equipment fault alarm may be raised.

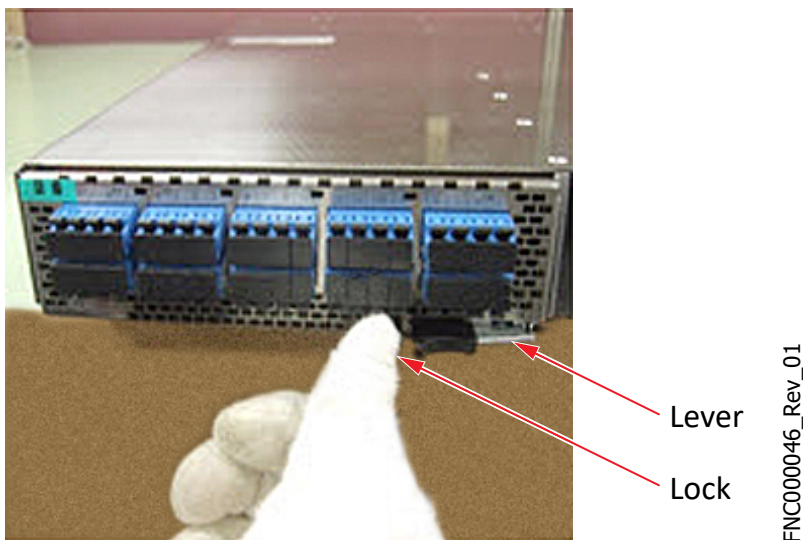


Figure 112
Inserting PIU

Step 19

Clean the optical connectors on the optical fibers.



Caution: Optical connector performance is affected by dust. Clean all connectors before reconnecting the cables. Keep the ends of fiber-optic cables covered when the cables are not in use.

Step 20

Remove the optical connector covers, if necessary, and carefully press the LC connector tab on each of the I/O fiber-optic cables to carefully install each cable into its correct connector on the PIU.

Step 21

Are the applicable port LEDs on the PIU lit green?

If YES:

Continue with next step.

If NO:

Call Fujitsu at 1-800-USE-FTAC (1-800-873-3822) for technical assistance.

Step 22

If not already done, log on the NE.

Step 23

Retrieve alarms.

Command:

```
> show alarms alarm-list
```

Step Result:

Clear any alarms and standing conditions currently being reported on the blade related to the replacement of the PIU. If unable to clear the alarms, call Fujitsu at 1-800-USE-FTAC (1-800-873-3822) for technical assistance.

✓ **This task is complete.**

8.3.3

Replace SFP Module

Prerequisites:

- Replacement SFP module is available.
- User must be logged on as fujitsu/1finity or have Level-4 or Level-5 privileges assigned to perform this procedure.

This procedure lists the steps required for replacing SFP modules installed in PIUs. This procedure is limited to replacing a module with the exact same module type.



Danger: This equipment contains high-power optical laser devices that transmit light at a dangerous level. Never look into the end of a fiber-optic cable because permanent eye damage or blindness could occur if the laser light is present. Follow local safety precautions regarding fiber-optic cables.



Danger: Never handle exposed fiber with bare hands or touch it to bare skin. Fiber fragments can enter the skin and are difficult to detect and remove. Follow local safety precautions regarding fiber.



Hint: Install modules of the same type above and below each other in the ports of the PIUs (for example, install a 1000Base-LX SFP module in port 5 and port 6). Using different types of modules

above and below each other can cause the modules to snag during removal because of different extraction lever designs and clearances.



Caution: Do not remove the wrong fiber connections. Removing a fiber connection that is in use disrupts service.

Step 1

Retrieve alarms.

Command:

```
> show alarms alarm-list
```

Step 2

Clear any alarms and standing conditions currently being reported on the NE. After these alarms and standing conditions are cleared, continue with next step.

Step 3

Obtain and inspect the replacement module for possible damage or debris. If necessary, record the serial and issue numbers.

Step 4

Inspect the fiber connections to the module installed in the PIU. Verify that each fiber is clearly marked or labeled to identify the correct connection location.

Step 5

Is each fiber clearly marked to show its corresponding connector?

If YES:

Continue with next step.

If NO:

Mark or label fibers according to local practice and continue with the next step.

Step 6

Carefully and slowly press the LC connector tab on the fiber-optic cable of a module to release and remove the fiber connectors from the module.



Caution: Do not remove the wrong fiber connections. Removing a fiber connection that is in use disrupts service.

Step 7

Carefully grasp the latch and slowly remove the module from the PIU.

Step 8

Install temporary optical connector covers into the optical connectors on the module.

Step 9

Carefully install the replacement module into the same port of the PIU.

Step 10

Clean the optical connectors on the optical fibers.



Caution: Optical connector performance is affected by dust. Clean all connectors before reconnecting the cables. Keep the ends of fiber-optic cables covered when the cables are not in use.

Step 11

Remove the optical connector covers, if necessary, and carefully press the LC connector tab on each of the I/O fiber-optic cables to carefully install each cable into the connector.

Step 12

Are the applicable port LEDs on the PIU lit green?

If YES:

Continue with next step.

If NO:

Call Fujitsu at 1-800-USE-FTAC (1-800-873-3822) for technical assistance.

Step 13

Retrieve alarms.

Command:

```
> show alarms alarm-list
```

Step Result:

Clear any alarms and standing conditions currently being reported on the NE related to the replacement of the module. If unable to clear the alarms, call Fujitsu at 1-800-USE-FTAC (1-800-873-3822) for technical assistance.

✓ **This task is complete.**

8.3.4

Replace Blade in Rack

Prerequisite:

- Power cables removed from the power terminal adapter (AC00-SD11) on the rear of the blade
- An appropriate ESD wrist strap worn by user and properly grounded to avoid ESD damage during installation or removal
- All fibers removed from the blade and appropriately labeled
- If used, all RJ-45 cables removed from the blade and appropriately labeled

- If used, all PIUs removed from the blade
- A recent database backup for the blade is available for Main blade replacement

Note: If no database backup is available, the blade needs to be completely reprovisioned. No database backup is required for a tributary blade replacement.

- Screwdriver set (including straight-slot and Phillips head)



Warning: After the blade is powered up, power down within 15 minutes causes database corruption, which leads to blade start-up failure and requires a return to factory for reprogramming.



Warning: The blade is heavy. When handling the blade, position two hands in the center at all times. Moving or lifting the blade with one hand or without assistance (if required) may result in personal injury, damage to the equipment, or both.

Step 1

If the Main blade is being replaced and the blade has the USB flash drive (provides secondary storage of the OSL and the database) installed, remove the USB flash drive and set aside.

Step 2

Verify the replacement blade is the same blade type as the blade being replaced.

Step 3

Verify the software load on the replacement blade is the same as the blade being replaced.

Step 4

Unlatch the fiber restraint gates on each side of the blade.

Note: Slide the fiber restraint gate slightly inward to open the gate.

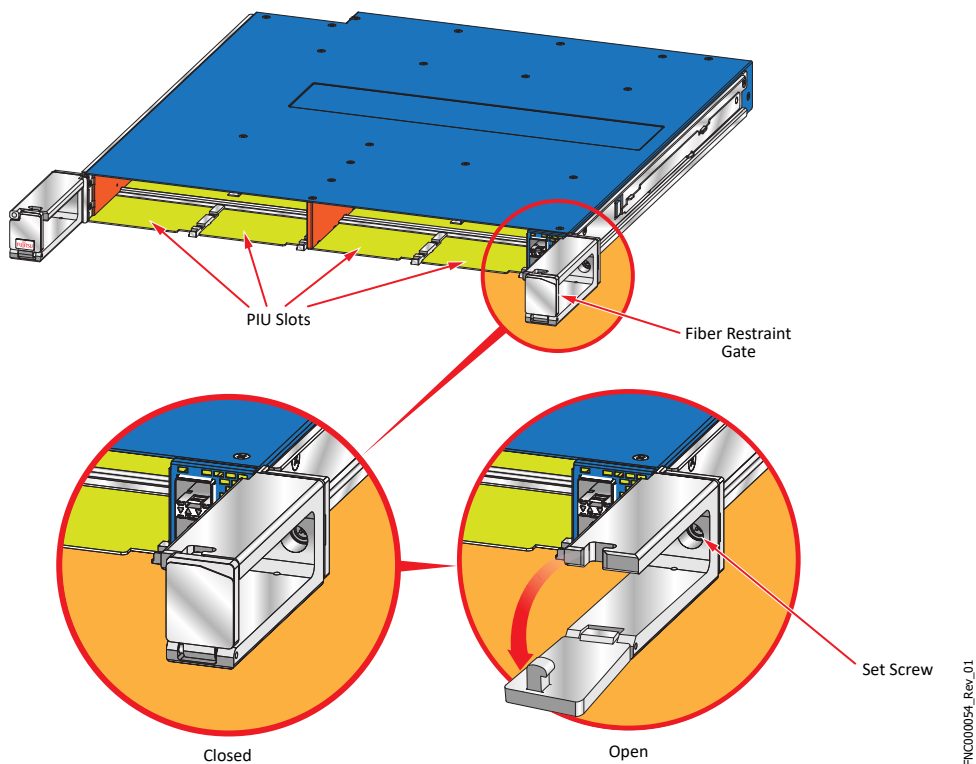


Figure 113
Unlatching Fiber Restraint Gate

Step 5

After opening the fiber restraint gates, carefully loosen the set screws on the left-hand and right-hand sides of the blade, using a Phillips head screwdriver, to detach from the rack.

Note: The set screws are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 6

Close both fiber restraint gates after loosening the set screws and disengaging the blade from the rack.

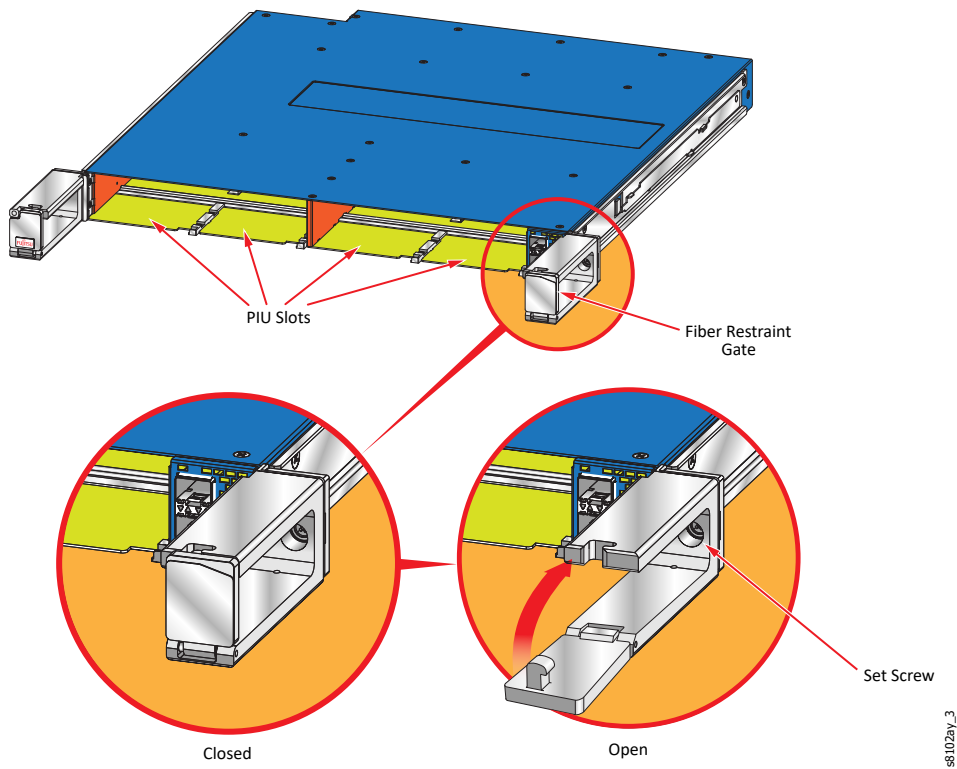


Figure 114
Latching Fiber Restraint Gate

Step 7

At the front of the blade, use the fiber management arms to slowly pull the blade out of the rack to the locked position.

At the locked position, the locks on both sides of the blade stop the blade at approximately halfway out.



Attention: Do not touch the optical fibers in adjoining slots. Make sure fibers from other blades do not obstruct the blade being removed.

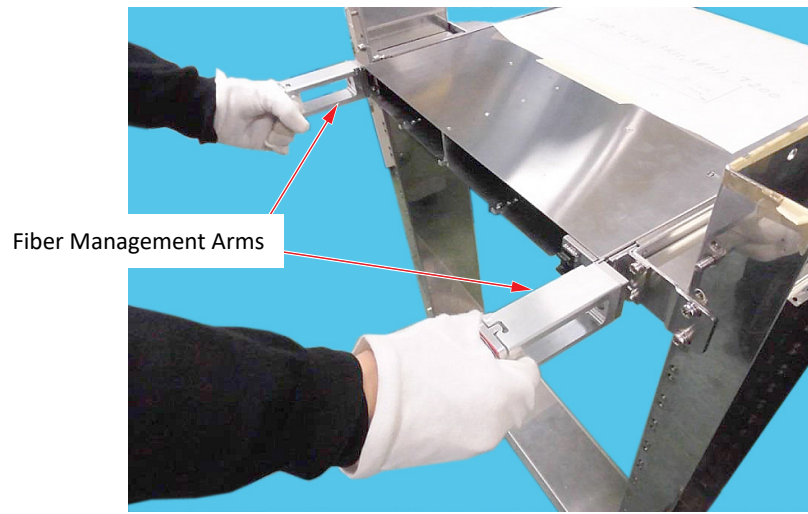


Figure 115
Pulling Blade from Rack

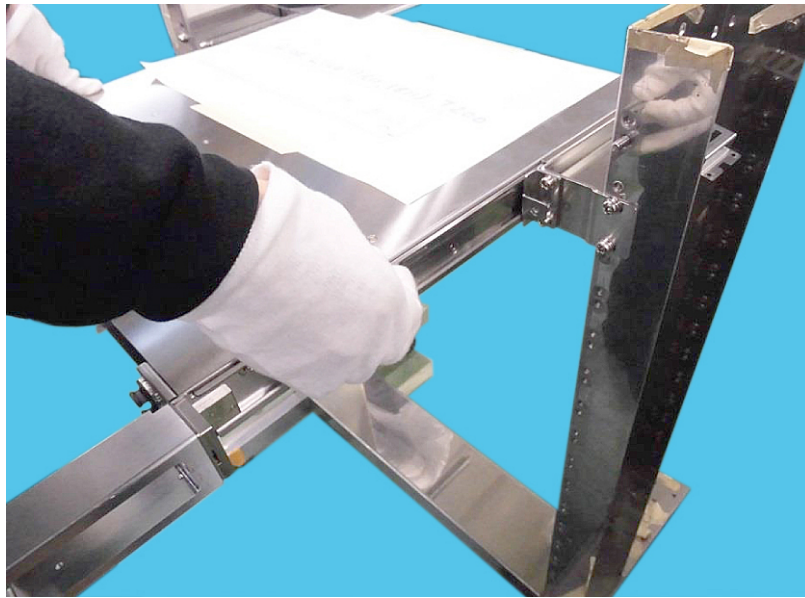


Figure 116
Locked Position



Warning: The blade is heavy; always use two hands and be prepared to support the entire weight of the blade after releasing the blade from the locked position.

Step 8

Find the locking tabs on each side of the blade.

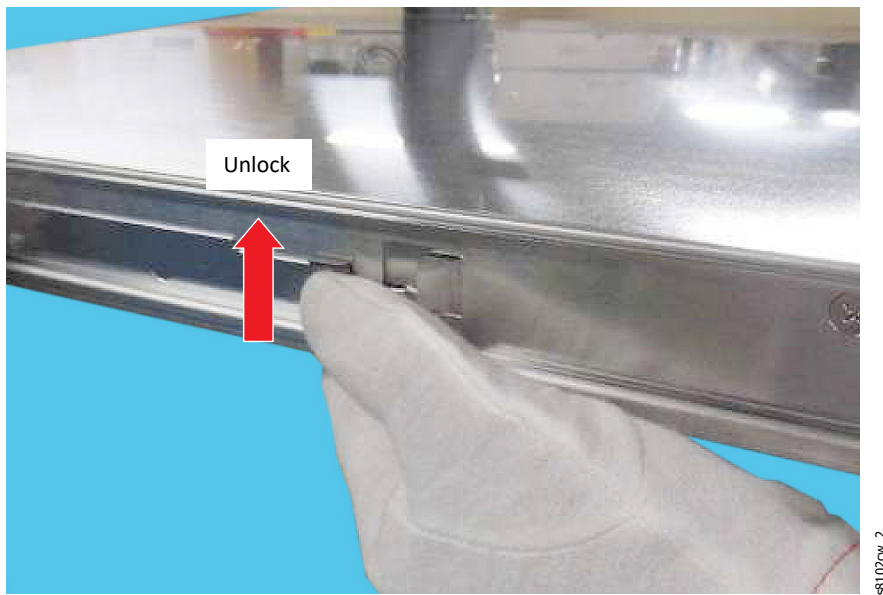


Figure 117
Unlocking Blade from Rack

Step 9

With the blade at the locked position, place one hand on each side of the blade.

Step 10

Press both locking tabs UP and slowly pull the blade to release. Continue pulling blade until fully removed from the rack.

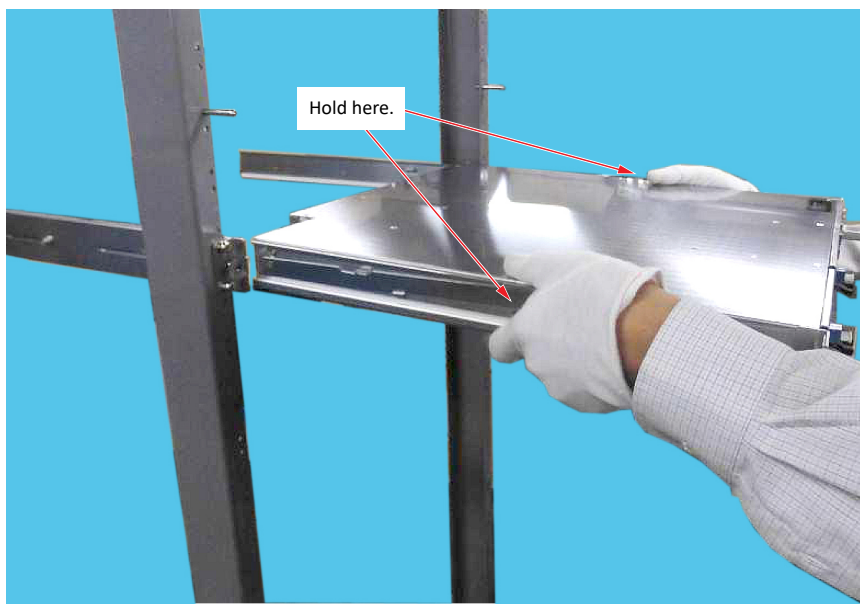


Figure 118
Removing Blade from Rack

Step 11

Store the blade in ESD-protective packaging.

Note: Follow local practice/procedures for repair or replacement of equipment, as applicable.

Step 12

Remove the protection cover from the power terminal adapter.

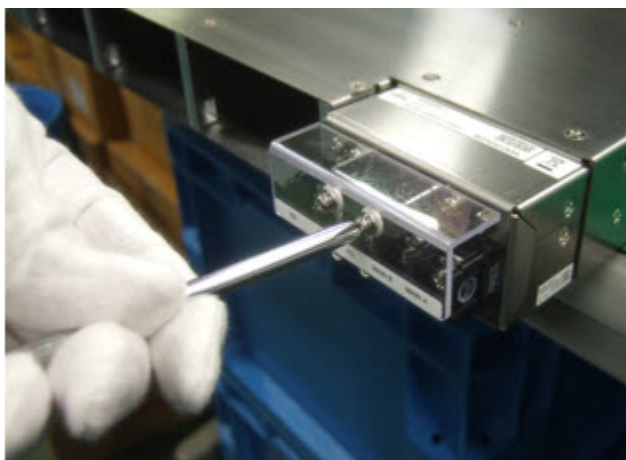


Figure 119
Protection Cover

Step 13

On a flat ESD-safe surface carefully loosen the attachment screws. Loosen the screws until they are fully disengaged from the blade.



Figure 120
Attachment Screws

Step 14

Confirm the screws have been fully disengaged from the blade.



Caution: The blade and power terminal adapter may be damaged if the screws are not fully disengaged from the blade.

Step 15

Slightly tilt the power terminal adapter forward away from the blade.



Figure 121
Tilting Power Terminal Adapter

Step 16

Pull the power terminal adapter up and away from the blade.

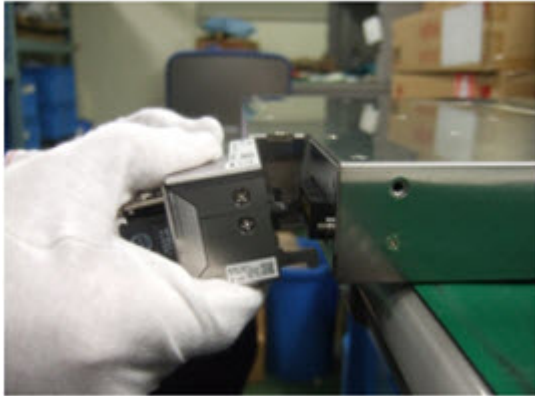


Figure 122

Pulling Power Terminal Adapter

Step 17

Before unpacking the replacement blade, inspect the shipping containers for signs of damage.

Note: Claims for shipping damage must be filed with the transportation company in accordance with their published procedure. Contact Fujitsu Sales Representative for any shipment questions.

Step 18

Unpack and carefully remove the replacement blade from its shipping container and place it on a clean, level, and ESD-safe surface. Use two hands as shown in the following figure to remove a blade unit from container.

Note: Leave the blade in the ESD-protective plastic bag until ready for installation.

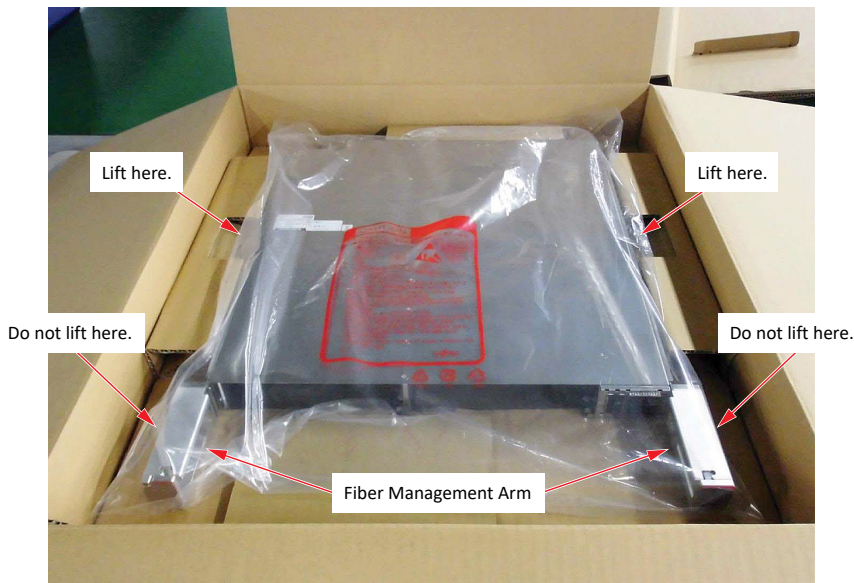


Figure 123
Removing Blade from Packing Container

Step 19

Inspect the equipment to ensure that it is free of any defects, damage, or other debris. If the equipment is damaged, follow local practices/procedures for repair or replacement of the damaged unit, as applicable.

Note: Use the checklist accompanying each shipment (if applicable) to verify that all items are present with the shipment.

Step 20

Record the serial number and issue number of the equipment in accordance with local record-keeping procedures, if necessary.

Step 21

Insert the hooks on the power terminal adapter into the holes on the rear of the blade.



Figure 124
Hooking Power Terminal Adapter

Step 22

Push the power terminal adapter until it meets the rear of the blade.



Figure 125
Pushing Power Terminal Adapter

Step 23

Press down slightly on the power terminal adapter to properly seat the adapter on the blade.

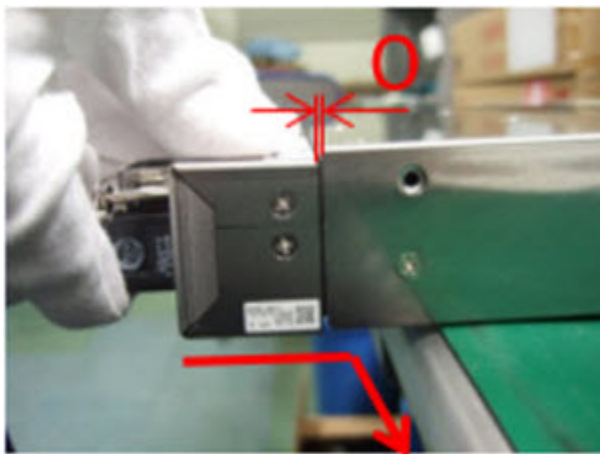


Figure 126
Seating Adapter

Step 24

Review the position of the power terminal adapter on the blade, verifying that no gap exists between the power terminal adapter and the blade and that the power terminal adapter is slightly below the top of the blade.



Figure 127
Correcting Power Terminal Adapter Placement

Step 25

On a flat ESD-safe surface, alternately tighten the attachment screws to secure the power terminal adapter onto the blade, careful not to overtighten them or tighten one screw completely before the other screw.



Caution: When tightening the two screws, alternate between them; do not overtighten either screw or tighten one completely before tightening the other.



Warning: The attachment screws mentioned in this step must be lined up straight in the holes and should easily turn, right up to the point of being tight. If any resistance occurs in the first turn, stop immediately, or risk damage to the blade (broken screws and screw holes on the blade).



Figure 128
Attachment Screws

Step 26

Replace the protection cover on the power terminal adapter.

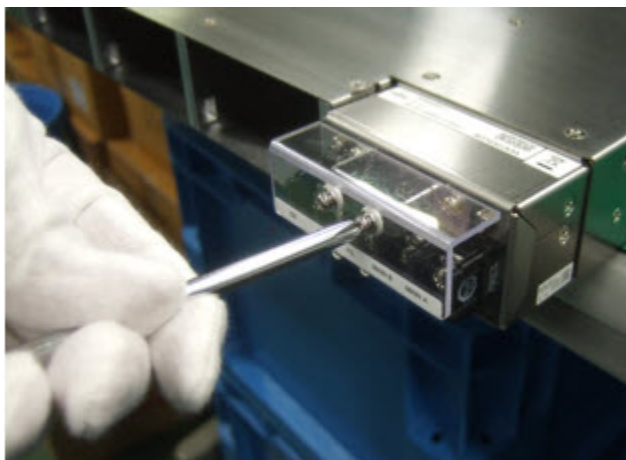


Figure 129
Attaching Protection Cover

Step 27

At the front of the rack lift and align the blade (inner rails) with the bracket tracks (outer rail) installed on the rack.

Note: Fujitsu recommends installing the blade without the PIUs installed.

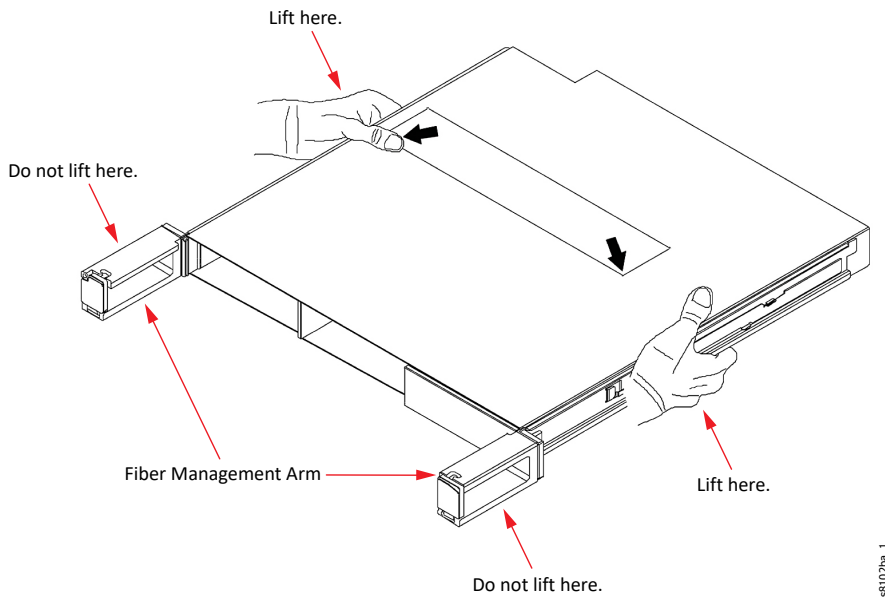


Figure 130
Proper Method to Lift Blade

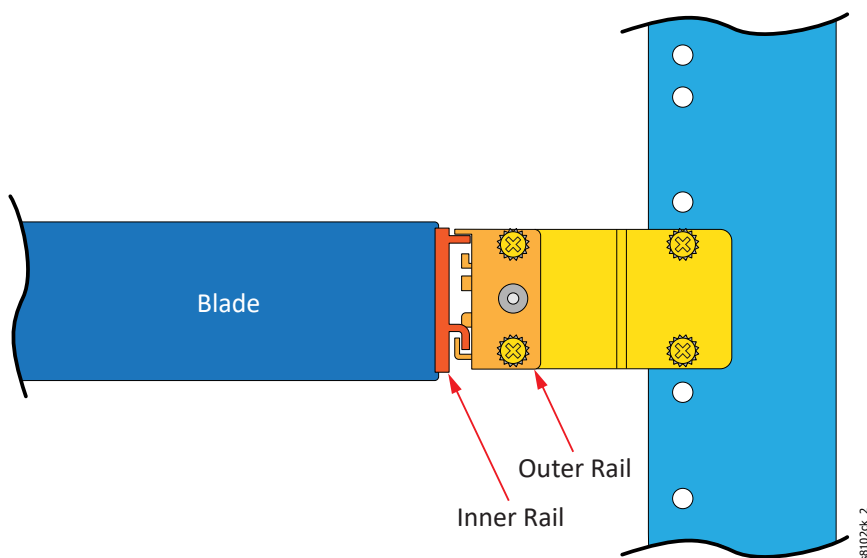


Figure 131
Blade and Bracket Alignment

Step 28

Insert the blade in the bracket track and slide it back until it locks in place.

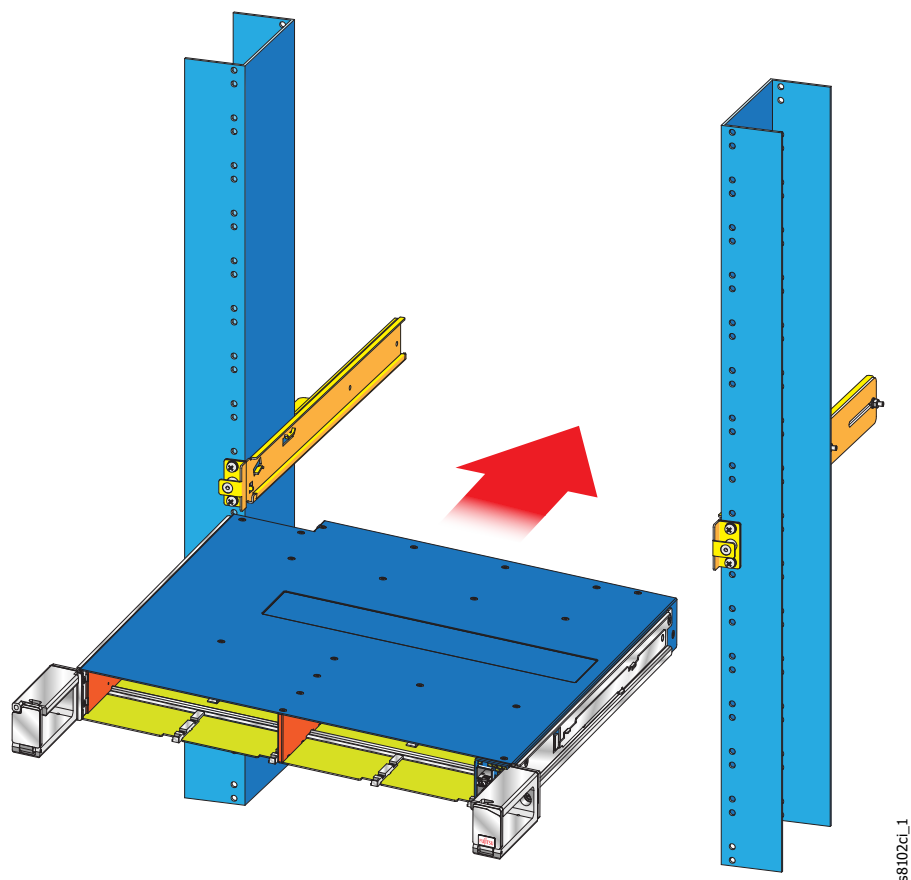


Figure 132
Installing Blade in 2-Post Rack

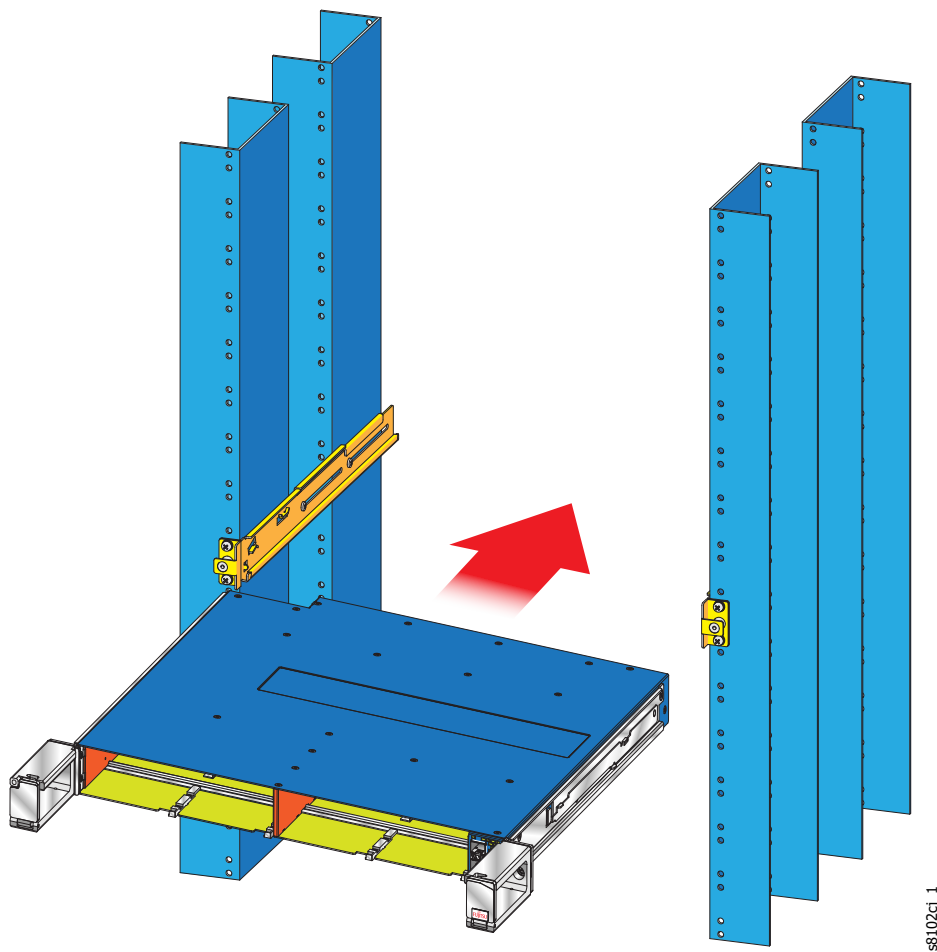


Figure 133
Installing Blade in 4-Post Rack

Step 29

Unlatch the fiber restraint gates on each side of the blade.

Note: Slide the fiber restraint gate slightly inward to open the gate.

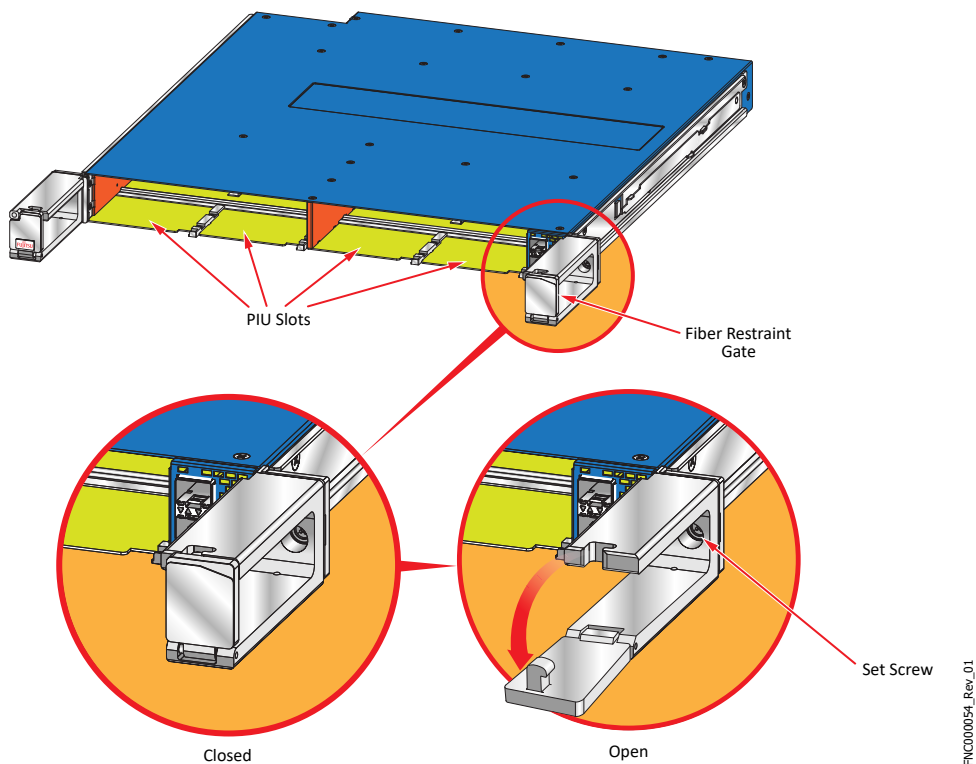


Figure 134
Unlatching Fiber Restraint Gate

Step 30

After opening the fiber restraint gates, secure the blade to the rack using the set screws on the left- and right-hand sides of the blade.

Note: The set screws for securing the blade are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 31

Replace the fiber-optic and RJ-45 cables that are previously removed.

Step 32

Verify that power is turned off at the source and install the power cables.

Step 33

Is the blade being installed a Main blade?

If YES:

Continue with next step.

If NO:

Proceed to [Step 39](#).

Step 34

Did the Main blade had a USB flash drive that was removed in [Step 1](#)?

If YES:

Continue with next step.

If NO:

Proceed to [Step 38](#).

Step 35

Insert USB flash drive in the replacement Main blade.

Step 36

Turn the power on at the source and wait for the blade to initialize.



Warning: After the blade is powered up, power down within 15 minutes causes database corruption, which leads to blade start-up failure and requires a return to factory for reprogramming.

Step 37

Verify that previous OSL and provisioning are restored to Main blade.

Step Result:

Procedure is complete.

Step 38

Is a database backup available for the blade being replaced?

If YES:

Proceed to [Restore Configuration Database](#) and continue with next step.

If NO:

Proceed to [Equipment Installation](#) and continue with next step.

Step 39

Turn the power on at the source and wait for the blade to initialize.

Step 40

Provision the tributary blade equipment.

Command:

```
> request set-shelfmode shelfid <x> shelfrole TRIB
```

Step Result:

Procedure is complete.



This task is complete.

8.3.5

Replace Tributary Blade with Tributary Auto Shelf ID Assignment Enabled

Prerequisites:

- Power cables removed from the power terminal adapter (AC00-SD11) on the rear of the blade
- An appropriate ESD wrist strap worn by user and properly grounded to avoid ESD damage during installation or removal
- All fibers removed from the blade and appropriately labeled
- If used, all RJ-45 cables removed from the blade and appropriately labeled
- If used, all PIUs removed from the blade
- Screwdriver set (including straight-slot and Phillips head)



Warning: The blade is heavy. When handling the blade, position two hands in the center at all times. Moving or lifting the blade with one hand or without assistance (if required) may result in personal injury, damage to the equipment, or both.

Step 1

Verify whether tributary auto shelf ID assignment is enabled.

Command:

```
% show system trib-shelfid-assignment
```

Step 2

Disconnect the tributary blade from Main blade.

Step 3

Clear any USI and shelf ID assigned to the blade and the DHCP configuration.

Command:

```
% request flush-shelf-prov-mode-blades
```

Step 4

Verify the replacement blade is the same blade type as the blade being replaced.

Step 5

Verify the software load on the replacement blade is the same as the blade being replaced.

Step 6

Unlatch the fiber restraint gates on each side of the blade.

Note: Slide the fiber restraint gate slightly inward to open the gate.

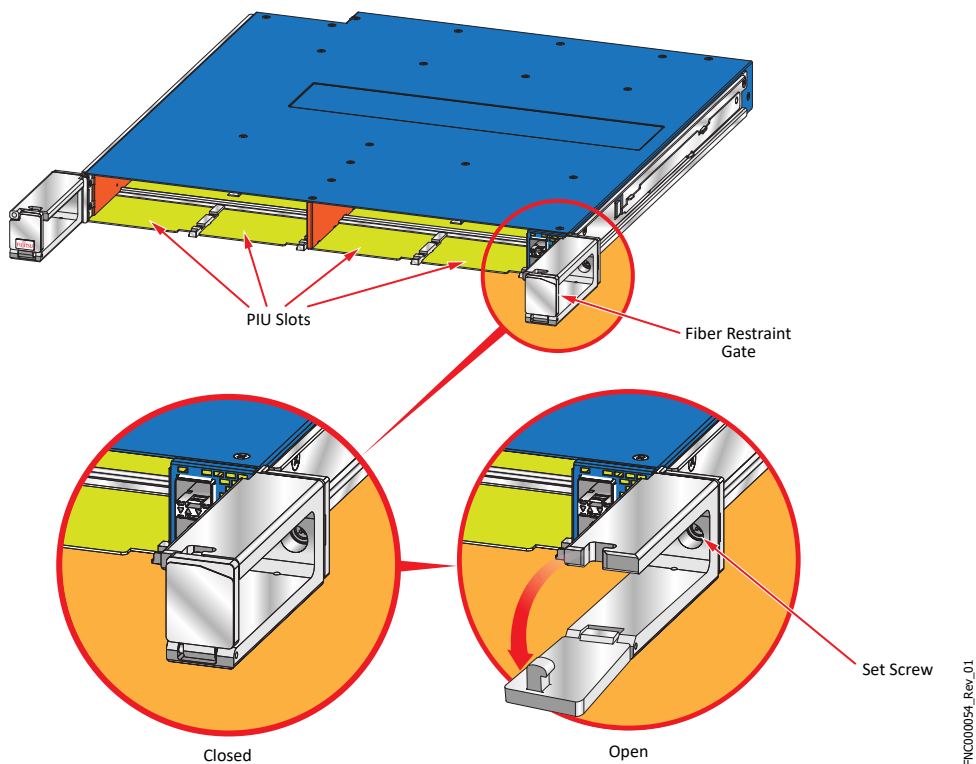


Figure 135
Unlatching Fiber Restraint Gate

Step 7

After opening the fiber restraint gates, carefully loosen the set screws on the left-hand and right-hand sides of the blade, using a Phillips head screwdriver, to detach from the rack.

Note: The set screws are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 8

Close both fiber restraint gates after loosening the set screws and disengaging the blade from the rack.

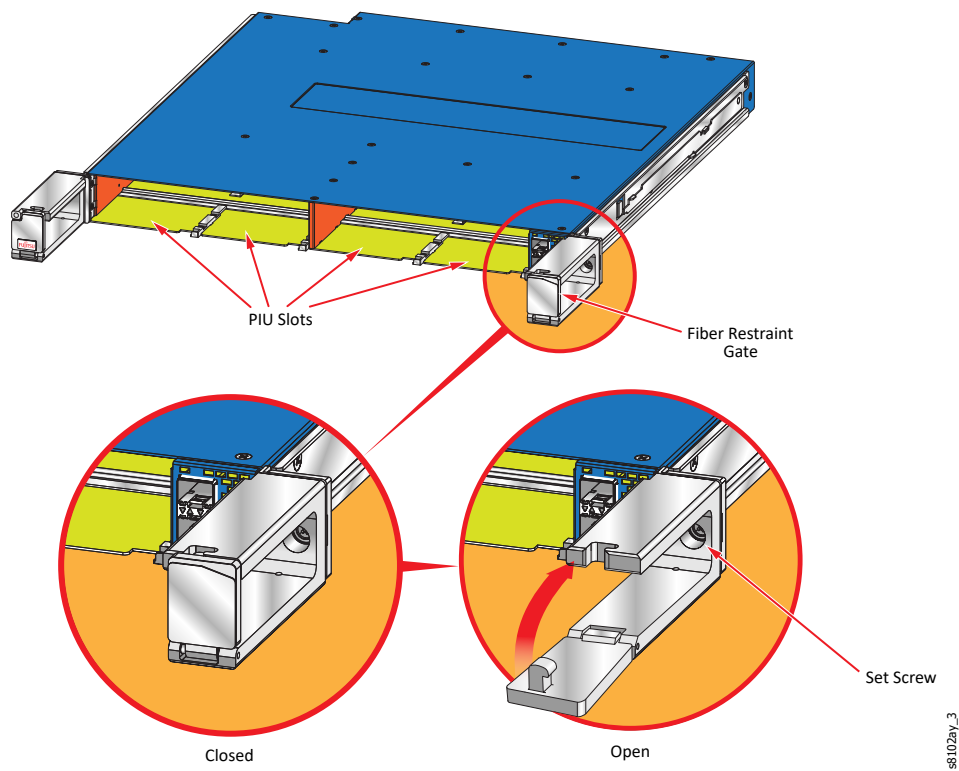


Figure 136
Latching Fiber Restraint Gate

Step 9

At the front of the blade, use the fiber management arms to slowly pull the blade out of the rack to the locked position.

At the locked position, the locks on both sides of the blade stop the blade at approximately halfway out.



Attention: Do not touch the optical fibers in adjoining slots. Make sure fibers from other blades do not obstruct the blade being removed.

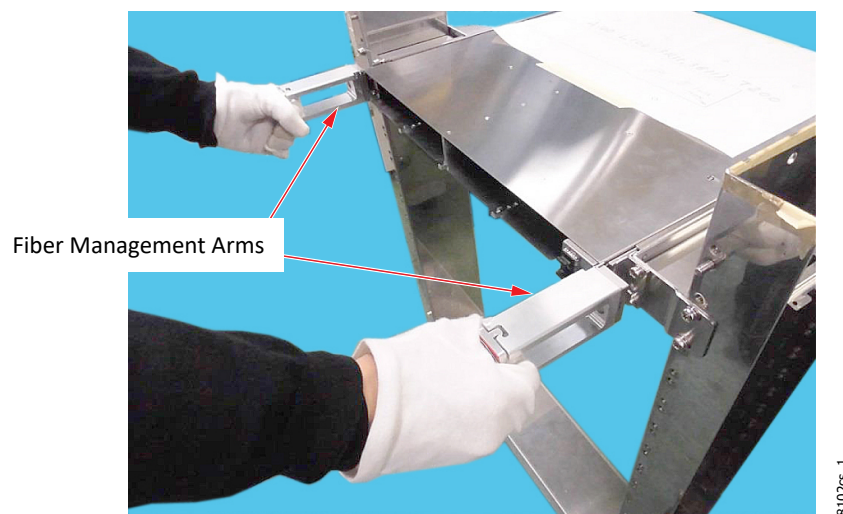


Figure 137
Pulling Blade from Rack

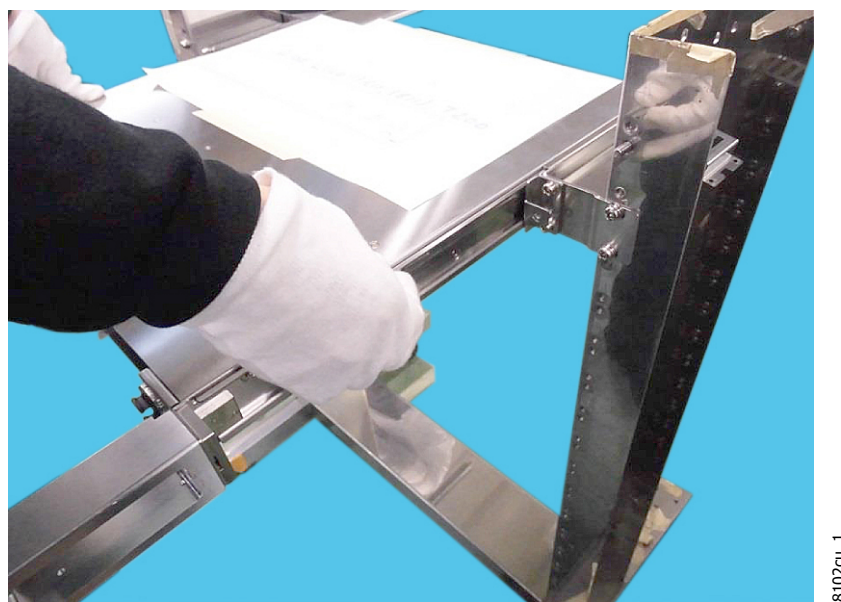


Figure 138
Locked Position



Warning: The blade is heavy; always use two hands and be prepared to support the entire weight of the blade after releasing the blade from the locked position.

Step 10

Find the locking tabs on each side of the blade.

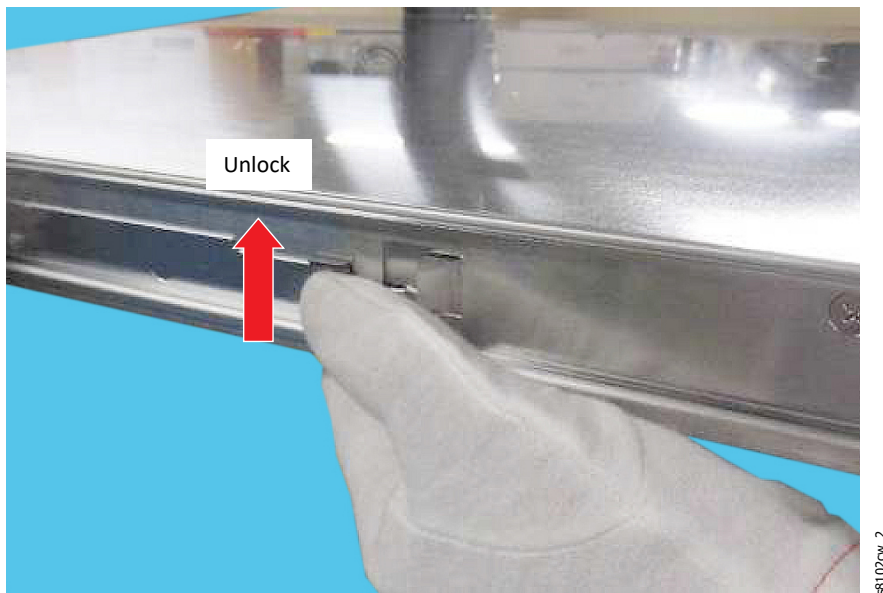


Figure 139
Unlocking Blade from Rack

Step 11

With the blade at the locked position, place one hand on each side of the blade.

Step 12

Press both locking tabs UP and slowly pull the blade to release. Continue pulling blade until fully removed from the rack.

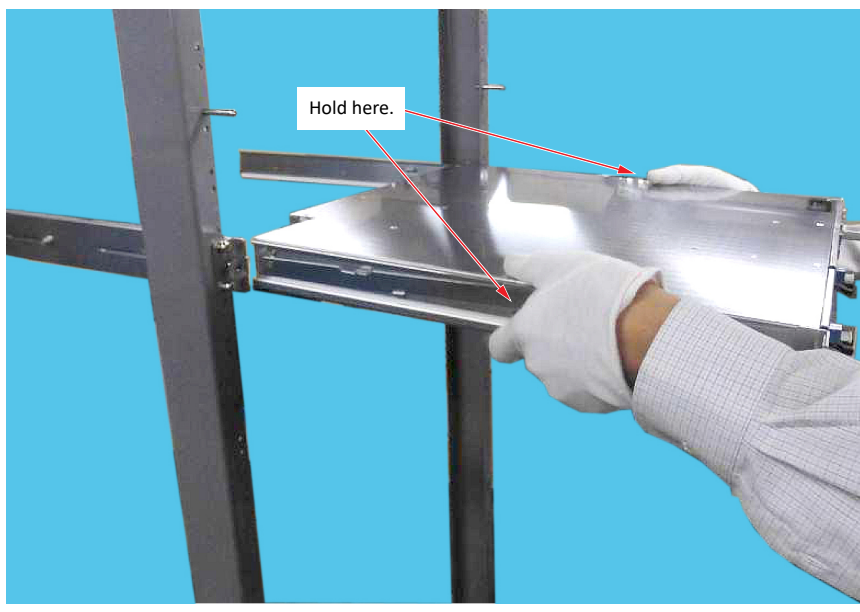


Figure 140
Removing Blade from Rack

Step 13

Store the blade in ESD-protective packaging.

Note: Follow local practice/procedures for repair or replacement of equipment, as applicable.

Step 14

Remove the protection cover from the power terminal adapter.

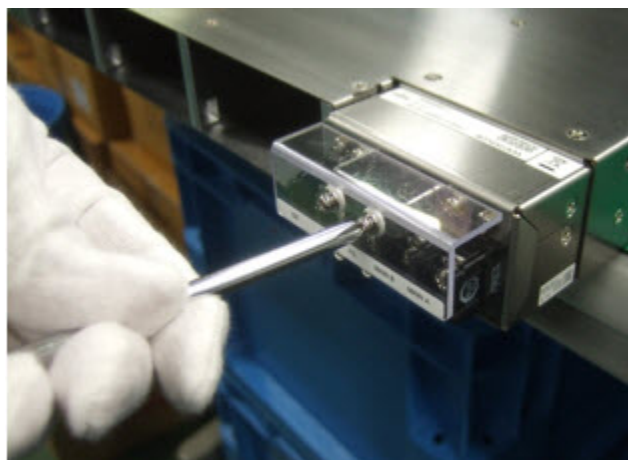


Figure 141
Protection Cover

Step 15

On a flat ESD-safe surface carefully loosen the attachment screws. Loosen the screws until they are fully disengaged from the blade.



Figure 142
Attachment Screws

Step 16

Confirm the screws have been fully disengaged from the blade.



Caution: The blade and power terminal adapter may be damaged if the screws are not fully disengaged from the blade.

Step 17

Slightly tilt the power terminal adapter forward away from the blade.



Figure 143
Tilting Power Terminal Adapter

Step 18

Pull the power terminal adapter up and away from the blade.

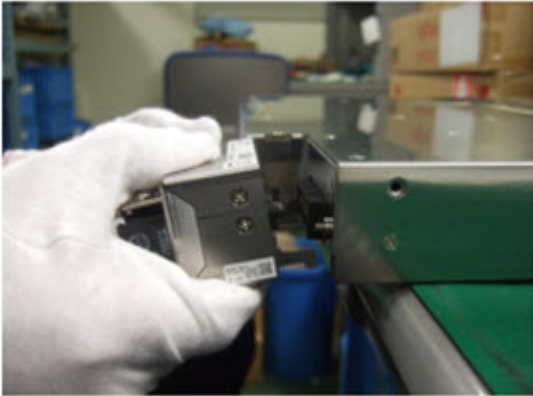


Figure 144

Pulling Power Terminal Adapter

Step 19

Before unpacking the replacement blade, inspect the shipping containers for signs of damage.

Note: Claims for shipping damage must be filed with the transportation company in accordance with their published procedure. Contact Fujitsu Sales Representative for any shipment questions.

Step 20

Unpack and carefully remove the replacement blade from its shipping container and place it on a clean, level, and ESD-safe surface. Use two hands as shown in the following figure to remove a blade unit from container.

Note: Leave the blade in the ESD-protective plastic bag until ready for installation.

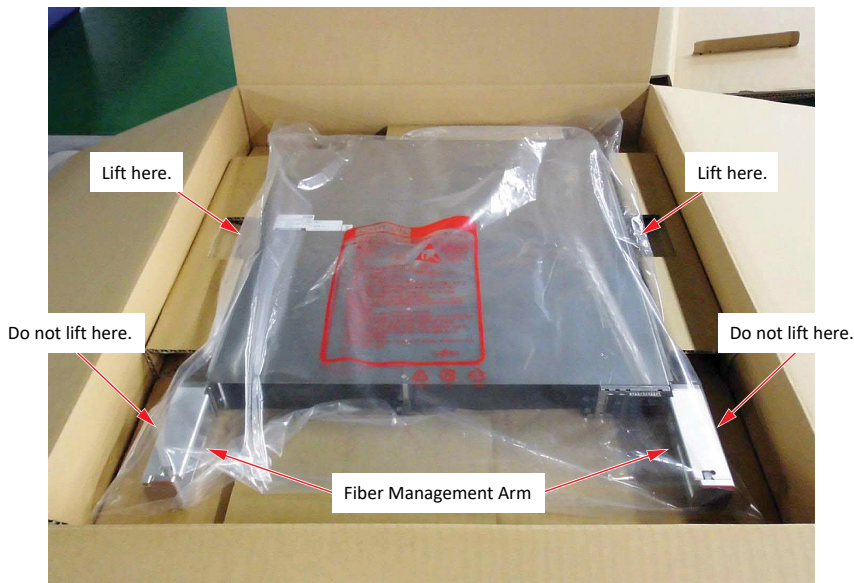


Figure 145
Removing Blade from Packing Container

Step 21

Inspect the equipment to ensure that it is free of any defects, damage, or other debris. If the equipment is damaged, follow local practices/procedures for repair or replacement of the damaged unit, as applicable.

Note: Use the checklist accompanying each shipment (if applicable) to verify that all items are present with the shipment.

Step 22

Record the serial number and issue number of the equipment in accordance with local record-keeping procedures, if necessary.

Step 23

Insert the hooks on the power terminal adapter into the holes on the rear of the blade.



Figure 146
Hooking Power Terminal Adapter

Step 24

Push the power terminal adapter until it meets the rear of the blade.



Figure 147
Pushing Power Terminal Adapter

Step 25

Press down slightly on the power terminal adapter to properly seat the adapter on the blade.

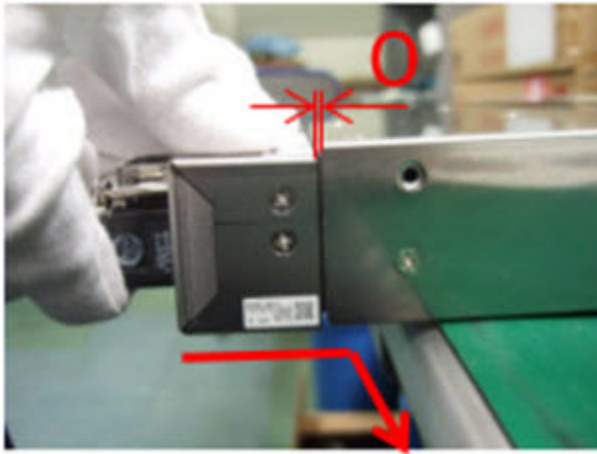


Figure 148
Seating Adapter

Step 26

Review the position of the power terminal adapter on the blade, verifying that no gap exists between the power terminal adapter and the blade and that the power terminal adapter is slightly below the top of the blade.



Figure 149
Correcting Power Terminal Adapter Placement

Step 27

On a flat ESD-safe surface, alternately tighten the attachment screws to secure the power terminal adapter onto the blade, careful not to overtighten them or tighten one screw completely before the other screw.



Caution: When tightening the two screws, alternate between them; do not overtighten either screw or tighten one completely before tightening the other.



Warning: The attachment screws mentioned in this step must be lined up straight in the holes and should easily turn, right up to the point of being tight. If any resistance occurs in the first turn, stop immediately, or risk damage to the blade (broken screws and screw holes on the blade).



Figure 150
Attachment Screws

Step 28

Replace the protection cover on the power terminal adapter.

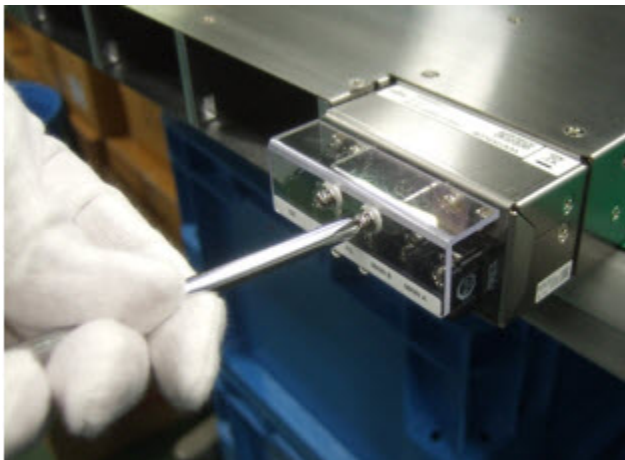


Figure 151
Attaching Protection Cover

Step 29

At the front of the rack lift and align the blade (inner rails) with the bracket tracks (outer rail) installed on the rack.

Note: Fujitsu recommends installing the blade without the PIUs installed.

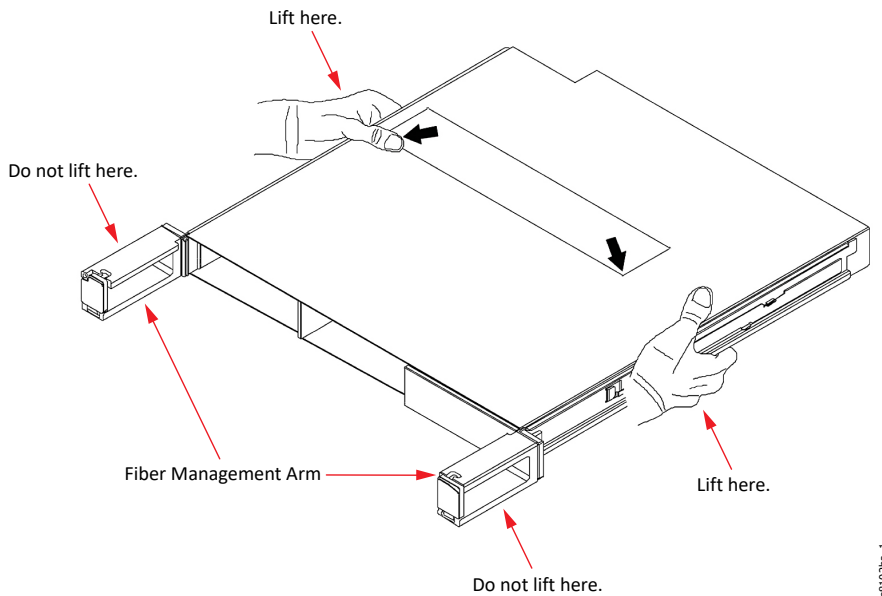


Figure 152
Proper Method to Lift Blade

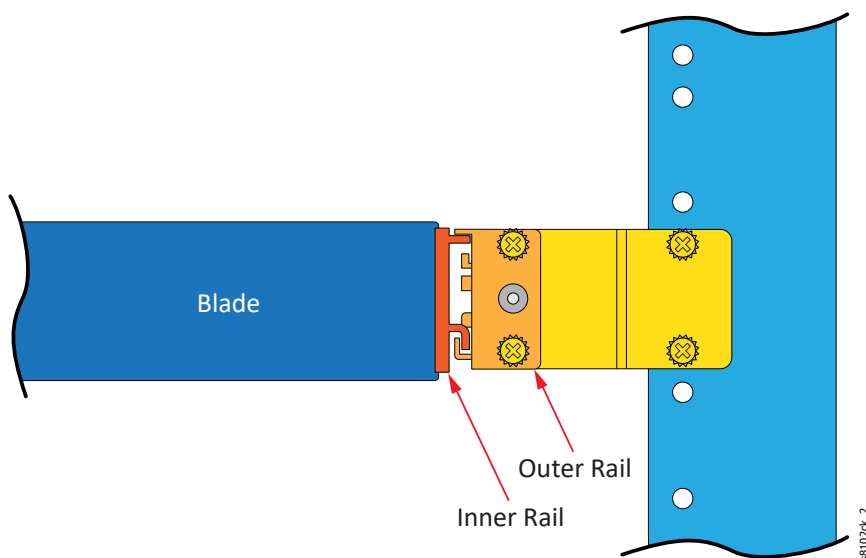


Figure 153
Blade and Bracket Alignment

Step 30

Insert the blade in the bracket track and slide it back until it locks in place.

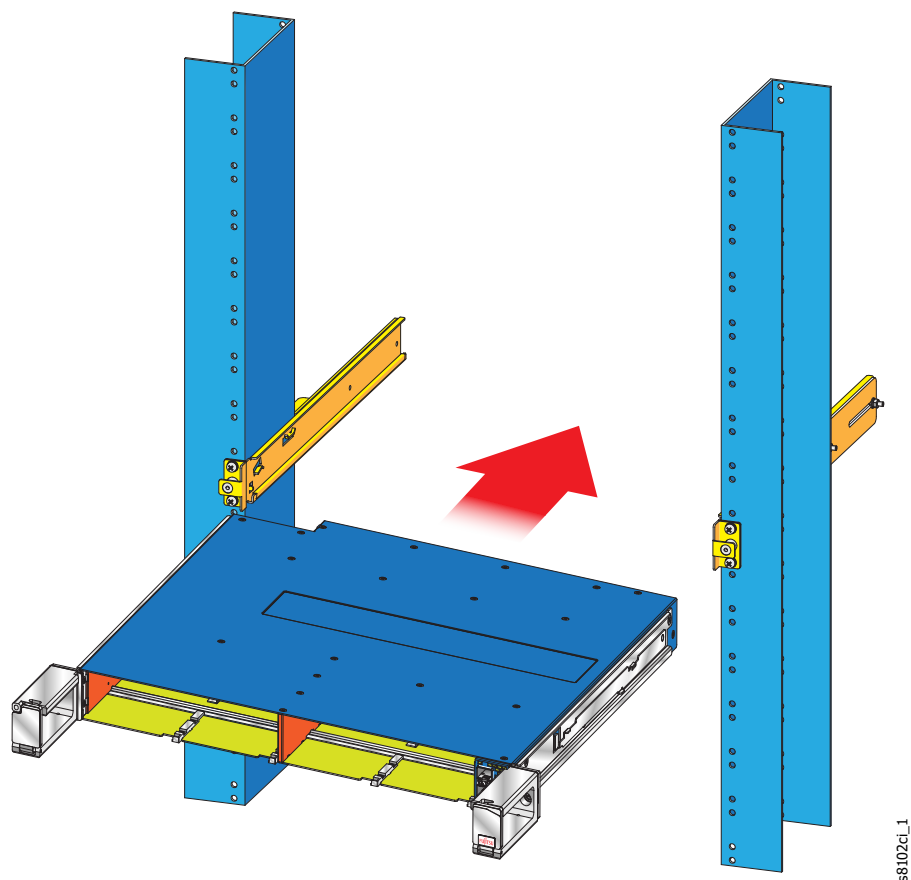


Figure 154
Installing Blade in 2-Post Rack

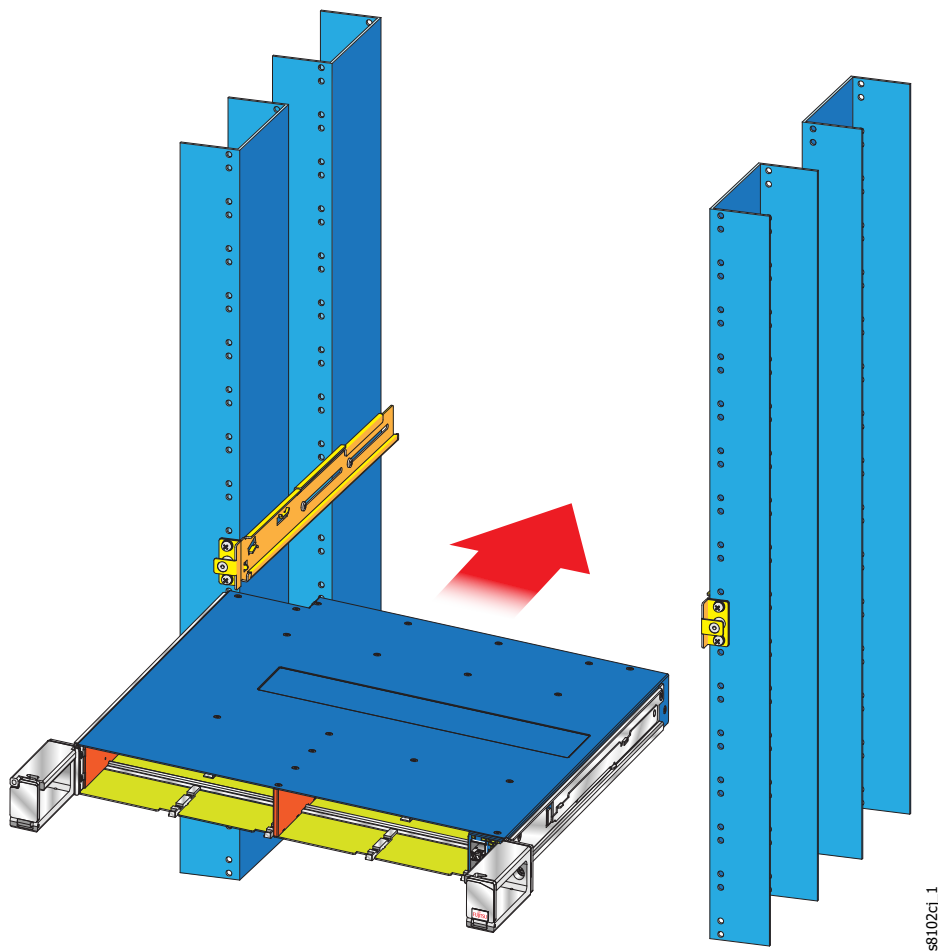


Figure 155
Installing Blade in 4-Post Rack

Step 31

Unlatch the fiber restraint gates on each side of the blade.

Note: Slide the fiber restraint gate slightly inward to open the gate.

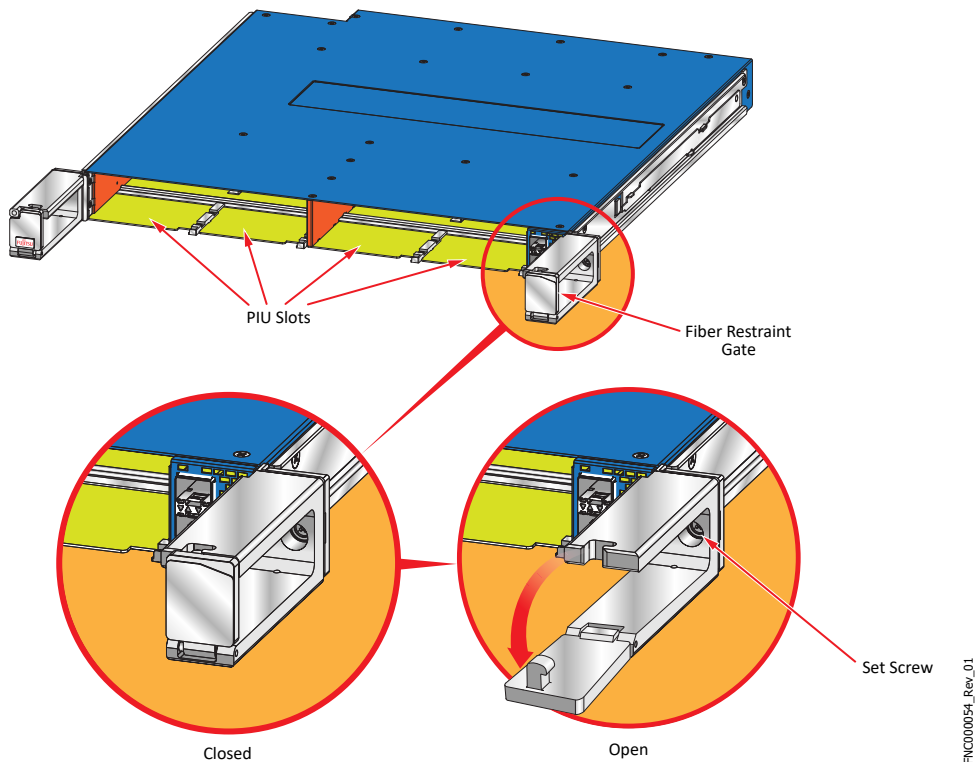


Figure 156
Unlatching Fiber Restraint Gate

Step 32

After opening the fiber restraint gates, secure the blade to the rack using the set screws on the left- and right-hand sides of the blade.

Note: The set screws for securing the blade are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 33

Replace the fiber-optic and RJ-45 cables that are previously removed.

Note: Do not plug in the interconnect LCN SFP modules.

Step 34

Verify that power is turned off at the source and install the power cables.

Step 35

Apply power to the tributary blade.

Step Result:

The tributary blade is in shelf provisioning mode and same release as the Main blade.

Step 36

Plug in all LCN SFP (to the Main and tributary blades) and connect the interconnect LAN cables.

Step 37

List the USI of the tributary blade.

Command:

```
% show shelf-prov-mode-blades
```

Output:

```
USI LBFJTU011725463R110100xxx shelf-id 0
```

Step 38

Set the desired shelf ID on the tributary blade manually.

Command:

```
% request set-shelf-id USI LBFJTU011725463R110100xxx shelf-id #
```

Step Result:

The tributary blade restarts and autoprovisions with the new shelf ID.

Step 39

After restart, retrieve the tributary blade information.

Commands:

```
% show inventory | tab
```

```
% show eqpt shelf #
```

✓ **This task is complete.**

8.3.6

Replace Blade in Housing

Prerequisites:

- An appropriate ESD wrist strap worn by user and properly grounded to avoid ESD damage during installation or removal
- All fibers removed from the blade and appropriately labeled
- If used, all RJ-45 cables removed from the blade and appropriately labeled
- If used, all PIUs removed from the blade
- A recent database backup for the blade is available
- Screwdriver set (including straight-slot and Phillips head)



Warning: The blade is heavy. When handling the blade, position two hands in the center at all times. Moving or lifting the blade with one hand or without assistance (if required) may result in personal injury, damage to the equipment, or both.

Step 1

If the Main blade is being replaced and the blade has the USB flash drive (provides secondary storage of the OSL and the database) installed, remove the USB flash drive and set aside.

Step 2

Verify the replacement blade is the same blade type as the blade being replaced.

Step 3

Verify the software load on the replacement blade is the same as the blade being replaced.

Step 4

Open the fiber restraint gates on each side of the blade being replaced.

Note: Slide the fiber restraint gate slightly inward to open the gate.

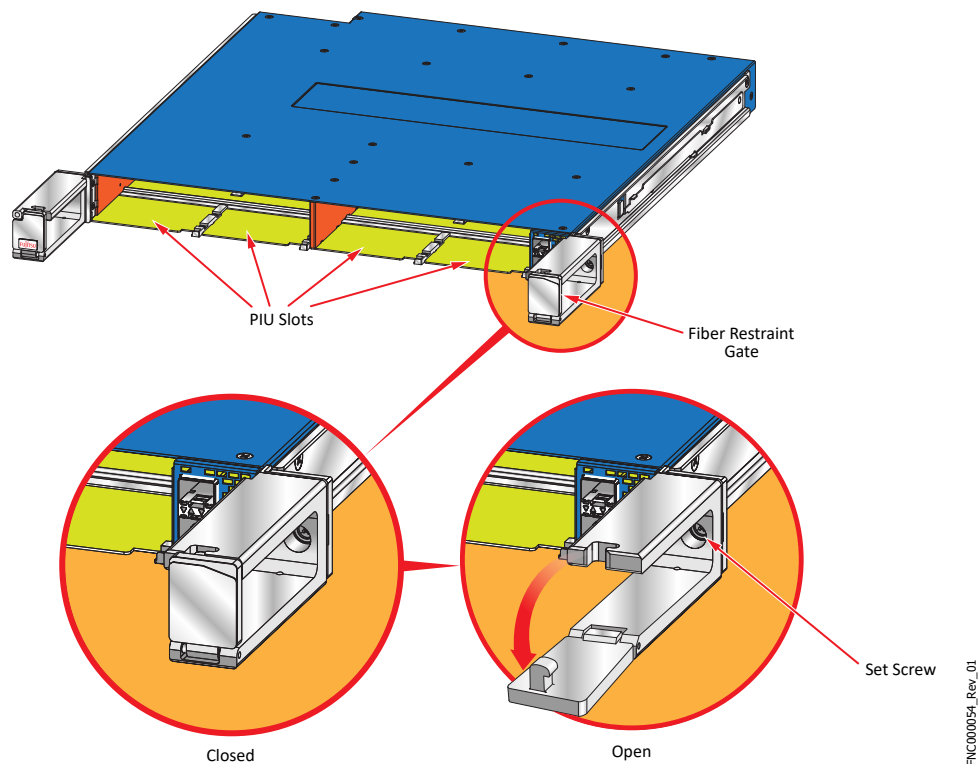


Figure 157
Unlatching Fiber Restraint Gate

Step 5

After opening the fiber restraint gates, carefully loosen the set screws on the left- and right-hand sides of the blade, using a Phillips head screwdriver, to detach from the Housing.

Note: The set screws are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 6

Close both fiber restraint gates after loosening the set screws and disengaging the blade from the Housing.

Step 7

At the front of the blade, use the fiber management arms to slowly pull the blade out of the Housing to the locked position.

At the locked position, the locks on both sides of the blade stop the blade at approximately halfway out.



Attention: Do not touch the optical fibers in adjoining slots. Make sure fibers from other blades do not obstruct the blade being removed.

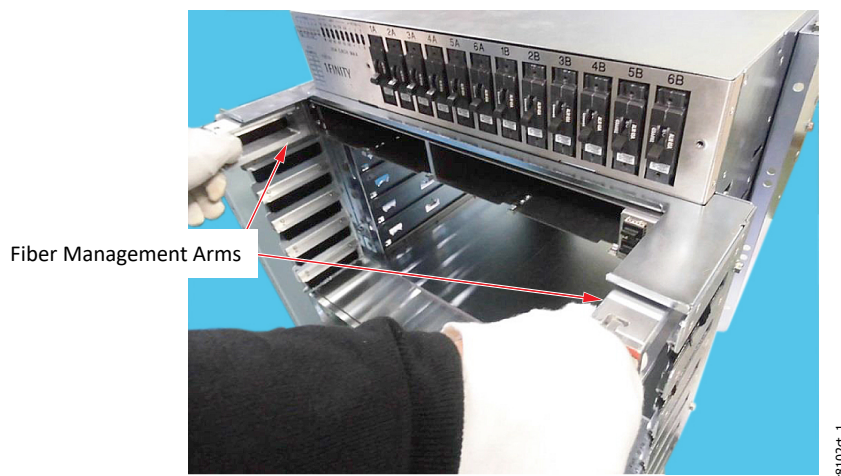


Figure 158
Pulling Blade from Housing



s8102cv_1

Figure 159
Locked Position



Warning: The blade is heavy; always use two hands and be prepared to support the entire weight of the blade after releasing the blade from the locked position.

Step 8

Find the locking tabs on each side of the blade.

Step 9

With the blade at the locked position, place one hand on each side of the blade.

Step 10

Press both locking tabs UP and slowly pull the blade to release. Continue pulling blade until fully removed from the Housing.

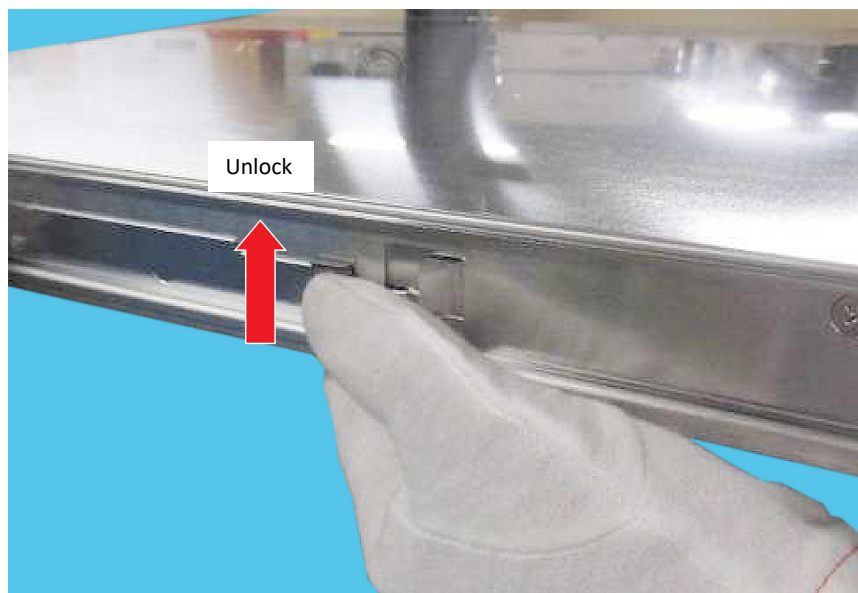


Figure 160
Unlocking Blade from Housing

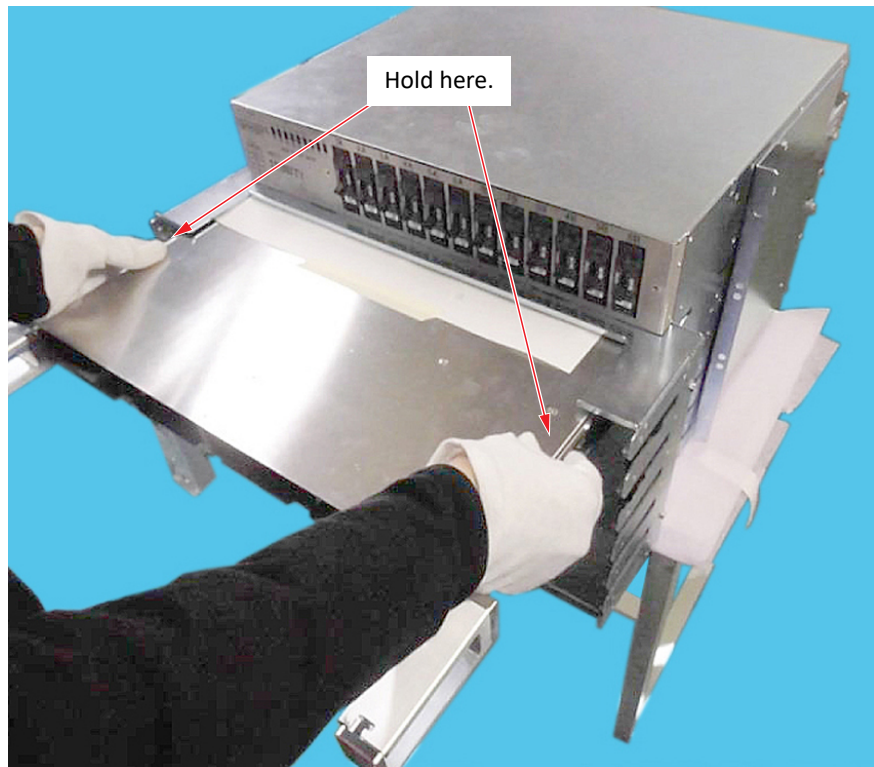


Figure 161
Removing Blade from Housing

Step 11

Store the blade in ESD-protective packaging.

Note: Follow local practice/procedures for repair or replacement of equipment, as applicable.

Step 12

Before unpacking the replacement blade, inspect the shipping containers for signs of damage.

Note: Claims for shipping damage must be filed with the transportation company in accordance with their published procedure. Contact Fujitsu Sales Representative for any shipment questions.

Step 13

Unpack and carefully remove the replacement blade from its shipping container and place it on a clean, level, and ESD-safe surface. Use two hands as shown in the following figure to remove a blade unit from container.

Note: Leave the blade in the ESD-protective plastic bag until ready for installation.

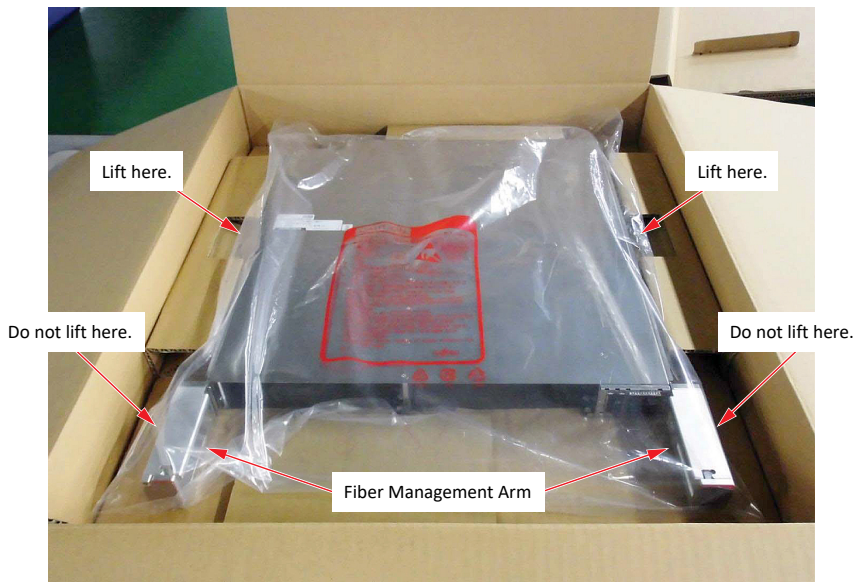


Figure 162
Removing Blade from Packing Container

Step 14

Inspect the equipment to ensure that it is free of any defects, damage, or other debris. If the equipment is damaged, follow local practices/procedures for repair or replacement of the damaged unit, as applicable.

Note: Use the checklist accompanying each shipment (if applicable) to verify that all items are present with the shipment.

Step 15

Record the serial number and issue number of the equipment in accordance with local record-keeping procedures, if necessary.

Step 16

Insert the blade into the appropriate slot of the 1FINITY Housing and slide it back until it locks in place.

Note: Fujitsu recommends installing the blade without the PIUs installed.

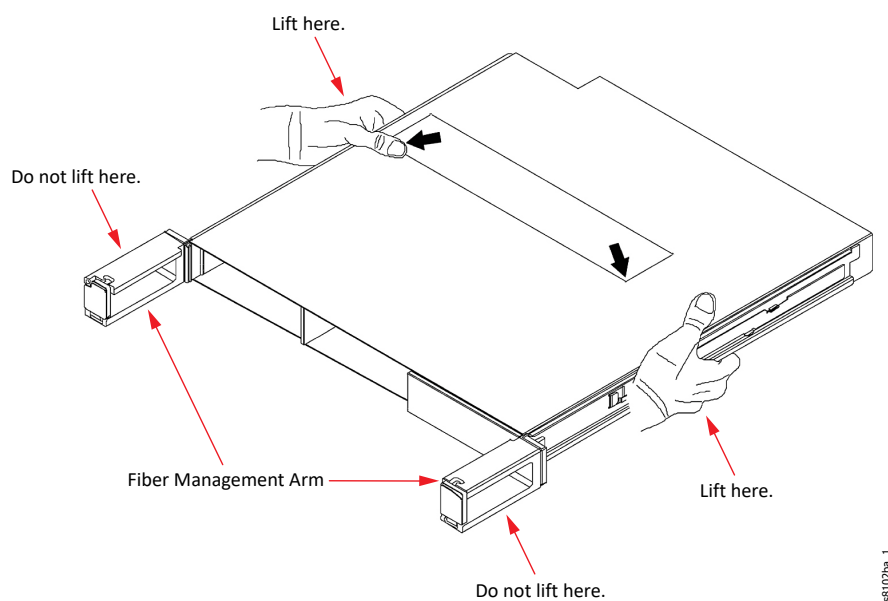
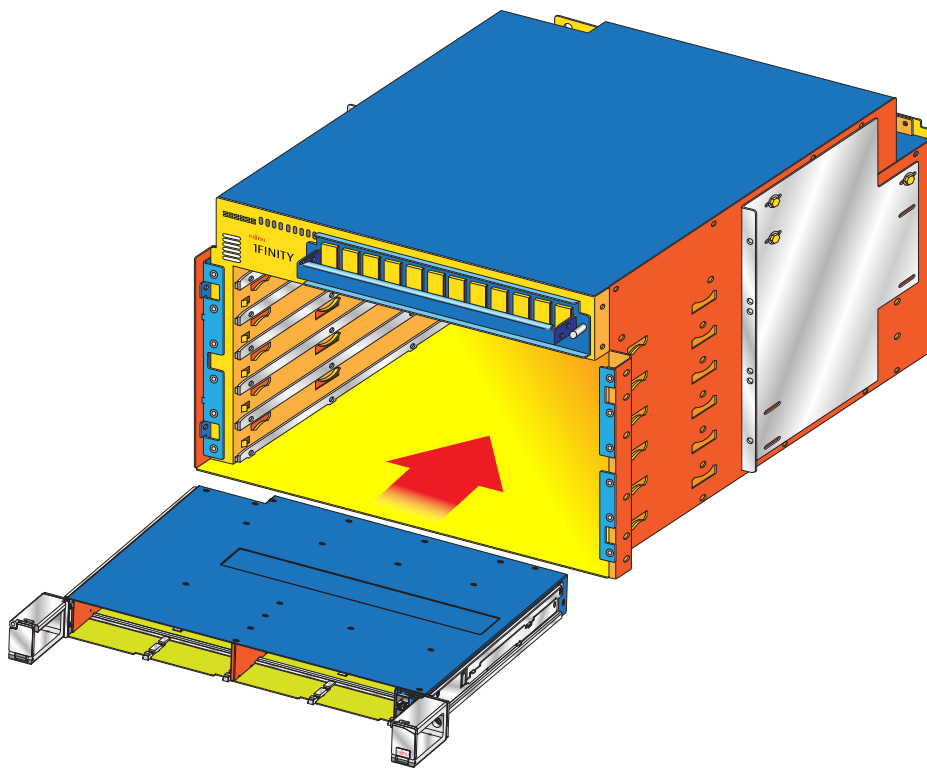


Figure 163
Proper Method to Lift Blade



s8102cm_1

Figure 164
Installing Blade in 1FINITY Housing

Step 17

Unlatch the fiber restraint gates on each side of the blade.

Note: Slide the fiber restraint gate slightly inward to open the gate.

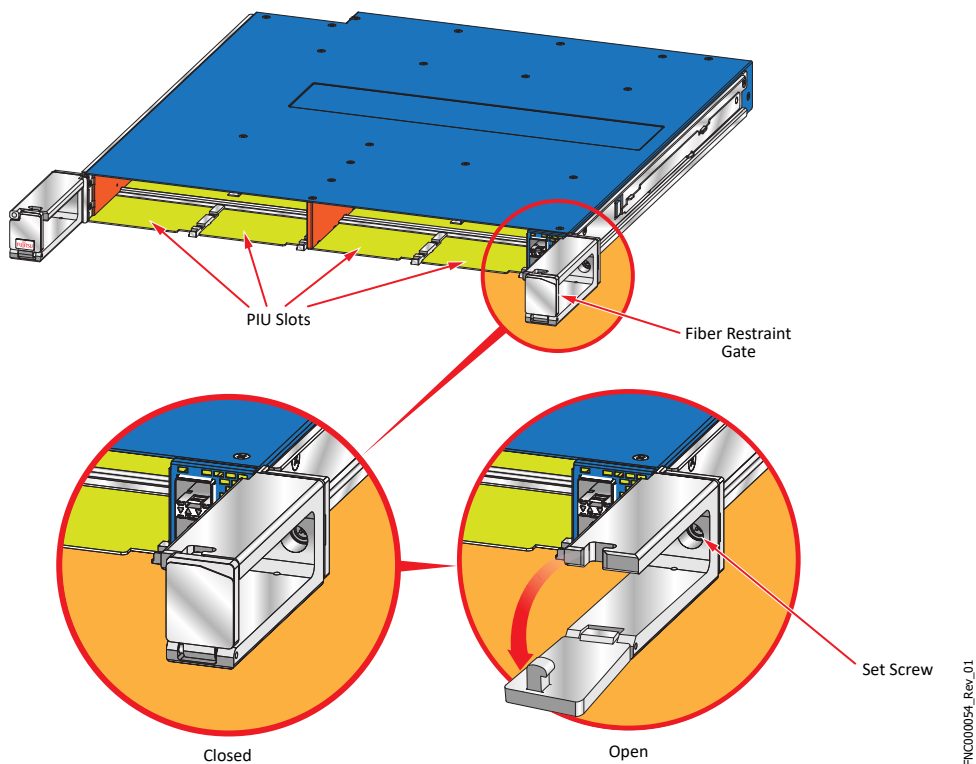


Figure 165
Unlatching Fiber Restraint Gate

Step 18

After opening the fiber restraint gates, secure the blade to the 1FINITY Housing using the set screws on the left- and right-hand sides of the blade.

Note: The set screws for securing the blade are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 19

Replace the fiber-optic and RJ-45 cables that are previously removed.

Step 20

Verify that power is turned off at the source and install the power cables.

Step 21

Is the blade being installed a Main blade?

If YES:

Continue with next step.

If NO:

Proceed to [Step 27](#).

Step 22

Did the Main blade had a USB flash drive that was removed in [Step 1](#)?

If YES:

Continue with next step.

If NO:

Proceed to [Step 26](#).

Step 23

Insert USB flash drive in the replacement Main blade.

Step 24

Turn the power on at the source and wait for the blade to initialize.



Warning: After the blade is powered up, power down within 15 minutes causes database corruption, which leads to blade start-up failure and requires a return to factory for reprogramming.

Step 25

Verify that previous OSL and provisioning are restored to Main blade.

Step Result:

Procedure is complete.

Step 26

Is a database backup available for the blade being replaced?

If YES:

Proceed to [Restore Configuration Database](#) and continue with next step.

If NO:

Proceed to [Equipment Installation](#) and continue with next step.

Step 27

Turn the power on at the source and wait for the blade to initialize.

Step 28

Provision the tributary blade equipment.

Command:

```
> request set-shelfmode shelfid <x> shelfrole TRIB
```

Step Result:

Procedure is complete.



This task is complete.

8.4 Upgrades

In this section:

8.4.1 Upgrade System Software and Firmware

8.4.1 Upgrade System Software and Firmware

Prerequisite:

- The user must be logged on as fujitsu/1finity or have Level-4 or Level-5 privileges assigned to perform this procedure.



Caution: Fujitsu highly recommends performing a database backup before upgrading system software.



Caution: System only supports upgrades from Release 2.5.2 to Release 19.1.1 and Release 2.6.2 to Release 19.1.1. Direct upgrades from previous releases are not supported.

This procedure describes how to use CLI commands in a CLI window to upgrade system software and FW. The FW needs to be upgraded if a new FW is part of the load.

Step 1

Transfer a software package to the `/var/ftp` directory of the blade.

Command:

```
> request transfer ftp action download local-file-path /var/ftp/1FINITY-LSeries-L100-19011.PGM remote-file-path ftp://anonymous:anonymous@192.168.1.2:21/1FINITY-LSeries-L100-19011.PGM
```

Note: From Release 19.1.1 or later, the structure of the PGM filename is changed from 1FINITY-L100-xxxxx.PGM to 1FINITY-LSeries-L100-xxxxx.PGM.

Step 2

Stage the upgrade.

Command:

```
> request sw-stage filename 1FINITY-LSeries-L100-19011.PGM
```

Step 3

Retrieve alarms to verify *softwareStageInProgress* alarm is raised.

Command:

```
> show alarms alarm-list
```

Note: Periodically repeat retrieving alarms to check alarm status until the *softwareStageInProgress* alarm is cleared.

Step 4

Activate the new software load.

Command:

```
> request sw-activate gissue 19.1.1 validationTimer 01-30-00
```

Step 5

After the blade reboots, cancel the validation timer to accept the new load.

Command:

```
> request cancel-validation-timer accept true
```

Step 6

Verify that the new software load is accepted for the active and backup banks.

Command:

```
> show sw-version
```

Step 7

Retrieve alarms to verify if *firmwareBackwardCompatibleAll* alarm is raised.

Command:

```
> show alarms alarm-list
```

Step 8

Execute FW update on each blade if a *firmwareBackwardCompatibleAll* alarm is raised.

Command:

```
> request fw-update shelf-id x force true overwrite false
```

Note: x represents the shelf-id of each individual blade.

Step Result:

A *firmwareInitInProgress* is reported on the target blade.

Step 9

Retrieve alarms to verify *firmwareInitInProgress* alarm is cleared.

Command:

```
> show alarms alarm-list
```

Note: Periodically repeat retrieving alarms to check alarm status until the *firmwareInitInProgress* alarm is cleared. The *firmwareBackwardCompatibleAll* alarm clears as well.



Caution: Waiting for *firmwareInitInProgress* alarm to clear may take 20...30 minutes depending on release upgrade path. Wait until alarm clears before proceeding with operations on system.

Note: After *firmwareInitInProgress* alarm is cleared, FW upgrade is complete.

✓ **This task is complete.**

8.5

Backup and Restore

In this section:

- 8.5.1 Back Up Configuration Database
- 8.5.2 Restore Configuration Database
- 8.5.3 Secondary Database via USB Flash Drive

8.5.1

Back Up Configuration Database

Prerequisites:

- Clear all alarms and standing conditions being reported on the blade.
- The user must be logged on as fujitsu/1finity or have Level-4 or Level-5 privileges assigned to perform this procedure.

Step 1

Copy the running database to a user-provided file in a given path.

Command:

```
> request db-backup filename /var/ftp/1FINITY-071716.dbs
```

Output:

```
status database Backup Complete to /var/ftp/1FINITY-071716.dbs  
[ok] [2016-07-17 14:23:26]
```



Attention: Database backup captures a snapshot of node provisioning database files (.DBS files) for future restoration in the event that database files in the node become corrupt. The .DBS files represent node provisioning at the time of backup, which may or may not match the current node provisioning.

Step 2

Transfer database from the `/var/ftp` directory of the blade to a user-provided remote file server.

Command:

```
> request transfer ftp action upload local-file-path /var/ftp/1FINITY-071716.dbs  
remote-file-path ftp://anonymous:anonymous@192.168.1.2/1FINITY-071716.dbs
```

✓ **This task is complete.**

8.5.2

Restore Configuration Database

Prerequisites:

- Clear all alarms and standing conditions being reported on the blade.
- The user must be logged on as fujitsu/1finity or have Level-4 or Level-5 privileges assigned to perform this procedure.

As part of database restore for the secondary database, when a new database is accepted, the CDSF content is updated. The user does not have access to the CDSF database which is distinctive from access to the NE database. The NE does not back up the CDSF storage as a part of NE database backup procedures to an external host.

The NE supports non-volatile storage of critical data that is distinctive from the data stored in the NE provisioning database. The storage facility used by the CDSF is distinct from the storage used by the NE database to such an extent that any corruption of the NE database does not cause a corruption of the data stored in the CDSF database.

Step 1

Retrieve the current alarms status on the blade.

Command:

```
> show alarms alarm-list
```

Step 2

Transfer the database to the `/var/ftp` directory of the blade.

Command:

```
> request transfer ftp action download local-file-path /var/ftp/1FINITY-XXXXXX.dbs
remote-file-path ftp://anonymous:anonymous@192.168.1.2/1FINITY-XXXXXX.dbs
```

Step 3

Restore the database.

Command:

```
> request db-restore filename 1FINITY-XXXXXX.dbs sysNameCheck true
```

Output:

```
status database Restore Complete from /var/ftp/1FINITY-071716.dbs
[ok] [2016-07-20 14:23:26]
```

Step 4

Activate the database.

Command:

```
> request db-activate rollBackTimer 00-20-00
```

Output:

```
status database Activate Complete
[ok] [2016-01-07 14:26:26]
```

Note: The blade restarts when a new database is activated.

Step 5

Cancel the rollback timer.

Command:

```
> request cancel-rollback-timer accept true
```

Output:

```
status Cancel Rollback Timer Complete
[ok][2016-01-07 14:29:56]
```

Step 6

Retrieve the alarms status on the blade.

Command:

```
> show alarms alarm-list
```

✓ **This task is complete.**

8.5.3

Secondary Database via USB Flash Drive

L100 Blade supports secondary database via USB flash drive. The USB flash drive provides secondary storage of the OSL and database in real time. Secondary storage of the database helps to reduce database corruption. Database corruption is raised when both the primary and secondary databases are corrupted; otherwise, the NE recovers from the valid database. If the NE detects that the primary database is corrupted, the NE automatically transfers and recovers the database using the secondary database stored in the USB flash drive. If the NE detects that the secondary database is corrupted, the Main blade automatically overwrites and backs up the secondary storage with the primary database.

Note: USB flash drive is supported on the Main blade only (L100 Blade). USB flash drive is not supported for tributary blades (L100 or L110 Blade).

■ The USB flash drive for secondary database can either be:

- Fujitsu preferred supplier
- User-supplied own USB flash drive

■ Recommended:

- Operating temperature: 0...50 °C
- NAND type: SLC
- Flammability: UL94-V0 or higher
- Regulations: VCCI/CE/FCC/Class B, RoHS, ESD

Note: Depends on the user safety/environment requirements.

■ USB flash drive requirements:

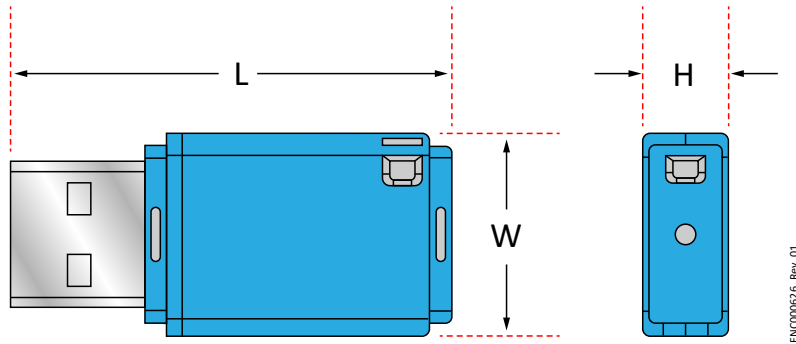


Figure 166
USB Flash Drive 2.0 Dimensions

- Interface: USB 2.0 compatible
- Capacity: 4...32 GB

Note: Minimum 4 GB is required and larger than 32 GB can be used by design, but maximum 32 GB is proved in testing.

- Minimum R/W speed: Read 20 Mb/s, write 10 Mb/s
- Maximum dimensions (W x L x H): 21.5 x 45 x 10 mm

Enable/Disable Secondary Database

The USB flash drive is enabled or disabled using the following CLI commands. The default value of `usb-secondary` is `enable`.

Enable the USB flash drive secondary database.

Command:

```
% set system database usb-secondary enable
```

Disable the USB flash drive secondary database.

Command:

```
% set system database usb-secondary disable
```

Save the changes.

Command:

```
% commit
```

Install USB Flash Drive

This procedure describes how to install USB flash drive.

Step 1

Plug in USB flash drive.

■ Factory shipped (Fujitsu) USB flash drive:

- NE verifies the encryption password and is good to continue.

■ Non-factory shipped (non-Fujitsu) USB flash drive:

- NE verifies the encryption password and fails. An *invalidUSB* alarm is raised.
- Format and encrypt the USB flash drive.

Command:

```
> request format-usb
```

Step Result:

After the USB flash drive is formatted and encrypted to Fujitsu format, the *invalidUSB* alarm clears and the OSL file automatically downloads to the USB flash drive. Software sync alarm *softwareSyncInProgress* is raised when copying OSL to USB flash drive and cleared when complete. Later the database file from primary database is downloaded to the USB flash drive.

Note: Database manager runs a 10-minute timer and detects if a change between primary database to secondary database exists, copies primary database to secondary database location (USB flash drive). After databases are in sync, *noSecondaryDatabase* alarm is cleared.

Step 2

View files saved to the USB flash drive. If user has no need to view, procedure is complete.

Command:

```
> request show-file filename /mnt/secondary/var/shared/confd/backup
```

Step 3

Optionally, remove the USB flash drive.

Note: User can keep the USB flash drive plugged into the shelf if required.

Command:

```
> request remove-usb
```

Step Result:

A *UsbSafeToRemove* event is raised when USB flash drive can be safely removed.

✓ **This task is complete.**

Restore Secondary Database from USB Flash Drive

This procedure describes how to use CLI commands in a L100 CLI window to restore the secondary database through the USB flash drive in a new or used L100 Blade. The procedure applies to the Main L100 Blade only. The tributary blades do not have database and are provisioned through the Main L100 Blade.

Note: The user must be logged on as fujitsu or have Level-4 or Level-5 privileges assigned to perform this procedure.

Step 1

Is a faulty blade being replaced?

If YES:

Continue with next step.

If NO:

Proceed to [Step 5](#).

Step 2

Power down the faulty L100 Blade.

Step 3

Remove the USB flash drive with the secondary database from the faulty L100 Blade.

Step 4

Uninstall the faulty L100 Blade from the rack. Refer to [Remove Blade from Rack](#) for more information.

Step 5

Install the replacement L100 Blade, but do not power it up. Refer to [Initial System Turn Up](#) for more information.

Note: To restore the secondary database on USB flash drive to the replacement blade, the replacement blade must be in shelf provisioning mode. If replacing a faulty blade with a previously used L100 Blade, place the blade into shelf provisioning mode before inserting the USB flash drive.

Step 6

Is the replacement blade a new L100 Blade from the factory that is already in shelf provisioning mode?

If YES:

Proceed to [Step 12](#).

If NO:

Continue with next step.

Step 7

Power up the replacement L100 Blade.

Step 8

Log on to the LMP and check the shelf is in shelf provisioning mode.

Command:

```
> show alarms alarm-list | tab
```

Output:

```
alarms alarm-list number-of-alarms 3
alarms alarm-list last-changed 2019-08-06T20:27:01.616938+00:00
```

LAST				LAST IS			
PERCEIVED				SERVICE			
NOTIFICATION	CIRCUIT						
RESOURCE	ALARM TYPE	ID		ALARM TYPE	QUALIFIER	LAST	STATUS
CHANGE		SEVERITY	LAST ALARM				
TEXT				AFFECTING	ENABLED	ID	

COM	noSecondaryDatabase			not-applicable	near-end		
2019-08-06T20:26:56.405191+00:00		minor		No Secondary			
Database				false	true	-	
COM	shelfProvisioningMode			not-applicable	near-end		
2019-08-06T20:27:01.569386+00:00		warning		Shelf Provisioning			
Mode		false		true	-		
COM	databaseLockedShelfProvModePresent			not-applicable	near-end		
2019-08-06T20:27:01.616938+00:00		warning		Database System Locked: Shelf			
Provisioning Mode Present		false	true				

If YES:

Proceed to [Step 10](#).

If NO:

Continue with next step.

Step 9

Set the replacement L100 Blade to shelf provisioning mode.

Command:

```
> request set-shelfmode reset
```

Note: The shelf reboots and the user needs to log back onto the blade.

Step 10

Retrieve alarms to verify the blade is in shelf provisioning mode.

Command:

```
> show alarms alarm-list | tab
```

Step 11

Power down the replacement L100 Blade.

Step 12

Connect the USB flash drive containing the database to be restored to the replacement L100 Blade.

Step 13

Plug in all the fibers and SFPs. Refer to [Configure IP Connection to DCN](#) for more information.

Step 14

Power up the replacement L100 Blade.

Step Result:

The USB flash drive automatically updates the database in the replaced L100 Blade.

✓ **This task is complete.**

Additional Instructions:

The user may need to do a firmware update or clear interconnect fails to the tributary blades.

9

Equipment Removal

In this chapter:

- 9.1 Remove Filler Panel
- 9.2 Remove SFP Module
- 9.3 Remove Plug-In Unit
- 9.4 Remove Fan Unit
- 9.5 Remove Power Cables
- 9.6 Remove Blade from Rack
- 9.7 Remove Blade from Housing

The procedures described in this section are used to remove L100 Blades, PIUs, pluggable modules, and other required equipment in 19- or 23-inch, 2- or 4-post racks or the 1FINITY Housing.

The following tools and equipment are required:

- Common hand tools
- DMM

9.1

Remove Filler Panel

The following links lead to related information:

[Install Filler Panel](#)



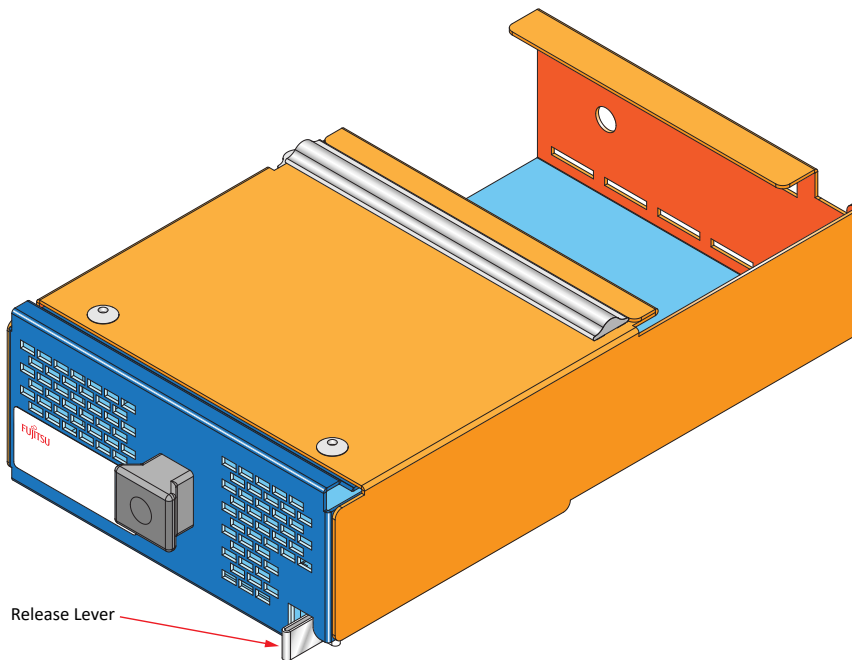
Warning: All unused PIU slots in the blade require a filler panel; otherwise, improper airflow may occur and result in equipment damage.

Step 1

Locate the filler panel to be removed.

Step 2

Press the release latch, and slowly and carefully remove the filler panel using the handle located at the front of the panel.



ss102ar_2

Figure 167
Removing Filler Panel



This task is complete.

9.2

Remove SFP Module

The following links lead to related information:

[Install SFP Module](#)



Danger: Never look into the end of a fiber-optic cable. Permanent eye damage or blindness can occur very quickly if the laser light is present. Follow local safety precautions regarding fiber.



Danger: Never handle exposed fiber with bare hands or touch it to bare skin. Fiber fragments can enter the skin and are difficult to detect and remove. Follow local safety precautions regarding fiber.



Caution: Unplugging a SFP module can be traffic-affecting. Make sure traffic is not dependent on the module before removing.

Step 1

Locate the SFP module to be removed.

Step 2

Inspect the fiber connections to (if any) the module to be removed. Verify that each fiber is clearly marked or labeled to identify the correct connection location.

Note: Fiber marking or labeling depends on local practice.

Step 3

Is each fiber clearly marked to show its corresponding connector?

If YES:

Continue with next step.

If NO:

Mark or label fibers according to local practice and continue with next step.

Step 4

Disconnect the fiber-optic cables from the module to be removed.



Caution: Do not remove the wrong fiber connections. Removing a fiber connection that is in use disrupts service.

Step 5

Lift the release lever on the module to release it from the PIU, and slowly and carefully pull module straight out.



Warning: Because the surface of the module may be hot to touch, use caution when removing the module.

Step 6

Set the module aside for paperwork processing, if required.

Step 7

If a new module is to be installed in the same position, follow the appropriate installation procedure.



This task is complete.

Additional Instructions:

If the SFP module is replaced, record the serial number and issue number of the replacement module.

9.3

Remove Plug-In Unit

The following links lead to related information:

[Install Plug-In Unit](#)

Prerequisites:

- Any fiber connections to the PIU have been removed and properly marked or labeled according to local practice.
- Replacement PIU or filler panel is available, if the blade is active.



Warning: PIUs can be damaged by ESD during installation or removal or whenever they are contacted. Wear an ESD wrist strap with a minimum resistance of 1 megohm that is properly secured to the ESD grounding wrist strap connection post to prevent possible PIU damage. Before using the strap, check it for opens, shorts, and the minimum resistance value. If the strap does not pass these checks, do not use it.



Caution: Do not remove the wrong fiber connections. Removing a fiber connection that is in use disrupts service.

Note: Removal/installation instructions involving an active blade are typically maintenance-related activities.

Step 1

Locate the PIU to be removed.

Step 2

Prepare slot for PIU removal.

Command:

```
> request prepare-removal shelf-id 1 slot-id x
```

Note: In this example, x = 1...4.

Step Result:

The LED turns red for any port with an installed pluggable module, but traffic remains UP.

Step 3

Ensure that an appropriate ESD wrist strap is worn and is properly grounded onto the ESD wrist strap grounding post on the chassis.

Step 4

Gently pull open the locking latch until the unit is released from the rear of the slot.

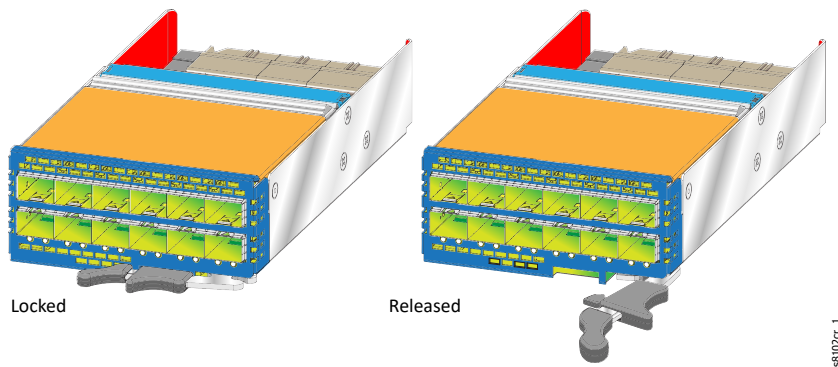


Figure 168
Releasing PIU

Step 5

Carefully and slowly remove the PIU from the blade.

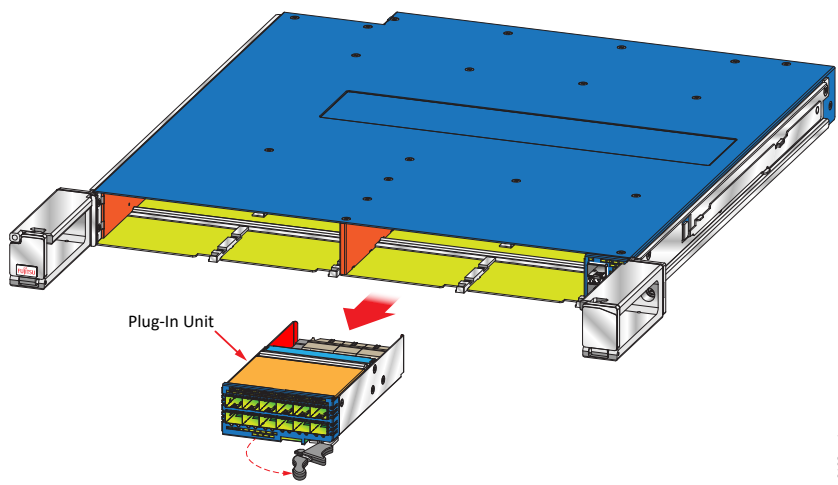


Figure 169
Removing PIU

Step 6

After PIU is removed, inspect the PIU for any damage or debris. Carefully remove any debris. If the PIU is damaged, follow local practice/procedures for repair or replacement of a damaged PIU, as applicable.

Step 7

Store the PIU in ESD-protective packaging.

Step 8

If a new PIU is to be installed in the same position, follow the appropriate PIU installation procedure. Wait 15 seconds before inserting the new PIU in the same position.

Note: If any PIU slots on an active blade are unoccupied, install filler panels in the unused slots.

✓ **This task is complete.**

Additional Instructions:

If the PIU is replaced, record the serial number and issue number of the replacement unit.

9.4

Remove Fan Unit

The following links lead to related information:

[Install Fan Unit](#)



Warning: Fan units are required to provide sufficient airflow (heat removal) through the blade. If a Fan unit is removed, reinstalled, or replaced, reinsert the Fan unit within 3 minutes; otherwise, the blade and service PIUs could be damaged because of overheating. The operating temperature is 5...40 °C.



Warning: PIUs can be damaged by ESD during installation or removal or whenever they are contacted. Wear an ESD wrist strap with a minimum resistance of 1 megohm that is properly secured to the ESD grounding wrist strap connection post to prevent possible PIU damage. Before using the strap, check it for opens, shorts, and the minimum resistance value. If the strap does not pass these checks, do not use it.

Step 1

Locate the Fan unit to be removed.

Note: If the Fan unit is being replaced because of a failure, FAN LEDs on rear-left side of the blade can identify which Fan unit has failed.

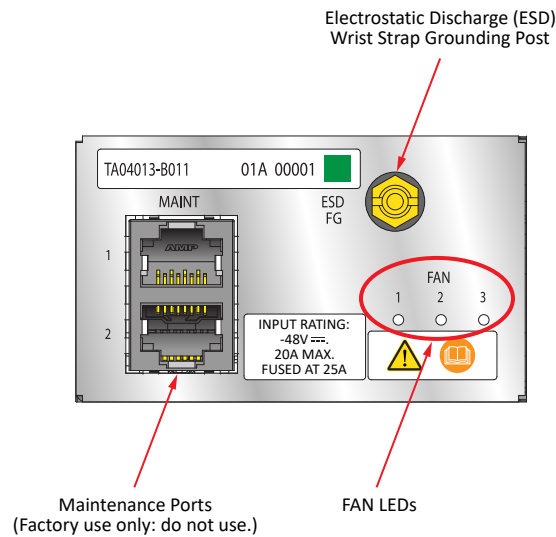


Figure 170
FAN LEDs

Step 2

Remove the Fan unit by performing the following substeps:

- a) Locate the release lever on the lower left-side corner of the Fan unit to be removed.
- b) Press the release lever to the right, and carefully and slowly pull on the cross bar to remove the Fan unit from blade.

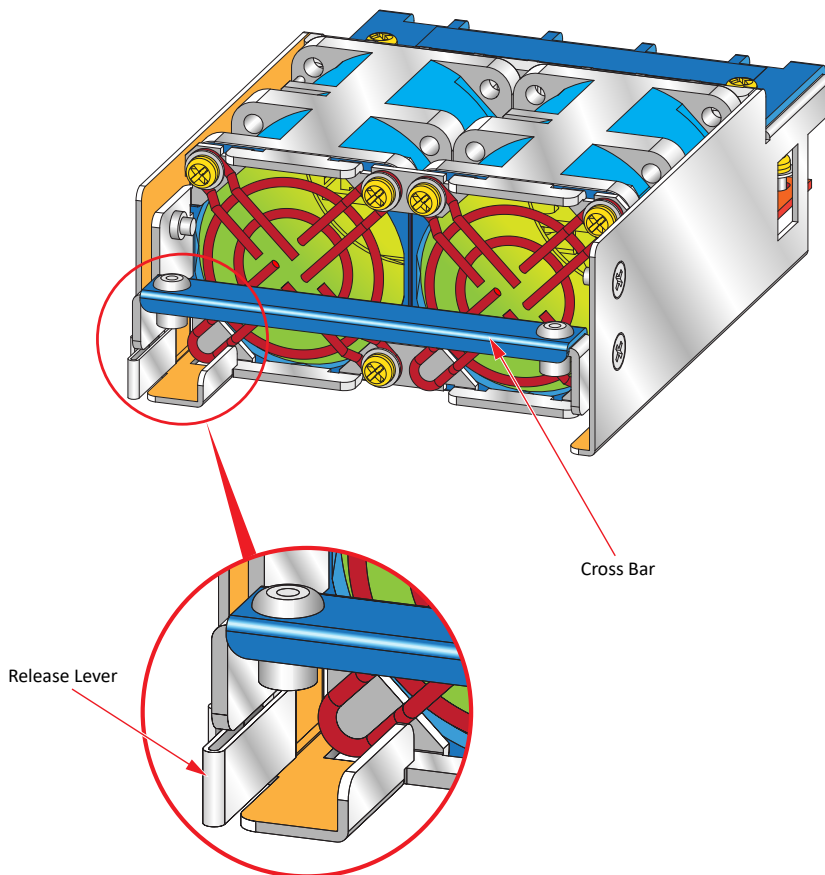


Figure 171
Removing Fan Unit

Step 3

Set the Fan unit aside for paperwork processing, if required.

Step 4

If a new Fan unit is installed in the same position, follow the appropriate Fan installation procedure.

✓ **This task is complete.**

Additional Instructions:

If the Fan unit is defective, follow local practice/procedures for repair or replacement of the equipment, as applicable.

9.5

Remove Power Cables

The following links lead to related information:

[Install Power Cables](#)

Prerequisites:

- Power to the blade has been turned OFF
- Screwdriver set (including straight-slot and Phillips head)
- DMM



Danger: Make sure both A and B power feeds are turned off, at the source, before starting this procedure. Use properly insulated tools and perform all safety precautions in accordance with local safety practices when working on or around power terminals, power cables, and power connectors. Contacting power terminals, uninsulated conductors, cable wiring, or any connector that could be part of a live electrical circuit can result in death or serious personal injury.



Attention: This procedure is not applicable if using the 1FINITY Housing option.

Step 1

If not already done, remove the protection cover from the power terminal adapter.

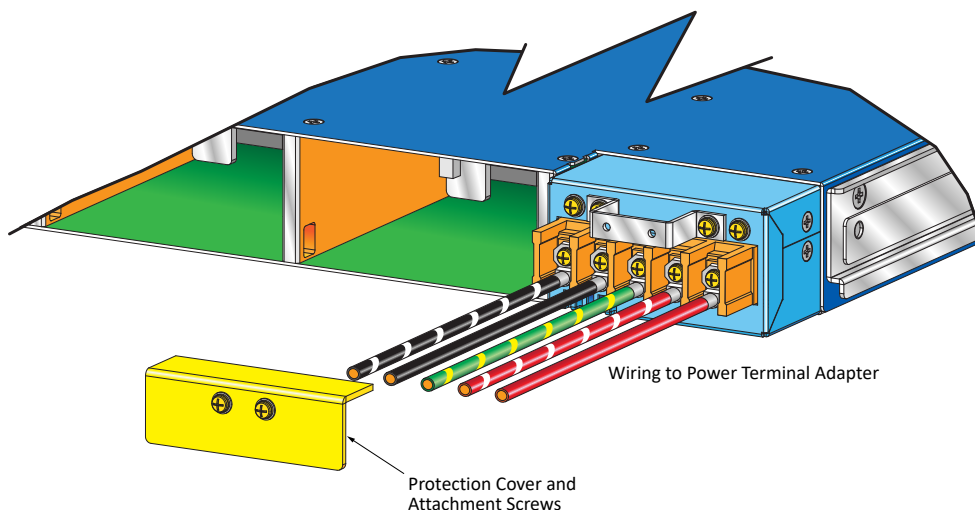


Figure 172
Removing Protection Cover



Danger: Verify power to the blade has been turned OFF and follow all local electrical safety procedures when working with electrical circuits to prevent possible serious injury or death.

Step 2

Remove the power cables from the power terminal adapter by performing the following substeps:

Note: The SYS LED indicator on the front of the blade and PWR LED on the rear of the blade are both turned OFF when power to the blade is shut off.

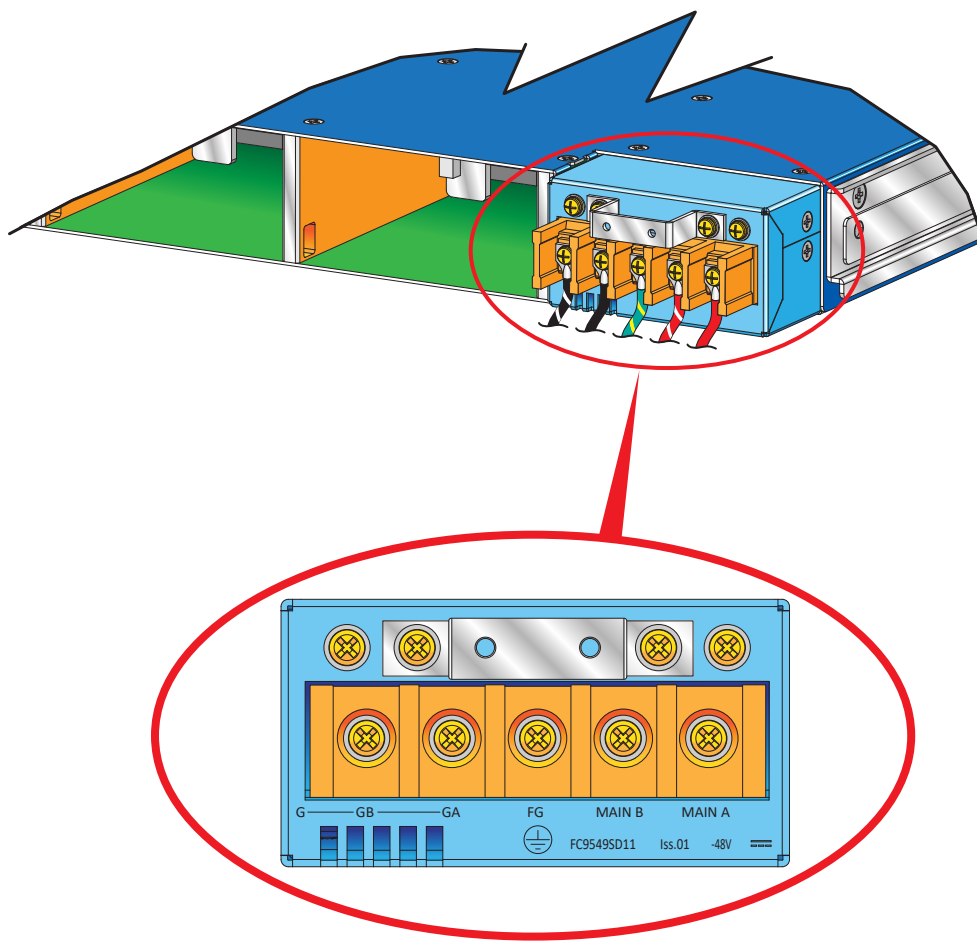


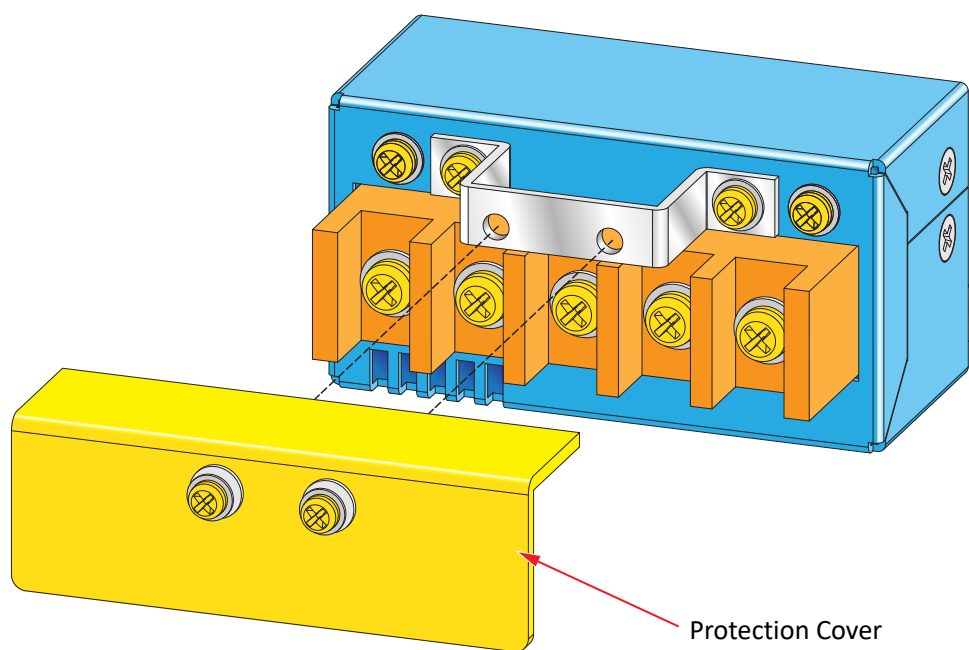
Figure 173
Removing Power Cables

- Detach the power feed B cable (red/white) from the terminal marked MAIN B on the power terminal adapter.
- Detach the ground cable (black/white) for the B power feed from the terminal marked GB on the power terminal adapter.
- Detach the power feed A cable (red) from the terminal marked MAIN A on the power terminal adapter.

- d) Detach the ground cable (black) for the A power feed from the terminal marked GA on the power terminal adapter.
- e) Detach the frame ground cable (green/yellow) from the terminal marked FG on the power terminal adapter.

Step 3

Replace the protection cover.



s8102ah_2

Figure 174
Replacing Protection Cover

✓ **This task is complete.**

9.6

Remove Blade from Rack

Prerequisites:

- Power cables have been removed from the power terminal adapter (AC00-SD11) on the rear of the blade.
- An appropriate ESD wrist strap is worn and is properly grounded to avoid ESD damage during installation, removal, or whenever equipment is handled.
- All fibers have been removed from the blade.
- PIUs have been removed from the blade.
- Screwdriver set (including straight-slot and Phillips head)



Warning: The blade is heavy. When handling the blade, position two hands in the center at all times. Moving or lifting the blade with one hand or without assistance (if required) may result in personal injury, damage to the equipment, or both.

Step 1

Unlatch the fiber restraint gates on each side of the blade.

Note: Slide the fiber restraint gate slightly inward to open the gate.

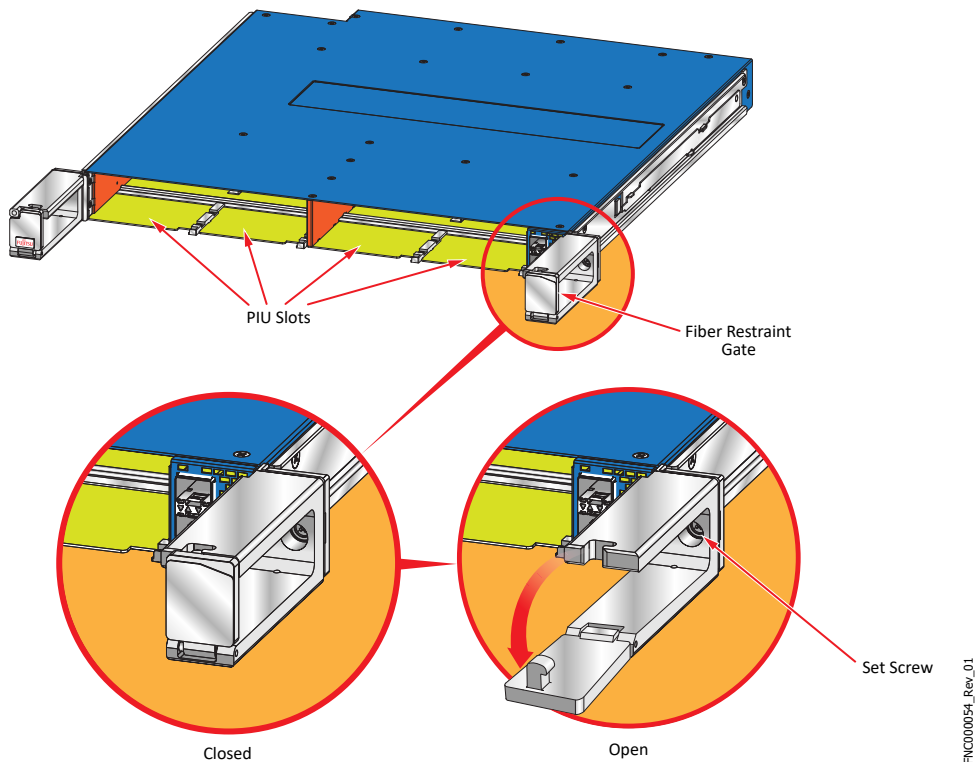


Figure 175
Unlatching Fiber Restraint Gate

Step 2

After opening the fiber restraint gates, carefully loosen the set screws on the left- and right-hand sides of the blade, using a Phillips head screwdriver, to detach from the rack.

Note: The set screws are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 3

Close both fiber restraint gates after loosening the set screws and disengaging the blade from the rack.

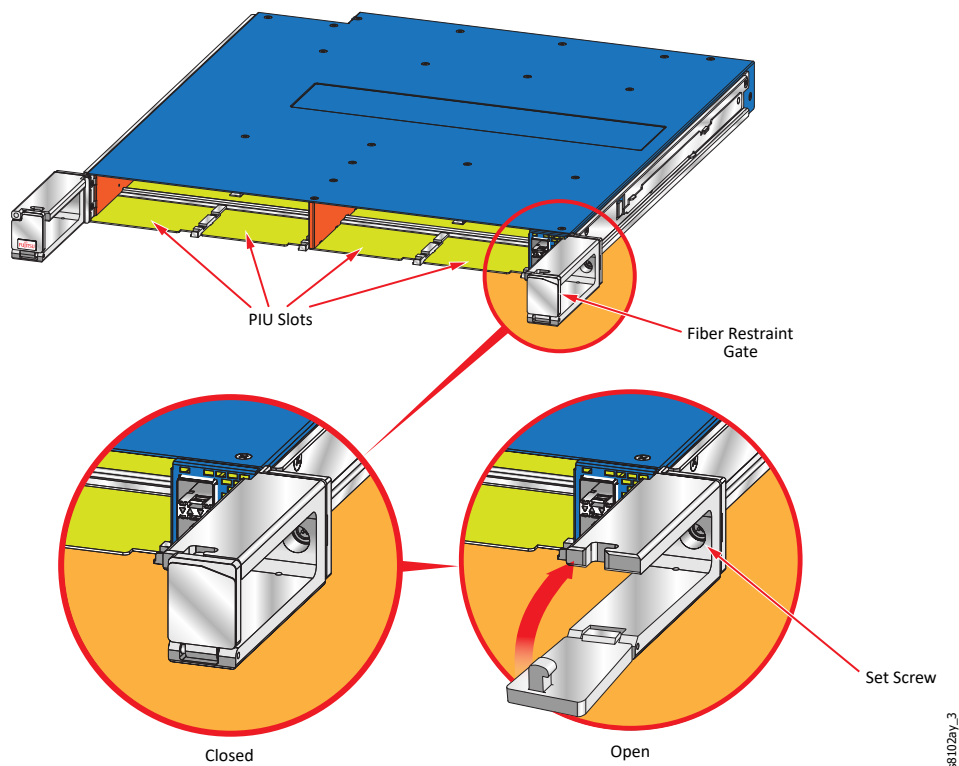


Figure 176
Latching Fiber Restraint Gate

Step 4

At the front of the blade, use the fiber management arms to slowly pull the blade out of the rack to the locked position.

At the locked position, the locks on both sides of the blade stop the blade at approximately halfway out.



Attention: Do not touch the optical fibers in adjoining slots. Make sure fibers from other blades do obstruct the blade being removed.

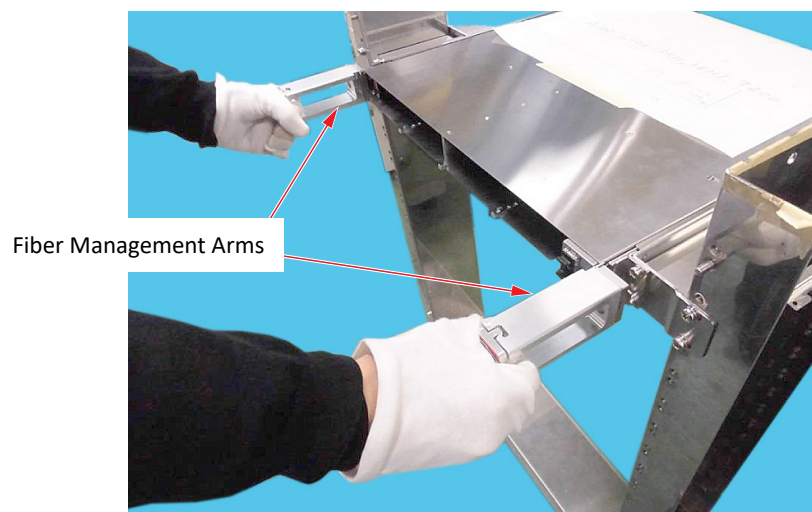


Figure 177
Pulling Blade from Rack

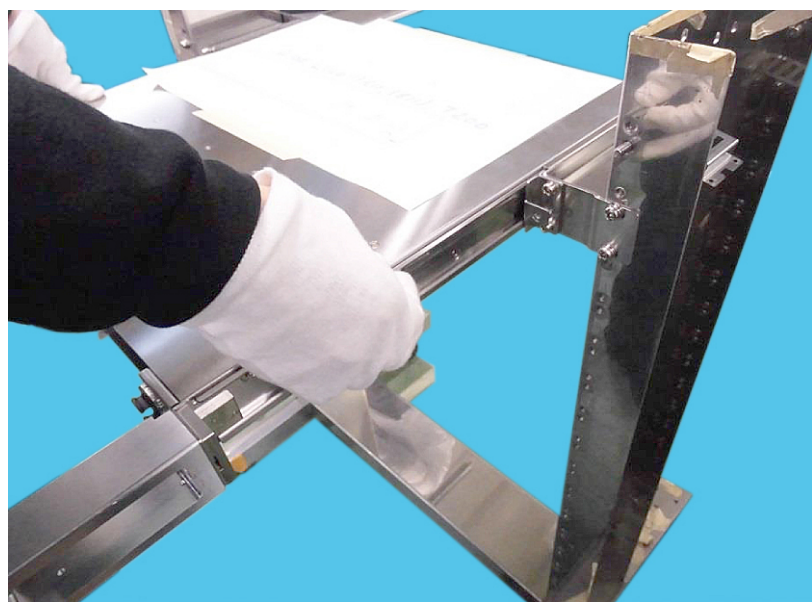


Figure 178
Locked Position

Step 5



Warning: The blade is heavy; always use two hands and be prepared to support the entire weight of the blade after releasing the blade from the locked position.

Remove the blade by performing the following substeps:

- a) Find the locking tabs on each side of the blade.
- b) With the blade at the locked position, place one hand on each side of the blade.
- c) Press both locking tabs UP and slowly pull the blade to release. Continue pulling blade until fully removed from the rack.

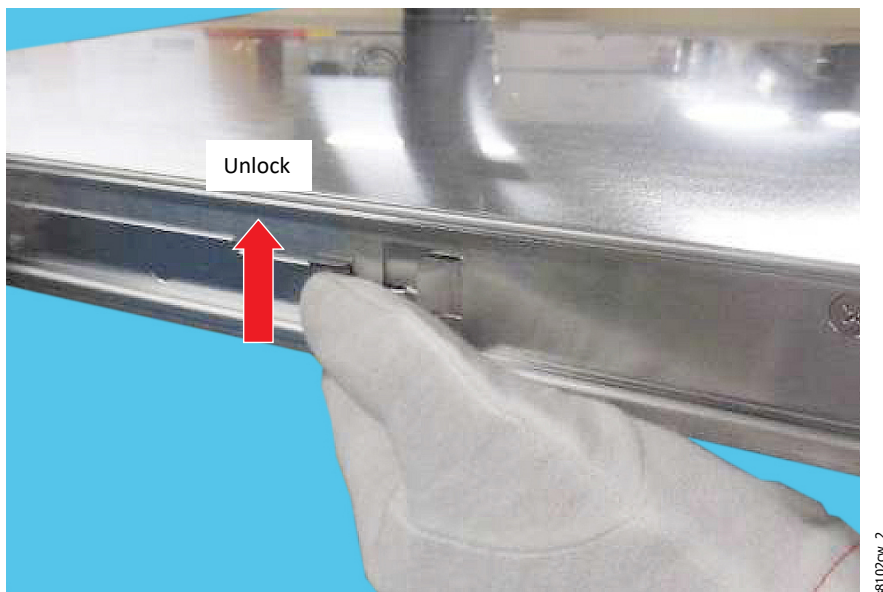


Figure 179
Unlocking Blade from Rack

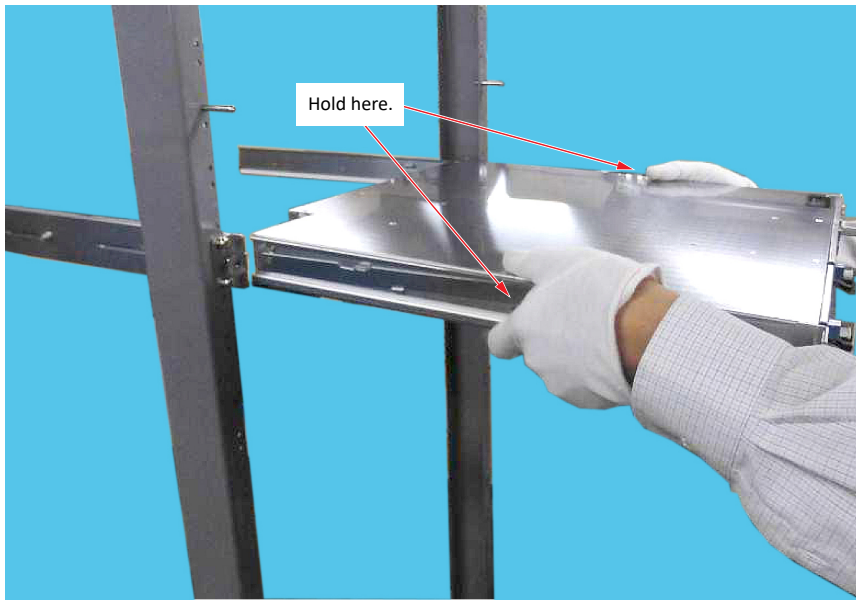


Figure 180
Removing Blade from Rack

Step 6

Store the blade in ESD-protective packaging.

✓ **This task is complete.**

Additional Instructions:

Follow local practice/procedures for repair or replacement of equipment, as applicable.

9.7

Remove Blade from Housing

Prerequisites:

- An appropriate ESD wrist strap is worn and is properly grounded to avoid ESD damage during installation, removal, or whenever equipment is handled.
- All fibers have been removed from the blade.
- PIUs have been removed from the blade.
- Screwdriver set (including straight-slot and Phillips head)



Warning: The blade is heavy. When handling the blade, position two hands in the center at all times. Moving or lifting the blade with one hand or without assistance (if required) may result in personal injury, damage to the equipment, or both.

Step 1

Open the fiber restraint gates on each side of the blade.

Note: Slide the fiber restraint gate slightly inward to open the gate.

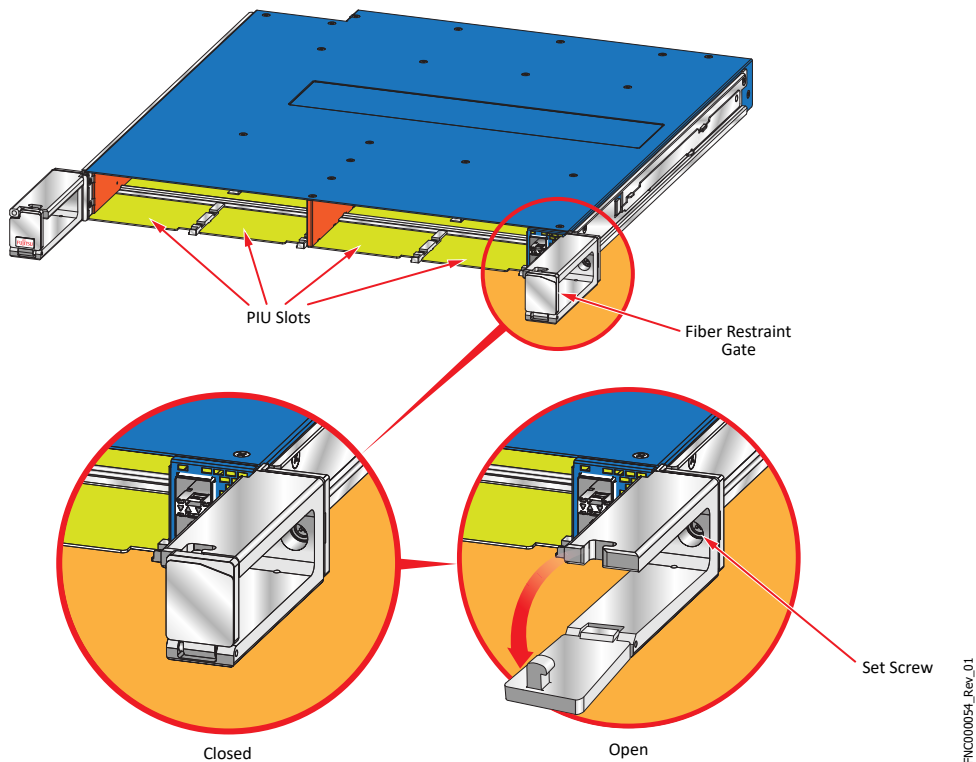


Figure 181
Unlatching Fiber Restraint Gate

Step 2

After opening the fiber restraint gates, carefully loosen the set screws on the left- and right-hand sides of the blade, using a Phillips head screwdriver, to detach it from the Housing.

Note: The set screws are accessible after opening the fiber restraint gates on the left- and right-hand sides of the blade.

Step 3

Close both fiber restraint gates after loosening the set screws and disengaging the blade from the Housing.

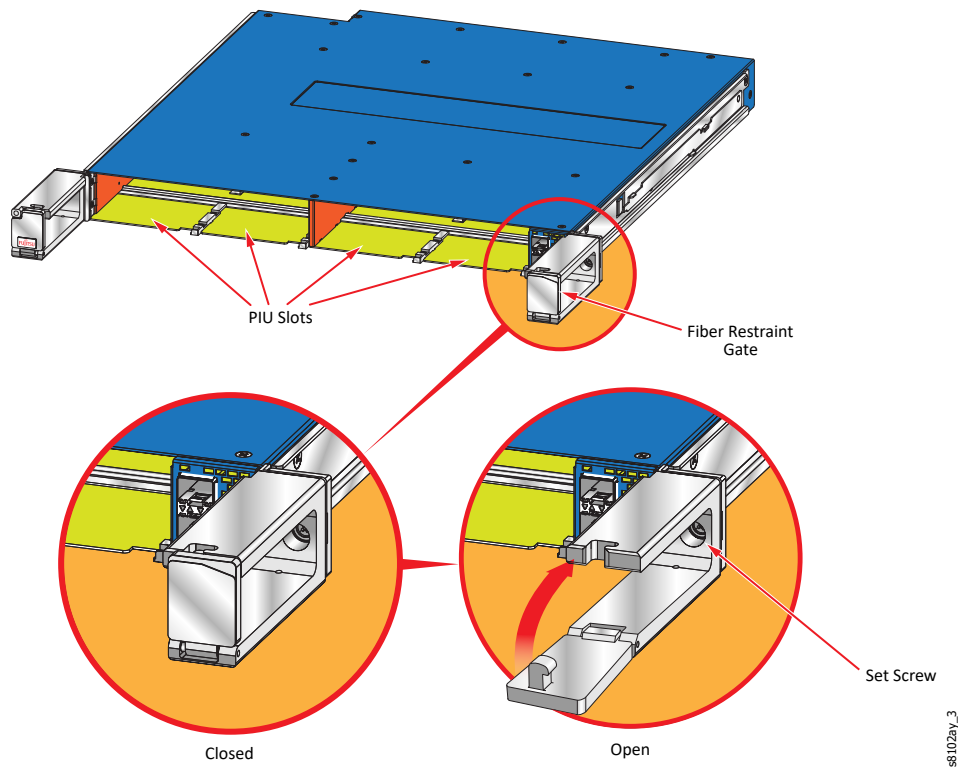


Figure 182
Latching Fiber Restraint Gate

Step 4

At the front of the blade, use the fiber management arms to slowly pull the blade out of the Housing to the locked position.

At the locked position, the locks on both sides of the blade stop the blade at approximately halfway out.



Attention: Do not touch the optical fibers in adjoining slots. Make sure fibers from other blades do not obstruct the blade being removed.

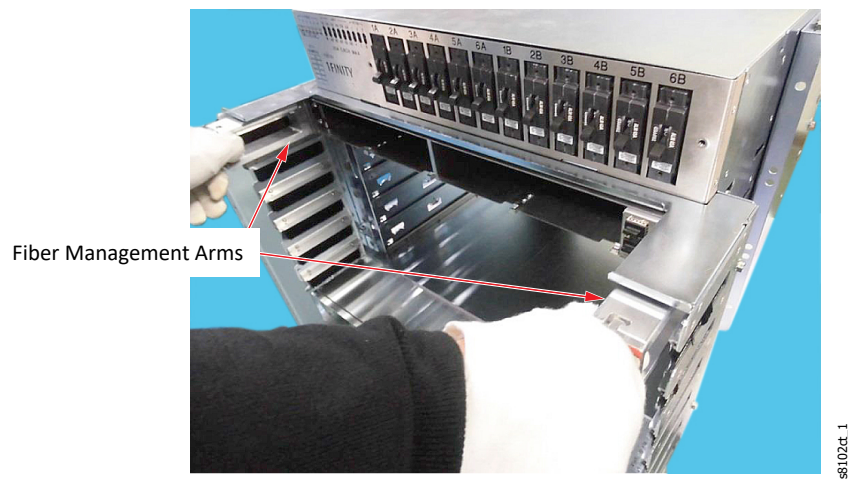


Figure 183
Pulling Blade from Housing



Figure 184
Locked Position

Step 5



Warning: The blade is heavy; always use two hands and be prepared to support the entire weight of the blade after releasing the blade from the locked position.

Remove the blade by performing the following substeps:

- a) Find the locking tabs on each side of the blade.
- b) With the blade at the locked position, place one hand on each side of the blade.
- c) Press both locking tabs UP and slowly pull the blade to release. Continue pulling blade until fully removed from the Housing.

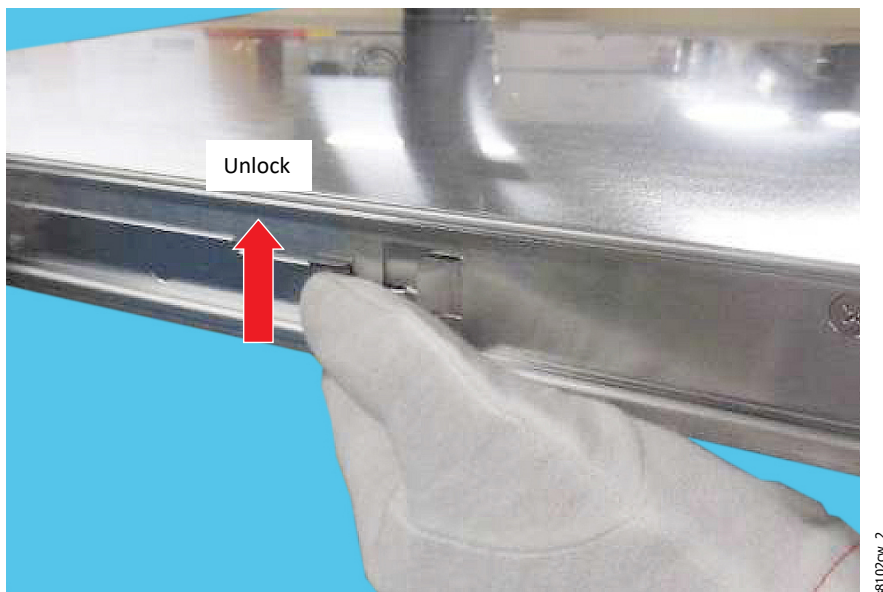
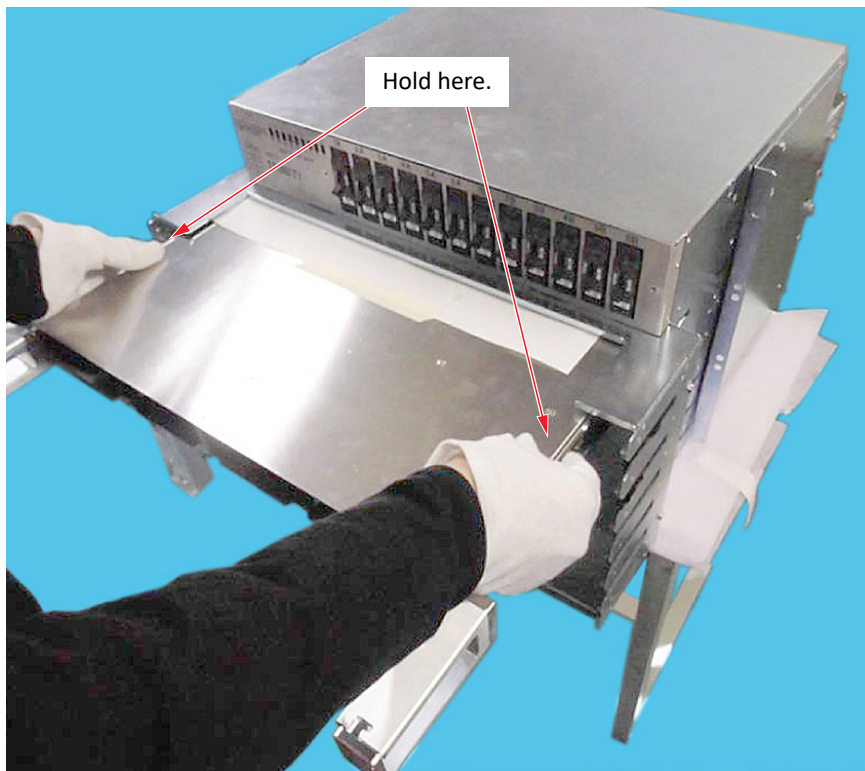


Figure 185
Unlocking Blade from Housing



s8102dk_1

Figure 186
Removing Blade from Housing

Step 6

Store the blade in ESD-protective packaging.

✓ **This task is complete.**

Additional Instructions:

Follow local practice/procedures for repair or replacement of equipment, as applicable.

A

Load Configuration Operation

In this appendix:

- A.1 Load Configuration Using NETCONF
- A.2 Load Configuration Using CLI
- A.3 Limitations on Load Configuration

Introduction

- Fujitsu software provides a faster and flexible way of restoring the database.
- Load configuration feature allows the user to load a user-editable configuration file in TEXT and XML formats to the system.
- The load configuration feature has many advantages over the traditional database restore.
- The traditional database restore invokes system restart and the database is locked until the user accepts the new database. Besides, the database is in a binary format, that is, the database is not in a user readable format.
- Using the load configuration feature, the user is able to make modifications to the database configuration file (TEXT or XML) before the database is being restored on the system.

Supported Operations

This release supports NETCONF with XML and CLI.

Interface	XML	TEXT
NETCONF	Merge, replace, delete, remove, create	Not supported
CLI	Demo only	Demo only

The following operations are supported in the load configuration process:

Load merge	Merge the content of the configuration file with the current configuration
-------------------	--

Load replace	Replace the content of the configuration file for the corresponding parts of the current configuration
---------------------	--

A.1

Load Configuration Using NETCONF

In this section:

- A.1.1 Save Configuration File
- A.1.2 Edit XML Configuration File

A.1.1

Save Configuration File

Save the configuration XML file in configuration mode.

Command:

```
> run show configuration | display xml | save FUJITSU-L100_config.xml
```

A.1.2

Edit XML Configuration File

Add Header

Load Replace

Edit the configuration file to add the header for load replace operation.

Example:

```
<edit-config>
  <default-operation>replace</default-operation>
  <target>
    <candidate/>
  </target>
  <test-option>set</test-option>
<config xmlns="http://tail-f.com/ns/config/1.0">
  ..... configuration saved in previous step
</config>
</edit-config>
```

Commit Changes

Save the changes.

Command:

```
<commit/>
```

Add Header

Load Merge

Edit the configuration file to add the header for load merge operation.

Example:

```
<edit-config>
    <default-operation>merge</default-operation>
    <target>
        <candidate/>
    </target>
    <test-option>set</test-option>
<config xmlns="http://tail-f.com/ns/config/1.0">
    ..... configuration saved in previous step, section 3.1
</config>
</edit-config>
```

Commit Changes

Save the changes.

Command:

```
<commit/>
```

Add Header

Load Override

Edit the configuration file to add the header for load override operation.

Example:

```
<edit-config>
    <default-operation>override</default-operation>
    <target>
        <candidate/>
    </target>
    <test-option>set</test-option>
<config xmlns="http://tail-f.com/ns/config/1.0">
    ..... configuration saved in previous step, section 3.1
</config>
</edit-config>
```

Commit Changes

Save the changes.

Command:

```
<commit/>
```

Load Configuration from File

Add the header by loading the configuration file for merge or replace operation.

Example:

```
<edit-config>
  <default-operation>merge</default-operation>
  <target>
    <candidate/>
  </target>
  <test-option>set</test-option>
  <url>sftp://fujitsu:lfinit@<IP ADDRESS>/home/fujitsu/L100-load-config.xml</url>
</edit-config>
```

Commit Changes

Save the changes.

Command:

```
<commit/>
```

A.2

Load Configuration Using CLI

In this section:

- A.2.1 Save Configuration File
- A.2.2 Edit TEXT Configuration File

A.2.1

Save Configuration File

Save the configuration TEXT file in configuration mode.

Command:

```
> run show configuration | save FUJITSU-L100_config.txt
```

A.2.2

Edit TEXT Configuration File

Add Header

Load Replace

Edit the configuration file to add the header for load replace operation.

Commands:

```
% load replace FUJITSU-L100_config.txt
```

```
% commit
```

Load Merge

Edit the configuration file to add the header for load merge operation.

Commands:

```
%load merge FUJITSU-L100_config.txt
```

```
% commit
```

A.3

Limitations on Load Configuration

In this section:

- A.3.1 Fabric and Port-Connection
- A.3.2 SNMPv3
- A.3.3 Fujitsu Optical Interfaces
- A.3.4 RollBack
- A.3.5 Load Configuration with Large Database
- A.3.6 LLDP

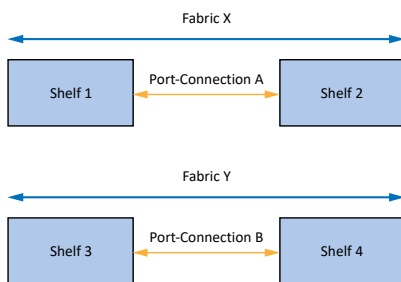
A.3.1

Fabric and Port-Connection

Different Configuration File from Running Database

If the running configuration consists of fabric and port connections and the configuration is different from the configuration file, the user needs to delete all fabric- and port-connection entities from the database first and execute the `load replace` operation.

Running Config:



Config File:

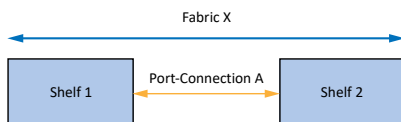


Figure 187

Restoring Configuration

In the preceding figure, the user restores the configuration file onto the system. The user needs to delete fabric X, fabric Y, port-connection A, and port-connection B before executing the `load replace` operation.

The following procedure is only required when the configuration file is different from the running configuration.

Example:

```
<edit-config>
  <target>
    <candidate/>
  </target>
  <config xmlns="http://tail-f.com/ns/config/1.0">
    <fabric xmlns="urn:fujitsu:params:xml:ns:yang:fabric" operation="remove">
    </fabric>
    <port-connection xmlns="urn:fujitsu:params:xml:ns:yang:eqpt:fcon"
      xmlns:nc="urn:ietf:params:xml:ns:netconf:base:1.0"
      nc:operation="remove">
      <connection-id>PORT-ADD-1</connection-id>
    </port-connection>
    <port-connection xmlns="urn:fujitsu:params:xml:ns:yang:eqpt:fcon"
      xmlns:nc="urn:ietf:params:xml:ns:netconf:base:1.0"
      nc:operation="remove">
      <connection-id>PORT-DROP-1</connection-id>
    </port-connection>
  </config>
</edit-config>

..... new configuration

<?xml version='1.0' encoding='utf-8'?>
  <rpc xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="25" >
    <commit/>
  </rpc>
```

Port-Connection Mode

When the configuration file is the same as running configuration, the user moves the line/port-connection/mode before the shelf line if the port-connection mode is colorless or AWG.

Example (Before):

```
<port-connection xmlns="urn:fujitsu:params:xml:ns:yang:eqpt:fcon">
  <connection-id>COLORLESS</connection-id>
  <shelf>1</shelf>
  <slot>0</slot>
  <subslot>0</subslot>
  <port>C3</port>
  <mode>colorless</mode>
</port-connection>
```

Example (After):

```
<port-connection xmlns="urn:fujitsu:params:xml:ns:yang:eqpt:fcon">
  <connection-id>COLORLESS</connection-id>
  <mode>colorless</mode>
  <shelf>1</shelf>
  <slot>0</slot>
  <subslot>0</subslot>
  <port>C3</port>
</port-connection>
```

Equipment/Interface and Alarm-Severity-Assignment

If the user provisions resource (equipment/interface) and alarm-severity-assignment at single commit, the operation fails. The user needs to provision resource and commit the resource first, and the user needs to provision alarm-severity-assignment.

Port-Connection Direction

If the user saves the configuration file in the current release, user deletes the port-connection `<direction></direction>` from the configuration file before executing the `load replace` command. The direction field is moved to operation side in the next release.

Example:

```
<port-connection xmlns="urn:fujitsu:params:xml:ns:yang:eqpt:fcon">
  <connection-id>PORT-ADD</connection-id>
  <src-shelf>2</src-shelf>
  <src-slot>1</src-slot>
  <src-subslot>0</src-subslot>
  <src-port>E1</src-port>
  <dst-shelf>1</dst-shelf>
  <dst-slot>0</dst-slot>
  <dst-subslot>0</dst-subslot>
  <dst-port>C1</dst-port>
  <mode>normal</mode>
  <direction>UNI</direction>      #direction need to be deleted
</port-connection>
```

Fabric Connection

When the user does a load replace on an existing fabric with changes in attributes like xcon-id, src-id, dst-if, and dst-och, the validation fails if the xcon-id is not alphabetically bigger than the xcon-id being replaced.

For example, consider a fabric connection with values (xcon-id=ABC, src-if=A, dst-if=B) is already provisioned in the database.

The following commands fail validation at commit because the *dst-if* is changed to C.

Commands:

```
% delete fabric och-connection ABC

% set fabric och-connection ABC src-if A dst-if C

% commit
```

The following commands are another failure example of which the user only changes the *xcon-id*.

Commands:

```
% delete fabric och-connection ABCDEF

% set fabric och-connection ABC src-if A dst-if B

% commit
```

In this case, the new *xcon-id=ABC* is alphabetically smaller than the previously created fabric with *xcon-id=ABCDEF*. As a result, the fabric *xcon-id=ABC* is executed first, but the validation fails at commit.



Important: Fujitsu recommends to prepend *z* and *YEAR-MONTH-DAY-HR* to the actual *xcon-id* when the user creates the fabric.

Command:

```
% set fabric och-connection z-2019-12-05-13-ADD-FROM-SHELF2-C2 src-if
```

A.3.2 SNMPv3

In case of SNMPv3, user provisioning is present in the system. The save configuration file does not contain the authentication and privacy passwords. The user enters the actual text password in the configuration file before executing the *load merge* or *load replace* command.

Example:

```
[-|#] <user>
  <username>authuser1</username>
  <authentication-protocol>md5</authentication-protocol>
  <authentication-password>admin12345</authentication-password>
  <privacy-protocol>des</privacy-protocol>
  <privacy-password>admin12345</privacy-password>
</user>
```

A.3.3 Fujitsu Optical Interfaces

If the user saves the configuration file in previous release, the user needs to delete *<band-capability></band-capability>* and *<amp-type></amp-type>* from the configuration data file before the execution of load configuration feature.

In this release both fields are moved to operation side.

Example:

```
<interface>
  <name>ots-1/0/0/E1</name>
  <type xmlns:ianaift="urn:ietf:params:xml:ns:yang:iana-if-
type">ianaift:opticalTransport</type>
  <enabled>true</enabled>
  <ots xmlns="urn:fujitsu:params:xml:ns:yang:interface:ot">
    <admin-status>up</admin-status>
    <fiber-type>smf</fiber-type>
    <degree-number>1</degree-number>
    <operation-mode>ope-mode-base</operation-mode>
    <amp-type>edfa</amp-type> #amplifier type need to be deleted
    <band-capability>lband</band-capability> #band capability need to be
deleted
  </ots>
</interface>
```

A.3.4 RollBack

The user cannot rollback to the state before the ILF installation (autoP of ILFs).

A.3.5 Load Configuration with Large Database

When load configuration with a large database is in progress and if the user executes `show | compare` before the load configuration commits to the database (a window of about 15 seconds), the user session displays *Error: internal error*. The user needs to retry the `show | compare` later.

A.3.6 LLDP

LLDP Disable Setting

To ensure LLDP settings are applied correctly, the *protocols* block needs to be placed after the *interfaces* block for load configuration.

For load configuration operation, LLDP *disable* setting must be in a specific order in configuration file. The LLDP *disable* should be after the supporting *eth* interface.

Example:

```
<interfaces xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces">
  <interface>
    ...
  </interface>
</interfaces>
<protocols xmlns="urn:fujitsu:params:xml:ns:yang:protocols">
  <protocol>
    <name>lldp</name>
    <type xmlns:lldp="urn:fujitsu:Params:xml:ns:yang:fujitsu-
lldp">lldp:fujitsuLldp</type>
    <lldp-instance xmlns="urn:fujitsu:Params:xml:ns:yang:fujitsu-lldp">
      <global-config>
        <adminStatus>enable</adminStatus>
        <msgTxInterval>30</msgTxInterval>
        <msgTxHoldMultiplier>4</msgTxHoldMultiplier>
      </global-config>
      <port>
        <ifName>eth-1/0/0/SC-E1</ifName>
        <adminStatus>disable</adminStatus>
      </port>
    </lldp-instance>
  </protocol>
</protocols>
```

Deleting protocols container to delete configured LLDP causes an error. In this release, this container is the only protocols container and does not allow it to be deleted. The workaround is to disable LLDP explicitly and leave the protocols container present.

Example:

```
<edit-config xmlns:lldp="urn:fujitsu:Params:xml:ns:yang:fujitsu-lldp"
  message-id="1">
  <target>
    <candidate/>
  </target>
  <config>
    <protocols xmlns="urn:fujitsu:params:xml:ns:yang:protocols">
      <protocol>
        <name>lldp</name>
        <lldp-instance xmlns="urn:fujitsu:Params:xml:ns:yang:fujitsu-lldp">
          <global-config>
            <adminStatus>disable</adminStatus>
          </global-config>
        </lldp-instance>
      </protocol>
    </protocols>
  </config>
</edit-config>
```

B

Frequency Wavelength

Table 59
Frequency Wavelength Table

Frequency (THz)	Wavelength (nm)
196.10	1528.77
196.05	1529.16
196.00	1529.55
195.95	1529.94
195.90	1530.33
195.85	1530.72
195.80	1531.12
195.75	1531.51
195.70	1531.90
195.65	1532.29
195.60	1532.68
195.55	1533.07
195.50	1533.47
195.45	1533.86
195.40	1534.25
195.35	1534.64
194.50	1541.35
195.25	1535.43
195.20	1535.82
195.15	1536.22

Table 59 (Cont.)
Frequency Wavelength Table

Frequency (THz)	Wavelength (nm)
195.10	1536.61
195.05	1537.00
195.00	1537.40
194.95	1537.79
194.90	1538.19
194.85	1538.58
194.80	1538.98
194.75	1539.37
194.70	1539.77
194.65	1540.16
194.60	1540.56
194.55	1540.95
194.50	1541.35
194.45	1541.75
194.40	1542.14
194.35	1542.54
194.30	1542.94
194.25	1543.33
194.20	1543.73
194.15	1544.13
194.10	1544.53
194.05	1544.92
194.00	1545.32
193.95	1545.72
193.90	1546.12
193.85	1546.52
193.80	1546.92
193.75	1547.32

Table 59 (Cont.)
Frequency Wavelength Table

Frequency (THz)	Wavelength (nm)
193.70	1547.72
193.65	1548.11
193.60	1548.51
193.55	1548.91
193.50	1549.32
193.45	1549.72
193.40	1550.12
193.35	1550.52
193.30	1550.92
193.25	1551.32
193.20	1551.72
193.15	1552.12
193.10	1552.52
193.05	1552.93
193.00	1553.33
192.95	1553.73
192.90	1554.13
192.85	1554.54
192.80	1554.94
192.75	1531.51
192.70	1555.75
192.65	1556.15
192.60	1556.55
192.55	1556.96
192.50	1557.36
192.45	1557.77
192.40	1558.17
192.35	1558.58

Table 59 (Cont.)
Frequency Wavelength Table

Frequency (THz)	Wavelength (nm)
192.30	1558.98
192.25	1559.39
192.20	1559.79
192.15	1560.20
192.10	1560.61
192.05	1561.01
192.00	1561.42
191.95	1561.83
191.90	1562.23
191.85	1562.64
191.80	1563.05
191.75	1563.45
191.70	1563.86
191.65	1564.27
191.60	1564.68
191.55	1565.09
191.50	1565.50
191.45	1565.90
191.40	1566.31
191.35	1566.72

C

Glossary Terms and Definitions

This chapter describes the product glossary terms and definitions.

2R3C	Two-Rate, Three-Color
AAA	Authentication, Authorization, and Accounting
ACE	Access Control Entry
ACL	Access Control List
ACO	Analog Coherent Optics
ACT	Active
ACTVST	Active Valid Signal Timer
AES	Advanced Encryption Standard
AGC	Automatic Gain Control
AINS	Automatic In-Service
AIS	Alarm Indication Signal
ALC	Automatic Line Control
ALS	Automatic Laser Shutdown
ANSI	American National Standards Institute
AO	Autonomous Output
AOC	Active Optical Cable
APR	Automatic Power Reduction
APS	Automatic Protection Switching
APSD	Automatic Power Shutdown
AVP	Attribute-Value Pair
B/W or BW	Bandwidth
B2B	Back-to-Back
BDI-O	Backward Detect Indication-Optical Layer
BIP	Bit Interleaved Parity

CAC	Connection Admission Control
CBN	Common Bonding Network
CBS	Committed Burst Size
CC	Continuity Check
CCINT	CCM Interval
CCM	Continuity Check Message
CD	Colorless, Directionless
CDC	Colorless, Directionless, Contentionless
CE	European Conformity (Conformité Européene)
CEP	Customer Edge Port
CFM	Connectivity Fault Management
CFP4	C Form-Factor Pluggable 4
CI	Communications Integrator
CIR	Committed Information Rate
CLI	Command Line Interface
CLRULE	Classification Rule
CM	Color Mode
CNP	Customer Network Port
CO	Central Offices
COE	Carrier-Oriented Ethernet
COM	Communication
CoS	Class of Service
CPE	Customer Premises Environment
CR	Critical
CSP	Critical Security Parameter
CVID	C-Tagged VLAN ID
C-VLAN	Customer Virtual Local Area Network
DAPI	Destination Access Point Identifier
DB	Database
DC-I	DC Isolated
DCN	Data Communications Network
DE	Drop Eligibility
DGD	Differential Group Delay

DHCP	Dynamic Host Configuration Protocol
DIP	Dynamic IP
DMM	Digital Multimeter
DNS	Domain Name Server
DOM	Domain
DP-QAM	Dual-Polarization Quadrature Amplitude Modulation
DP-QPSK	Dual-Polarization Quadrature Phase Shift Keying
DSCP	Differentiated Services Code Point
DST	Daylight Saving Time
EBS	Excess Burst Size
EBWPFL	Egress Bandwidth Profile
ECMP	Equal-Cost Multi-Path Routing
EHT	Hold-Off Timer
EIR	Excess Information Rate
EMC	Electromagnetic Compatibility
EOS	Ethernet Over SONET
EPL	Ethernet Private Line
ERP	Ethernet Ring Protection
ESD	Electrostatic Discharge
ETH	Ethernet Facility
HBER	High Bit Error Rate
IBWPFL	Ingress Bandwidth Profile
ICP	Internet Content Provider
LCN	Local Communication Network
LLDP	Link Layer Discovery Protocol
LMP	Local Management Port
MCN	Management Communication Network
MONTYPE	Monitor Type
NEM	NE Management
OAM	Operations, Administration, and Management
OCH	Optical Channel
OTDR	Optical Time-Domain Reflectometer
OTU	Optical Transmission Unit

P2P	Point-to-Point
PIU	Plug-In Unit
PM	Performance Monitoring
RF	Remote Fault
ROADM	Reconfigurable Optical Add/Drop Multiplexer
ROB	ROADM on a Blade
RPC	Remote Procedure Call
SDN	Software Defined Network
SNMP	Simple Network Management Protocol
SOAM	Service OAM
S-VLAN	Service Virtual Local Area Network
TM	Traffic Management
UNI	User Network Interface
vMCU	virtual Management and Control Unit
VNE	Virtual Network Element
VOA	Variable Optical Attenuator
WRED	Weighted Random Early Detection
WSS	Wavelength Selectable Switch
YANG	Yet Another Next Generation
ZTP	Zero Touch Provisioning

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